Point-Set Topology

— ***EYNTKA*** **Series** —

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August 22, 2026

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Introduction

A metric measures distance. From that notion comes most of what analysis defined so far: convergence, continuity, completeness, compactness, and so forth. The reader arriving here has spent at least a course on that machinery and has good reason to think distance is the starting point when generalizing to more abstract spaces.

It is not. Consider $\mathbb{R}^\omega$, the space of all real sequences, with the notion of coordinate-wise convergence (i.e. convergence at each coordinate). No metric presents itself, and the ones that can be manufactured are visibly artificial. Consider further the continuous functions on $[0, 1]$ with pointwise convergence, or the space of all functions $\mathbb{R} \to \mathbb{R}$; again the natural convergence is there and the distance is not. Worse, in such spaces sequences stop being adequate. There are sets whose closure contains points that no sequence from the set reaches, and maps that carry every convergent sequence to a convergent sequence without being continuous.

What survives, in every one of these examples, is the notion of a *neighbourhood*. One can still say when a point is surrounded by a "region", still say that a function does not tear things apart, still say that a set contains everything it approaches. Topology is the systematization of these concepts.

The axioms of a topology look arbitrary at first sight. The subject reached this point given the problem they solve. In the 1870s Cantor, studying where trigonometric series can fail to converge, was driven to classify the exceptional sets of points: he introduced limit points and the derived set, and found that the structure of a subset of $\mathbb{R}$ was itself a subject. Fréchet, in his 1906 thesis, extracted from this the first abstract framework in which convergence made sense without coordinates, and gave the axioms for what is now called a metric space. It was Hausdorff, in the *Grundzüge der Mengenlehre* of 1914, who took the step of dropping the metric and axiomatizing neighbourhoods directly. The open-set formulation used in this book settled over the following two decades, in the hands of Kuratowski, Sierpiński, Urysohn and Tychonoff, once it became clear that unions and finite intersections were the only closure conditions the theory needed[1].

That history explains the asymmetry in the axioms, which is otherwise the first thing a reader finds

[1] The word is older than any of this. It is from the Greek $\tau o \pi o \varsigma$ ('place, location') and $\lambda o \gamma o \varsigma$ ('study'), introduced by Listing in 1847 for a subject that then had nothing like the axioms above.

strange. Arbitrary unions of open sets are open, but only *finite* intersections are open. Without this restriction $\bigcap_n \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$ would be open, every singleton with it, and the topology on $\mathbb{R}$ would collapse to the power set, which distinguishes nothing. Infinite flexibility on one side and finite restriction on the other is what leaves a space with any shape at all.

Once spaces are defined without reference to distance, the question becomes which of their properties belong to the space itself. Recall from linear algebra that two vector spaces count as the same when an isomorphism joins them, and a property worth the name is one that any such isomorphism transports. Topology adopts the same discipline with its own notion of sameness. A *homeomorphism* is a bijection that is continuous in both directions, and spaces related by one are indistinguishable by any means the topology can express.

A property is then *topological* when every space homeomorphic to one having it has it too. Connectedness is topological, and this book proves it: a continuous image of a connected space is connected, so a homeomorphism carries connectedness both ways. Boundedness is not. The interval $(0, 1)$ is bounded and $\mathbb{R}$ is not, yet the two are homeomorphic, so boundedness belongs to the metric and not to the space[2]. Deciding which side of that line a given property falls on is the work of the next several chapters.

The familiar picture of a homeomorphism as a rubber-sheet deformation is worth holding lightly. It is a good guide to easy cases and it misleads in two. The first is that homeomorphism does not require the deformation to happen anywhere: the two circles below sit in $\mathbb{R}^3$ and are homeomorphic, each being a circle, yet neither can be moved to the other through $\mathbb{R}^3$ without cutting.

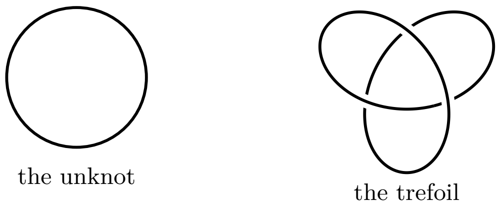

Figure 1: Two circles in $\mathbb{R}^3$. They are homeomorphic, since each is a circle, but no motion of $\mathbb{R}^3$ carries one to the other. Homeomorphism compares the spaces themselves, not the way they sit in an ambient space.

The second is gluing, which the rubber-sheet picture doesn't match: identifying the two endpoints of an interval produces a circle, and section 2.4 makes that construction precise.

There is a common slogan in the topology, that it is *flexible* geometry, and it is worth saying exactly how far it holds. Defining a "geometric space" is a harder task than it first seems[3], but a

[2] It is preserved by *isometries*, the maps that preserve distance exactly, which is a far stronger requirement than homeomorphism.

[3] No single definition covers the cases. Klein's Erlangen programme of 1872 proposed one: a geometry is a space with a transitive group of transformations, and the geometry *is* the study of what that group leaves invariant. It accounts for the classical geometries elegantly, Euclidean, affine, projective and hyperbolic differing only in the group. But it excludes too much. A Riemannian manifold typically has no symmetries at all, its isometry group being trivial, so Klein's criterion would deny it is a geometry [5]. Riemann's competing conception, from his lecture of 1854, puts the structure infinitesimally on a manifold instead and covers exactly those cases, at the price of presupposing the manifold; Cartan reconciled the two in the 1920s with spaces that are infinitesimally Kleinian. Each later widening of

good mostly accurate idea of a geometry is a topological space together with some rigid structure imposed on it. This rigidification structure cuts down the maps one is allowed to call symmetries, i.e. forcing "structure" (ex. distances are preserved, angles are preserved, curvature is preserved, etc.). A metric is the case already in hand: it turns a space into a geometric object, and its symmetries are not all the homeomorphisms, but only the "isometries" (maps preserving distance exactly: $d(f(x), f(y)) = d(x, y)$). Other structures behave the same way, each selecting its own smaller class of admissible maps[4]. In that direction the slogan is right: topology is what remains when no such structure is added, and it therefore admits the largest class of maps.

Where the slogan fails is in suggesting that every topological space is a geometry with its rigidity removed. Most are not: the co-finite topology on an uncountable set, or the two-point space $\{0, 1\}$ whose open sets are $\emptyset$, $\{1\}$ and the whole space, carry no metric, sit inside no manifold, and are not loosened versions of anything geometric (orbifold, scheme, (algebraic, Deligne, etc.) stacks, derived equivalents, etc.). So the layer this book studies is not geometry made flexible – not all spaces are "geometrizable". Overall, it is the layer on which geometry is imposed, and it is strictly larger than the collection of spaces that can carry any "geometry".

Chapter 1 sets up spaces, continuity, and homeomorphism, together with the vocabulary every later argument runs on: bases, closed sets, closure and limit points, the Hausdorff condition, and the countability axioms. Chapter 2 builds new spaces from old, by subspaces, products, limits over directed systems, and quotients, and then shows that all four are instances of two constructions, the coarsest topology making a family of maps continuous and the finest; metric spaces reappear here as one source of topologies among others, and a bonus section locates the notion of uniformity that sits between metric and topological.

Chapter 3 introduces the key topological invariances of the book. Connectedness and compactness are developed in full, the second through to Tychonoff's theorem, and nets and filters repair the failure of sequences that motivated the subject in the first place. Chapter 4 asks how well a space separates its points and closed sets, and finds that mild-looking hypotheses give a great deal: Urysohn's lemma manufactures continuous functions out of nothing but normality, and with second countability it makes a space metrizable, classifying exactly the spaces on which distance is definable. Chapter 5 carries this into the tool geometry runs on, the partition of unity, and ends with the largest compactification a space admits.

Chapter 6 is a prelude to *Algebraic Topology* [7]. Everything before it distinguishes spaces by properties; here a space is assigned a *group*, and the distinction becomes algebraic. Covering spaces do the computing, the circle gets its fundamental group, and Brouwer's fixed point theorem falls out as a consequence. Chapter 7 closes with spaces carrying a compatible group law, the objects on which the analysis and arithmetic parts of this series are built.

A reader who finishes will be able to take two topological spaces and decide, with proof, whether they are the same using the fundamental tools of topology. The reader will hold the point-set language in which functional analysis, measure theory, differential and algebraic topology, and the theory of Lie and profinite groups are all written.

"space" has broken the previous definition again. What survives is a family resemblance, not a definition, which is why what follows is offered as a serviceable idea rather than a criterion.

[4] A Riemannian metric does it for smooth manifolds [5], a structure group for bundles [9]; a symplectic form is another such choice. In each case the rigidifying object is added *on top of* an underlying topology, and the automorphisms shrink accordingly.

0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7⋆; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

1

Introduction to Topology

This chapter covers the definition of a topology, with examples, how to compare topologies, and the notions that arise as soon as the definition is in place: closure, interior, countability, and the rest.

1.1 Basic Definitions and Constructions

The definition of a topology that eventually settled looks, at first sight, very far from any geometric notion. The abstraction went in the direction of giving an object a notion of *localness*. Localness is not distance: the two merge in well-behaved spaces, as metric spaces will show in section 2.6, but they come apart in general, and the finite-complement topology below is the first place they do. What localness records is how a space behaves on its different parts, so that properties of the whole can be recovered from properties of the pieces. Using local information to reach global information is one of the recurring moves in mathematics, and topology is the setting in which it is made precise.

Definition 1.1.1: Topological Space

Given a set X, a *topological space* is a pair $(X, \mathcal{T}_X)$ where $\mathcal{T}_X \subseteq \mathcal{P}(X)$ is a set of subsets of X such that:

1. $\emptyset$ and X are in $\mathcal{T}_X$
2. $\mathcal{T}_X$ is closed under arbitrary unions
3. $\mathcal{T}_X$ is closed under finite intersections

An element of $\mathcal{T}_X$ is called *open*. A set $S \subseteq X$ is *closed* if its complement is in $\mathcal{T}_X$, that is if $S^c \in \mathcal{T}_X$.

So $Y \subseteq X$ is open exactly when $Y \in \mathcal{T}_X$. Better names for open and closed might have been "local" and "co-local", which reflect the intuition more faithfully in the stranger topologies; but the vocabulary inherited from the Euclidean case is fixed, and there is no point relitigating it now.

Example 1.1.2: Topological Spaces

1. For *any* set X, the collection $\mathcal{T}_X = \{\emptyset, X\}$ is a topology, called the *trivial* or *indiscrete topology*. In terms of localness every point sits in the same single neighbourhood, so no *chunk* or *sub-part* of the object can be isolated: only the whole is ever visible, and the points have no local properties beyond belonging to it.

2. For any set X, the full power set $\mathcal{T}_X = \mathcal{P}(X)$ is a topology, the *discrete topology*. Here localness is as sharp as it can be, the smallest local level being the individual points. Since unions of points are again open, every subset is open. This is the opposite extreme: every level of detail is resolved at the same time.

3. For any set X, take $\mathcal{T}_X = \{Y \subseteq X \big| Y = \emptyset \text{ or } |X \setminus Y| < \infty\}$. This is the *co-finite* or *finite-complement topology*.

   1.1.3 Note Why not simply call U open when it is finite? Because an arbitrary union of finite sets need not be finite. Passing to complements repairs this: unions *shrink* the complement $(X \setminus (U_1 \cup U_2))$, keeping it finite, while intersections are restricted to finitely many, which is exactly what the axioms require.

4. (For a reader who has done some analysis.) Take $n \geq 1$ and $X = \mathbb{R}^n$, with

$$\mathcal{T}_X = \{\text{Euclidean open sets in } \mathbb{R}^n\}$$

   written $(\mathbb{R}^n)_{std}$, $(\mathbb{R}^n)_{euc}$, or simply $\mathbb{R}^n$ where the abuse of notation is harmless.

   1.1.4 Note In analysis this topology is normally defined through open balls. There is no metric here to supply one. A metric on $\mathbb{R}^n$ *can* be defined, and it does induce this topology, but that comes later in section 2.6. For now the topology is just a collection of sets.

5. The set $\mathbb{C}$ with all open discs, exactly as for $\mathbb{R}^2$. The two are topologically equivalent, which is the precise sense in which they are the same geometric object: every topological property of $\mathbb{R}^2$ is a property of $\mathbb{C}$.[a] What separates them is *algebraic*: $\mathbb{C}$ carries a multiplication that $\mathbb{R}^2$ does not. Definition 1.2.13 makes the sense of "topologically equivalent" precise.

6. Let X be any set and fix $x_0 \in X$. Then $\mathcal{T}_X = \{Y \subseteq X \big| Y = \emptyset \text{ or } x_0 \in Y\}$ is a topology, the *particular point topology*.

7. Let X carry a total order $<$, so that for $a \neq b$ either $a < b$ or $b < a$, and $a < b < c$ implies $a < c$. Writing

   (a) $(a, b) \coloneqq \{c \in X : a < c, c < b\}$

   (b) $(-\infty, a) \coloneqq \{c \in X : c < a\}$

(c) $(a, \infty) \coloneqq \{c \in X : a < c\}$

the unions of such sets, together with X itself, form a topology: the *order topology*. It is treated properly in definition 1.3.13, once bases are available, and it supplies several of the examples that separate the properties studied later.

8. The topologies on a three-point set, in full:

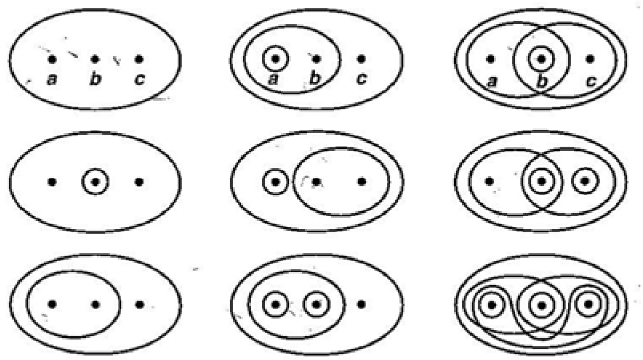

Figure 1.1: All topologies on a three-element set.

[a]The two do come apart once more structure is added. The differential structures available on $\mathbb{C}$ give it a geometry different from that of $\mathbb{R}^2$, which is the subject of Riemann surfaces and of symplectic geometry.

1.1.5 Note One technical point. An arbitrary intersection of open sets *may* happen to be open; what the axioms withhold is the guarantee. For instance $\bigcap_{i\in\mathbb{N}}(-i, 1) = (-1, 1)$, which is open in the Euclidean topology. That example is a little contrived, but genuine ones appear with *product spaces* in section 2.2.

To make the notion of a local property concrete, take $\mathbb{R}$ with the standard topology and $x \in \mathbb{R}$. The set $(x-1, x+1)$ is open and contains x, and the usual phrasing is that $(x-1, x+1)$ *is a neighbourhood of* x. Neighbourhoods are not unique: the point x has uncountably many, one for each $\epsilon \in \mathbb{R}_{>0}$ in the family $(x-\epsilon, x+\epsilon)$. Open neighbourhoods are the starting point for everything a topology does. This is not to say the points are irrelevant, since the underlying set constrains what topologies are available, but what distinguishes one topology from another is its open sets.

Example 1.1.6: Open Neighbourhoods on a Sphere

Consider a sphere and some topologies on it. The trivial topology permits only the entire sphere to be seen: not points, not subsets, only the object that results from all of them. The discrete topology goes to the other extreme, treating every subset as a local property. These are the two ends of a scale of precision, and every other topology on the sphere sits somewhere between them, each corresponding to a different way of looking locally.

For a topology strictly between the two, embed the sphere in $\mathbb{R}^3$ centred at the origin and take the slices cut by the xy-plane as it travels up the z-axis, together with all unions and finite intersections of them. A different choice intersects every open set of $\mathbb{R}^3$ with the sphere, and gives the topology the sphere inherits as a subspace.

Notice the asymmetry in the axioms: open sets survive arbitrary unions, while closed sets survive only finite ones. Allowing both operations to be arbitrary collapses the theory. Take the open sets $\left(\frac{-1}{n}, \frac{1}{n}\right)$ in Euclidean $\mathbb{R}$. Their intersection over all $n \in \mathbb{N}$ is $\{0\}$, and the same construction at any other point makes every singleton open. Arbitrary unions of singletons then give every subset, so the topology collapses to $\mathcal{P}(\mathbb{R})$. Topologies that were distinguishable by their local behaviour would become identical. Restricting intersections to finitely many is what removes that wiggle room, and it is the first illustration of a pattern worth watching: admitting infinite operations gives flexibility, and past a certain point the flexibility destroys the structure it was meant to describe.

There is a subject built on exactly the opposite trade. In *measure theory* a σ-algebra is closed under *countable* unions and intersections alike, the countability restriction playing the role that finiteness plays here. Those spaces are the foundation of real analysis and of the Lebesgue integral.

With topologies in hand, the most basic relation between two of them is whether one contains the other.

Definition 1.1.7: Finer And Coarser

Let X be a set, and $(X, \mathcal{T}_1), (X, \mathcal{T}_2)$ be two topologies. Then

1. $(X, \mathcal{T}_1)$ is *finer* than $(X, \mathcal{T}_2)$ if $\mathcal{T}_1 \supseteq \mathcal{T}_2$
2. $(X, \mathcal{T}_1)$ is *coarser* than $(X, \mathcal{T}_2)$ if $\mathcal{T}_1 \subseteq \mathcal{T}_2$

Example 1.1.8: Finer and Coarser Topologies

1. The trivial topology is coarser than every other topology on the same set.
2. The discrete topology is finer than every other topology on the same set.
3. Take $\mathbb{R}$ generated by open balls and $\mathbb{R}$ generated by open squares. Each is finer *and* coarser than the other: they contain one another, and so define the same localness.

Two topologies need not be *comparable* at all: neither need be finer nor coarser than the other.

Example 1.1.9: Incomparable Topologies

Take $X = \{a, b, c\}$, and $\mathcal{T}_1 = \{\{a\}, \{b, c\}, X, \emptyset\}$, as well as $\mathcal{T}_2 = \{\{a, b\}, \{c\}, \emptyset, X\}$. Both are topologies, and $\mathcal{T}_1 \nsubseteq \mathcal{T}_2$ while $\mathcal{T}_1 \nsupseteq \mathcal{T}_2$.

The next example takes longer to absorb and repays the time: the two topologies it introduces reappear throughout the book as counterexamples.

Example 1.1.10: The Lower-Limit and K-Topologies

Take $\mathbb{R}$ as the underlying set and let the topology be generated by the intervals $[a, b)$; generation is made precise in section 1.3, and for now it means taking all such intervals and all their unions. This is the *lower-limit topology*, written $\mathbb{R}_l$. It is *strictly finer* than the standard topology: every open interval (a, b) is the union $\bigcup_n [a + \frac{1}{n}, b)$, so the standard topology is contained in it, while $[0, 1)$ is not a union of open intervals, since any open interval containing 0 also contains points below 0.

Now take all open intervals $A = \{(a,b) | a < b\}$, and in addition set $K = \{\frac{1}{n} | n \in \mathbb{N}\}$ and adjoin all sets of the form $B = (a,b) \setminus K$. Together $A \cup B$ generates the *K-topology*, denoted $\mathbb{R}_K$. It contains A, hence all of $\mathcal{T}_{\mathbb{R}}$, and also sets such as $(-1,1) \setminus K$. That set is not open in the standard topology: around the point 0 every standard open set contains points of K, so no open interval about 0 fits inside it. So $\mathbb{R}_K$ is strictly finer than $\mathbb{R}_{\text{euc}}$.

Although $\mathcal{T}_l$ and $\mathcal{T}_K$ are both finer than $\mathcal{T}_{euc}$, they are *incomparable*. The set $[0,1)$ lies in $\mathcal{T}_l$ but not in $\mathcal{T}_K$, by the argument already given for the standard topology, since the sets $(a,b) \setminus K$ only remove points and never supply a left endpoint. Conversely $(-1,1) \setminus K \in \mathcal{T}_K$ but $(-1,1) \setminus K \notin \mathcal{T}_l$: a basic set $[a,b)$ containing 0 contains an interval to the right of 0, and every such interval meets K. Had the half-open intervals been closed on the *right* instead, this second obstruction would not arise.

This is what it means for topologies to carry different local structure. Not every topology defines the same notion of localness, and two of them can define notions that cannot be compared at all.

The last construction of this section builds a new space out of a subset of an old one. It is placed here, ahead of continuity, because the very first rules for building continuous maps refer to it; section 2.1 returns to it with the tools of the intervening sections.

Definition 1.1.11: Subspace Topology

Let X be a topological space and Y a subset of X. The *subspace topology* on Y is

$$\mathcal{T}_Y = \{U \cap Y | U \text{ is open in } X\}$$

1.1.12 Note Y itself need not be open in X. When it is, a set open in Y is also open in X, since it is then an intersection of two open sets; without that hypothesis the two notions of openness genuinely differ. The interval $[0,1]$ is open in itself as a subspace of $\mathbb{R}$, but not open in $\mathbb{R}$.

Example 1.1.13: Subspaces

1. $\mathbb{Z} \subseteq \mathbb{R}$ inherits the discrete topology, since $(n - \frac{1}{2}, n + \frac{1}{2}) \cap \mathbb{Z} = \{n\}$.
2. $[0,1] \subseteq \mathbb{R}$ inherits a topology whose basic open sets include half-open intervals such as $[0, \frac{1}{2})$, which arise as $(-1, \frac{1}{2}) \cap [0,1]$.
3. $\mathbb{Q} \subseteq \mathbb{R}$ inherits a topology in which the set $\{q \in \mathbb{Q} \mid q^2 < 2\}$ is clopen, since its boundary points in $\mathbb{R}$ are irrational.

Exercise 1.1.1

1. Let $X = \{a,b,c\}$. Decide which of the following are topologies on X, and for those that fail, name the axiom violated: $\mathcal{T}_1 = \{\emptyset, \{a\}, \{a,b\}, X\}$; $\mathcal{T}_2 = \{\emptyset, \{a\}, \{b\}, X\}$; $\mathcal{T}_3 = \{\emptyset, \{a,b\}, \{b,c\}, \{b\}, X\}$.

2. Let $\mathcal{T}$ and $\mathcal{T}'$ be topologies on the same set X. Show that $\mathcal{T} \cap \mathcal{T}'$ is a topology, and give an example on a three-point set showing that $\mathcal{T} \cup \mathcal{T}'$ need not be.

3. Verify that the co-finite topology of example 1.1.2 really is a topology on any set X, and determine for which X it coincides with the discrete topology.

4. Let X be uncountable and let $\mathcal{T} = \{\emptyset\} \cup \{U \subseteq X \mid X \setminus U \text{ is countable}\}$, the *co-countable* topology. Show that $\mathcal{T}$ is a topology, and that it is strictly coarser than the discrete topology and strictly finer than the co-finite topology.

5. Let $Y \subseteq X$ carry the subspace topology of definition 1.1.11. Show that if $Z \subseteq Y$, the topology Z inherits from Y agrees with the one it inherits from X.

1.2 Continuity

Continuity began as the promise that a curve could be drawn without lifting pen from paper. It was made precise by the epsilon-delta definition, which extends to maps $f \colon \mathbb{R}^n \to \mathbb{R}^m$. Neither formulation survives the passage to spaces with no notion of distance, so a more abstract definition is needed.

The algebraic analogy is a good one: continuous maps are to topology what homomorphisms are to algebra. A homomorphism guarantees that the target retains some of the structure of the source. The same holds here. A continuous map between spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ says that $\mathcal{T}_Y$ carries over some of the "localness" recorded by $\mathcal{T}_X$.

Definition 1.2.1: Continuous Functions

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. Then $f : X \to Y$ is said to be *continuous* if

$$\forall U \in \mathcal{T}_Y,\ f^{-1}(U) \in \mathcal{T}_X$$

1.2.2 Note Functions still map *points* of the underlying set, not open sets. The definition is phrased in terms of sets, but the points have not gone anywhere, and problems are usually solved by chasing them.

Example 1.2.3: Continuous Functions

1. The familiar functions of a first calculus course: $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$, or $f(x) = \sin(x)$, and so on. Note that $f(x) = \log(x)$ is continuous only once the domain is restricted to the positive reals.[a]

2. Take the domain and co-domain both to be $\mathbb{R} \times \mathbb{R}$, but with different topologies. In the domain the open sets are generated by $(a, b) \times \mathbb{R}$, so the first coordinate is Euclidean and the second is the whole line; in the co-domain the roles are reversed, with sets $\mathbb{R} \times (a, b)$. These two topologies are *incomparable*. Nevertheless the map

$$f(x, y) = (0, x)$$

is continuous. Taking any basic open set $\mathbb{R} \times (a, b)$ of the co-domain,

$$f^{-1}\big(\mathbb{R} \times (a, b)\big) = (a, b) \times \mathbb{R}$$

which is open in the domain. Since the basic open set was arbitrary, every open set has open pre-image.

3. A constant map $f(x) = y_0$ is *always* continuous. If V is open in Y with $y_0 \in V$ then $f^{-1}(V) = X$; if $y_0 \notin V$ then $f^{-1}(V) = \emptyset$. Both are open.

[a] Or, in the complex setting, on $\mathbb{C} \setminus \{0\}$.

Why preimages rather than images? The choice is between preserving the whole topology of the *domain* while missing sets in the co-domain, and preserving the whole topology of the *co-domain* while missing sets in the domain. The second is more useful. Working with a continuous function, every open set of the co-domain has a counterpart in the domain, namely its preimage, and no care is needed about whether some $U \subseteq X$ satisfies $f(U) = V$. The asymmetry is structural: $f^{-1}(Y) = X$ always holds, whereas $f(X) = Y$ need not, since continuous maps need not be surjective. Since the usual aim is to learn about a space by mapping into it, the co-domain is the side worth privileging.

Maps whose images of open sets are open deserve a name of their own.

Definition 1.2.4: Open Maps

Let $f : X \to Y$ be a map such that if $U \subseteq X$ is an open set, then $f(U) \subseteq Y$ is an open set. Then f is an *open map*.

The two conditions are independent, and it is worth seeing both failures.

Example 1.2.5: Continuous but not open

Take

$$f : \mathbb{R} \to \mathbb{R} \qquad f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

For a basic open (a, b): if $a > 0$ then $f^{-1}((a, b)) = (a, b)$, and if $a \leq 0$ then $f^{-1}((a, b)) = (-\infty, b)$. Both are open, the second because $(-\infty, b) = \bigcup_{n \in \mathbb{N}}(-n, b)$, so f is continuous. But $(-1, 1)$ maps onto $[0, 1)$, which is *not* open in $\mathbb{R}$.

Example 1.2.6: Open, but not Continuous

Let $f : \mathbb{R} \to \mathbb{R}$ be the identity, with the Euclidean topology on the domain and the discrete topology on the co-domain. Every image is open, since every subset of the co-domain is open. But the pre-image of $\{0\}$ is $\{0\}$, which is not open in the Euclidean topology.

Open maps matter because they control the continuity of the inverse.

Proposition 1.2.7: An Open Continuous Bijection Has Continuous Inverse

Let $f : X \to Y$ be a continuous bijection. Then f^{-1} is continuous if and only if f is an open map.

***Proof*:**
For a bijection the preimage under f^{-1} of a set $U \subseteq X$ is its image under f:

$$(f^{-1})^{-1}(U) = \left\{y \in Y \;\middle|\; f^{-1}(y) \in U\right\} = f(U)$$

So f^{-1} is continuous exactly when $f(U)$ is open for every open U, which is the definition of an open map (definition 1.2.4).

Continuity is also stable under moving either topology in the right direction.

Proposition 1.2.8: Refining the Domain and Coarsening the Co-Domain

Let $f : X \to Y$ be continuous. Then f remains continuous if X is given a finer topology, or Y a coarser one.

***Proof*:**
Let $\mathcal{T}'_X \supseteq \mathcal{T}_X$. For $V \in \mathcal{T}_Y$ the set $f^{-1}(V)$ lies in $\mathcal{T}_X$ by continuity, hence in $\mathcal{T}'_X$; so f is continuous for the finer domain topology.

Let $\mathcal{T}'_Y \subseteq \mathcal{T}_Y$. Every $V \in \mathcal{T}'_Y$ is also in $\mathcal{T}_Y$, so $f^{-1}(V)$ is open; the coarser co-domain topology simply asks fewer sets to be checked. In both cases the defining condition still holds, as we sought to show.

This does not make the domain and co-domain comparable to each other. They need not be.

Example 1.2.9: Continuous Function Between *Incomparable* Topologies

Take

$$T_1 = \{\emptyset, [a,b], [a,b]^c, \mathbb{R}\}, \qquad T_2 = \{\text{Euclidean topology}\}$$

and the function with $f([a,b]) = 0$ and $f([a,b]^c) = 1$. It is continuous. For any open $U \in T_2$ there are four cases:

1. if $0, 1 \notin U$, then $f^{-1}(U) = \emptyset$
2. if $0 \in U$ but $1 \notin U$, then $f^{-1}(U) = [a,b]$
3. if $1 \in U$ but $0 \notin U$, then $f^{-1}(U) = [a,b]^c$
4. if $0, 1 \in U$, then $f^{-1}(U) = \mathbb{R}$

All four are open in T_1, so f is continuous even though T_1 and T_2 are not comparable. What decides continuity is the function, not a relation between the two topologies.

The reason the localness intuition feels strained here is that f is very coarse: it collapses entire regions of the domain onto two points, and those two points sit inside uncountably many open sets of the co-domain. This is the topological version of a familiar algebraic phenomenon, where a homomorphism relates structures that have little else in common.

Constructing a nontrivial continuous map between two strange spaces raises the question of what is possible at all. The answer depends on the topologies, and the following example shows how tightly they can constrain matters.

Example 1.2.10: Limits of Continuous Functions Between Spaces
Take
$$\mathcal{T}_2 = \{\emptyset, (-\infty, 2) \cup (2, \infty), (-1, 1), (-\infty, 2) \cup (2, \infty) \cup (-1, 1), \mathbb{R}\}$$
and $f : \mathbb{R} \to \mathbb{R}$ with $\mathcal{T}_2$ on the domain, the Euclidean topology on the co-domain, and f not constant, a constant map being continuous for trivial reasons. Being non-constant, f takes at least two values, and every open set containing one of them must have open preimage. Suppose $f((-1, 1)) = 0$, and suppose the remaining points go to 1. The question is which set B can be sent to 1.

An open V containing 1 must have open preimage; if it contains both 0 and 1 the preimage is a union of open sets and so causes no trouble. Had some $x \in (-1, 1)$ been sent elsewhere, say to -1, the preimage of a small open set around 0 would fail to be open. So the difficulty is concentrated in a single point: where can 1.5 go? Sending it to 0 or to 1 puts it in the preimage of one of the two sets above, which is then not open. Sending it to a fresh value, say -2, is no better: the Euclidean open set $(-3, 3)$ would then have preimage $\{-2\} \cup (-1, 1)$, which is not in $\mathcal{T}_2$. There is nowhere for 1.5 to go, and no non-constant continuous map exists.

The obstruction disappears once the open sets of the domain cover $\mathbb{R}$, where covering excludes the use of $\mathbb{R}$ itself, since otherwise every topology covers trivially.[a] Replacing $(-1, 1)$ by $[-2, 2]$ in the topology above and sending it to 0, with the complement going to 1, does give a continuous function.

[a]This is unrelated to open covers and to covering maps, both of which appear later.

A further limitation survives even then: a continuous map out of a space with a finite topology has finite image. If an open set of the domain were split, part going to one point and part to another, an open set containing just one of those points would have a preimage that is not open. So what a continuous map can do depends heavily on the topology the domain carries. The trivial topology always admits the constant maps, but richer targets may be out of reach.

The discussion has emphasised the domain, but both sides matter. The co-domain supplies the sets whose preimages must be open, and so governs how finely the domain must be resolved. That dependence runs through the function itself: a finite topology on the domain forces a finite image, which remains possible even when the two topologies are incomparable.

The opposite arrangement is also instructive. Give the domain the discrete topology; then every preimage is open, and every function out of it is continuous whatever the co-domain. Give the *co-domain* the discrete topology instead and the situation inverts: there are now so many open sets to pull back that few functions survive.

That second case is worth a moment, because it picks out the maps that work universally. A function continuous into the discrete topology stays continuous into any coarser topology on the same set, and every topology is coarser than the discrete one. So these are exactly the functions continuous for *every* topology on the co-domain. This is an instance of a *universal property* in Top, the category of topological spaces.

With the definition in place, there are standard ways of building continuous functions, used constantly in what follows.

Proposition 1.2.11: Rules for constructing continuous functions

The following all produce continuous maps. Items (2), (4) and (5) refer to a *subspace*, meaning a subset carrying the subspace topology of definition 1.1.11.

1. *Constant function:* if $f : X \to Y$ maps all of X to a single point $y_0 \in Y$, then f is continuous.
2. *Inclusion:* if A is a subspace of X, the inclusion $i : A \to X$ is continuous.
3. *Composition:* if $f : X \to Y$ and $g : Y \to Z$ are continuous, then $g \circ f : X \to Z$ is continuous.
4. *Restricting the domain:* if $f : X \to Y$ is continuous and A is a subspace of X, then $f|_A : A \to Y$ is continuous.
5. *Restricting or expanding the range:* let $f : X \to Y$ be continuous. If Z is a subspace of Y containing $f(X)$, the map $g : X \to Z$ obtained by restricting the range is continuous. If Z has Y as a subspace, the map $h : X \to Z$ obtained by expanding the range is continuous.
6. *Local formulation of continuity:* $f : X \to Y$ is continuous if X is a union of open sets U_α with $f|_{U_\alpha}$ continuous for each α.

***Proof*:**

1. For V open in Y, $f^{-1}(V)$ is X if $y_0 \in V$ and $\emptyset$ otherwise; both are open.
2. For V open in X, $i^{-1}(V) = V \cap A$, which is open in A by the definition of the subspace topology.
3. For W open in Z, $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$, and $g^{-1}(W)$ is open by continuity of g, so its preimage under f is open by continuity of f.
4. $f|_A = f \circ i$ is a composite of continuous maps by the previous two items.
5. For V open in the codomain, the preimage under the restricted map is the preimage under f intersected with the relevant subspace, which is open in that subspace; when the

codomain is enlarged or cut down to a subspace containing the image, preimages of relative open sets agree with preimages of open sets of the ambient space.

6. If X is covered by open sets on each of which f restricts to a continuous map, then for V open the set $f^{-1}(V)$ meets each member of the cover in a set open in that member, hence open in X, and $f^{-1}(V)$ is the union of these.

Each item follows directly from the definition of continuity, as we sought to show.

1.2.12 Note Restricting a continuous map to a subspace is continuous, but the converse traffic does not flow: a map that is continuous on X need not induce a continuous map *out of* a subspace into some third space.[a] The diagram to keep in mind is

$$\begin{tikzcd} A \arrow[r, "i"] \arrow[rr, bend right, "f"'] & X \arrow[r, "f"] & Y \end{tikzcd}$$

where f is continuous from X to Y and i is the inclusion.

[a]A worked instance is at this discussion.

Requiring that open sets map to open sets as well as pull back to open sets makes the two spaces almost indistinguishable. Such a map is called *bi-continuous*. Adding bijectivity closes the gap completely: the open sets of the domain then correspond exactly to those of the co-domain. A bi-continuous bijection is a *homeomorphism*.

Definition 1.2.13: Homeomorphism

If $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ are topological spaces, a map $f : X \to Y$ is a *homeomorphism* if

1. f is bijective
2. $f^{-1}(\mathcal{T}_Y) = \mathcal{T}_X$, that is, f is continuous in both directions

$(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ are *homeomorphic* if a homeomorphism between them exists.

A homeomorphism is the strongest relation two topological spaces can bear. It is the isomorphism of the subject.[1] Every topological property proved in this book is invariant under homeomorphism, which is what makes such properties worth proving: a property held by one space is held by every space homeomorphic to it.

Example 1.2.14: Homeomorphism

$\mathbb{R}^2$ and $\mathbb{C}$ are topologically the same, by the map

$$\mathbb{R}^2 \to \mathbb{C} \qquad (x, y) \mapsto x + iy$$

If $f : X \to Y$ is a homeomorphism then so is $f^{-1} : Y \to X$. Informally, X and Y are homeomorphic when there is a bijection between the underlying sets that induces a bijection between the topologies, which is the content of the next corollary.

[1]Literally so: homeomorphisms *are* the isomorphisms of the category Top.

Corollary 1.2.15: A Homeomorphism Matches The Two Topologies

If $f\colon X \to Y$ is a homeomorphism, then $U \mapsto f(U)$ is a bijection from $\mathcal{T}_X$ onto $\mathcal{T}_Y$, with inverse $V \mapsto f^{-1}(V)$.

***Proof*:**
Condition (2) of definition 1.2.13 reads $f^{-1}(\mathcal{T}_Y) = \mathcal{T}_X$. The inclusion $\subseteq$ is continuity, giving $f^{-1}(V)$ open for V open. The inclusion $\supseteq$ says every open $U \subseteq X$ is $f^{-1}(V)$ for some open V, and then $f(U) = V$ by bijectivity, so f is an open map. Bijectivity of f gives $f^{-1}(f(U)) = U$ and $f(f^{-1}(V)) = V$, so the two assignments are mutually inverse maps between $\mathcal{T}_X$ and $\mathcal{T}_Y$, as we sought to show.

A slightly weaker notion is used constantly: a map that is a homeomorphism *onto its image*.

Definition 1.2.16: Embedding

Let X, Y be topological spaces. If $f : X \to f(X)$ is a homeomorphism, where $f(X)$ carries the subspace topology, then f is an *embedding*.

Exercise 1.2.1

1. Describe all continuous maps out of a discrete space, and all continuous maps into a space carrying the trivial topology.
2. Show that the identity $\mathbb{R} \to \mathbb{R}_l$ is not continuous while $\mathbb{R}_l \to \mathbb{R}$ is, where $\mathbb{R}_l$ is the lower-limit topology of example 1.1.10.
3. Is a composite of open maps open?
4. Must an open map be continuous, or a continuous map open? Settle both with maps between two-point spaces.

1.3 Generating Topologies (Basis)

The topology on $\mathbb{R}$ is rarely described by listing its members: a set such as $(1, 2) \cup (10, 11)$ is open, but it is not what comes to mind. What comes to mind is the collection of open intervals, and the rest of the topology is assembled from those by taking unions. That smaller collection is a *basis*, and the pattern is general: a well-chosen subfamily of open sets *generates* the whole topology.

Definition 1.3.1: Basis Of Topology

Let X be a set. A set $\beta \subseteq \mathcal{P}(X)$ is a basis if

1. $\forall x \in X$, $\exists B_0 \in \beta$ such that $x \in B_0$
2. $\forall B_1, B_2 \in \beta$ and $x \in B_1 \cap B_2$, there exists $B_3 \in \beta$ such that $x \in B_3 \subseteq B_1 \cap B_2$

1.3.2 Note This is not the notion of basis from linear algebra, where the representation of a vector in terms of the basis is *unique*. Here no uniqueness is claimed or wanted: a point typically lies in many basis elements, and a given open set is typically a union of them in many ways.

The two conditions are exactly what is needed to build a topology from β.

Lemma 1.3.3: Basis Generating A Topology

Let β be a basis. Then

$$\mathcal{T}_\beta \overset{\text{def}}{=} \left\{U \subseteq X \middle| \forall x \in U, \exists B_0 \in \beta \text{ such that } x \in B_0 \subseteq U\right\}$$

is a topology on X.

***Proof*:**
The set $\emptyset$ lies in $\mathcal{T}_\beta$ vacuously, and $X \in \mathcal{T}_\beta$ by the first basis condition, which supplies for each $x \in X$ a basis element containing it.

For arbitrary unions, let $U = \bigcup_\alpha U_\alpha$ with each $U_\alpha \in \mathcal{T}_\beta$, and take $x \in U$. Then $x \in U_\alpha$ for some α, so there is a basis element B_x with $x \in B_x \subseteq U_\alpha \subseteq U$. Since x was arbitrary, $U \in \mathcal{T}_\beta$.

For finite intersections, first take two sets $U_1, U_2 \in \mathcal{T}_\beta$ and some $x \in U_1 \cap U_2$. There are basis elements with $x \in B_1 \subseteq U_1$ and $x \in B_2 \subseteq U_2$, so $x \in B_1 \cap B_2$, and the second basis condition supplies B_3 with $x \in B_3 \subseteq B_1 \cap B_2 \subseteq U_1 \cap U_2$. Hence $U_1 \cap U_2 \in \mathcal{T}_\beta$.

The general case follows by induction on n. The case $n = 1$ is trivial; assuming $U_1 \cap \cdots \cap U_{n-1} \in \mathcal{T}_\beta$, intersecting with U_n is the two-set case just treated, giving $U_1 \cap \cdots \cap U_n \in \mathcal{T}_\beta$, as we sought to show.

Since this is *the* construction of a topology from a basis, it gets a name.

Definition 1.3.4: Topology Generated With Respect To β

Let β be a basis. Then

$$\mathcal{T}_\beta \overset{\text{def}}{=} \left\{U \subseteq X \middle| \forall x \in U, \exists B_0 \in \beta \text{ such that } x \in B_0 \subseteq U\right\}$$

is the topology generated by β

1.3.5 Note *Every* topology has a basis, since a topology is a basis for itself: both conditions of definition 1.3.1 hold trivially when $\beta = \mathcal{T}_X$. The content of the definition is therefore not existence but economy, in finding a small β that generates a large $\mathcal{T}_X$.

The working picture is that a basis is a collection whose unions are exactly the topology wanted.

Example 1.3.6: Topologies Generated by Basis

1. $\beta = \{\{x\} \,|\, x \in X\}$ generates the discrete topology
2. $\beta = \{X\}$ generates the trivial topology
3. $\beta = \{U \times V \mid U, V \subseteq \mathbb{R},\ \text{open}\}$ generates the standard topology on $\mathbb{R}^2$. More generally,

$$\beta_n = \{U_1 \times \cdots \times U_n \mid \text{open } U_i \subseteq \mathbb{R}, 1 \leq i \leq n\}$$

generates the standard topology on $\mathbb{R}^n$; the verification is lemma 1.3.9 applied in both directions, against the basis of open balls

$$\beta'_n = \{B_\epsilon(a) \mid \epsilon > 0,\ a \in \mathbb{R}^n\}$$

Each ball is a union of open boxes and each box a union of balls, so the two bases generate the same topology. This is the first instance of a space carrying *different* bases; in general there is no minimal one.

Why insist that a basis be closed under the second condition rather than asking for a collection whose unions *and* intersections give the topology? Both are usable, and the difference is only how much structure is assumed up front. A basis already accommodates intersections, so theorems proved from it need to check less; a sub-basis assumes nothing, which often makes a topology easier to construct in the first place.

Definition 1.3.7: Sub-basis

Let X be a set. A *sub-basis* for a topology on X is any collection $\mathcal{S} \subseteq \mathcal{P}(X)$. The collection of all finite intersections of members of $\mathcal{S}$, with the empty intersection taken to be X itself,

$$\mathcal{B} = \{S_1 \cap \cdots \cap S_n \mid n \geq 0,\ S_1, \ldots, S_n \in \mathcal{S}\}$$

is then a basis. The topology having $\mathcal{B}$ as a basis is the topology *generated by* $\mathcal{S}$; it is the coarsest topology on X in which every member of $\mathcal{S}$ is open.

Every basis is in particular a sub-basis, and generating a topology from a sub-basis differs only by the intermediate step of forming finite intersections.

1.3.1 Simplification With Basis

Lemma 1.3.8: Smallest Topology Containing β

Let β be a basis. Then $\mathcal{T}_\beta$ is the *smallest* topology on X in which every element of β is open.

***Proof*:**
Every $B \in \beta$ lies in $\mathcal{T}_\beta$, since each of its points x satisfies $x \in B \subseteq B$. So $\mathcal{T}_\beta$ is a topology in which the members of β are open.

Let $\mathcal{T}$ be any other such topology and let $U \in \mathcal{T}_\beta$. For each $x \in U$ there is $B_x \in \beta$ with $x \in B_x \subseteq U$, so $U = \bigcup_{x \in U} B_x$ is a union of members of β, each open in $\mathcal{T}$, hence $U \in \mathcal{T}$. Therefore $\mathcal{T}_\beta \subseteq \mathcal{T}$, completing the proof.

Comparing two topologies is easier through their bases than through their open sets.

Lemma 1.3.9: Simplification Due To Basis

Let X be a set and let β, β' be bases for X generating $\mathcal{T}, \mathcal{T}'$ respectively. Then

$$\mathcal{T} \text{ is finer than } \mathcal{T}' \quad \Leftrightarrow \quad \forall B_0' \in \beta',\ \forall x \in B_0',\ \exists B_0 \in \beta \text{ with } x \in B_0 \subseteq B_0'$$

***Proof*:**
Necessity. Suppose $\mathcal{T}$ is finer than $\mathcal{T}'$. Let $B_0' \in \beta'$ and $x \in B_0'$. Then $B_0' \in \mathcal{T}' \subseteq \mathcal{T}$, so B_0' is open in $\mathcal{T}$, and since β generates $\mathcal{T}$ there is $B_0 \in \beta$ with $x \in B_0 \subseteq B_0'$.

Sufficiency. Suppose the stated condition holds and let $U \in \mathcal{T}'$. For $x \in U$ choose $B_0' \in \beta'$ with $x \in B_0' \subseteq U$, then $B_0 \in \beta$ with $x \in B_0 \subseteq B_0' \subseteq U$. So U is a union of members of β and hence lies in $\mathcal{T}$. Therefore $\mathcal{T}' \subseteq \mathcal{T}$, completing the proof.

A basis also shortens continuity proofs.

Lemma 1.3.10: Continuity With Basis

Let X, Y be topological spaces and let β be a basis for the topology on Y. Then $f : X \to Y$ is continuous if and only if $f^{-1}(B)$ is open in X for every $B \in \beta$.

***Proof*:**
Necessity is immediate, since each $B \in \beta$ is open in Y.

Sufficiency. Let $V \subseteq Y$ be open. By definition of the generated topology, V is a union of basis elements, say $V = \bigcup_\alpha B_\alpha$. Preimages commute with unions, so

$$f^{-1}(V) = \bigcup_\alpha f^{-1}(B_\alpha)$$

which is a union of open sets and hence open. So f is continuous, as we sought to show.

For $\mathbb{R}^n$ with the Euclidean topology this recovers the epsilon-delta definition.

Definition 1.3.11: ϵ-δ Definition of Continuity

A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous if

$$\forall a \in \mathbb{R}^n,\ \forall \epsilon > 0,\ \exists \delta > 0 \text{ s.t. } \forall x,\ \|x - a\| < \delta \Rightarrow \|f(x) - f(a)\| < \epsilon$$

The point a is quantified first, so δ may depend on it; requiring one δ to serve every a is the stronger condition of definition 2.6.15.

Reading this against lemma 1.3.10: the clause "for all ϵ" ranges over the basis elements of the co-domain, namely the open balls, and the conclusion asserts that the preimage of such a ball contains a ball about each of its points, which is exactly the statement that the preimage is open. The two definitions therefore say the same thing, with the ϵ-δ formulation naming the basis explicitly.

Conversely, given a topology, how is a basis found? Every topology is a basis for itself, so the question is really how to find a smaller one.

Lemma 1.3.12: Finding Basis When Given A Topology

Let X be a topological space. Suppose that $\mathcal{C}$ is a collection of open sets of X such that for each open set U of X, and each $x \in U$, there is an element $C \in \mathcal{C}$ such that $x \in C \subseteq U$. Then $\mathcal{C}$ is a basis for the topology of X

***Proof*:**
First, $\mathcal{C}$ is a basis in the sense of definition 1.3.1: taking $U = X$ shows every $x \in X$ lies in some $C \in \mathcal{C}$, and if $x \in C_1 \cap C_2$ with $C_1, C_2 \in \mathcal{C}$, then $C_1 \cap C_2$ is open, so the hypothesis supplies $C_3 \in \mathcal{C}$ with $x \in C_3 \subseteq C_1 \cap C_2$.

Let $\mathcal{T}_{\mathcal{C}}$ be the topology it generates. Every $C \in \mathcal{C}$ is open in X by hypothesis, so $\mathcal{T}_{\mathcal{C}} \subseteq \mathcal{T}_X$. Conversely if $U \in \mathcal{T}_X$ then for each $x \in U$ there is $C \in \mathcal{C}$ with $x \in C \subseteq U$, so U is a union of members of $\mathcal{C}$ and lies in $\mathcal{T}_{\mathcal{C}}$. The two topologies agree, as we sought to show.

Bases are the standard working device for topologies, and most later arguments are carried out on a basis rather than on the full collection of open sets.

One family of topologies is most naturally described by a basis rather than by listing its open sets, and it will supply several of the examples that separate the properties studied later in this book. Any set carrying an order has a canonical topology built from its intervals, generalizing the way the open intervals generate the standard topology on $\mathbb{R}$.

Definition 1.3.13: Order Topology

Let X be a set with more than one element carrying a *simple order* $<$: a relation that is irreflexive, transitive, and satisfies trichotomy (for $x \neq y$ exactly one of $x < y$, $y < x$ holds). Write (a, b), $[a, b]$, $[a, b)$ and $(a, b]$ for the sets of $x \in X$ satisfying the corresponding inequalities against a and b. Let $\mathcal{B}$ be the collection of all sets of the following three kinds:

1. all open intervals $(a, b) = \{x \in X \big| a < x < b\}$;
2. all intervals $[a_0, b)$, where a_0 is the smallest element of X (if one exists);
3. all intervals $(a, b_0]$, where b_0 is the largest element of X (if one exists).

Then $\mathcal{B}$ is a basis for a topology on X, called the *order topology*.

The second and third families are present only to cover the endpoints: without them a smallest element would lie in no basis element at all, and $\mathcal{B}$ would fail to cover X. When X has neither a smallest nor a largest element, the open intervals alone suffice. That $\mathcal{B}$ satisfies the two conditions of definition 1.3.1, and so generates a topology by lemma 1.3.3, is immediate: the three kinds together cover X, and the intersection of two intervals is again an interval of one of the three kinds, possibly empty.

Example 1.3.14: Order Topologies

1. On $\mathbb{R}$ with its usual order, the order topology is exactly the standard topology, since $\mathbb{R}$ has no smallest or largest element and the open intervals are the standard basis.
2. On $\mathbb{Z}_+$ the order topology is the *discrete* topology. The one-point set $\{n\}$ equals the basis element $(n-1, n+1)$ for $n > 1$, and $\{1\} = [1, 2)$ is of the second kind, so every singleton is open.
3. On $\{1, 2\} \times \mathbb{Z}_+$ in the dictionary order (compare first coordinates, break ties by the second) the order topology is *not* discrete. Every point is isolated except $(2, 1)$. A basis element containing $(2, 1)$ is either an interval (a, b) with $a < (2, 1)$, forcing $a = (1, n)$ for some n, or the initial interval $[(1, 1), b)$, since $(1, 1)$ is the smallest element. In the first case the interval contains every $(1, m)$ with $m > n$; in the second it contains the whole first row. Either way it meets the first row infinitely often, so $(2, 1)$ is a limit point of that row and the space is not discrete.

The third example is the standard illustration that an order topology can be discrete at all but one point, and it is the smallest ordered set exhibiting the phenomenon that reappears, on an uncountable scale, in the ordered spaces used later to separate paracompactness from compactness.

Exercise 1.3.1

1. Verify that $\beta = \{[a, b) \mid a < b \text{ in } \mathbb{R}\}$ satisfies the two conditions of definition 1.3.1, and identify the topology it generates.
2. Show that the intervals with *rational* endpoints, $\beta_{\mathbb{Q}} = \{(p, q) \mid p < q \text{ in } \mathbb{Q}\}$, generate the standard topology on $\mathbb{R}$. Conclude that $\mathbb{R}$ has a countable basis.

3. Let $\mathcal{S} = \{(-\infty, a) \mid a \in \mathbb{R}\} \cup \{(b, \infty) \mid b \in \mathbb{R}\}$. Which topology on $\mathbb{R}$ does $\mathcal{S}$ generate as a subbasis (definition 1.3.7)?

4. Let $\beta \subseteq \beta'$ be bases on the same set X. Show that $\mathcal{T}_\beta \subseteq \mathcal{T}_{\beta'}$, and give an example where the inclusion is strict and one where it is an equality with $\beta \neq \beta'$.

5. Let X be finite. Show that the topology on X has a unique smallest basis, namely the collection of *minimal* open sets $U_x = \bigcap\{U \in \mathcal{T}_X \mid x \in U\}$.

1.4 Closed Sets

The axioms were stated with *arbitrary* unions and *finite* intersections. The opposite convention, finite unions and arbitrary intersections, defines the same subject. What matters is that *one* of the two operations is restricted to finitely many; which one is a matter of bookkeeping.

This section develops that duality. Defining a topology one way determines it the other way, and properties expressed in one language, borders, connectedness, closure, interior, translate into the other. Everything rests on the identity

$$\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} (A_i)^c$$

which turns arbitrary unions of open sets into arbitrary intersections of their complements. Those complements deserve a name.

Definition 1.4.1: Closed Sets

Let X be a set and $\mathcal{T}_X$ be a topology on X. A set $A \subseteq X$ is closed if its complement $A^c = X \setminus A$ is open

1.4.2 Note Nothing forces the choice of open sets over closed ones as the primitive notion; the two carry the same information and differ only in presentation. Some arguments are shorter in one language than the other, which is the practical reason for keeping both available.

Example 1.4.3: Closed Sets

1. $[a, b]$ is closed in $\mathbb{R}$, being the complement of $(-\infty, a) \cup (b, \infty)$. Every one-point set is closed in $\mathbb{R}$; identifying the complementary open set is a good check of the definition.

2. In any topological space $\emptyset$ and X are closed, being complements of one another and both open. In the trivial topology these are the only closed sets. In the discrete topology every set is closed, since every set is open.

3. Arbitrary intersections of closed sets are closed, and finite unions of closed sets are closed; this is theorem 1.4.4 below.

The pattern of finite unions and arbitrary intersections is exactly the mirror of the axioms, which is what “co-open” means here. A topology can be specified from that side instead.

Theorem 1.4.4: Thinking Of Topology In Terms Of Closed Sets

Let X be a topological space. Then the following conditions hold:

1. $\emptyset$ and X are closed
2. arbitrary intersections of closed sets are closed
3. finite unions of closed sets are closed

***Proof*:**
Each item is the De Morgan dual of a topology axiom.

1. $\emptyset = X \setminus X$ and $X = X \setminus \emptyset$, and both X and $\emptyset$ are open, so both are closed.
2. Let $\{C_\alpha\}$ be closed. Then $X \setminus \bigcap_\alpha C_\alpha = \bigcup_\alpha (X \setminus C_\alpha)$ is a union of open sets, hence open, so the intersection is closed.
3. Let $C_1, \ldots, C_n$ be closed. Then $X \setminus \bigcup_{i=1}^n C_i = \bigcap_{i=1}^n (X \setminus C_i)$ is a finite intersection of open sets, hence open, so the union is closed.

The three axioms are recovered from these by complementation, as we sought to show.

Neither set of axioms is preferable on mathematical grounds. One of the two operations must be restricted, since otherwise $\bigcap_n \left(\frac{-1}{n}, \frac{1}{n}\right) = \{0\}$ would be open and, as shown in section 1.1, the topology would collapse to the power set. Convention settles on arbitrary unions and finite intersections.

Continuity is one of the notions that translates directly.

Theorem 1.4.5: Continuity And Closed Sets

The function $f : X \to Y$ is continuous if and only if $f^{-1}(V)$ is closed for every closed $V \subseteq Y$

***Proof*:**
The whole content is the set-theoretic identity

$$f^{-1}(Y \setminus V) = X \setminus f^{-1}(V)$$

which holds because $x \in f^{-1}(Y \setminus V)$ says $f(x) \notin V$, that is $x \notin f^{-1}(V)$.

Suppose f is continuous and $V \subseteq Y$ is closed. Then $Y \setminus V$ is open, so $f^{-1}(Y \setminus V) = X \setminus f^{-1}(V)$ is open, making $f^{-1}(V)$ closed. Conversely, suppose preimages of closed sets are closed and let $U \subseteq Y$ be open. Then $Y \setminus U$ is closed, so $X \setminus f^{-1}(U)$ is closed and $f^{-1}(U)$ is open, completing the proof.

One further rule is used often enough to carry its own name.

Lemma 1.4.6: Pasting Lemma

Let $X = A \cup B$, where A and B are closed in X. Let $f : A \to Y$ and $g : B \to Y$ be continuous. If $f(x) = g(x)$ for every $x \in A \cap B$, then f and g combine to give a continuous function $h : X \to Y$, defined by setting $h(x) = f(x)$ if $x \in A$ and $h(x) = g(x)$ if $x \in B$

***Proof*:**
The function h is well defined: the two formulas agree on $A \cap B$ by hypothesis, and $A \cup B = X$, so every point receives exactly one value.

For continuity, let $C \subseteq Y$ be closed. Then

$$h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$$

Now $f^{-1}(C)$ is closed in A by continuity of f, and A is closed in X, so $f^{-1}(C)$ is closed in X; the same argument applies to $g^{-1}(C)$. A union of two closed sets is closed by theorem 1.4.4, so $h^{-1}(C)$ is closed. Since preimages of closed sets are closed, h is continuous by theorem 1.4.5, completing the proof.

With closed sets available, closed maps can be defined alongside open ones.

Definition 1.4.7: Closed Maps

Given topological spaces X, Y, a map $f : X \to Y$ is a *closed map* if the image of every closed set is closed

Continuity, openness and closedness are logically independent.

Example 1.4.8: Continuous, Open and Closed Maps
Example 1.2.5 is continuous but not open, and example 1.2.6 is open but not continuous. Further cases worth working out:

1. Example 1.2.5 is also *closed*.
2. Example 1.2.6 is *closed* as well as open.
3. The constant map $f : \mathbb{R} \to \mathbb{R}$, $f(x) = c$, is continuous and closed but not open, since $f((0,1)) = \{c\}$.
4. A map that is all three: any homeomorphism, for instance $x \mapsto x + 1$ on $\mathbb{R}$. A map that is none of the three: $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = 1/x$ for $x \neq 0$ and $f(0) = 0$. It is not continuous at 0; it is not open, since $f((-1,1)) = (-\infty,-1) \cup \{0\} \cup (1,\infty)$ has 0 as an isolated point; and it is not closed, since $[1,\infty)$ is closed while $f([1,\infty)) = (0,1]$ is not.

For a map that is continuous and open but not closed, take $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x,y) = x$, where $\mathbb{R}^2$ carries the topology generated by

$$\{S \times T \mid S, T \subseteq \mathbb{R} \text{ open}\}$$

and $\mathbb{R}$ the usual topology. The image of the closed set $\{(x,y) \mid xy = 1\}$ is $\mathbb{R} \setminus \{0\}$, which is not closed.

As already noted, $\emptyset$ and X are always both closed and open. Sets with that property are important enough to name.

Definition 1.4.9: Clopen

In a topological space X, $A \subseteq X$ is *clopen* if A is open *and* closed

Clopen sets return in force with connectedness, where their existence is precisely what a space fails to have when it is connected.

1.4.1 Closure

Since closed sets are stable under arbitrary intersections, every set has a smallest closed set containing it.

Definition 1.4.10: Closure

Let X be a topological space and $A \subseteq X$. The *closure* $\overline{A}$ is the smallest closed subset of X containing A:

$$\overline{A} = \bigcap_{A \subseteq V,\ V \text{ closed}} V$$

The intersection is over a nonempty family, since X itself is closed and contains A, and it is closed by theorem 1.4.4. For a closed set A the definition gives $A = \overline{A}$.

Two remarks. The set A is arbitrary and need not be open. And there is no dual construction using open sets and intersections, since openness is not preserved under arbitrary intersections; the dual instead uses unions.

Definition 1.4.11: Interior

Let X be a topological space and $A \subseteq X$. The *interior* $\operatorname{int}(A)$ is the largest open set contained in A:

$$\operatorname{int}(A) = \bigcup_{V \subseteq A,\ V \text{ open}} V$$

Both definitions are extremal, and it is the extremality rather than the formula that gets used.

Proposition 1.4.12: The Closure Is The Smallest Closed Superset

Let $A \subseteq X$. If B is closed and $A \subseteq B$, then $\overline{A} \subseteq B$.

***Proof*:**
The closure $\overline{A}$ is the intersection of all closed sets containing A, and B is one of them, so the intersection is contained in B, as we sought to show.

Corollary 1.4.13: The Interior Is The Largest Open Subset

Let $A \subseteq X$. If B is open and $B \subseteq A$, then $B \subseteq \operatorname{int}(A)$.

***Proof*:**
The interior $\operatorname{int}(A)$ is the union of all open sets contained in A, and B is one of them, so B is contained in that union, completing the proof.

Example 1.4.14: Closure and Interior

1. $\overline{(0,1)} = [0,1]$ in $\mathbb{R}$.
2. $\operatorname{int}([0,1]) = (0,1)$ in $\mathbb{R}$.

Both follow from the definitions once it is checked that no smaller closed set contains $(0,1)$ and no larger open set sits inside $[0,1]$.

The definition is an intersection over closed sets, but closure admits a purely local description in terms of open sets, which is how it is used in practice.

Theorem 1.4.15: Neighborhood Intersection Property

$x \in \overline{A} \Leftrightarrow$ every open set U containing x satisfies $U \cap A \neq \emptyset$

***Proof*:**
Necessity. Suppose $x \in \overline{A}$ and let U be open with $x \in U$. If $U \cap A = \emptyset$ then $X \setminus U$ is a closed set containing A, so $\overline{A} \subseteq X \setminus U$ by proposition 1.4.12, contradicting $x \in U \cap \overline{A}$.

Sufficiency. Suppose every open $U \ni x$ meets A, and suppose $x \notin \overline{A}$. Then $U = X \setminus \overline{A}$ is open and contains x, so it meets A; but $A \subseteq \overline{A}$, so $U \cap A = \emptyset$, a contradiction. Hence $x \in \overline{A}$, completing the proof.

This is the sense in which closure is a border. The condition is automatic for points of A itself, since $x \in U \cap A$ whenever $x \in A$. Its force is on the points of $\overline{A} \setminus A$: such a point cannot be separated from A by any open set, so it lies arbitrarily close to A in the only sense the topology can express. When A is closed there are no such points; the interesting case is when $\overline{A}$ is strictly larger.

The border intuition has limits. Since $\mathbb{R}$ is closed, $\overline{\mathbb{R}} = \mathbb{R}$, and $\mathbb{R}$ has no border in any intuitive sense. Coarse topologies distort the picture further.

Example 1.4.16: Huge Closure
Let
$$T_{\text{large closure}} = \{\emptyset, (-1,1), (-\infty,2) \cup (2,\infty), (-\infty,2) \cup (-1,1) \cup (2,\infty), \mathbb{R}\}$$

The closed sets are then $\mathbb{R}$, $[-2,2]$, $(-\infty,-1]\cup[1,\infty)$ and $\emptyset$, so

$$\overline{(0,1)} = [-2,2]$$

The border is enormous. A better intuition is therefore to think of closure in terms of which open sets sit near the *edge* of A: here the topology has so few open sets that nothing local happens near the edge at all, and the closure is forced to be large.

Whatever is added in passing from A to $\overline{A}$ is itself closed when A is open. Indeed $X\setminus\overline{A}$ is open, hence so is $(X\setminus\overline{A})\cup A$, and its complement is exactly $\overline{A}\setminus A$.

The mechanism behind a large closure is that the border admits no open set at all, and that is forced by the definition rather than by the choice of example.

Corollary 1.4.17: A Border Contains No Nonempty Open Set

Let $A\subseteq X$ be open. Then no nonempty open set is contained in $\overline{A}\setminus A$.

***Proof*:**
Suppose U is open, nonempty, and $U\subseteq\overline{A}\setminus A$. Then $U\cap A=\emptyset$ by construction. But choosing $x\in U$ gives $x\in\overline{A}$, so theorem 1.4.15 applied to the open set U containing x forces $U\cap A\neq\emptyset$, a contradiction. Hence no such U exists, completing the proof.

Equivalently, the closure of a set is the smallest enlargement for which no open set fits inside the border.

Finally, continuity interacts with closure in one direction only.

Theorem 1.4.18: Continuity and Closure

Let $f:X\to Y$ be continuous. Then $f(\overline{A})\subseteq\overline{f(A)}$ for every $A\subseteq X$

***Proof*:**
Let $y\in f(\overline{A})$, say $y=f(x)$ with $x\in\overline{A}$, and let V be open with $y\in V$. Then $f^{-1}(V)$ is open by continuity and contains x, so by theorem 1.4.15 it meets A: pick $a\in A\cap f^{-1}(V)$. Then $f(a)\in f(A)\cap V$, so every open set around y meets $f(A)$, and theorem 1.4.15 gives $y\in\overline{f(A)}$, as we sought to show.

1.4.2 Limit Points

Limit points are a closely related notion, treated here in general and connected to sequences in section 1.5.1.

Definition 1.4.19: Limit Points

Let X be a topological space, and $A \subseteq X$. A point $x \in X$ is a *limit point* of A if $\forall U \subseteq X$ open with $x \in U$, then $U \cap (A \setminus \{x\}) \neq \emptyset$. Write $\lim(A)$ for the set of limit points of A.

The quantifier is over $x \in X$, not $x \in A$: a limit point need not belong to the set. This is what makes it possible for a set to approach its edge without containing it. The difference from closure is the deletion of x itself, which rules out the trivial case where x meets A only in x.

Example 1.4.20: Limit Points

Let $X = \mathbb{R}$ with the standard topology.

1. $A_1 = [0, 1]$ has limit points $[0, 1]$.
2. $A_2 = (0, 1)$ has limit points $[0, 1]$.
3. $A_3 = \{0\}$ has no limit points, whereas $\overline{A_3} = A_3$.
4. $A_4 = \left\{\frac{1}{n} : n \in \mathbb{Z}_+\right\}$ has the single limit point 0, whereas $\overline{A_4} = A_4 \cup \{0\}$.
5. If $\mathbb{R}$ instead carries the trivial topology, the only open set containing a point is $\mathbb{R}$ itself, so $x \in \lim(A)$ exactly when $A \setminus \{x\} \neq \emptyset$. Hence $\lim(A) = \mathbb{R}$ whenever A has at least two points, while $\lim(\{a\}) = \mathbb{R} \setminus \{a\}$ and $\lim(\emptyset) = \emptyset$.

For a reader who knows sequences, a limit point is the limit of a sequence in A that does not use the point itself, which is why constant sequences are excluded. That is why 1 is not a limit point of A_4 above. The interesting case is again a point *outside* A, where the condition says there are points of A arbitrarily near it. Capturing what happens in a neighbourhood of a point is the recurring theme of the subject.

Closure and limit points differ exactly by the constant case: closure admits points that meet A only in themselves, limit points do not. The precise relation is the following.

Theorem 1.4.21: Relating $\overline{A}$ With $\lim(A)$

$$\overline{A} = A \cup \lim(A)$$

***Proof*:**

The inclusion $\supseteq$. Certainly $A \subseteq \overline{A}$. If $x \in \lim(A)$ and U is open with $x \in U$, then U meets $A \setminus \{x\}$ and in particular meets A, so $x \in \overline{A}$ by theorem 1.4.15.

The inclusion $\subseteq$. Let $x \in \overline{A}$ and suppose $x \notin A$. For any open $U \ni x$, theorem 1.4.15 gives a point of $U \cap A$, and that point differs from x since $x \notin A$. So $U \cap (A \setminus \{x\}) \neq \emptyset$ and $x \in \lim(A)$. Hence every point of $\overline{A}$ lies in A or in $\lim(A)$, as we sought to show.

One more notion is read off the closure, and it is used from section 1.6 onwards.

Definition 1.4.22: Dense Subset

A subset $A \subseteq X$ is *dense* in X if $\overline{A} = X$. Equivalently, by theorem 1.4.15, every nonempty open subset of X meets A.

Example 1.4.23: Dense Subsets

$\mathbb{Q}$ is dense in $\mathbb{R}$, since every open interval contains a rational. So is $\mathbb{R} \setminus \mathbb{Q}$. In a discrete space the only dense subset is the whole space, since each singleton is open. In the trivial topology on a set with at least one point, every nonempty subset is dense.

Limit points still include much of the interior of a set. Isolating the edge alone requires one more definition.

Definition 1.4.24: Border of Set

$$\partial(A) = \overline{A} \setminus \operatorname{int}(A) = X \setminus (\operatorname{int}(A) \cup \operatorname{int}(X \setminus A)) = \overline{A} \cap \overline{X \setminus A}$$

Exercise 1.4.1

1. In $\mathbb{R}$ compute $\overline{\mathbb{Q}}$, $\operatorname{int}(\mathbb{Q})$, $\partial\mathbb{Q}$, and the same three for $\mathbb{Z}$.
2. Show that $\overline{A \cup B} = \overline{A} \cup \overline{B}$ for any two subsets, and that $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$ with the inclusion sometimes strict.
3. Show that A is clopen if and only if $\partial A = \emptyset$.
4. Let X be infinite with the co-finite topology. Compute $\overline{A}$ for A finite and for A infinite, and deduce which subsets are dense.
5. Show that $\operatorname{int}(A \cap B) = \operatorname{int}(A) \cap \operatorname{int}(B)$, and find subsets of $\mathbb{R}$ with $\operatorname{int}(A) \cup \operatorname{int}(B) \neq \operatorname{int}(A \cup B)$.

1.5 Hausdorff Spaces

Hausdorff spaces have already appeared in passing. The condition is that any two distinct points can be surrounded by disjoint open sets: each point gets a neighbourhood of its own, and the two do not meet. This imposes a genuine amount of localness, particularly on spaces with uncountably many points. The behaviour of sequences is what motivated the axiom historically, and it is the cleanest way in.

Definition 1.5.1: Hausdorff Space

A topological space X is called *Hausdorff* if for all pairs of distinct points $x_1, x_2 \in X$, there exist open sets U_1, U_2 such that $x_1 \in U_1$, $x_2 \in U_2$ and $U_1 \cap U_2 = \emptyset$.

Figure 1.2: Two points of a Hausdorff space separated by disjoint open sets.

The axiom says the space is fine enough to put room between any two points.

Example 1.5.2: Hausdorff Spaces

1. $\mathbb{R}$ is Hausdorff: distinct $x < y$ are separated by $(x-\epsilon, x+\epsilon)$ and $(y-\epsilon, y+\epsilon)$ for $\epsilon = \frac{y-x}{2}$.
2. The discrete topology is Hausdorff, taking singletons.
3. The trivial topology is *not* Hausdorff once $|X| \geq 2$, the only nonempty open set being X.
4. $\mathbb{R}^n$ is Hausdorff, and so is the unit circle $S^1 \subseteq \mathbb{R}^2$ with its subspace topology.
5. If $\mathcal{T}$ is Hausdorff and $\mathcal{T}'$ is finer than $\mathcal{T}$, then $\mathcal{T}'$ is Hausdorff, since the separating sets of $\mathcal{T}$ remain open in $\mathcal{T}'$.

Example 1.5.3: A Space That Is Not Hausdorff
Take the topology on $\mathbb{R}$ generated by the intervals $(-n, n)$ for $n \in \mathbb{N}$, together with $\emptyset$ and $\mathbb{R}$. Every nonempty basic open set contains 0, so any two open sets meet, and no two points can be separated; the points -0.5 and 0.5 already fail. Enlarging the family to all $(-r, r)$ with $r \in \mathbb{R}_{>0}$ changes nothing, since the sets are still nested: 2 and 4 remain inseparable.

The condition also has a closed-set form, which is a useful exercise in translating between the two languages.

Lemma 1.5.4: Hausdorff In Terms Of Closed Sets

A space X is Hausdorff if and only if for all $x_1 \neq x_2$ there exist closed sets $M_1, M_2 \subseteq X$ with

1. $M_1 \cup M_2 = X$
2. $x_1 \notin M_1$
3. $x_2 \notin M_2$

***Proof*:**
The two formulations correspond under $U_i = X \setminus M_i$, which is a bijection between closed sets and open sets by definition 1.4.1. Under it, $x_i \notin M_i$ becomes $x_i \in U_i$, and

$$M_1 \cup M_2 = X \iff X \setminus (M_1 \cup M_2) = \emptyset \iff U_1 \cap U_2 = \emptyset$$

by De Morgan. So a pair of closed sets with the three listed properties is exactly a pair of disjoint open neighbourhoods of x_1 and x_2, which is the condition of definition 1.5.1.

Lemma 1.5.5: Finite Sets In Hausdorff Spaces

If X is Hausdorff and $A \subseteq X$ is finite, then A is closed

***Proof*:**
Since finite unions of closed sets are closed by theorem 1.4.4, it suffices to treat singletons. Fix $x \in X$. For each $g \neq x$ the Hausdorff property supplies an open U_g with $g \in U_g$ and $x \notin U_g$. Then

$$\bigcup_{g \neq x} U_g = X \setminus \{x\}$$

the inclusion $\subseteq$ because no U_g contains x, and $\supseteq$ because each $g \neq x$ lies in its own U_g. The left side is open, so $\{x\}$ is closed, as we sought to show.

1.5.6 Note The converse fails: finite sets being closed does not force Hausdorffness. In $\mathbb{R}$ with the finite-complement topology every finite set is closed, yet any two nonempty open sets meet, since their complements are finite. The weaker condition that finite sets are closed is the T_1 axiom, and chapter 4 places both in a hierarchy.

1.5.1 Sequences

Definition 1.5.7: Sequence

Let X be a set. Then a *sequence* is a function $f : \mathbb{N}_{>0} \to X$ that assigns to each positive natural number an element of X. The element $f(n)$ is often denoted x_n, and f itself $(x_n)_{n=1}^{\infty}$, or (x_n), or simply x_n where the context is clear.

Definition 1.5.8: Convergent sequence

Let X be a topological space, and $a_n \subseteq X$. This sequence *converges* to $a \in X$ if for every open set U with $a \in U$ there exists N such that $a_n \in U$ for all $n \geq N$.

Nothing in that definition forces a limit to be unique. Hausdorffness is exactly what does.

Lemma 1.5.9: Hausdorff Spaces Means At Most 1 Element When Converges

If X is Hausdorff, every sequence converges to *at most* 1 element.

***Proof*:**
Suppose $a_n \to x$ and $a_n \to y$ with $x \neq y$. Hausdorffness gives disjoint open $U \ni x$ and $V \ni y$. Convergence supplies N_1 with $a_n \in U$ for $n \geq N_1$, and N_2 with $a_n \in V$ for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$ the point a_n lies in $U \cap V = \emptyset$, which is impossible. So $x = y$, completing the proof.

The condition is sufficient but not necessary. Give $\mathbb{R}$ the co-countable topology, whose open sets are $\emptyset$ together with the complements of countable sets. It is not Hausdorff, since any two nonempty open sets meet (their complements are countable and $\mathbb{R}$ is not). Yet a sequence (a_n) converging to a must be eventually equal to a: the set $\mathbb{R} \setminus \{a_n \mid a_n \neq a\}$ is open and contains a, so all but finitely many terms lie in it, and those terms equal a. An eventually constant sequence has one limit, so limits are unique here without the axiom.

Continuity also interacts well with sequences.

Proposition 1.5.10: Continuity With Respect To Sequences

Let $f : X \to Y$ be continuous and let $(x_n)_{n=1}^{\infty}$ converge in X. Then $(f(x_n))_{n=1}^{\infty}$ converges in Y: continuity preserves convergent sequences.

***Proof*:**
Suppose $x_n \to x$ and let V be open with $f(x) \in V$. Then $f^{-1}(V)$ is open and contains x, so there is N with $x_n \in f^{-1}(V)$ for all $n \geq N$, whence $f(x_n) \in V$ for all $n \geq N$. So $f(x_n) \to f(x)$, completing the proof.

The converse fails: preserving convergent sequences does not make a function continuous. Two repairs are available, and both are carried out later.

1. Strengthen the domain. First countability (definition 1.6.3) is what is needed, and proposition 1.6.8 carries it out for closure; second countability suffices because it implies first countability (proposition 1.6.6). Compactness does *not* help: it says nothing about neighbourhood bases.
2. Generalize sequences to *nets*, which index by a directed set rather than by $\mathbb{N}$ and carry exactly the extra flexibility needed. Nets are treated in section 3.3.

Exercise 1.5.1

1. Show that a finite Hausdorff space carries the discrete topology.
2. Show that Hausdorffness is inherited by subspaces, and that the co-finite topology on an infinite set is T_1 but not Hausdorff.
3. Let $f, g \colon Y \to X$ be continuous with X Hausdorff. Show that $\{y \in Y \mid f(y) = g(y)\}$ is closed in Y, and deduce that two continuous maps into a Hausdorff space agreeing on a dense subset are equal.

4. Show that in the co-finite topology on an infinite set, a sequence with pairwise distinct terms converges to *every* point, so limits are as far from unique as a topology can make them. Explain why this is an instance of lemma 1.5.9 rather than a counterexample to its converse, and identify what a counterexample to the converse would have to look like.

1.6 Countability Axioms

Different sizes of infinity have already mattered repeatedly. It is often useful that a topology be generated by the smallest one, a countable family, and that condition has a name.

Definition 1.6.1: Second-Countability

A topological space X is called *second-countable* if it has a countable basis.

Example 1.6.2: Second Countability

$\mathbb{R}$ is second-countable: the open intervals with rational endpoints form a countable basis, since every open set is a union of them and there are countably many such intervals.

Second countability is a strong restriction, and it returns in chapter 4, where it upgrades regularity to normality, and again in chapter 5, where together with local compactness it forces paracompactness. It has three weaker relatives, each obtained by asking for countability of a different thing: of a neighbourhood base at a point, of a dense set, or of a subcover.

Definition 1.6.3: First-Countability

A space X is *first-countable* if every $x \in X$ has a countable *neighbourhood base*: a countable family of neighbourhoods of x such that every neighbourhood of x contains one of them.

Definition 1.6.4: Separability

A space X is *separable* if it has a countable dense subset.

Definition 1.6.5: Lindelöf Space

A space X is *Lindelöf* if every open cover of X has a countable subcover.

Second countability implies all three, and the proofs are the same argument run three ways: a countable basis is a countable supply of open sets, and each condition asks for a countable supply of something a basis can produce.

Proposition 1.6.6: Second-Countable Spaces Are First-Countable, Separable and Lindelöf

A second-countable space is first-countable, separable and Lindelöf.

***Proof*:**
Let $\mathcal{B} = \{B_1, B_2, \dots\}$ be a countable basis.

First-countable. For $x \in X$ the subfamily $\{B_n \mid x \in B_n\}$ is countable, and every neighbourhood of x contains an open set containing x, hence contains some basis element containing x.

Separable. Discard any empty B_n and choose $x_n \in B_n$ for each remaining index. The set $D = \{x_n\}$ is countable, and it is dense: a nonempty open U contains some nonempty B_n, hence contains x_n, so every nonempty open set meets D and $\overline{D} = X$ by theorem 1.4.15.

Lindelöf. Let $\mathcal{A}$ be an open cover. Let N be the set of indices n for which $B_n \subseteq A$ for some $A \in \mathcal{A}$, and for each $n \in N$ choose such an A_n. Then $\{A_n \mid n \in N\}$ is a countable subfamily of $\mathcal{A}$, and it covers: given $x \in X$, some $A \in \mathcal{A}$ contains x, and some B_n satisfies $x \in B_n \subseteq A$, so $n \in N$ and $x \in B_n \subseteq A_n$.

None of the three implications reverses in general, and the Sorgenfrey line of example 4.2.6 is separable, Lindelöf and first-countable while failing to be second-countable. For metric spaces the situation collapses.

Proposition 1.6.7: Separable Metric Spaces Are Second-Countable

A metrizable space is second-countable if and only if it is separable.

***Proof*:**
One direction is proposition 1.6.6. Conversely let d induce the topology and let $\{d_n\}$ be a countable dense set. The balls $B(d_n, q)$ with q a positive rational form a countable family; they are a basis. Let U be open and $x \in U$, and take $\epsilon > 0$ with $B(x, \epsilon) \subseteq U$. Density supplies d_n with $d(x, d_n) < \epsilon/3$, and a rational $q \in (\epsilon/3, \epsilon/2)$ gives $x \in B(d_n, q)$; moreover any $y \in B(d_n, q)$ has $d(x, y) \leq d(x, d_n) + d(d_n, y) < \epsilon/3 + \epsilon/2 < \epsilon$, so $B(d_n, q) \subseteq B(x, \epsilon) \subseteq U$. Hence the family is a basis.

First-countability is exactly the hypothesis under which sequences are adequate, which is the deficiency section 3.3 repairs without it.

Proposition 1.6.8: Sequences Suffice in First-Countable Spaces

Let X be first-countable and $A \subseteq X$. Then $x \in \overline{A}$ if and only if some sequence in A converges to x.

***Proof*:**
A sequence in A converging to x forces every neighbourhood of x to meet A, so $x \in \overline{A}$ by theorem 1.4.15.

Conversely let $x \in \overline{A}$ and let $\{N_k\}$ be a countable neighbourhood base at x. Replacing N_k by $N_1 \cap \cdots \cap N_k$, which is again a neighbourhood, assume the base is decreasing. Each N_k meets A by theorem 1.4.15; choose $a_k \in N_k \cap A$. Given any neighbourhood M of x, some $N_j \subseteq M$, and then $a_k \in N_k \subseteq N_j \subseteq M$ for all $k \geq j$. So $a_k \to x$.

Exercise 1.6.1

1. Show that every metrizable space is first-countable, and that $\mathbb{R}_l$ is first-countable. (Metric spaces are the prerequisite notion of [1]; they are set up again in section 2.6.)
2. Let X be uncountable with the discrete topology. Determine which of first-countable, second-countable, separable and Lindelöf it satisfies.
3. Show that a second-countable space has at most countably many pairwise disjoint nonempty open sets, and deduce that every discrete subspace of a second-countable space is countable.
4. Show that a subspace of a second-countable space is second-countable.

2

Constructing Topologies

This chapter covers how to take a product of topologies and still have a topology, and how to define a sub-topology: two more of the fundamental ways of building new spaces from old.

2.1 Subspace (or Induced) Topologies

The subspace topology of definition 1.1.11 was set up in section 1.1, because continuity and embeddings already needed it. What was not developed there is how it interacts with the other apparatus of chapter 1: with bases, with closed sets, and with iterated subspaces. That is the business of this section, and it is also the template for every construction that follows, each of which will be tested against the same list.

Theorem 2.1.1: Basis For Y Relative To X

1. if β is a basis for the topology on X, then $\left\{B \cap Y \middle| B \in \beta\right\}$ is a basis for the subspace topology on Y
2. Let $X \supseteq Y \supseteq Z$, let X carry a topology $\mathcal{T}_X$, and let $\mathcal{T}_Y$ be the induced topology on Y. Then the topologies induced on Z from X and from Y agree.

***Proof*:**

1. Let U be open in X and $y \in U \cap Y$. Since β is a basis there is $B \in \beta$ with $y \in B \subseteq U$, whence $y \in B \cap Y \subseteq U \cap Y$. So every open set of the subspace topology is a union of sets of the form $B \cap Y$, and each such set is open in Y by definition of the subspace topology. Hence $\{B \cap Y \middle| B \in \beta\}$ is a basis for it.

2. An open set of the topology induced on Z from X has the form $U \cap Z$ with U open in X. Since $Z \subseteq Y$ we have $U \cap Z = (U \cap Y) \cap Z$, and $U \cap Y$ is open in Y, so the set is open in the topology induced from Y. Conversely an open set of the topology induced from Y has the form $V \cap Z$ with $V = U \cap Y$ open in Y, and then $V \cap Z = U \cap Z$ is induced from X. The two topologies therefore have the same open sets.

This establishes both claims, as we sought to show.

The construction is also compatible with products, which are treated in section 2.2: taking subspaces and taking products commute.

Proposition 2.1.2: Product Of Subspace Topologies

Let X, Y be topological spaces, and $A \subseteq X$, $B \subseteq Y$. The subspace topology on $A \times B$ is the same as the product topology of the subspace topologies on A and B.

***Proof*:**
Both topologies on $A \times B$ have a basis, and it suffices to see the two bases agree. A basis element of the product of the subspace topologies has the form $(U \cap A) \times (V \cap B)$ with U open in X and V open in Y. A basis element of the subspace topology inherited from $X \times Y$ has the form $(U \times V) \cap (A \times B)$. These are the same set, since

$$(U \times V) \cap (A \times B) = (U \cap A) \times (V \cap B)$$

So the two topologies have a common basis and therefore coincide, as we sought to show.

2.1.3 Convention A subset of $\mathbb{R}$ written without further comment, such as $\left\{\frac{1}{n} \middle| n \in \mathbb{N}\right\}$, always carries the subspace topology inherited from $\mathbb{R}$. The same convention applies to subsets of any space already under discussion.

As with the ambient space, the subspace topology can be described from the closed side.

Lemma 2.1.4: Subspace Topology And Closed Sets

Let $Y \subseteq X$ carry the subspace topology. Then $A \subseteq Y$ is closed in Y if and only if there is a $V \subseteq X$ closed in X with $A = V \cap Y$

***Proof*:**
Necessity. Suppose A is closed in Y. Then $Y \setminus A$ is open in Y, so $Y \setminus A = U \cap Y$ for some U open in X. Put $V = X \setminus U$, closed in X. Then $V \cap Y = Y \setminus (U \cap Y) = Y \setminus (Y \setminus A) = A$.

Sufficiency. Suppose $A = V \cap Y$ with V closed in X. Then $X \setminus V$ is open in X, so $(X \setminus V) \cap Y$ is open in Y, and it equals $Y \setminus A$. Hence A is closed in Y, completing the proof.

Closedness is transitive down a chain of subspaces, provided each step is closed in the one above.

Corollary 2.1.5: Transitivity of Closedness in Subspaces

Let X be a topological space. Suppose $Y \subseteq X$ is closed and $A \subseteq Y$ is closed in Y. Then A is closed in X.

***Proof*:**
Since A is closed in Y, lemma 2.1.4 gives a closed $V \subseteq X$ with $A = V \cap Y$. Both V and Y are closed in X, and a finite intersection of closed sets is closed by theorem 1.4.4, so A is closed in X, as we sought to show.

Exercise 2.1.1

1. Let $Y = [0,1] \subseteq \mathbb{R}$. Which of $[0, \frac{1}{2})$, $(\frac{1}{2}, 1]$, $\{0\}$ and $(\frac{1}{3}, \frac{2}{3})$ are open in Y? Which are closed in Y?
2. Show that if Y is open in X, then a subset of Y is open in Y if and only if it is open in X; and that the same holds with "closed" throughout when Y is closed in X.
3. Give an example of $Z \subseteq Y \subseteq X$ with Z closed in Y but not closed in X, and explain why this does not contradict corollary 2.1.5.
4. Call $x \in A$ *isolated* in A if $\{x\}$ is open in the subspace A. Show that $\mathbb{Q} \subseteq \mathbb{R}$ has no isolated points, and exhibit infinitely many clopen subsets of $\mathbb{Q}$.
5. Show that a subspace of a discrete space is discrete, and that a subspace of a space with the trivial topology has the trivial topology.

2.2 Product Topology

Products are the second standard way of building spaces: several topologies are combined into one on the Cartesian product of their underlying sets. The open sets are generated by the obvious candidates, the rectangles.

Definition 2.2.1: Product Topology basis

Let X and Y be topological spaces. The *product topology* on $X \times Y$ is generated by the basis

$$\beta = \left\{ S \times T \middle| S \subseteq X \text{ open}, T \subseteq Y \text{ open} \right\}$$

Corollary 2.2.2: The Rectangles Form a Basis

The collection β of definition 2.2.1 satisfies the two conditions of definition 1.3.1, so it generates a topology.

***Proof*:**
The first condition holds because $X \times Y \in \beta$. For the second, rectangles are closed under intersection:

$$(S_1 \times T_1) \cap (S_2 \times T_2) = (S_1 \cap S_2) \times (T_1 \cap T_2)$$

since a point lies in the left side exactly when each coordinate lies in both of its constraints. The right side is again a rectangle with open sides, so it is a member of β containing the given point, completing the proof.

The same definition extends to any finite number of factors.

Definition 2.2.3: Product Topology on n Topologies

The *product topology* on $X_1 \times \cdots \times X_n$ is generated by the basis

$$\beta = \left\{ S_1 \times \cdots \times S_n \middle| S_i \subseteq X_i \text{ open} \right\}$$

2.2.4 Warning Subsets of $X \times Y$ need careful handling. A subset $A \subseteq X \times Y$ is generally *not* of the form $U \times V$. The diagonal $A = \{(x, x) \mid x \in X\} \subseteq X \times X$ is the standard case: it contains (x, x) and (y, y) but not (x, y), whereas any product $U \times V$ containing the first two must contain the third. This matters most when taking subspaces of a product.

Example 2.2.5: Product Topologies

1. $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is the familiar case, though the basis here consists of open rectangles rather than open balls:

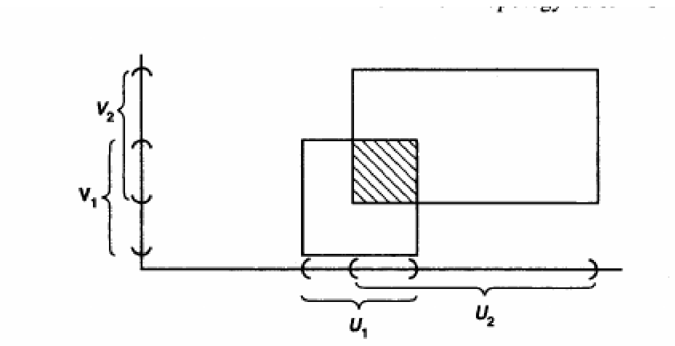

Figure 2.1: A basic open set of the product topology on $\mathbb{R}^2$ is a rectangle, not a ball.

The two bases generate the same topology, since every ball contains a rectangle about each of its points and conversely; section 2.6 makes this precise.

2. Take $\mathcal{T} = \{\emptyset, \{a\}, \{a, b\}\}$ on $X = \{a, b\}$. Then the product topology on $X \times X$ is

$$\{\emptyset, \{(a, a)\}, \{(a, a), (a, b)\}, \{(a, a), (b, a)\}, \{(a, a), (a, b), (b, a)\}, X \times X\}$$

The second-to-last set is a union of two basis elements and is not itself a rectangle, which illustrates the warning above.

Having built the product, the natural question is what characterises it. The answer is that it is the smallest topology making the two coordinate projections continuous.

Theorem 2.2.6: Projection And Product Topologies

The product topology on $X \times Y$ is the coarsest topology for which π_X and π_Y are continuous

***Proof*:**
Write $\pi_X\colon X \times Y \to X$ for $\pi_X(u,v) = u$. If $U \subseteq X$ is open then $\pi_X^{-1}(U) = U \times Y$, a basis element, so π_X is continuous; the same argument applies to π_Y.

For minimality, let $\mathcal{T}'$ be any topology on $X \times Y$ making both projections continuous. Then $\mathcal{T}'$ contains $\pi_X^{-1}(U) = U \times Y$ and $\pi_Y^{-1}(V) = X \times V$ for all open U, V, hence contains their intersection $U \times V$. So $\mathcal{T}'$ contains every basis element of the product topology, and therefore contains the product topology itself, as we sought to show.

Categorically, this is the universal property of the product transported from sets to spaces: the projections are the limit cone, and the defining diagram

$$\begin{tikzcd} & \prod_j X_j \arrow[d, "\pi_\alpha"] \\ Y \arrow[ur, "f"] \arrow[r, "f_\alpha"'] & X_\alpha \end{tikzcd}$$

is required to commute with continuous maps rather than mere functions.

Lemma 2.2.7: π_I Is An Open Map

Each projection $\pi_i\colon \prod_j X_j \to X_i$ is an open map.

***Proof*:**
Let $U \subseteq \prod_j X_j$ be open and let $x_i \in \pi_i(U)$, so $x_i = \pi_i(x)$ for some $x \in U$. Choose a basic open set $\prod_j U_j \subseteq U$ containing x, with U_j open and $U_j = X_j$ for all but finitely many j. Then U_i is open in X_i, contains x_i, and satisfies $U_i = \pi_i\left(\prod_j U_j\right) \subseteq \pi_i(U)$, the equality because every coordinate of the basic set is nonempty. So $\pi_i(U)$ contains an open neighbourhood of each of its points and is open, as we sought to show.

2.2.8 Warning Projections are open but need not be *closed*. Take $X = Y = \mathbb{R}$ and let $\Gamma = \{(x, 1/x) \mid x > 0\}$, the positive branch of the hyperbola. It is closed in $\mathbb{R}^2$, being the preimage of $\{1\}$ under the continuous map $(x,y) \mapsto xy$ intersected with the closed half-plane $x \geq 0$ minus the origin; yet $\pi_X(\Gamma) = (0, \infty)$ is open and not closed. Compactness of the discarded factor repairs this: if Y is compact then π_X is a closed map, which is proved in chapter 3.

Theorem 2.2.6 says a map *into* a product is controlled by its coordinates. The reverse question, whether a map *out of* a product is controlled by its restrictions in each variable, has a negative answer: separate continuity in each variable does not imply joint continuity. The function $f(x, y) = xy/(x^2 + y^2)$ with $f(0, 0) = 0$ is continuous in each variable separately but not at the origin, since along $y = x$ it takes the constant value $\frac{1}{2}$.

Two further traps are worth naming explicitly.

1. The product topology creates a two-dimensional grid: the sets $(1, 0)$ and $(0, 1)$, read as intervals in separate factors, say nothing about the point $(1, 1)$.
2. An open $U \subseteq \mathbb{R}^2$ need not be a single rectangle $V \times W$. It is a *union* of rectangles, which is a weaker statement.

Example 2.2.9: Open Sets That Are Not Rectangles

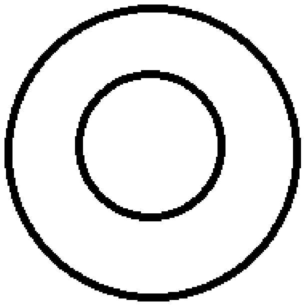

Figure 2.2: An open set of the product topology that is not a product $V \times W$.

The region shown is open in $\mathbb{R}^2$ but is an arbitrary union of rectangles rather than one of them. The closed case is similar: the diagonal $V = \{(x, x) \mid x \in \mathbb{R}\}$ is closed, being the preimage of $\{0\}$ under the continuous map $(x, y) \mapsto x - y$, yet it is not a product of closed sets with more than one point. It is a union of the closed singletons $\{(x, x)\}$, but arbitrary unions of closed sets need not be closed, so that observation alone does not establish closedness.

Proposition 2.2.10: Product Of Hausdorff Spaces

Let X and Y be nonempty. Then X and Y are Hausdorff if and only if $X \times Y$ is Hausdorff.

***Proof*:**
Necessity. Suppose X and Y are Hausdorff and let $(x_1, y_1) \neq (x_2, y_2)$ in $X \times Y$. The points differ in at least one coordinate; say $x_1 \neq x_2$, the other case being symmetric. Choose disjoint open $U_1 \ni x_1$ and $U_2 \ni x_2$ in X. Then $U_1 \times Y$ and $U_2 \times Y$ are open, disjoint because $U_1 \cap U_2 = \emptyset$, and contain the two points.

Sufficiency. Suppose $X \times Y$ is Hausdorff, and recall the standing assumption that X and Y are nonempty. Fix $y_0 \in Y$ and let $x_1 \neq x_2$ in X. Then $(x_1, y_0) \neq (x_2, y_0)$, so there are disjoint open

W_1, W_2 in $X \times Y$ containing them. Shrinking to basis elements, choose $U_i \times V_i \subseteq W_i$ open with $(x_i, y_0) \in U_i \times V_i$. The sets U_1 and U_2 are open in X and contain x_1 and x_2; they are disjoint, since a common point x would give $(x, y_0) \in (U_1 \times V_1) \cap (U_2 \times V_2) \subseteq W_1 \cap W_2$, which is empty. Hence X is Hausdorff, and the argument for Y is symmetric, completing the proof.

Proposition 2.2.11: Subspace Of A Hausdorff Space Is Hausdorff

Let X be Hausdorff and let $A \subseteq X$ carry the subspace topology. Then A is Hausdorff.

***Proof*:**
Let $a_1 \neq a_2$ in A. Regarded as points of X they are distinct, so there are disjoint open U_1, U_2 in X with $a_i \in U_i$. By the definition of the subspace topology (section 2.1) the sets $U_1 \cap A$ and $U_2 \cap A$ are open in A, they contain a_1 and a_2, and they are disjoint because $U_1 \cap U_2 = \emptyset$, as we sought to show.

2.2.1 The Infinite Case

The definitions above were stated for finitely many factors, and the restriction was deliberate. Infinitely many factors admit two competing generalizations, and they do not agree.

Definition 2.2.12: Cross product of infinite topologies (Box Topology)

Let $\mathcal{I}$ index topological spaces X_i. The *box topology* on $\prod_{i \in \mathcal{I}} X_i$ has as basis all sets

$$\prod_{i \in \mathcal{I}} U_i$$

with U_i open in X_i for every i.

Definition 2.2.13: Cross product of infinite topologies (Product Topology)

Let $\mathcal{I}$ index topological spaces X_i. The *product topology* on $\prod_{i \in \mathcal{I}} X_i$ has as basis all sets $\prod_{i \in \mathcal{I}} U_i$ where U_i is open in X_i for each i, and $U_i = X_i$ for all but finitely many i.

2.2.14 Note The two definitions coincide when $\mathcal{I}$ is finite, since the finiteness condition is then vacuous; this is why definition 2.2.3 needed no such clause. For infinite $\mathcal{I}$ they differ, and the box topology is strictly finer: every basic product-open set is box-open, but $\prod_n (-1, 1) \subseteq \mathbb{R}^\omega$ is box-open and not product-open.

Which of the two deserves the name is settled by the universal property. The product topology retains it and the box topology does not: a map into a box product can have all its coordinate functions continuous without being continuous itself. Theorem 2.2.6 generalizes as follows.

Theorem 2.2.15: Continuous Maps On Product Topologies

Let $\prod_{i\in I} X_i$ carry the product topology and let $f\colon Y \to \prod_{i\in I} X_i$ have coordinates $f(y) = (f_i(y))_{i\in I}$, so that $f_i = \pi_i \circ f$:

$$\begin{tikzcd}
& \prod_{i\in I} X_i \arrow[d, "\pi_i"] \\
Y \arrow[ur, "f"] \arrow[r, "f_i"] & X_i
\end{tikzcd}$$

Then f is continuous if and only if every f_i is continuous.

***Proof*:**
Necessity. If f is continuous then so is each $f_i = \pi_i \circ f$, being a composite of continuous maps: the projections π_i are continuous by theorem 2.2.6.

Sufficiency. Suppose every f_i is continuous. The product topology of definition 2.2.13 is generated by the subbasis consisting of the sets $\pi_i^{-1}(U)$ with $i \in I$ and U open in X_i, so it suffices to check that the preimage under f of each such subbasis element is open: a map is continuous as soon as preimages of subbasis elements are open, since preimage commutes with unions and finite intersections. Now

$$f^{-1}\big(\pi_i^{-1}(U)\big) = (\pi_i \circ f)^{-1}(U) = f_i^{-1}(U)$$

which is open in Y because f_i is continuous. Hence f is continuous, completing the proof.

Example 2.2.16: Failure on the Box Topology
Consider $\mathbb{R}^\omega$, the countable product of copies of $\mathbb{R}$, and let $f\colon \mathbb{R} \to \mathbb{R}^\omega$ be

$$f(t) = (t, t, t, \dots)$$

Each coordinate function is $f_n(t) = t$, which is continuous, so f is continuous for the product topology by theorem 2.2.15. It is *not* continuous for the box topology. Take the box-open set

$$V = \prod_n \left(\frac{-1}{n}, \frac{1}{n}\right)$$

If $f^{-1}(V)$ were open in $\mathbb{R}$ it would contain an interval $(-\delta, \delta)$ about 0, giving $f\big((-\delta,\delta)\big) \subseteq V$ and hence, applying π_n,

$$(-\delta, \delta) \subseteq \left(\frac{-1}{n}, \frac{1}{n}\right)$$

for every n. This fails as soon as $n > 1/\delta$. So $f^{-1}(V) = \{0\}$, which is not open, and f is not continuous.

This is the same phenomenon met in section 1.1: allowing infinitely many constraints at once gives flexibility, and here it gives too much. Requiring all but finitely many factors to be the whole space is precisely what keeps the universal property intact.

Several results survive for both topologies, since their proofs use only basis elements. Closure is one of them.

Proposition 2.2.17: Closure Of A Product Is Product Of Closure

Let $\{X_j\}_{j\in J}$ be topological spaces, let $\prod_{j\in J} X_j$ carry the product topology, and let $A_j \subseteq X_j$ for each j. Then

$$\overline{\prod_{j\in J} A_j} = \prod_{j\in J} \overline{A_j}$$

***Proof*:**

The inclusion $\prod \overline{A_j} \subseteq \overline{\prod A_j}$. Let $x = (x_j) \in \prod_j \overline{A_j}$ and let U be a basic open set of the product topology containing x, so $U = \prod_j U_j$ with U_j open and $U_j = X_j$ for all but finitely many j. For each j the point x_j lies in $\overline{A_j}$, so $U_j \cap A_j \neq \emptyset$ by theorem 1.4.15; choose $a_j \in U_j \cap A_j$. Then $(a_j) \in U \cap \prod_j A_j$, so U meets the product, and theorem 1.4.15 places x in its closure.

The reverse inclusion. Let $x \in \overline{\prod_j A_j}$ and fix an index k. Let U_k be open with $x_k \in U_k$. Then $\pi_k^{-1}(U_k)$ is open and contains x, so it meets $\prod_j A_j$; a point of the intersection has k-th coordinate in $U_k \cap A_k$. Hence every open set around x_k meets A_k, so $x_k \in \overline{A_k}$ by theorem 1.4.15. As k was arbitrary, $x \in \prod_j \overline{A_j}$, completing the proof.

Exercise 2.2.1

1. Show that $\pi_1 \colon \mathbb{R}^2 \to \mathbb{R}$ is open but not closed, and exhibit a closed set whose image is not closed.
2. Show that the box topology on $\mathbb{R}^\omega$ is strictly finer than the product topology by exhibiting a set open in one and not the other, and show that the two agree on $\mathbb{R}^n$ for every finite n.
3. Show that $\prod_i A_i$ is closed in $\prod_i X_i$ whenever each A_i is closed in X_i, and that the corresponding statement for open sets fails in the product topology.
4. Show that a product of *nonempty* spaces is Hausdorff if and only if every factor is Hausdorff, and explain where nonemptiness is used.

2.3 Limits over Directed Systems

A product treats its factors as unrelated. Very often they are not: the factors come with maps between them, and the objects of interest are the families of points that respect those maps. A real number is determined by its decimal truncations, each of which determines the shorter ones; a p-adic integer is determined by its residues modulo $p, p^2, p^3, \ldots$, each of which determines the earlier ones. The limit of such a system is the construction that turns a family of approximations into a single space, and it is the reason products are worth having in the infinite case at all.

The construction is the categorical *limit* of a diagram indexed by a directed set. The index set has to be one along which approximations can be compared, which is exactly the directed condition.

2.3.1 Terminology The older literature, and much of the arithmetic literature still, calls the object constructed below a *projective limit* or an *inverse limit*, and writes it $\varprojlim$. This series uses the bare word *limit* and names the shape of the diagram separately, so what follows is "the limit of an inverse system" rather than "the inverse limit". The dual construction, over a directed system with maps pointing forwards, is the *colimit*, written $\varinjlim$; it is not needed here.

Definition 2.3.2: Directed Set

A *directed set* is a pair $(J, \leq)$ where $\leq$ is a reflexive, transitive relation on J such that for all $i, j \in J$ there exists $k \in J$ with $i \leq k$ and $j \leq k$.

Definition 2.3.3: Inverse System

An *inverse system* of topological spaces over a directed set $(J, \leq)$ consists of a space X_j for each $j \in J$ together with a continuous map

$$f_{ij} \colon X_j \to X_i \qquad \text{whenever } i \leq j$$

subject to $f_{ii} = \mathrm{id}_{X_i}$ and $f_{ij} \circ f_{jk} = f_{ik}$ whenever $i \leq j \leq k$. The maps f_{ij} are the *bonding maps*.

The bonding maps point *backwards*, from the finer index to the coarser one: a residue modulo p^3 determines a residue modulo p^2, not the other way round. An element of the limit is a coherent choice across the whole system.

Definition 2.3.4: Limit of an Inverse System

Let $\{X_j, f_{ij}\}$ be an inverse system over $(J, \leq)$. Its *limit* is the subspace

$$\varprojlim_{j \in J} X_j = \left\{ (x_j) \in \prod_{j \in J} X_j \;\middle|\; f_{ij}(x_j) = x_i \text{ whenever } i \leq j \right\}$$

of the product, carrying the subspace topology of the product topology, with the maps π_i obtained by restricting the projections.

Two things deserve emphasis. The limit is a subspace of the product, so everything already proved about products applies to it; and the topology is the *product* topology, not the box topology, which is what makes the next result true.

Proposition 2.3.5: Universal Property of the Limit

Let $\{X_j, f_{ij}\}$ be an inverse system and let Y be a space. Suppose given continuous maps $g_j \colon Y \to X_j$ which are *compatible*, meaning $f_{ij} \circ g_j = g_i$ whenever $i \leq j$. Then there is a unique continuous map

$$g\colon Y \to \varprojlim_j X_j \qquad \text{with} \qquad \pi_j \circ g = g_j \text{ for every } j$$

***Proof*:**
Uniqueness is forced: any such g must satisfy $g(y) = (g_j(y))_{j \in J}$, since the j-th coordinate of $g(y)$ is $\pi_j(g(y)) = g_j(y)$.

For existence, define $g(y) = (g_j(y))_j$. This lands in the limit: compatibility gives $f_{ij}(g_j(y)) = g_i(y)$ for all $i \leq j$, which is exactly the coherence condition. As a map into the product, g is continuous by theorem 2.2.15, since each coordinate function g_j is continuous; and a continuous map into the product whose image lies in a subspace is continuous into that subspace with its subspace topology (section 2.1). Composing with the restricted projections returns g_j, so g has the required property.

The property determines the limit up to a unique homeomorphism, by the usual argument: two spaces with the same universal property receive unique maps to each other whose composites must, by uniqueness, be the identities.

Example 2.3.6: Limits over Directed Systems

1. *The p-adic integers.* Fix a prime p, let $J = \mathbb{N}$ with its usual order, let $X_n = \mathbb{Z}/p^n\mathbb{Z}$ with the discrete topology, and let $f_{mn} \colon \mathbb{Z}/p^n\mathbb{Z} \to \mathbb{Z}/p^m\mathbb{Z}$ for $m \leq n$ be reduction. The limit is $\mathbb{Z}_p$, the ring of p-adic integers: a p-adic integer is precisely a coherent sequence of residues. The topology it inherits is the one induced by the p-adic metric.

2. *The Cantor set.* Take $X_n = \{0,1\}^n$ discrete, with f_{mn} truncation to the first m coordinates. The limit is $\{0,1\}^{\mathbb{N}}$, homeomorphic to the Cantor set.

3. *A constant system.* If every $X_j = X$ and every bonding map is id_X, the limit is the diagonal copy of X inside X^J, so nothing new is produced. Such limits are interesting exactly when the bonding maps forget information.

4. *Real numbers from truncations.* Let $X_n = \mathbb{Z}$ with $f_{mn}(a) = \lfloor a/10^{n-m} \rfloor$. A coherent sequence records the digits of a real number, which is the sense in which $\mathbb{R}$ is assembled from its truncations.

Such limits are the standard vehicle for profinite objects: profinite groups, profinite completions, and the Galois groups of infinite extensions are all limits of finite discrete systems, and the topology they carry is the one defined here. The property that makes them usable is that a limit of nonempty compact spaces is nonempty, which needs compactness and so waits for theorem 3.2.45.

Exercise 2.3.1

1. Show that $(\mathbb{N}, \leq)$ and the set of finite subsets of a set S ordered by inclusion are directed, and that $(\mathbb{Z}, |)$ under divisibility is directed while $(\mathbb{Z}, <)$ is not reflexive and so is not.
2. Let every bonding map f_{ij} be a homeomorphism. Show that $\varprojlim_j X_j$ is homeomorphic to any one X_i.
3. Exhibit an inverse system of nonempty spaces whose limit is empty.
4. Show that the limit of an inverse system of Hausdorff spaces is Hausdorff, without assuming compactness.
5. Show that $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is uncountable, by exhibiting an injection from $\{0, 1\}^{\mathbb{N}}$ into it.

2.4 Quotient Topology

Subspaces are the sub-objects of topology; quotients are the dual construction. The category Top behaves unusually here. In Grp a quotient requires a *congruence relation*, an equivalence relation compatible with the algebraic structure, so that $[x \cdot y] = [x] \cdot [y]$ is well defined. No such compatibility is needed for topological quotients: an arbitrary equivalence relation suffices. That flexibility has a price. Quotients interact badly with the other constructions, and neither subspaces nor products of quotient maps are quotient maps without extra hypotheses; theorem 2.4.9 gives the hypothesis that repairs the first case and theorem 3.4.7 the one that repairs the second.

2.4.1 Basic Properties and Definitions

The guiding picture is gluing. Identifying the two endpoints of an interval gives a circle; identifying opposite edges of a rectangle in the appropriate way gives a torus; collapsing the whole boundary of a disc to a point gives a sphere. The task is to say what topology the glued object should carry.

That is governed by the map performing the gluing.

Definition 2.4.1: Quotient Map

Let X and Y be topological spaces and let $p : X \to Y$ be surjective. Then p is a *quotient map* if

$$\text{a subset } U \subseteq Y \text{ is open in } Y \text{ if and only if } p^{-1}(U) \text{ is open in } X$$

The condition is strictly stronger than continuity, which supplies only the implication from U open to $p^{-1}(U)$ open. The converse direction is the content: an open preimage forces the set downstairs to have been open. The condition is sometimes called *strong continuity*. Read geometrically, room to move in the glued space must already have been present upstairs.

A quotient map is close to a homeomorphism, and is one exactly when it is injective.

The closed-set formulation follows by the usual complementation $p^{-1}(Y \setminus B) = X \setminus p^{-1}(B)$: a surjective continuous p is a quotient map if and only if a set is closed in Y precisely when its preimage is closed.

A second description is often more usable. Given $f : X \to Y$ and $A \subseteq X$, call A *saturated* with respect to f if A contains every fibre it meets, equivalently $f^{-1}(f(A)) = A$. Then p is a quotient map if and only if it is continuous, surjective, and carries saturated open sets to open sets, or equivalently saturated closed sets to closed sets.

Two convenient sufficient conditions: a surjective continuous map that is *open*, or one that is *closed*, is a quotient map. Neither is necessary, and quotient maps exist that are neither open nor closed, since the requirement on preimages is genuinely weaker than the requirement on images.

The analogy with quotient groups is close. Continuity is the structure-preserving half; the extra condition ensures the map does more than project onto a smaller space, by forcing every open set upstairs that is saturated to descend.

Example 2.4.2: Gluing

1. Let $p : [0,1] \cup [2,3] \to [0,2]$, both subspaces of $\mathbb{R}$, glue the two intervals:

$$p(x) = \begin{cases} x & \text{for } x \in [0,1] \\ x-1 & \text{for } x \in [2,3] \end{cases}$$

   It is surjective, which is the gluing. It is closed: a closed basic set $[a,b]$ of the domain has image a closed interval. It is continuous, since the preimage of a closed interval of the co-domain is a closed interval or a union of two of them. Being closed, continuous and surjective, it is a quotient map.

   It is not *open*: the set $[0,1]$ is open in the subspace $[0,1] \cup [2,3]$ but its image $[0,1]$ is not open in $[0,2]$. Removing the gluing point destroys the quotient property. On $[0,1) \cup [2,3]$ the map is still continuous and surjective, but $[2,3]$ is open in the domain and saturated, while $p([2,3]) = [1,2]$ is not open in $[0,2]$.

2. Projections are quotient maps. The map $\pi_1 : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is surjective, continuous and open by lemma 2.2.7.

   It is not closed: $C = \{(x,y) \mid xy = 1\}$ is closed in $\mathbb{R}^2$ while $\pi_1(C) = \mathbb{R} \setminus \{0\}$ is not closed. Restricting to $C \cup \{(0,0)\}$ gives a continuous surjection onto $\mathbb{R}$ for which $\{(0,0)\}$ is open in the subspace and saturated, yet has non-open image.

3. More generally, a retraction is a quotient map. Let $A \subseteq X$ and let $r : X \to A$ be continuous with $r(a) = a$ for every $a \in A$.

***Proof*:**
It suffices to prove the general statement: if $r : X \to Y$ and $f : Y \to X$ are continuous with $r \circ f = \mathrm{id}_Y$, then r is a quotient map.

Such an r is surjective, since $r(f(y)) = y$ for every y, and it is continuous by hypothesis. Let $U \subseteq Y$ have $r^{-1}(U)$ open. Continuity of f makes $f^{-1}(r^{-1}(U))$ open, and

$$f^{-1}(r^{-1}(U)) = (r \circ f)^{-1}(U) = \mathrm{id}^{-1}(U) = U$$

so U is open. Hence r is a quotient map. For a retraction onto A the inclusion $f : A \to X$, $f(a) = a$, is the required right inverse, as we sought to show.

4. Let $\pi_1 : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be the first projection, let $A = \{(x, y) \mid x \geq 0 \text{ or } y = 0\}$, and let $q : A \to \mathbb{R}$ restrict π_1. Then q is a quotient map that is neither open nor closed.

***Proof*:**
The map q is surjective, since $(x, 0) \in A$ for every x, and continuous as a restriction.

It is not open: the set $A \cap \big((-1, 1) \times (-1, 1)\big)$ is open in A and contains $(-\frac{1}{2}, 0)$, but every neighbourhood of $-\frac{1}{2}$ in $\mathbb{R}$ meets points whose only preimages in A lie on the line $y = 0$; the image is $(-1, 1)$, which happens to be open here, so instead take the open set $A \cap \big((-1, 0) \times \mathbb{R}\big)$, whose image is $(-1, 0)$. It is not closed: the closed set $\{(x, 1/x) \mid x > 0\} \subseteq A$ has image $(0, \infty)$.

That q is nonetheless a quotient map follows because it admits a continuous right inverse, $s(x) = (x, 0)$, which lands in A for every x and satisfies $q \circ s = \mathrm{id}_{\mathbb{R}}$; the previous item then applies, completing the proof.

A quotient map determines a topology, and it is the only one that works.

Definition 2.4.3: Quotient Topology

If X is a space, A a *set*, and $p : X \to A$ a surjective map, then there is exactly one topology $\mathcal{T}$ on A for which p is a quotient map, namely

$$\mathcal{T} = \big\{U \subseteq A \;\big|\; p^{-1}(U) \text{ is open in } X\big\}$$

It is the *quotient topology* induced by p.

That this collection is a topology follows because preimage commutes with unions and intersections, and that it is the unique one is immediate from the defining condition.

Example 2.4.4: Quotient Topology

Take $p : \mathbb{R} \to \{a, b, c\}$ with

$$p(x) = \begin{cases} a & \text{if } x > 0 \\ b & \text{if } x < 0 \\ c & \text{if } x = 0 \end{cases}$$

The induced quotient topology is

$$\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{a, b, c\}\}$$

since $p^{-1}(\{a\}) = (0, \infty)$ and $p^{-1}(\{b\}) = (-\infty, 0)$ are open, while any set containing c has preimage containing 0, which is open only if the set is everything.

The same construction is usually presented through partitions, which is where the gluing picture becomes literal.

Definition 2.4.5: Quotient Space

Let X be a topological space and let X^* be a partition of X into disjoint subsets whose union is X. Let $p : X \to X^*$ be the surjection carrying each point of X to the element of X^* containing it. With the quotient topology induced by p, the space X^* is the *quotient space* of X.

The categorical reading explains why this is the right topology. A partition gives a surjection of sets $X \to X^*$; the question is which topologies on X^* make it continuous. The indiscrete topology certainly does, since only $\emptyset$ and X^* must be pulled back. The useful question is how *fine* a topology can be taken while preserving continuity, and the answer is exactly the quotient topology: declaring U open when $p^{-1}(U)$ is open admits every set it can, and anything finer would contain a set whose preimage is not open. So the quotient topology is the finest topology making p continuous, which is the dual statement to theorem 2.2.6 describing the product topology as the coarsest making the projections continuous.

The upshot is that a surjection out of X determines the topology downstream.

Example 2.4.6: More Quotient Topologies

1. Let X be the closed unit disc
$$\left\{(x, y) \;\middle|\; x^2 + y^2 \leq 1\right\}$$
in $\mathbb{R}^2$, and let X^* consist of the one-point sets $\{(x, y)\}$ with $x^2 + y^2 < 1$ together with the single set $S^1 = \left\{(x, y) \;\middle|\; x^2 + y^2 = 1\right\}$. Collapsing the boundary circle to a point gives a quotient space homeomorphic to the 2-sphere
$$S^2 = \left\{(x, y, z) \;\middle|\; x^2 + y^2 + z^2 = 1\right\}$$
The homeomorphism is induced by any continuous surjection $X \to S^2$ that is injective on the open disc and collapses the boundary, for instance wrapping each radius onto a meridian; theorem 2.4.8 then supplies the induced map and corollary 3.2.20 upgrades it to a homeomorphism, X^* being compact and S^2 Hausdorff.

2. Let X be the square $[0,1] \times [0,1]$ and partition it into the singletons of interior points, the pairs $\{(x,0),(x,1)\}$ and $\{(0,y),(1,y)\}$ along the edges, and the four corners

$$\{(0,0),(1,0),(1,1),(0,1)\}$$

as a single set. The quotient space is homeomorphic to the torus.

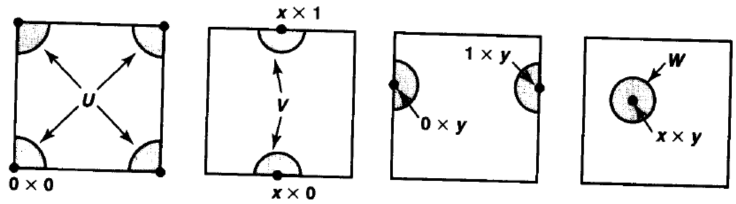

Figure 22.5

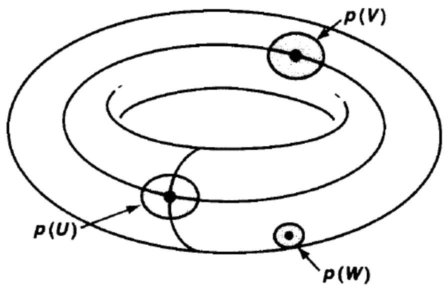

Figure 2.3: The torus as a quotient of the square, with opposite edges identified.

2.4.7 Note Since partitions correspond to equivalence relations, quotient maps can be specified relationally, which is usually the most convenient form: take a surjection f and define it by $f(a) = f(b) \Leftrightarrow a\,R\,b$. For instance, if $f : \mathbb{R} \to S$ is a surjection of sets with $f(a) = f(b) \Leftrightarrow a - b \in \mathbb{Q}$, the induced quotient topology on S is the trivial topology. Indeed a nonempty saturated open $U \subseteq \mathbb{R}$ contains a full $\mathbb{Q}$-coset, hence is dense; its complement is closed and also saturated, so if that complement were nonempty it would likewise contain a dense coset and, being closed, would equal $\mathbb{R}$, contradicting $U \neq \emptyset$. So $U = \mathbb{R}$.

Products came with a criterion for continuity of a map *into* a product. The dual statement for quotients is a criterion for continuity of a map *out of* a quotient.

Theorem 2.4.8: Universal Property Of Quotient Maps

Let $p : X \to Y$ be a quotient map, let Z be a space, and let $g : X \to Z$ be constant on each fibre $p^{-1}(\{y\})$. Then g induces a unique map $f : Y \to Z$ with $f \circ p = g$. The induced f is continuous if and only if g is continuous, and f is a quotient map if and only if g is a quotient map.

$$\begin{tikzcd} X \arrow[d, "p"'] \arrow[dr, "g"] & \\ Y \arrow[r, dotted, "f"'] & Z \end{tikzcd}$$

***Proof*:**

Existence and uniqueness of f. Since p is surjective and g is constant on each fibre $p^{-1}(\{y\})$, setting $f(y)$ to be the common value of g on $p^{-1}(\{y\})$ is well defined, and it is the only assignment satisfying $f \circ p = g$.

Continuity of f. Let $W \subseteq Z$ be open. Then $p^{-1}(f^{-1}(W)) = (f \circ p)^{-1}(W) = g^{-1}(W)$, which is open because g is continuous. Since p is a quotient map, a set of Y is open exactly when its preimage is open, so $f^{-1}(W)$ is open. Hence f is continuous.

Conversely, if f is continuous then $g = f \circ p$ is continuous as a composite, so continuity of g and of f are equivalent.

The quotient-map clause. Suppose f is a quotient map. Then $g = f \circ p$ is a composite of quotient maps, hence a quotient map, by the identity $p^{-1}(f^{-1}(U)) = g^{-1}(U)$: the set U is open iff $f^{-1}(U)$ is open iff $g^{-1}(U)$ is open.

Conversely suppose g is a quotient map. It is surjective, and $g = f \circ p$, so f is surjective. Let $U \subseteq Z$ have $f^{-1}(U)$ open in Y. Continuity of p makes $p^{-1}(f^{-1}(U)) = g^{-1}(U)$ open in X, and since g is a quotient map this forces U open in Z. Together with continuity of f, already established, this makes f a quotient map, completing the proof.

The practical value is that maps out of a glued space are built upstairs. To produce a continuous $f\colon S^1 \to Z$, it suffices to produce a continuous $g\colon I \to Z$ taking the same value at both endpoints, since g is then constant on the fibres of the quotient map $p\colon I \to S^1$ and descends.

$$\begin{tikzcd} I \arrow[d, "p"'] \arrow[dr, "g"] & \\ S^1 \arrow[r, dotted, "f"'] & Z \end{tikzcd}$$

2.4.2 Relationship With Previous Concepts

Quotient maps preserve much less than one might hope. Injectivity is what a homeomorphism has and a quotient map lacks, and most of the failures trace back to that.

Subspaces are the first casualty. If $p : X \to Y$ is a quotient map and $A \subseteq X$ a subspace, the restriction $p : A \to p(A)$ need not be a quotient map. The following recovers the situation under hypotheses.

Theorem 2.4.9: Subspaces And Quotient Maps

Let $p : X \to Y$ be a quotient map, let A be a subspace of X saturated with respect to p, and let $q : A \to p(A)$ be the restriction of p.

1. If A is open or closed in X, then q is a quotient map.
2. If p is an open map or a closed map, then q is a quotient map.

***Proof*:**

1. Suppose A is open (the closed case is symmetric). Let $V \subseteq p(A)$ with $q^{-1}(V)$ open in A; since A is open in X, $q^{-1}(V)$ is open in X. Saturation gives $p^{-1}(V) = q^{-1}(V)$, because every point of X mapping into $V \subseteq p(A)$ already lies in A. So $p^{-1}(V)$ is open in X, and since p is a quotient map V is open in Y, hence open in $p(A)$. The converse implication is continuity of q, so q is a quotient map.
2. Suppose p is open (the closed case is symmetric). Let $V \subseteq p(A)$ with $q^{-1}(V)$ open in A, say $q^{-1}(V) = U \cap A$ with U open in X. Then $p(U \cap A) = V$ by saturation, and $V = p(U) \cap p(A)$: the inclusion $\subseteq$ is clear, and conversely a point $p(u) \in p(A)$ with $u \in U$ has some preimage in A, which by saturation forces $u \in A$. Since p is open, $p(U)$ is open in Y, so V is open in $p(A)$. Again the converse is continuity of q.

In both cases q is a quotient map, as we sought to show.

Composition behaves well, directly from $p^{-1}(q^{-1}(U)) = (q \circ p)^{-1}(U)$: a composite of quotient maps is a quotient map. Products do not: the product of two quotient maps need not be a quotient map.

Separation also degrades. A quotient of a Hausdorff space need not be Hausdorff, and the failure is not exotic.

Example 2.4.10: A Hausdorff Space With Non-Hausdorff Quotient

Take $X = \mathbb{R}$, which is Hausdorff, and partition it into $\mathbb{Q}$-cosets, so that $x \sim y$ exactly when $x - y \in \mathbb{Q}$. As shown in the note above, the quotient carries the trivial topology, which is not Hausdorff once the quotient has more than one point, and it does, since $\mathbb{R}/\mathbb{Q}$ is uncountable.

A smaller example is the line with two origins: take two copies of $\mathbb{R}$ and identify x in one with x in the other for every $x \neq 0$. The two origins cannot be separated, since every neighbourhood of each contains a punctured interval about 0, and the two punctured intervals meet.

The collapse is not gradual either: a quotient of a Hausdorff space can fail even the weakest separation axiom. Let $S = \{0, 1\}$ carry the Sierpiński topology $\{\emptyset, \{1\}, S\}$, and define $p\colon \mathbb{R} \to S$ by $p(x) = 1$ for $x > 0$ and $p(x) = 0$ for $x \leq 0$. Then $p^{-1}(\{1\}) = (0, \infty)$ is open and $p^{-1}(\{0\}) = (-\infty, 0]$ is not, so a subset of S is open exactly when its preimage is: p is a quotient map. Its domain is Hausdorff and its image is not even T_1, since $\{1\}$ is not closed.

Exercise 2.4.1

1. Show that $\mathbb{R}/\mathbb{Q}$ with the quotient topology is the trivial topology.
2. Show that a continuous surjection that is open, or closed, is a quotient map.
3. Show that a composite of quotient maps is a quotient map.
4. Let $p\colon X \to Y$ be a quotient map. Show that Y is discrete if and only if every fibre of p is open.
5. A map $f\colon X \to Y$ is a *local homeomorphism* if every $x \in X$ has an open neighbourhood carried homeomorphically onto an open subset of Y. Let L be the quotient of two disjoint copies of $\mathbb{R}$ obtained by identifying the two copies of each *nonzero* real. Show that the natural map $L \to \mathbb{R}$ is a local homeomorphism, that L is not Hausdorff, and conclude that a local homeomorphism transports Hausdorffness in neither direction.

2.5 Weak and Final Topologies

The subspace, product and quotient topologies are three instances of two general constructions. Given a family of maps, one can ask for the coarsest topology making them all continuous, or for the finest. The first is the *initial* or *weak* topology, the second the *final* topology, and between them they subsume everything built so far.

Definition 2.5.1: Weak [initial] Topology

Let $F = \{f_\alpha : X \to Y_\alpha\}_{\alpha\in A}$ be a collection of maps from a set X to topological spaces Y_α. The coarsest topology on X making every f_α continuous is the *weak topology* generated by F. It has as subbasis the sets $f_\alpha^{-1}(U)$ for $U \subseteq Y_\alpha$ open and $\alpha \in A$.

The dual question, which topology on X is the *finest* making all the maps continuous, is uninteresting: the discrete topology always works. Minimality is what carries content, and it is well posed because the intersection of any family of topologies making the maps continuous again makes them continuous, so a coarsest one exists. Equivalently, if $\mathcal{T}'$ is any topology making every member of F continuous, then $\mathcal{T} \subseteq \mathcal{T}'$ for the weak topology $\mathcal{T}$.

Example 2.5.2: Weak Topology

1. Let $\{X_\alpha\}_{\alpha\in A}$ be topological spaces and $X = \prod_{\alpha\in A} X_\alpha$, with $\Pi = \{\pi_\alpha : X \to X_\alpha\}$ the projections. The weak topology generated by Π is *precisely* the product topology, which is the content of theorem 2.2.6.
2. Let $A \subseteq X$ and let $\iota : A \to X$ be the inclusion. The weak topology generated by ι is the *subspace* topology, since $\iota^{-1}(U) = U \cap A$.
3. Let X be a normed space over a topological field $\mathbb{F}$, so that the norm induces a metric and hence a topology. Let $X^* := \mathrm{Hom}(X, \mathbb{F})$ be the dual, the continuous linear maps $X \to \mathbb{F}$.

> The weak topology generated by X^* is the *weak topology* of functional analysis. A net converges in it, $x_\alpha \to x$, exactly when $f(x_\alpha) \to f(x)$ for every $f \in X^*$, written $x_\alpha \xrightarrow{w} x$. For infinite-dimensional X it is strictly coarser than the norm topology.

The construction imposes no conditions on the maps or the target spaces, and that is where its usefulness lies: the family F can be chosen to suit the problem. Topologies on function spaces are built this way, on spaces of smooth functions on a compact set, of bounded continuous functions, or of bounded functions, and such spaces are the basic objects of functional analysis.

Reversing the direction of the maps gives the dual construction.

Definition 2.5.3: Final Topologies

Let Y be a set, let $\{X_\alpha\}_{\alpha\in A}$ be topological spaces, and let $F = \{f_\alpha : X_\alpha \to Y\}$ be maps into Y. The finest topology on Y making every f_α continuous is the *final topology*. Explicitly, $U \subseteq Y$ is open if and only if $f_\alpha^{-1}(U)$ is open for every $\alpha \in A$.

Existence again follows from a closure property, this time because the collection of such U is itself a topology. The asymmetry between the two constructions is that a map is easier to make continuous by coarsening its codomain and refining its domain, so minimality is the interesting condition upstream and maximality downstream.

> **Example 2.5.4: Final Topologies**
> Let X be a topological space, let X^* be a partition of X, and let $p : X \to X^*$ be the associated surjection. The final topology generated by p is the *quotient topology* of definition 2.4.3.

The terminology is slightly lopsided. The initial topology is called weak, but the final topology is not called strong, because that name is already taken: on a normed space the *strong topology* is the one generated by the balls $B_r(x)$ for $x \in X$ and $r > 0$, that is, the norm topology itself. The contrast between strong and weak belongs properly to functional analysis, where a norm topology is preferred when available but often is not, as on spaces of smooth functions or general infinite-dimensional vector spaces. The weak topology is then constructed instead, typically from a countable family of seminorms; the systematic treatment belongs to [10], with [6] as the standard external account.

Exercise 2.5.1

1. Show that the weak topology generated by a single injective map $f\colon X \to Y$ is $\{f^{-1}(U) \mid U \text{ open in } Y\}$, and that it makes f an embedding onto its image.
2. Show that the weak topology generated by the empty family of maps is the trivial topology, and the final topology generated by the empty family is the discrete topology.
3. Let $F = \{f_\alpha\colon X \to Y_\alpha\}$ and give X the weak topology. Show that a map $g\colon Z \to X$ is continuous if and only if every $f_\alpha \circ g$ is continuous, and identify this as the pattern shared by theorem 2.2.15 and theorem 2.4.8.
4. The subspace topology is a weak topology. Is it also a *final* topology, for the inclusion or for any family of maps into A? Settle the question for $A = \{0\} \subseteq \mathbb{R}$.

2.6 Metric Spaces

A metric space is a different kind of object from a topological space. A topology is a set X together with a distinguished family of subsets $\mathcal{T}_X$; a metric space is a set X together with a distance function d_X. The two live in different categories.[1] They are studied together because every metric induces a topology, so metric spaces can be treated by topological means, and because most spaces met in analysis arise this way.

The translation is lossy in one direction. Every metric space is Hausdorff as a topological space, which is proved below, but purely metric notions do not survive a homeomorphism. Boundedness is one: a continuous bijection can carry a bounded space onto an unbounded one. Completeness is another: $(-1, 1)$ is homeomorphic to $\mathbb{R}$, yet the first is incomplete and the second complete. Such properties belong to the metric, not to the topology it induces.

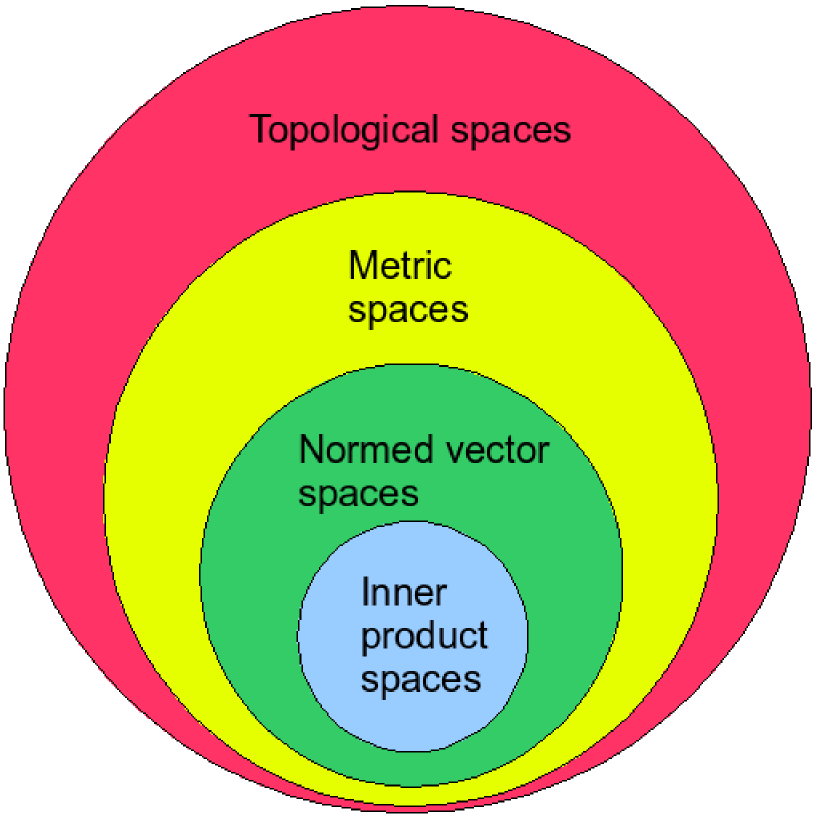

Figure 2.4: The hierarchy of spaces: metrizable inside Hausdorff inside topological.

[1]Metric spaces are the objects of Met, topological spaces those of Top.

2.6.1 Basic Definitions and Examples

Definition 2.6.1: Metric

Let X be a set. A *metric* on X is a function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

such that

1. $d(x, y) = d(y, x)$
2. $d(x, y) + d(y, z) \geq d(x, z)$
3. $d(x, y) = 0$ if and only if $x = y$

2.6.2 Note The biconditional in the third axiom is doing real work. Dropping the reverse implication would admit $d(x, y) = 0$ for all x, y, which induces the indiscrete topology; that space is not Hausdorff, and the Hausdorff property proved below would fail.

For $\epsilon > 0$ write

$$B_d(x, \epsilon) = \left\{ y \middle| d(x, y) < \epsilon \right\}$$

Definition 2.6.3: Metric Topology

If d is a metric on X, the *metric topology* on X is the topology generated by the basis

$$\mathcal{B} = \left\{ B_d(x, \epsilon) \middle| x \in X, \epsilon > 0 \right\}$$

That this is the right topology to attach to a metric has a clean characterization: it is the coarsest topology on X making $d\colon X \times X \to \mathbb{R}$ continuous. So the passage from metric to topology is not a convention but the minimal choice making the distance function itself topological.

Lemma 2.6.4: Metric topology is Hausdorff

Every metric topology is Hausdorff.

***Proof*:**
Let $x_1 \neq x_2$ and put $r = d(x_1, x_2)$, which is positive by the third metric axiom. Take $\epsilon = r/2$. If some y lay in $B_\epsilon(x_1) \cap B_\epsilon(x_2)$ then the triangle inequality would give

$$r = d(x_1, x_2) \leq d(x_1, y) + d(y, x_2) < \tfrac{r}{2} + \tfrac{r}{2} = r$$

which is impossible. So the two balls are disjoint open sets separating x_1 from x_2, as we sought to show.

A topological space may be induced by many different metrics, and it is often the topology rather than the particular distance function that matters. Two terms keep the distinction.

Definition 2.6.5: Metrizable Topologies

A topological space X is *metrizable* if there exists a metric on X inducing its topology.

Definition 2.6.6: Metric Space

A *metric space* is a metrizable topological space together with a *specific* metric d inducing its topology, written (X, d).

2.6.7 Note Elsewhere a metric space is often defined as a set with a metric, with no topology mentioned. The two accounts agree: such a set carries the induced topology and is metrizable by construction.

Example 2.6.8: Metric Spaces

Several metrics on $\mathbb{R}^2$, all of which induce the standard topology except where noted.

1. $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, the Euclidean metric.
2. $|x_1 - x_2| + |y_1 - y_2|$, the taxicab metric.
3. $\max(|x_1 - x_2|, |y_1 - y_2|)$, the supremum metric.
4. $\min\left(1, \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}\right)$, the standard bounded metric of theorem 2.6.17.
5. $d(x, y) = 0$ if $x = y$ and 1 otherwise, the discrete metric, which induces the *discrete* topology rather than the standard one.

The last is worth contrasting with the fourth: both are bounded by 1, but truncating a metric preserves its topology while flattening it to a two-valued function does not.

A related observation: adding a constant gap to a metric collapses it to the discrete one. For the Euclidean metric on $\mathbb{R}$ and fixed $\epsilon > 0$, set

$$\overline{d}_\epsilon(x, y) = \begin{cases} d(x, y) + \epsilon & x \neq y \\ 0 & x = y \end{cases}$$

Then $B_{\overline{d}_\epsilon}(x, \epsilon/2) = \{x\}$, so every singleton is open and the induced topology is discrete.

Lemma 2.6.9: Subspace Of Metric Space Is Metric Space

If (X, d) is a metric space and $Y \subseteq X$, then $(Y, d|_Y)$ is a metric space, and the subspace topology on Y coincides with the metric topology induced by $d|_Y$

***Proof*:**
The restriction $d|_Y$ inherits symmetry, the triangle inequality, and vanishing exactly on the diagonal, so it is a metric.

For the topologies, it suffices to compare bases. For $g \in Y$ and $\epsilon > 0$,

$$B_d(g, \epsilon) \cap Y = B_{d|_Y}(g, \epsilon)$$

since both sides consist of the points of Y within ϵ of g. The left side is a basic open set of the subspace topology by theorem 2.1.1, and the right side a basic open set of the metric topology of $d|_Y$, so the two topologies have a common basis, as we sought to show.

Definition 2.6.10: Continuity For Metric Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces. Then $f : X \to Y$ is continuous if and only if

$$\forall x \in X,\ \forall \epsilon > 0,\ \exists \delta > 0 \text{ such that } d_X(x, x') < \delta \Rightarrow d_Y(f(x), f(x')) < \epsilon$$

This is the epsilon-delta definition of definition 1.3.11, now with the metric supplying the balls directly.

Some notions depend on the metric rather than on the topology it induces.

Definition 2.6.11: Diameter

For $A \subseteq X$, the diameter of A is

$$\operatorname{diam}(A) = \sup\{d(a_1, a_2) \mid a_1, a_2 \in A\}$$

Definition 2.6.12: Bounded

A metric space (X, d) is *bounded* if $\operatorname{diam}(X) < \infty$

A second such notion measures how far a point sits from a whole set. It will be used in several later arguments, because it converts a set into a real-valued function without any choices.

Definition 2.6.13: Distance To A Set

Let (X, d) be a metric space and $A \subseteq X$ nonempty. The *distance from x to A* is

$$d(x, A) = \inf\{d(x, a) \mid a \in A\}$$

Lemma 2.6.14: The Distance Function Is Continuous

Let $A \subseteq X$ be nonempty. Then for all $x, y \in X$,

$$|d(x, A) - d(y, A)| \leq d(x, y)$$

so $x \mapsto d(x, A)$ is a continuous map $X \to \mathbb{R}$. Moreover $d(x, A) = 0$ if and only if $x \in \overline{A}$.

***Proof*:**
For any $a \in A$ the triangle inequality gives $d(x, a) \leq d(x, y) + d(y, a)$, and taking the infimum over a on the left,

$$d(x, A) \leq d(x, y) + d(y, a)$$

for every $a \in A$. Now taking the infimum over a on the right yields $d(x, A) \leq d(x, y) + d(y, A)$, that is, $d(x, A) - d(y, A) \leq d(x, y)$. Swapping x and y gives the reverse inequality, and the two together give the stated bound on the absolute value. Continuity follows at once: given $\epsilon > 0$, the choice $\delta = \epsilon$ works at every point, so the criterion of definition 2.6.10 is met.

For the last claim, $d(x, A) = 0$ says that every $\epsilon > 0$ admits an $a \in A$ with $d(x, a) < \epsilon$, which is exactly the statement that every ball about x meets A. Since the balls form a basis for the metric topology, theorem 1.4.15 makes that equivalent to $x \in \overline{A}$, as claimed.

Definition 2.6.15: Uniform Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is *uniformly continuous* if

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that } \forall x, x' \in X,\ \ d_X(x, x') < \delta \Rightarrow d_Y(f(x), f(x')) < \epsilon$$

The only difference from definition 2.6.10 is the order of the quantifiers: ordinary continuity allows δ to depend on both ϵ and the point x, while uniform continuity requires a single δ that works everywhere at once. That is why the notion has no purely topological formulation. Speaking of one δ serving all points requires comparing distances measured at different places, and a topology alone provides no such comparison. Two metrics inducing the same topology can disagree about which maps are uniformly continuous, which settles the matter.

Reading the quantifiers off the definition gives one implication immediately.

Proposition 2.6.16: Uniform Continuity Implies Continuity

Every uniformly continuous map between metric spaces is continuous, and the converse fails in general.

***Proof*:**
Let f be uniformly continuous and fix $x \in X$ and $\epsilon > 0$. The definition supplies a δ valid for all pairs of points, and in particular for the pairs (x, x'), so the criterion of definition 2.6.10 holds at x. As x was arbitrary, f is continuous.

For the converse, take $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$, which is continuous. Fix $\epsilon = 1$ and suppose some $\delta > 0$ worked for every pair. Choose x with $x > 1/\delta$ and set $x' = x + \delta/2$. Then $d(x, x') < \delta$

while

$$|f(x') - f(x)| = \left| x\delta + \frac{\delta^2}{4} \right| > x\delta > 1 = \epsilon$$

so no such δ exists and f is not uniformly continuous.

That boundedness is metric rather than topological is shown by the following, which makes every metric space bounded without changing its topology.

Theorem 2.6.17: Bounding Any Metric Spaces

Let (X, d) be a metric space and define the *standard bounded metric* $\overline{d}(x, y) = \min\{d(x, y), 1\}$. Then $\overline{d}$ is a metric and induces the same topology on X as d.

***Proof*:**

That $\overline{d}$ is a metric is immediate except for the triangle inequality. If either $d(x, y) \geq 1$ or $d(y, z) \geq 1$ then the right side of $\overline{d}(x, z) \leq \overline{d}(x, y) + \overline{d}(y, z)$ is at least 1 and the left side at most 1. Otherwise both terms equal the corresponding d-values, and $\overline{d}(x, z) \leq d(x, z) \leq d(x, y) + d(y, z) = \overline{d}(x, y) + \overline{d}(y, z)$.

For the topologies, note that for $\epsilon < 1$ the two balls agree: $B_{\overline{d}}(x, \epsilon) = B_d(x, \epsilon)$, since $\overline{d}(x, y) < \epsilon < 1$ holds exactly when $d(x, y) < \epsilon$. Every d-ball contains a d-ball of radius less than 1 about each of its points, and likewise for $\overline{d}$, so the two collections of balls generate the same topology, completing the proof.

The border of a set in a metric space is always thin, in a sense the earlier discussion of closure could not make precise without a distance.

Theorem 2.6.18: Border Of Set In Metric Space

Let A be a nonempty open set in the metric space X. Then $x \in \overline{A} \setminus A$ if and only if

$$\forall \epsilon > 0,\ B_\epsilon(x) \cap A \neq \emptyset \ \wedge \ B_\epsilon(x) \cap (X \setminus A) \neq \emptyset$$

***Proof*:**

Throughout, theorem 1.4.15 is used in the form: $x \in \overline{A}$ if and only if every open set containing x meets A.

Necessity. Let $x \in \overline{A} \setminus A$ and let $\epsilon > 0$. Since $B_\epsilon(x)$ is an open set containing x and $x \in \overline{A}$, it meets A. It also meets $X \setminus A$, since x itself lies there.

Sufficiency. Suppose the condition holds. Every open $U \ni x$ contains some $B_{\epsilon_0}(x)$, which meets A by hypothesis, so U meets A and $x \in \overline{A}$. And $B_\epsilon(x)$ meets $X \setminus A$ for every ϵ; were $x \in A$, openness of A would give an ϵ with $B_\epsilon(x) \subseteq A$, a contradiction. So $x \notin A$, completing the proof.

The content is that a border cannot be thick. If some ball $B_\epsilon(x)$ were contained in $\overline{A} \setminus A$, it would be an open set around a point of $\overline{A}$ missing A entirely, contradicting theorem 1.4.15. This is the metric refinement of corollary 1.4.17.

2.6.2 Product Spaces and Metrics

Given two metric spaces, which metric on $X \times Y$ induces the product topology? The following one does.

Definition 2.6.19: product metric

Let (X, d_X), (Y, d_Y) be metric spaces. Then

$$d_{X\times Y}((x_1, y_1), (x_2, y_2)) = \max\big(d_X(x_1, x_2), d_Y(y_1, y_2)\big)$$

Lemma 2.6.20: Metric Product Space

1. $(X \times Y, d_{X\times Y})$ is a metric space
2. Its metric topology coincides with the product topology

***Proof*:**

1. Positivity and symmetry are inherited from d_X and d_Y, and $d_{X\times Y}$ vanishes exactly when both coordinates agree. For the triangle inequality, each of $d_X(x_1, x_3) \le d_X(x_1, x_2) + d_X(x_2, x_3)$ and the analogous Y-inequality is bounded by $d_{X\times Y}((x_1, y_1), (x_2, y_2)) + d_{X\times Y}((x_2, y_2), (x_3, y_3))$, and the maximum of two quantities each bounded by that sum is bounded by it.
2. The ball $B_{d_{X\times Y}}((x, y), \epsilon)$ equals $B_{d_X}(x, \epsilon) \times B_{d_Y}(y, \epsilon)$, directly from the definition of the maximum. These products form a basis for the product topology, and conversely any basic product open set $U \times V$ containing (x, y) contains such a square once ϵ is taken smaller than the two radii available in U and V. So the two topologies have a common basis.

Both claims hold, as we sought to show.

Theorem 2.6.21: Metrizability Of Product Spaces

Let $(X_n)_{n\in\mathbb{Z}_+}$ be nonempty topological spaces. Then every X_n is metrizable if and only if $X = \prod_{n\in\mathbb{Z}_+} X_n$ is metrizable in the product topology.

***Proof*:**
Necessity. Suppose each X_n is metrizable, with metric d_n; replacing d_n by its standard bounded version (theorem 2.6.17) we may assume $d_n \le 1$. Define

$$D(x, y) = \sup_{n\in\mathbb{Z}_+} \frac{d_n(x_n, y_n)}{n}$$

That D is a metric follows coordinatewise. It is finite and bounded by 1 because each $d_n \le 1$; symmetry is immediate; the triangle inequality passes through the supremum, since for each n

$$\frac{d_n(x_n, z_n)}{n} \le \frac{d_n(x_n, y_n)}{n} + \frac{d_n(y_n, z_n)}{n} \le D(x, y) + D(y, z)$$

and taking the supremum on the left gives $D(x,z) \leq D(x,y) + D(y,z)$; and $D(x,y) = 0$ forces $d_n(x_n, y_n) = 0$ for every n, hence $x = y$.

The two topologies agree. *Product ⊆ metric:* let $U = \prod_n U_n$ be basic open, with $U_n = X_n$ for all n outside a finite set F, and let $x \in U$. For each $n \in F$ pick ϵ_n with $B_{d_n}(x_n, \epsilon_n) \subseteq U_n$, and set $\epsilon = \min_{n \in F} \epsilon_n / n$. If $D(x,y) < \epsilon$ then for each $n \in F$,

$$\frac{d_n(x_n, y_n)}{n} \leq D(x,y) < \epsilon \leq \frac{\epsilon_n}{n}$$

so $d_n(x_n, y_n) < \epsilon_n$ and $y_n \in U_n$; the coordinates outside F are unconstrained, so $y \in U$. Hence $B_D(x, \epsilon) \subseteq U$.

Metric ⊆ product: given $B_D(x, \epsilon)$, choose N with $1/N < \epsilon/2$ and let

$$U = \left(\prod_{n<N} B_{d_n}\left(x_n, \tfrac{\epsilon}{2}\right) \right) \times \prod_{n \geq N} X_n$$

a basic product open set containing x. For $y \in U$ the coordinates below N contribute less than $\epsilon/2$ each, and those from N on contribute at most $1/n \leq 1/N < \epsilon/2$, since $d_n \leq 1$. So $D(x,y) \leq \epsilon/2 < \epsilon$ and $U \subseteq B_D(x, \epsilon)$.

Sufficiency. Suppose X is metrizable. Each X_n embeds in X as a slice, obtained by fixing a point in every other coordinate, and that slice carries the subspace topology; a subspace of a metrizable space is metrizable by lemma 2.6.9. Hence each X_n is metrizable, completing the proof.

Exercise 2.6.1

1. Show that the discrete metric $d(x,y) = 1$ for $x \neq y$ induces the discrete topology, and that a metric space is discrete as a topological space if and only if every point is isolated.
2. Show that d and $\overline{d} = \min\{d, 1\}$ have the same convergent sequences with the same limits, while $\overline{d}$ is always bounded. Which of the two statements is topological?
3. Show that $d(x, A) = 0$ does not imply $x \in A$, and characterize the sets A for which it does.
4. Show that $\mathbb{R}^\omega$ with the box topology is not first-countable, and deduce that it is not metrizable. Contrast this with theorem 2.6.21.

2.7⋆ Uniform Spaces

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

The section exists because a particular word keeps appearing without a home. *Uniform* continuity, *uniform* convergence, and completeness are all spoken of freely, yet none of them is a topological property, and treating them as metric properties gets the answer wrong in a way worth being precise about. There is a structure they belong to, and locating it settles a question that otherwise recurs indefinitely.

Start with the diagnosis. A metric does two jobs at once. It says which points are near a *given* point, and that job is topological: only the induced topology matters, and theorem 2.6.17 showed that replacing d by $\min\{d, 1\}$ changes nothing about it. It also says when two points are within ϵ of each other *uniformly across the space*, so that nearness at one place can be compared with nearness at another. That second job is invisible to the topology, and it is what uniform continuity, Cauchy sequences and completeness all use.

The invisibility is easy to demonstrate.

Example 2.7.1: Completeness Is Not Topological

The spaces $\mathbb{R}$ and $(0, 1)$ are homeomorphic, by the map of section 3.1.4. With their usual metrics $\mathbb{R}$ is complete and $(0, 1)$ is not, since the sequence $1/n$ is Cauchy in $(0, 1)$ with no limit there. So completeness is not preserved by homeomorphism and is therefore not a topological property. Total boundedness fails the same test: $(0, 1)$ is totally bounded and $\mathbb{R}$ is not.

The structure that records the missing information is a distinguished family of subsets of $X \times X$, each to be read as "the pairs of points that are close to within some tolerance".

Definition 2.7.2: Uniformity

A *uniformity* on a set X is a collection $\mathcal{U}$ of subsets of $X \times X$, called *entourages*, such that

1. every $U \in \mathcal{U}$ contains the diagonal $\Delta = \{(x, x) \mid x \in X\}$;
2. if $U \in \mathcal{U}$ and $U \subseteq V \subseteq X \times X$ then $V \in \mathcal{U}$;
3. if $U, V \in \mathcal{U}$ then $U \cap V \in \mathcal{U}$;
4. if $U \in \mathcal{U}$ then $U^{-1} = \{(y, x) \mid (x, y) \in U\} \in \mathcal{U}$;
5. for every $U \in \mathcal{U}$ there is $V \in \mathcal{U}$ with $V \circ V \subseteq U$, where

$$V \circ V = \{(x, z) \mid \exists\, y \text{ with } (x, y) \in V \text{ and } (y, z) \in V\}$$

The pair $(X, \mathcal{U})$ is a *uniform space*.

Read $(x, y) \in U$ as "x and y are U-close". The five conditions then say: every point is close to itself; a weaker requirement than a legitimate one is legitimate; two tolerances can be imposed at once; closeness is symmetric; and every tolerance can be halved, which is condition (5) and is the triangle inequality in disguise.

Example 2.7.3: The Uniformity of a Metric

Let (X, d) be a metric space and for $\epsilon > 0$ put

$$U_\epsilon = \{(x, y) \in X \times X \mid d(x, y) < \epsilon\}$$

The collection $\mathcal{U}_d$ of all subsets of $X \times X$ containing some U_ϵ is a uniformity. Conditions (1) to (4) are immediate from the metric axioms, and (5) holds with $V = U_{\epsilon/2}$, since $d(x, y) < \epsilon/2$ and $d(y, z) < \epsilon/2$ give $d(x, z) < \epsilon$. This is the sense in which condition (5) encodes the triangle inequality.

Every uniformity induces a topology, so uniform spaces sit between metric spaces and topological spaces exactly as advertised.

Proposition 2.7.4: The Topology of a Uniformity

Let $(X, \mathcal{U})$ be a uniform space and for $x \in X$ and $U \in \mathcal{U}$ write $U[x] = \{y \mid (x, y) \in U\}$. Then

$$\mathcal{T}_{\mathcal{U}} = \{O \subseteq X \mid \forall x \in O\ \exists U \in \mathcal{U} \text{ with } U[x] \subseteq O\}$$

is a topology on X, and for a metric space $\mathcal{T}_{\mathcal{U}_d}$ is the metric topology of definition 2.6.3.

***Proof*:**
Both $\emptyset$ and X satisfy the condition vacuously and trivially. If $\{O_\alpha\} \subseteq \mathcal{T}_{\mathcal{U}}$ and $x \in \bigcup O_\alpha$ then $x \in O_\beta$ for some β, and the entourage witnessing that works for the union. If $O_1, O_2 \in \mathcal{T}_{\mathcal{U}}$ and $x \in O_1 \cap O_2$, take U_1, U_2 with $U_i[x] \subseteq O_i$; then $U_1 \cap U_2$ is an entourage by condition (3) and $(U_1 \cap U_2)[x] \subseteq O_1 \cap O_2$. So $\mathcal{T}_{\mathcal{U}}$ is a topology.

For a metric, $U_\epsilon[x] = B_d(x, \epsilon)$, so the condition on O says precisely that every point of O has a ball around it inside O, which is the metric topology.

Entourages give neighbourhoods, which is the fact every later argument uses.

Lemma 2.7.5: Entourage Sections Are Neighbourhoods

Let $(X, \mathcal{U})$ be a uniform space, $U \in \mathcal{U}$ and $x \in X$. Then $U[x]$ is a neighbourhood of x in $\mathcal{T}_{\mathcal{U}}$: it contains an open set containing x.

***Proof*:**
Put

$$O = \{y \in X \mid \exists W \in \mathcal{U} \text{ with } W[y] \subseteq U[x]\}$$

Then $x \in O$, taking $W = U$. Also $O \subseteq U[x]$: if $y \in O$ with $W[y] \subseteq U[x]$, then $y \in W[y]$ because W contains the diagonal, so $y \in U[x]$.

Finally O is open. Let $y \in O$, with $W[y] \subseteq U[x]$, and choose $W' \in \mathcal{U}$ with $W' \circ W' \subseteq W$. For $z \in W'[y]$ and $w \in W'[z]$, the pairs (y, z) and (z, w) lie in W', so $(y, w) \in W' \circ W' \subseteq W$ and $w \in W[y] \subseteq U[x]$. Thus $W'[z] \subseteq U[x]$, so $z \in O$. Hence $W'[y] \subseteq O$, which is the condition for $O \in \mathcal{T}_{\mathcal{U}}$.

2.7.6 Which topologies arise Not every topology comes from a uniformity. The ones that do are exactly the *completely regular* spaces, treated in section 5.4; the easy direction is that a uniform space is completely regular, and the converse produces a uniformity from the family of continuous maps to $[0, 1]$. So the uniformizable spaces are a proper class between the metrizable and the general, and every compact Hausdorff space is among them.

With the structure named, the notions that had no home acquire one, and each is now visibly a statement about entourages rather than about distances.

Definition 2.7.7: Uniform Continuity and Cauchy Filters

Let $(X,\mathcal{U})$ and $(Y,\mathcal{V})$ be uniform spaces.

1. $f\colon X \to Y$ is *uniformly continuous* if for every $V \in \mathcal{V}$ there is $U \in \mathcal{U}$ with $(f \times f)(U) \subseteq V$.
2. A filter $\mathcal{F}$ on X is *Cauchy* if for every $U \in \mathcal{U}$ there is $A \in \mathcal{F}$ with $A \times A \subseteq U$.
3. X is *complete* if every Cauchy filter converges.
4. f is a *uniform homeomorphism* if it is a bijection and both f and f^{-1} are uniformly continuous.

For a metric space (1) is the ϵ-δ condition of definition 2.6.15, since $(f \times f)(U_\delta) \subseteq V_\epsilon$ says exactly that $d(x, x') < \delta$ forces $d(f(x), f(x')) < \epsilon$ with δ independent of the point. Condition (2) says a Cauchy filter contains sets of arbitrarily small diameter, which for the filter of tails of a sequence is the Cauchy condition. Completeness and total boundedness are invariants of uniform homeomorphism and not of homeomorphism, which is precisely what example 2.7.1 showed.

2.7.1 The Right Morphisms for Metric Spaces

The framework answers a question that is otherwise hard to pose. A category is its objects together with its morphisms, and choosing the morphisms decides what counts as an invariant. For metric spaces there are three defensible choices, and they are not interchangeable.

1. *Continuous maps.* The isomorphisms are homeomorphisms, so the metric is entirely forgotten and one is really doing topology on metrizable spaces. Completeness is not an invariant.
2. *Uniformly continuous maps.* The isomorphisms are uniform homeomorphisms. Completeness, total boundedness and the Cauchy condition all become invariants, while the actual numerical distances do not: d and $\min\{d, 1\}$ are uniformly homeomorphic. This is the category in which completions exist and are unique, and it is the one whose objects are really uniform spaces wearing a metric.
3. *Maps with $d(f(x), f(y)) \leq d(x, y)$*, the short or 1-Lipschitz maps. The isomorphisms are the isometries, so the numbers are preserved exactly. This choice is the categorically best behaved of the three: it has products and coproducts, and it realizes a metric space as a category enriched over the ordered monoid $([0, \infty], \geq, +)$, with the triangle inequality becoming composition.

There is no single right answer, and that is the useful conclusion. Uniform continuity is the right morphism exactly when completeness is the thing one wants preserved, which is why analysis reaches for it and why the completion of a metric space is a construction in the uniform category rather than the topological or the isometric one.

2.7.2 Pseudometrics and Sources of Uniformities

A pseudometric is a metric with the axiom $d(x, y) = 0 \Rightarrow x = y$ weakened to $d(x, x) = 0$ (i.e. a point is always 0-distance from itself, but $x \neq y$ does not imply $d(x, y) \neq 0$). It still generates entourages by the recipe of example 2.7.3, and a *family* of pseudometrics generates a uniformity by taking finite intersections of the resulting sets. Every uniformity arises this way, which is the precise sense in which uniform spaces are "metric spaces without a single metric":

$$\text{one pseudometric} \rightsquigarrow \text{one tolerance}; \qquad \text{a family} \rightsquigarrow \text{a uniformity}$$

A uniformity has a countable base of entourages exactly when it comes from a single pseudometric, so metrizability of a uniformity is a countability condition, and the reason general uniform spaces are needed at all is that many natural families of tolerances are uncountable.

Example 2.7.8: Sources of Uniformities

1. *Metric spaces*, by example 2.7.3.
2. *Topological groups.* A group with a topology, in the sense of chapter 7, carries a canonical uniformity: for each neighbourhood N of the identity, take $U_N = \{(x, y) \mid x^{-1}y \in N\}$. This is why completeness makes sense for a topological group with no metric in sight, and it is the uniformity in which completions of such groups are built.
3. *Compact Hausdorff spaces.* On such a space continuity and uniform continuity coincide (theorem 3.2.33), and only one compatible uniformity exists, so for compact Hausdorff spaces the distinction this section draws collapses entirely.
4. *Function spaces.* On $C(X, Y)$ with Y uniform, the sets $\{(f, g) \mid (f(x), g(x)) \in V \text{ for all } x \in K\}$, indexed by K compact and V an entourage of Y, generate the uniformity of uniform convergence on compact sets. Its topology is the compact-open topology of definition 3.4.1, which is what theorem 3.4.4 proves in the case $K = X$.
5. *Products.* An arbitrary product of uniform spaces carries a product uniformity, generated by finitely many coordinate conditions; the induced topology is the product topology, not the box topology.

Families of pseudometrics appear wherever a topology is specified by a family of seminorms rather than a norm, which is the standard situation in the theory of topological vector spaces, and again where a measure induces the pseudometric $\mu(A \bigtriangleup B)$ on measurable sets and completeness of the resulting space is the statement that limits of measurable sets are measurable. Both belong to later books; what they share is the mechanism described here.

The summary is the one the section opened with. "Uniform" is not a metric word and not a topological word. It names the layer between them, it is generated by families of pseudometrics, and completeness lives there. Whether that layer deserves a full theory is a separate question, and the usual verdict is that it does not, because the cases that arise in practice are metric spaces, topological groups and compact Hausdorff spaces, each of which can be handled directly. The concept is worth having anyway, for the same reason a definition is worth having whenever a word is being used without one.

Exercise 2.7.1

1. Identify which metric axiom each of the five conditions of definition 2.7.2 encodes, and say which metric axiom is encoded by none of them.
2. Show that a uniformly continuous map is continuous for the induced topologies, and that the

identity $(\mathbb{R}, d) \to (\mathbb{R}, \overline{d})$ is a uniform homeomorphism while $\mathbb{R} \to (0,1)$, $t \mapsto \frac{1}{2}(1 + \frac{t}{1+|t|})$, is a homeomorphism that is not one.

3. Show that the discrete uniformity, generated by the single entourage Δ, induces the discrete topology, and that every map out of it is uniformly continuous.

4. Show that $\mathcal{T}_{\mathcal{U}}$ is Hausdorff if and only if $\bigcap_{U \in \mathcal{U}} U = \Delta$.

5. Show that a uniformly continuous map carries Cauchy filters to Cauchy filter bases, and explain why this is what makes a completion functorial.

3

Topological Invariances

The study now turns to topological properties that are *invariant*: those which, whenever a homeomorphism joins two spaces, are present in both or in neither. The chapter makes one detour from that theme, to metric spaces, taken after the Hausdorff condition, since they present a very interesting example.

3.1 Connectedness

Connectedness is another property which is between points and the whole thing which talks about the nature of the geometry. The property records whether an open set can be "broken down further" into open sets. If a set is connected, then relative to that open set it cannot be broken into a disjoint union of two (or more [1]) other open sets. A disconnected example is a pair of disjoint open balls $B_1(-10)$ and $B_1(10)$ in the Euclidean topology. Similarly, each respective balls are connected in the Euclidean topology.

It might be tempting to then say that it gives some idea of contiguity between points. However, that is not necessarily the case. In a topology outlined below, $B_r(x)$ is *always* disconnected for every r and x with *any* underlying set with the exception of singletons. The topologies in which this holds is not even necessarily topologies that are intuitively disconnected. In the Euclidean topology balls are connected, but adjoining the closures of balls to the open sets makes every set disconnected.

3.1.1 Basic Properties And Definitions

Clopen sets return here, to say when a set *is* clopen and what that implies.

[1] There is no real point in saying *or more*: taking the union of all but one of the sets returns to the two-set case.

Lemma 3.1.1: separation of sets regarding clopen sets

Suppose $\mathcal{T}_X$ be a topological space such that $X = A \cup B$ and $A \cap B = \emptyset$, where A and B are also arbitrary sets. Then the following are Equivalent:

1. A is clopen
2. B is clopen
3. A, B are closed
4. A, B are open

***Proof*:**
Throughout, $B = X \setminus A$ and $A = X \setminus B$, since A and B partition X.

(1) $\Rightarrow$ (4): if A is clopen then A is open, and $B = X \setminus A$ is open because A is closed. (4) $\Rightarrow$ (3): if both are open then each is the complement of the other, hence each is closed. (3) $\Rightarrow$ (2): if both are closed then B is closed and $B = X \setminus A$ is open, so B is clopen. (2) $\Rightarrow$ (1): symmetric to (1) $\Rightarrow$ (4) with the roles exchanged, giving that A is clopen. The four conditions are therefore equivalent, as we sought to show.

Clopen sets were introduced earlier, with $\emptyset$ and X as the "trivial" examples, followed by the separation lemma. There is an intuition in saying that a space X is completely disjoint (meaning $\overline{A} \cap B = A \cap \overline{B} = \emptyset$) if X is as described in that lemma. That condition is the basis of the definition of connectedness.

Definition 3.1.2: Connected

Let X be a topology, and take the set $A \subseteq X$. A is *connected* if and only if the only clopen subsets of A are $\emptyset$ and A.

One way to read this is as a "mandatory" overlap between the open sets of A: the intuition for connectedness should really come from the topology rather than the underlying set [2].

Example 3.1.3: Connected Sets

1. The interval $I = (0, 1)$ is connected in the Euclidean topology. Suppose, for contradiction, that $I = A \cup B$ with $A \cap B = \emptyset$ and A, B open (equivalently clopen, by lemma 3.1.1). Then A and B must be unions of open intervals. One of them must contain the points (0.., so let that be in A. Since A is the union of open intervals, there is a sub-interval in A such that $(0, \epsilon) \in A$, and some points $(\epsilon, ..$ are not in A (since B cannot be non-empty). Say $(\epsilon, \epsilon + \delta) \subseteq B$. This would imply that $\epsilon \notin A$ and $\epsilon \notin B$, and there are no singleton intervals. So no such decomposition exists and I is connected. A more formal proof of this fact is theorem 3.1.20.

[2] naturally, sometimes the underlying set (like $\mathbb{R}$) has properties that are a great influence to a topology (like Euclidean topology), but then the topology was chosen to have those properties, so again it comes down to the topology

2. $X = (0,1) \cup (2,3) \subset \mathbb{R}$. Then A is clopen in X, and so disconnected.

3. The trivial topology is connected.

4. the discrete topology is totally disconnected (only singletons are connected)

5. Let X be any infinite set with the finite-complement topology. Then if $A \subseteq X$ is infinite, it is connected (try taking $A \cup B = A, A \cap B = \emptyset$, A, B open)

6. A space is said to be *totally* disconnected if the only connected components are the open sets. Is the discrete topology totally disconnected? Are there others?

7. Is $\mathbb{R}_l$ connected? What sets are connected?

So for a disconnected $X = A \cup B$ with $A \cap B = \emptyset$ and A, B clopen, the pieces A and B are sort of independent from each other. Notice that $\forall c \subseteq X$, c is open if and only if $c \cap A$ is open and $c \cap B$ is open. In that sense the openness of the two pieces is independent; neither constrains the other. The same holds for closed sets.

Lemma 3.1.4: Connectedness and independence

Let $X = A \sqcup B$ with A and B clopen. Then a set $C \subseteq X$ is open if and only if both $C \cap A$ and $C \cap B$ are open, and closed if and only if both $C \cap A$ and $C \cap B$ are closed.

***Proof*:**
Suppose $X = A \sqcup B$ with A, B clopen. If $C \subseteq X$ is open then $C \cap A$ and $C \cap B$ are open, being intersections of open sets. Conversely if both $C \cap A$ and $C \cap B$ are open then so is their union, which is C since $A \cup B = X$. The same argument with closed sets in place of open ones gives the closed statement, completing the proof.

Connectedness looks intuitive at first, but it depends heavily on the topology, and the weird topological examples to get a better feel of connectivity:

Example 3.1.5: Super Clopen
Take

$$\mathcal{T}_{\text{super-clopen}} = \langle \{ \dots, [-3,2], (2,1), [1,0], (0,1)[1,2], (2,3), \dots \} \rangle$$

then the entire real line is covered in open sets. Now consider $[1,2]$. This set is open. by definition. The complement of it $(-\infty, 1) \cup (2, \infty)$ is the union of all other elements, so it is *also* open, and therefore $[1,2]$ is also closed. Therefore it is clopen. In fact, every open set in this topology is clopen. Therefore, the union of any open sets will produce a set which has more than the trivial clopen subset in the subspace topology, and more generally, any set $A \subseteq X$ would have more than the trivial clopen subsets unless A is element of the basis of $\mathcal{T}_{\text{super-clopen}}$ topology. So this topology has a lot of connected components, but anything bigger than that is not connected.

There is an even more extreme example of this, where only single point sets are connected; such a topology is called *totally disconnected.* The discrete topology is an obvious one, but much subtler ones exist! Will will show that a topology generated by intervals (a, b) are connected. But, $\mathbb{R}_l$

(the lower-limit topology, generated by $[0,1)$ intervals) is totally disconnected. The decomposition $(-\infty,0)\cup[0,\infty)$ is a disconnection, showing that $\mathbb{R}_l$ is disconnected, and the same argument generalizes to an arbitrary subset of $\mathbb{R}_l$. Any geometric intuition about connectedness therefore depends on the topology it was acquired in.

The disconnectedness above might look like an artefact of a contrived topology. This need not be the case:

Example 3.1.6: Making the Euclidean Topology a Bit Finer
Take $\mathbb{R}$ as the underlying space. Take this as the basis for a topology:

$$\mathcal{T}_{\text{extremely disconnected}}\langle\{y\in X\big|d(x,y)<\epsilon,\forall\epsilon\}\rangle\cup\langle\{y\in X\big|d(x,y)\leq\epsilon,\forall\epsilon\}\rangle$$

That is, the set of all epsilon balls around every point, and the set of epsilon balls *with* their closure in the Euclidean topology. This set is strictly finer than the Euclidean topology, since every basis element of $\mathcal{T}_{eu}$, is in $\mathcal{T}_{\text{super disconnected}}$, but not conversely. Take an arbitrary interval $I=(a,b)$ and let $r=b-a$. Intervals are connected in $\mathbb{R}$. However, consider

$$S=\left(a+\frac{r}{10},b-\frac{r}{10}\right)\subseteq I$$

Take the complement of it in the subspace topology

$$S^c=\left(a+\frac{r}{10},b-\frac{r}{10}\right)^c=\left(a,a+\frac{r}{10}\right]\cup\left[b-\frac{r}{10},b\right)$$

which is open in I because $I\cap\left(\left[a-1,a+\frac{r}{10}\right]\cup\left[b-\frac{r}{10},b+1\right]\right)=S^c$, so S is clopen, and so is a nontrivial clopen subset of I: the interval I is *not* connected. The same construction splits any ball in half.

This space has a name: a space in which the closure of every open set is open is called *extremely disconnected*!

Familiar topologies can also fail, when the underlying set forces the intersection between the topology and the set to be disconnected:

Example 3.1.7: Eucdliean Topology on $\mathbb{Q}$
(from tutorial, put some place appropriate). Note that $(\mathbb{Q},\mathcal{T}_{Eu})$ is totally disconnected (only singletons are connected.) Take any $A\subseteq\mathbb{Q}$ with $|A|\geq 2$, say containing two points $x<y$. There is then some $\epsilon\in\mathbb{R}$ such that $x<\epsilon<y$. Then $((-\infty,\epsilon)\cap\mathbb{Q})\cup(\epsilon,\infty)$. Then for any A now, it will be the union of two open sets, meaning that it is there must be more than 1 trivial clopen set.

In all these cases the gap between the pieces was small, which might suggest that all connected spaces have some idea os closeness in common. This need not be the case:

Example 3.1.8: Connected space with large gaps
Let

$$I^2=I\amalg I,\qquad \mathcal{T}_{I^2}=\langle\{\{(c,0)\}\cup\{(d,1)\}\mid(c,d)\subseteq I\}\rangle$$

that is, I^2 consists of the points $(a,0)$ and $(b,1)$ for $a,b\in I$, and the topology is generated by the intervals who "mirror" each other on both sides. Note that this topology strictly coarser than the subspace topology traces of the product topology. The construction "glues" two open sets together so that what was be two clopen sets in the relative subspace will not be the only clopen set, making it connected.

In particular:

$$\{((a,b),0),(0,(a,b))\} \qquad a \neq 0$$

is connected.

Note also that some of the spaces mentioned before are Hausdorff, while others are not. So "oddities" in connectedness will not be repaired by assuming the space is Hausdorff.

A related exercise:

Example 3.1.9: Not Homeomorphic

Connectedness distinguishes spaces that no earlier invariant could separate. The device is to delete a point and look at what is left. Call x a *cut point* of a connected space X if $X \setminus \{x\}$ is disconnected. A homeomorphism $h : X \to Y$ restricts to a homeomorphism $X \setminus \{x\} \to Y \setminus \{h(x)\}$, so it carries cut points to cut points and non-cut points to non-cut points. The number of non-cut points is therefore a topological invariant. Three consequences follow.

1. *The intervals* $(0,1)$*,* $(0,1]$ *and* $[0,1]$ *are pairwise non-homeomorphic.* Every point of $(0,1)$ is a cut point, since removing x leaves $(0,x) \cup (x,1)$, a separation into two nonempty disjoint open sets. In $(0,1]$ the point 1 is not a cut point, because $(0,1)$ is connected, while every other point is as before. In $[0,1]$ both 0 and 1 fail to be cut points. So the three intervals have 0, 1 and 2 non-cut points respectively, and no two can be homeomorphic.

2. *Embeddings in both directions do not force a homeomorphism.* Take $X = (0,1)$ and $Y = [0,1]$. The inclusion $X \to Y$ is an embedding, and so is $g : Y \to X$ with $g(t) = (t+1)/3$, whose image is $[\frac{1}{3}, \frac{2}{3}]$ and which is a homeomorphism onto that image. Yet X and Y are not homeomorphic, by the previous item.

3. $\mathbb{R}^n$ *and* $\mathbb{R}$ *are not homeomorphic for* $n > 1$*.* Removing a point from $\mathbb{R}$ disconnects it. Removing a point p from $\mathbb{R}^n$ with $n \geq 2$ does not: given $x, y \neq p$, the segment from x to y either misses p, or else can be replaced by a two-segment path through any point off the line through x and y, which exists because $n \geq 2$. So $\mathbb{R}^n \setminus \{p\}$ is path-connected, hence connected by lemma 3.1.28, and no homeomorphism $\mathbb{R}^n \to \mathbb{R}$ can exist.

The last item shows $\mathbb{R}^n$ and $\mathbb{R}$ are not homeomorphic under the Euclidean topology. Separating $\mathbb{R}^n$ from $\mathbb{R}^m$ for other pairs $n \neq m$ is harder, and the same deletion trick refined by the fundamental group of chapter 6 handles the case $n = 2$: the punctured plane has fundamental group $\mathbb{Z}$, while $\mathbb{R}^m \setminus \{p\}$ is simply connected for $m \geq 3$.

3.1.2 Properties of Connected Spaces

Lemma 3.1.10: Subspace Topology And Disconectedness

Let $Y \subseteq X$ with $Y = A \cup B$, where A and B are nonempty and disjoint. If

$$\overline{A} \cap B = \emptyset \qquad \text{and} \qquad A \cap \overline{B} = \emptyset$$

then Y is disconnected. Both closures are taken in X.

The purpose of this lemma is that it never refers to the subspace topology on Y.

***Proof*:**
Consider $\overline{A} \cap Y$, which is closed in the subspace Y by lemma 2.1.4, being the intersection of Y with a closed set of X. It contains A, since $A \subseteq \overline{A}$ and $A \subseteq Y$. It also misses B:

$$B \cap (\overline{A} \cap Y) \subseteq B \cap \overline{A} = \emptyset$$

the inclusion because dropping the factor Y only enlarges the set, and the equality by hypothesis. Since $Y = A \cup B$, a subset of Y missing B lies in A, so $\overline{A} \cap Y \subseteq A$ and therefore $A = \overline{A} \cap Y$. Hence A is closed in Y, and the symmetric argument makes B closed in Y.

Each is then also open in Y, being the complement of the other. So A and B are nonempty disjoint clopen sets covering Y, which is a separation, and Y is disconnected, as we sought to show.

One further lemma before the constructions. It should be intuitive:

Lemma 3.1.11: Connected Sets Within Disconnected Spaces

Let $X = A \cup B$, $A \cap B = \emptyset$ A, B clopen. Let $Y \subseteq X$. Assume Y is connected. Then either

$$Y \subseteq A \qquad \text{or} \qquad Y \subseteq B$$

***Proof*:**
Since A is clopen in X, the set $Y \cap A$ is clopen in the subspace Y, being the intersection of Y with a set that is both open and closed. A connected space has no clopen subsets other than $\emptyset$ and itself, so $Y \cap A = \emptyset$ or $Y \cap A = Y$. In the first case $Y \subseteq X \setminus A = B$; in the second $Y \subseteq A$. These are the two alternatives, as we sought to show.

Corollary 3.1.12: Singleton's Are Always Connected

Given any topology and any set, singletons are connected

***Proof*:**
Let $\{x\}$ carry the subspace topology. A separation would express $\{x\}$ as a union of two disjoint nonempty open sets, which is impossible since $\{x\}$ has only one point and so no two disjoint nonempty subsets at all. Hence $\{x\}$ is connected, completing the proof.

Example 3.1.13: Connectedness Is Not Inherited by Subspaces
The space $\mathbb{R}$ is connected, but the subspace $A = [0,1] \cup [4,5]$ is not. In the subspace topology both pieces are open, since $[0,1] = A \cap (-1,2)$ and $[4,5] = A \cap (3,6)$, and each is the complement of the other in A, hence also closed. So A has a clopen subset other than $\emptyset$ and A, which is a separation.

However, there is one condition in which a subspace is guaranteed to be a connected space:

Proposition 3.1.14: Connectedness Survives Adding Limit Points

Let A be a connected subspace of X. If $A \subseteq B \subseteq \overline{A}$, then B is connected. That is, adjoining to A any collection of its limit points leaves it connected.

***Proof*:**
Suppose $B = C \sqcup D$ is a separation, with C, D nonempty and open in B. The set A is connected and $A \subseteq B$, so A lies entirely in C or entirely in D; say $A \subseteq C$. Then $\overline{A} \cap B \subseteq \overline{C} \cap B$, and since C is also closed in B (its complement D is open in B) we get $\overline{C} \cap B = C$. But $B \subseteq \overline{A}$, so $B = \overline{A} \cap B \subseteq C$, forcing $D = \emptyset$, a contradiction. Hence no separation exists and B is connected, completing the proof.

Proposition 3.1.15: Quotient Maps and Connectedness

Let $p : X \to Y$ be a quotient map. If each set $p^{-1}(\{y\})$ is connected, and Y is connected, then X is connected

***Proof*:**
Suppose $X = U \sqcup V$ is a separation into nonempty open sets. Each fibre $p^{-1}(\{y\})$ is connected, so it lies wholly in U or wholly in V; hence U and V are both saturated with respect to p. Then $p(U)$ and $p(V)$ are disjoint, nonempty, cover Y, and are open because $p^{-1}(p(U)) = U$ is open and p is a quotient map. This separates Y, contradicting its connectedness. So no separation of X exists, as we sought to show.

3.1.3 Constructing connected spaces

Proposition 3.1.16: points in common between connected sets

Let $\{A_i\}_{i \in I}$ be subsets of X which are connected. Assume $p \in \bigcap_{i \in I} A_i$. Then

$$Y = \bigcup_{i \in I} A_i$$

is connected

***Proof*:**
Suppose $Y = C \sqcup D$ is a separation into nonempty relatively open sets. The point p lies in one of them, say $p \in C$. Each A_i is connected and contained in Y, so A_i lies wholly in C or wholly in D; since $p \in A_i \cap C$, it must lie in C. Therefore $Y = \bigcup_i A_i \subseteq C$, forcing $D = \emptyset$, a contradiction. So Y is connected, completing the proof.

For example $A = \{(x, y) \in \mathbb{R}^2 \mid xy = 0\}$ is connected, by the previous lemma applied to the two axes,

The next proposition is the general-topology form of the Intermediate Value Theorem (theorem 3.1.23), which returns with the connectedness of the real line.

Proposition 3.1.17: Continuous Images of Connected Spaces Are Connected

Let $f : X \to Y$ be continuous. If X is connected, then $f(X)$ is connected in the subspace topology.

***Proof*:**
Suppose $f(X) = C \sqcup D$ were a separation, with C and D nonempty, disjoint and open in the subspace $f(X)$. Regarding f as a map onto $f(X)$, which is continuous by proposition 1.2.11(5), the sets $f^{-1}(C)$ and $f^{-1}(D)$ are open in X. They are disjoint, since C and D are; they are nonempty, since C and D are subsets of the image; and their union is X, since $C \cup D = f(X)$. That is a separation of X, contradicting connectedness. So no separation of $f(X)$ exists and $f(X)$ is connected, as we sought to show.

Theorem 3.1.18: Finite Product Of Connected Spaces

A finite product of connected spaces is connected

intuition

***Proof*:**
It suffices to treat a product of two spaces, the general finite case following by induction. Let X and Y be connected and fix $(a, b) \in X \times Y$. For each $x \in X$ the set

$$T_x = (\{x\} \times Y) \cup (X \times \{b\})$$

is a union of two connected sets, since $\{x\} \times Y$ is homeomorphic to Y and $X \times \{b\}$ to X, and they share the point (x, b); so T_x is connected by proposition 3.1.16. Every T_x contains (a, b), and $\bigcup_{x \in X} T_x = X \times Y$. Applying proposition 3.1.16 once more gives connectedness of the product, completing the proof.

3.1.4 Connectedness for The Real lines

Let $X = \mathbb{R}$ with the Euclidean topology. Some special properties of connectedness hold on this space The finite case bootstraps to the infinite one, provided the product topology is used.

Proposition 3.1.19: Arbitrary Products And Connectedness

An arbitrary product of connected spaces is connected in the product topology. In the box topology this fails: $\mathbb{R}^\omega$ is a product of connected spaces that is not connected.

***Proof*:**
The product topology. Fix a point $a = (a_j)$ in the product. For a finite set F of indices let X_F be the set of points agreeing with a outside F; it is homeomorphic to the finite product $\prod_{j\in F} X_j$, hence connected by theorem 3.1.18. All the X_F contain a, so their union Y is connected by proposition 3.1.16. Moreover Y is dense: any basic open set constrains only finitely many coordinates, so it meets the corresponding X_F. A set containing a connected dense subset is connected by proposition 3.1.14, so the product is connected.

The box topology. In $\mathbb{R}^\omega$ with the box topology, let A be the set of bounded sequences and B the set of unbounded ones. Both are nonempty and they partition the space. Given x, the box-open set $\prod_n (x_n - 1, x_n + 1)$ consists of sequences differing from x by less than 1 in each coordinate, so it lies entirely in A if $x \in A$ and entirely in B if $x \in B$. Hence A and B are both open, and they separate the space, as we sought to show.

Theorem 3.1.20: $[0, 1]$ Is Connected

The closed interval $I = [0, 1]$ is connected

***Proof*:**
This is Heine-Borel but in a different formulation.

As usual with connectedness proofs, we need to show there is only the trivial clopen sets. let $A \subseteq I$ be a clopen set by possibly replacing $A \leftrightarrow I \setminus A$. Without loss of generality, let $0 \in A$. Define

$$m := \sup(b \in I \big| [0, b] \subseteq A)$$

that is, m is the largest number such that $[0, m] \subseteq A$. We'll show that $m = 1$, showing that $A = I$, and thus I only have the trivial clopen sets, and so is connected

First, let's show that $m \in A$ (since the supremum doesn't *have* to be inside A). Assume then that $m \in I \setminus A$. Then $\exists \epsilon > 0$ such that $(m - \epsilon, m + \epsilon) \cap I \subseteq I \setminus A$. The $\cap I$ is because of the subspace topology (recall that it's open in I if U is an open set, then $U \cap I$ is open in I). This would imply that $\forall b \in (m - \epsilon, m + \epsilon)$, $[0, b] \not\subseteq A$. However, that's a contradiction by the way we defined m, since that would imply that $b \leq m - \epsilon$.

Next, we need to show that $m = 1$. Let's assume $m < 1$. Since A is open, then there exists a $\epsilon > 0$ such that $[m, m + \epsilon) \subseteq A$, which means that $[0, m + \frac{\epsilon}{2}] = [0, m] \cup [m, m + \frac{\epsilon}{2}] \subseteq A$, again a contradiction since m must be the largest such number.

Thus, $A = [0, 1]$

Corollary 3.1.21: any interval is connected

$[r_1, r_2]$ is connected, and so is (r_1, r_2), and so $[r_1, r_2)$ and $(r_1, r_2]$ is connected

***Proof*:**
The closed interval $[r_1, r_2]$ is homeomorphic to $[0, 1]$ by the affine map $t \mapsto r_1 + t(r_2 - r_1)$ when $r_1 < r_2$, and is a single point otherwise; theorem 3.1.20 gives connectedness of $[0, 1]$, and connectedness is preserved by continuous images (proposition 3.1.17), so $[r_1, r_2]$ is connected.

For the other three, note that each of (r_1, r_2), $[r_1, r_2)$ and $(r_1, r_2]$ contains (r_1, r_2) and is contained in its closure $[r_1, r_2]$. Writing $(r_1, r_2) = \bigcup_n [r_1 + \frac{1}{n}, r_2 - \frac{1}{n}]$ over all large n exhibits it as a union of connected sets with a common point, hence connected by proposition 3.1.16. Adding some or all of the two endpoints adds limit points, which preserves connectedness by proposition 3.1.14, as we sought to show.

Note that $\mathbb{R}$ is connected, since $\bigcup_{n\in\mathbb{N}}(-n, n) = \mathbb{R}$ is a union of connected sets sharing the point 0. Alternatively,

$$f(x) = \frac{1}{1-x} - \frac{1}{x}$$

since it is a homeomorphism between $(0, 1)$ and $\mathbb{R}$.

The same argument gives connectedness of $\mathbb{R}^n$.

The corollary says intervals are connected. The converse also holds, and pinning it down is what makes connectedness usable as a computational tool on the line.

Proposition 3.1.22: Connected Subsets Of $\mathbb{R}$ Are Intervals

A subset $C \subseteq \mathbb{R}$ is connected if and only if it is convex, that is, if and only if $a, b \in C$ and $a < c < b$ imply $c \in C$.

***Proof*:**
If C is convex it is one of the intervals, rays, $\emptyset$, a point, or $\mathbb{R}$ itself, all connected by corollary 3.1.21 and the remarks above.

Conversely, suppose C is not convex, so there are $a, b \in C$ and $c \notin C$ with $a < c < b$. Put

$$U = C \cap (-\infty, c), \qquad V = C \cap (c, \infty)$$

Both are open in the subspace C, they are disjoint, and they are nonempty because $a \in U$ and $b \in V$. Since $c \notin C$, their union is all of C. That is a separation of C, so C is not connected. Contrapositively, connected implies convex, as claimed.

The generalization of the first-year analysis theorem is now immediate.

Theorem 3.1.23: Intermediate Value Theorem (Ivt)

Let X be connected and $f : X \to \mathbb{R}$ continuous. If $a, b \in f(X)$ and y lies between a and b, then $y \in f(X)$: there is a $c \in X$ with $f(c) = y$.

***Proof*:**
The image $f(X)$ is connected by proposition 3.1.17, hence convex by proposition 3.1.22. Convexity applied to $a, b \in f(X)$ with y between them gives $y \in f(X)$, which is the assertion.

Example 3.1.24: Intermedaite Value Theorem
The following examples use tricks worth knowing:

1. Let $f : S^1 \to \mathbb{R}$ be continuous. Show that there exists a point $x, -x \in S^1$ such that $f(x) = f(-x)$

   ***Proof*:**
   Let's assume there does not exist a $x, -x \in S^1$ such that $f(x) = f(-x)$, that is, $f(x) \neq f(-x)$ for all $x \in \S^1$. Let A be the upper hemisphere $\cup(1, 0)$, and let B be the lower hemisphere $\cup(-1, 0)$. But, for every $x \in A$, $\exists y \in B$ such that $y = -x$. Then since S^1 is a subspace of a Hausdorff space, it is also Hausdorff, and so for every $x \in A$, we can find two open sets $U_{f(x)}$, $V_{f(-x)}$ such that $U_{f(x)} \cap V_{f(-x)} = \emptyset$. We can do this for every point, forming

   $$U = \bigcup_{a \in A} U_f(a) \qquad V = \bigcup_{b \in B} V_{f(b)}$$

   (Check that $U \cap V = \emptyset$, then take the pre-image, which would cover S^1, but still be disconnected)

2. Let $f : X \to X$ be continuous. Show that if $X = [0, 1]$, there is a point x such that $f(x) =$ x. The point is called a *fixed point* of f. What happens if $X = [0, 1)$ or $(0, 1)$?

   ***Proof*:**
   Let's say $x \neq f(x)$ for all $x \in [0, 1]$. The trick is to define this function: $g : [0, 1] \to [-1, 1]\,, g(x) = x - f(x)$. Then since $[0, 1]$ is connected, $g([0, 1])$ is connected. Since $0 \notin g([0, 1])$, then it cannot be that $g([0, 1]) = [-a, 0) \cup (0, b)$, since that would be a discontinuity,

   so it must either be in the lower or upper region. Without loss of generality, we can say that $g([0, 1]) \subseteq (0, 1]$. Here is then were we would use the fact that the limit point exists to pull an interesting conclusion!

3.1.5 Path Connectedness

In unfamiliar topologies, connectedness can produce results that do not match the Euclidean picture. It is still worth preserving the Euclidean notion in some form. Staying with that intuition, a space is connected when one never has to "leave" it to travel between two of its points. That reading is how the standard counter-examples were built: choose a topology which is disconnected to Euclidean eyes, yet whose separating sets fail to be open.

This notion is the one most often meant informally by “connected”, so it gets a name.

Definition 3.1.25: Path Connectedness

Let X be a topological space. A *path* in X is a continuous map $f : [0,1] \to X$. If every two points $a, b \in X$ are joined by a path, then X is *path-connected*.

3.1.26 Note Munkres lets f be from any closed interval to X, which is functionally the same, but makes book-keeping a lot harder

Example 3.1.27: Path Connected Sets

1. Given $\mathbb{R}^n$ with the Euclidean topology, then for every $x_0 \in \mathbb{R}^n$ $B(x_0)$ is a path-connected set.
2. The unit circle S^1 is path connected
3. A harder one, for a reader who knows some algebra: $GL_n(\mathbb{C})$ is path connected. (Hint: link to the identity matrix. A fuller hint: here)
   (a) Having done that one, show that $GL_n(\mathbb{R})$ is *not* path connected (which will come down to the invertibility matrices in $\mathbb{R}$ are split between $+$ and $-$)

Lemma 3.1.28: Path-Connected $\Rightarrow$ Connected

If X is path-connected, then it's connected

***Proof*:**
By contraposition: a disconnected space is not path-connected. Suppose $X = A \sqcup B$ is a separation, so A and B are nonempty, disjoint, open and cover X; both are then also closed, being complements of one another, hence clopen. Pick $a \in A$ and $b \in B$.

Let $f\colon [0,1] \to X$ be any continuous map with $f(0) = a$. The interval $[0,1]$ is connected by theorem 3.1.20, so $f([0,1])$ is connected by proposition 3.1.17, and lemma 3.1.11 places it entirely inside A or entirely inside B. It contains $a \in A$, so it lies in A and therefore misses b. Hence $f(1) \neq b$: no path joins a to b, and X is not path-connected, as we sought to show.

Path-connectedness is therefore stronger than connectedness. The converse does not hold:

Example 3.1.29: Topologist Sin-curve
The topologists sin curve union $\{0\} \times (-1,1)$ is the typical example of connected, but not path-connected set.

Let $S = \{(0,0)\} \cup \{(x, \sin(1/x)) | 0 < x < 1\}$ and $f = (f_1, f_2) : [0,1] \to S$ is path-connected. The argument shows that if $f(0) = (0,0)$, then $f(t) = (0,0)$ for all t.

For the sake of contradiction, assume that $f(t)$ is not always $(0,0)$. For convenience, remove any constant initial part of the interval, so that $0 = \sup\left\{t \middle| f([0,t]) = \{(0,0)\}\right\}$, replacing f by the consider the equivalent function which immediately leaves 0). By theorem 2.2.15, if f is continuous, then so are it's component functions, meaning f_1 and f_2 is continuous. Since f_2 is continuous, then $\forall \epsilon, \exists \delta$, s.t. $|t - 0| < \delta \Rightarrow |f_2(t) - 0| = |f_2(t)| < \epsilon$. Since the supremum of f_2 is 1, take $\epsilon = 1$ so that $|f_2(t)| < 1$ for all $t < \delta$. Since $\delta > 0$, there is can pick a t_0 such that $0 < t_0 < \delta$, and where $f_1(t) > 0$ (since this function is oscillating in a understandable manner for $t > 0$). Since f_1 is continuous *and* it's domain is an interval, then by the the intermediate value theorem (theorem 3.1.23) makes the image of an interval an interval. Thus, $f_1([0,t_0]) = [0, f_1(t_0)]$. Since $f_2(t) = \sin(1/f_1(t))$ for all t with $f_1(t) \neq 0$ and $\sin(1/x)$ aps $(0,\epsilon)$ onto $[-1,1]$ for all $\epsilon > 0$, it follows that $[-1,1]$ is in the image of f_2 restricted to $[0,t_0]$, contradicting the assumption that $t_0 < \delta$. Thus, there cannot exist a a non-constant path from $(0,0)$.

The topologist's sine curve is the example to keep in mind for path-connectedness. Note that taking $(-1,1)$, then it is path connected. This implies that it's not true that if A is path-connected, then $\overline{A}$ is path-connected, since the closure of the interval here will produce a non path-connected space. This is interesting, since it shows closure need not respect path-connectedness: an oscillation of unbounded frequency can obstruct any path to the limit point.[a]

At first, this might seem an unintuitive of a proof but the "finiteness" of this will come from the fact that $[-1,1]$ is something called *compact* in the Euclidean topology. This is important, for that "finiteness" of paths because of the compactness of closed interval in $\mathbb{R}$ will have great consequences.

[a] The domain being the compact set $[0,1]$, messing with infinities like this can break things!

For the converse, the space must be connected and *locally* path-connected. That a local hypothesis plus a mild extra condition yields a global conclusion is the pattern set out at the start of section 1.1: local properties can be assembled into global ones. Local path-connectedness returns in section 3.1.6.

Proposition 3.1.30: Product Of Path-Connected Spaces Is Path-Connected

If $\{X_i\}_{i\in I}$, are path connected spaces, then so is $\prod_{i\in I} X_i$

***Proof*:**
Give $\prod_{i\in I} X_i$ the product topology and take two points $x = (x_i)$ and $y = (y_i)$. Each X_i is path-connected, so for every i there is a continuous $\alpha_i \colon [0,1] \to X_i$ with $\alpha_i(0) = x_i$ and $\alpha_i(1) = y_i$; choosing one for each i uses the axiom of choice when I is infinite. Define

$$\alpha\colon [0,1] \to \prod_{i\in I} X_i, \qquad \alpha(t) = (\alpha_i(t))_{i\in I}$$

Its i-th component is α_i, which is continuous, so α is continuous by theorem 2.2.15. And $\alpha(0) = x$, $\alpha(1) = y$. Any two points of the product are therefore joined by a path, as we sought to show.

Proposition 3.1.31: Continuous Images of Path-Connected Spaces Are Path-Connected

Let $f\colon X \to Y$ be continuous. If X is path-connected, then $f(X)$ is path-connected.

***Proof*:**
Let $y_1, y_2 \in f(X)$, say $y_i = f(x_i)$. Since X is path connected there is a continuous $\alpha\colon [0,1] \to X$ with $\alpha(0) = x_1$ and $\alpha(1) = x_2$. Then $f \circ \alpha$ is continuous as a composite, starts at y_1 and ends at y_2, and takes values in $f(X)$. So any two points of $f(X)$ are joined by a path in $f(X)$, as we sought to show.

3.1.6 Local Connectedness

As noted in lemma 3.1.4, the open sets of a connected set do not decouple. That is made this concept more rigorous, by defining a way of looking at a space by separating it into it's connected components

Definition 3.1.32: Connected Components

Given X, define an equivalence relation on X by setting $x \sim y$ if there is a connected subspace of X containing both x and y. The equivalence classes are the *components* (or connected components) of X

Verifying it's an equivalence class is pretty simple

Theorem 3.1.33: Connected Components Of A Space

The components of X are connected disjoint subspace sof X whose union is X, such that each nonempty connected subspace of X intersects only one of them

***Proof*:**
Define $x \sim y$ if some connected subspace of X contains both. This is reflexive by corollary 3.1.12, symmetric by construction, and transitive: if $A \ni x, y$ and $B \ni y, z$ are connected then $A \cup B$ is connected by proposition 3.1.16, since they share y, and it contains x and z. The components are the equivalence classes, so they are disjoint and cover X.

Each class C is connected: fixing $x_0 \in C$, every $y \in C$ lies in some connected $A_y \ni x_0, y$, and $A_y \subseteq C$ because any point of A_y is related to x_0. So $C = \bigcup_{y \in C} A_y$ is a union of connected sets with the common point x_0, hence connected.

Finally, a nonempty connected $A \subseteq X$ meets only one class: if $x, y \in A$ then $x \sim y$, so all points of A lie in a single class, completing the proof.

Definition 3.1.34: Path-Connected Components

Given X, define an equivalence relation on X by setting $x \sim y$ if there is a path-connected subspace of X containing both x and y. The equivalence classes are the *path-components* of X

Verifying this equivalence relation requires a bit more work, but is worth going over.

The corresponding theorem holds for path-components:

Theorem 3.1.35: Path-Connected Components Of A Space

The path-components of X are connected disjoint subspace sof X whose union is X, such that each nonempty path-connected subspace of X intersects only one of them

***Proof*:**
Define $x \sim y$ if a path in X joins them. This is reflexive via constant paths, symmetric by reversal, and transitive by concatenation, so the path components are its equivalence classes and are therefore disjoint with union X.

Each class is path-connected by construction: any two of its points are joined by a path, and that path lies in the class since every point on it is joined to the endpoints. Finally, a nonempty path-connected $A \subseteq X$ has all its points mutually joined, so they lie in a single class, completing the proof.

Example 3.1.36: Connected, but not Path Connected

1. Take $\mathbb{Q}$ as a subspace of $\mathbb{R}$. Then since $\mathbb{Q}$ is totally disconnected, every component and path-component are just the singletons
2. Let $\overline{S}$ be the topologist sine-curve. Then $\overline{S}$ has 1 connected component, and 2 path-components: One is the curve S, while the other is eh vertical interval $V = 0 \times [-1, 1]$.

   Deleting from V all points with rational second coordinate gives a space that has only one connected component, but *uncountably many* path-components!

Definition 3.1.37: Locally Connected

A space X is *locally connected at x* if for every neighbourhood U of x there is a connected neighbourhood V of x contained in U. If X is locally connected at each of its points, it is said simply *locally connected*. Similarly, X is *locally path-connected at x* if for every neighborhood U of x, there is a path-connected neighbourhood V of x contained in U. If X is locally path connected at each of its points, then it is said to be *locally path connected*

Example 3.1.38: Locally Connected Sets
very interval in $\mathbb{R}$ is connected and path-connected, and hence locally connected/path-connected. The subspace $[0, 1) \cup (0, 1]$ is not connected, but locally connected. The topologists's sine curve is connected but not locally connected. The rationals $\mathbb{Q}$ are neither connected or locally connected

Theorem 3.1.39: Locally Connected If And Only If Open Components

A space X is locally connected if and only if for every open set U of X, each component of U is open in X

***Proof*:**
Necessity. Suppose X is locally connected, let U be open, let C be a component of U, and let $x \in C$. Local connectedness gives a connected open V with $x \in V \subseteq U$. Since V is connected and meets C, it lies in the component of x in U, which is C. So C contains a neighbourhood of each of its points and is open.

Sufficiency. Suppose every component of every open set is open. Given $x \in X$ and an open $U \ni x$, the component C of x in U is open, connected, and satisfies $x \in C \subseteq U$. So the connected open sets form a basis, which is local connectedness, completing the proof.

similarly

Theorem 3.1.40: Locally Path-Connected If And Only If Open Components

A space X is locally path-connected if and only if for every open set U of X, each path-component of U is open in X

***Proof*:**
Necessity. Suppose X is locally path-connected, let U be open, let C be a path component of U, and let $x \in C$. Local path-connectedness gives a path-connected open V with $x \in V \subseteq U$. Every point of V is joined to x by a path in $V \subseteq U$, so V lies in the path component of x in U, which is C. Hence C is a neighbourhood of each of its points and is open.

Sufficiency. Suppose every path component of every open set is open. Let $x \in X$ and let U be an open set containing x. The path component C of x in U is then open, path-connected by construction, and satisfies $x \in C \subseteq U$. So X has a basis of path-connected open sets, which is local path-connectedness, completing the proof.

The connectedness of $[0, 1]$ proved in theorem 3.1.20 used two features of $\mathbb{R}$ and nothing else: that between any two points there is a third, and that every bounded set has a least upper bound. Isolating those two properties identifies exactly the ordered sets whose intervals are connected, and gives a supply of connected spaces well beyond the real line.

Definition 3.1.41: Linear Continuum

A simply ordered set L with more than one element is a *linear continuum* if

1. L has the least upper bound property: every nonempty subset of L that is bounded above has a supremum in L;
2. for every $x < y$ in L there exists $z \in L$ with $x < z < y$.

A subset $C \subseteq L$ is *convex* if $x, y \in C$ and $x < z < y$ imply $z \in C$.

Theorem 3.1.42: Convex Subsets of a Linear Continuum are Connected

Let L be a linear continuum in the order topology. Then every convex subset of L is connected. In particular L itself, and every interval and ray in L, is connected.

***Proof*:**
Let $Y \subseteq L$ be convex and suppose, for contradiction, that $Y = A \sqcup B$ is a separation, so A and B are nonempty, disjoint, open in Y, and have union Y. Choose $a \in A$ and $b \in B$; without loss of generality $a < b$. Convexity of Y gives $[a, b] \subseteq Y$.

Step 1: locate a candidate cut point. Let

$$A_0 = A \cap [a, b], \qquad B_0 = B \cap [a, b]$$

so that $[a, b] = A_0 \sqcup B_0$ is a separation of $[a, b]$ in its subspace topology, with $a \in A_0$ and $b \in B_0$. The set A_0 is nonempty and bounded above by b, so by the least upper bound property it has a supremum $c = \sup A_0$, and $a \leq c \leq b$, whence $c \in [a, b] \subseteq Y$.

Step 2: rule out $c \in A_0$. Suppose $c \in A_0$. Then $c \neq b$, so $c < b$. Since A_0 is open in $[a, b]$, there is a basis element around c contained in A_0. It cannot be of the form $(e, b]$, since that would place b in A_0 rather than B_0; so it is $[a, d)$ or (e, d) with $d \leq b$, and in either case it contains an interval $[c, d)$ with $c < d \leq b$. By the second linear-continuum axiom there is z with $c < z < d$, and then $z \in A_0$ with $z > c$, contradicting that c is an upper bound of A_0.

Step 3: rule out $c \in B_0$. Suppose instead $c \in B_0$. Then $c \neq a$, so $a < c$. Since B_0 is open in $[a, b]$, there is a basis element around c contained in B_0. It cannot be of the form $[a, d)$, since that would place a in B_0 rather than A_0; so it is (e, d) or $(e, b]$ with $e < c$, and in either case some interval $(e, c]$ with $a \leq e < c$ lies in B_0. Now e is an upper bound of A_0: any $x \in A_0$ satisfies $x \leq c$, and $x \in (e, c]$ is impossible since that interval lies in B_0, so $x \leq e$. This contradicts that c is the *least* upper bound of A_0.

Since $[a, b] = A_0 \sqcup B_0$, the point c lies in one of the two, and both cases are impossible. So no separation of Y exists, completing the proof.

Taking $L = \mathbb{R}$ recovers theorem 3.1.20 and corollary 3.1.21 as the special case in which the least upper bound property is the completeness of the reals. The value of the general statement is that it applies to ordered sets built from ordinals, which are the source of the standard examples separating the covering properties studied in chapter 5.

Exercise 3.1.1

1. Show that $\mathbb{R}$ with the co-finite topology is connected, and that $\mathbb{Q} \subseteq \mathbb{R}$ is totally disconnected: its only connected subsets are the singletons and $\emptyset$.
2. Show that a space is connected if and only if every continuous map $X \to \{0,1\}$ into the two-point discrete space is constant.
3. Show that if X is connected and $f\colon X \to \mathbb{R}$ is continuous and takes both a positive and a negative value, then f vanishes somewhere.
4. Show that the components of a space are closed, and give a space whose components are not open.
5. Show that $\mathbb{R}^n$ and $\mathbb{R}$ are not homeomorphic for $n > 1$ by removing a point, and explain why the same argument does not separate $\mathbb{R}^2$ from $\mathbb{R}^3$.

3.2 Compactness

(cool results from Tao's blog here)

Compactness captures the idea that finitely many open sets suffice to describe a space. This is a strong property, and it removes many of the pathologies that come with uncountable spaces: a non-compact space can contain an uncountable discrete subspace, which is awkward to work with). My general way of thinking about compactness is that it gets rid of infinities, which can spoil some things.

3.2.1 Basic Properties And Definitions

Definition 3.2.1: Cover

A collection $\mathcal{A}$ of subsets of a space X is a *cover*, or a *covering*, of X if the union of the elements of $\mathcal{A}$ equals X. It is an *open covering* of X if its elements are open subset of X.

Definition 3.2.2: Compactness

A space X is *compact* if every open covering $\mathcal{A}$ of X contains a finite subcollection that also a cover

3.2.3 Note The definition is about the *space*, not about a subset of it; the relative version comes shortly.

Example 3.2.4: Compact Sets

1. The trivial topology is compact
2. finite spaces are compact
3. the real line is *not* compact ($\mathbb{R} = \bigcup_{n\in\mathbb{N}}(-n, n)$, so no finite subset of that union can cover $\mathbb{R}$)

4. $(0, 1]$ is *not* compact.

5. Adding open sets to a topology can break connectedness, as seen earlier. Same in compactness. Let $X = \mathbb{R}$. If $\mathcal{T} = \{\emptyset, X\}$, and $\mathcal{T}' = \mathcal{P}(X)$, then X is compact in $\mathcal{T}$ (since $\mathcal{T}$ is finite), but $\mathbb{R}$ is *not* compact in $\mathcal{T}'$ (since the set $\{x\}_{x\in\mathbb{R}}$ doesn't have a finite-subcover!) This shows how Compactness is a very topological property! The topology on a set can change whether a set is compact or not. Note that the converse is true: if $\mathcal{T}'$ is a finer topology than $\mathcal{T}$ and X is compact in $\mathcal{T}'$, then X is compact in $\mathcal{T}$. Let A be a cover for X from $\mathcal{T}$. Then A is also a cover for A in $\mathcal{T}'$. Then A admits a finite sub-cover. Then X is also compact in $\mathcal{T}$.

6. The next example hides an "infinity" that can be exploited to break compactness. Take $\mathbb{R}_K$, the K-topology, which is *finer* than the Euclidean topology: its basis consists of the usual open intervals together with all sets $(a, b) \setminus K$, where

$$K = \{\tfrac{1}{n} \mid n \in \mathbb{N}\}$$

The subspace $[0, 1]$ is compact in the Euclidean topology by theorem 3.2.24, but not in $\mathbb{R}_K$. To see it, cover $[0, 1]$ by the sets

$$U = (-1, 1] \setminus K, \qquad U_n = \left(\frac{1}{n+1}, \frac{1}{n-1}\right) \text{ for } n \geq 2, \qquad U_1 = \left(\frac{1}{2}, 2\right)$$

The set U is open in $\mathbb{R}_K$ by definition and catches everything outside K, including 0; each U_n is a Euclidean interval and catches the single point $1/n$ of K (and no other, since consecutive reciprocals are separated). So $\mathcal{C} = \{U\} \cup \{U_n\}_{n\in\mathbb{N}}$ covers $[0, 1]$. But no member other than U_n contains $1/n$, so every U_n is needed, and no finite subcollection can cover. The point 0 is the site of the failure: it can be separated from K by an open set, which leaves infinitely many points of K to be caught one at a time. That is exactly the kind of escape to infinity that compactness forbids.

Definition 3.2.5: Sub-Cover

Let $Y \subseteq X$ and let S be a collection of subsets of X. Then S *covers* Y if $\bigcup_{B\in S} B \supseteq Y$

Figure 3.1: subcover

Lemma 3.2.6: Compactness With Cub-Cover

Let $Y \subseteq X$. Then Y is compact if and only if *every* collection S of open sets covering Y has a finite subset which also open covers Y.

***Proof*:**
proof both ways:

($\Rightarrow$) We know that $Y = \bigcup_{B\in S}(B \cap Y)$. Therefore, there exists $S' \subseteq S$ such that $Y = \bigcup_{B\in S'}(B \cap Y)$, which means that $Y \subseteq \bigcup_{B\in S'} B$

($\Leftarrow$) Let T be an open cover of Y, so $T = \{S_i \cap Y, i \in I\}$ (I indexing set) for S_i an open set in X. And so

$$Y = \bigcup_{i\in I} T_i \subset \bigcup_{i\in I} S_i$$

And so $S = \{S_i, i \in I\}$ covers Y, and so by assumption there exists a finite $J \subseteq I$ such that $S' = \left\{S_j \middle| j \in J\right\}$ covers Y. so there's a T' which also covers Y.

3.2.2 Properties of Compact Spaces

Here are a couple of properties of compact spaces:

Theorem 3.2.7: closed subsets of compact spaces are compact

Let $Y \subseteq X$ be closed, and X be compact. Then Y is compact

***Proof*:**
Let $\mathcal{S}$ be a cover of Y by sets open in X; it is enough to treat covers of this form, since a set open in the subspace Y is the intersection with Y of a set open in X. Adjoin the complement,

$$\mathcal{T} = \mathcal{S} \cup \{X \setminus Y\}$$

which is open because Y is closed. Then $\mathcal{T}$ covers X: a point of Y is covered by $\mathcal{S}$, and a point outside Y by $X \setminus Y$. Compactness of X gives a finite subcover $\mathcal{T}' \subseteq \mathcal{T}$.

Discard $X \setminus Y$ from $\mathcal{T}'$ if it occurs, leaving a finite subfamily of $\mathcal{S}$. It still covers Y, since the discarded set contains no point of Y. So every open cover of Y has a finite subcover, and Y is compact, as we sought to show.

This is a powerful statement: a closed subspace of a compact space is *compact*. It also indicates how closely compactness and closedness are related, giving another way of thinking about closed sets in a compact space.

The converse is not necessarily true:

Example 3.2.8: Compact But Not Closed

1. Let $X = \{1, 2, 3\}$ with the trivial topology. Then $Y = \{1\}$ is compact, but *not* closed. Alternatively $\mathcal{T} = \{\emptyset, \{1\}, X\}$, in which case $\{1\}$ is compact, *open not closed*.
2. A little less trivial of an example. Let X be an infinite set with the finite-complement topology. Then every subset of X is compact (worth checking). [a]

[a] The cool thing about this example is that it is T_1, singletons being closed under that weaker condition, so it is an example which is almost Hausdorff, which will be the proper condition for the converse.

A condition for the converse appears once Hausdorffness is assumed.

The next theorem is the one that yields the Extreme Value Theorem once compactness in $\mathbb{R}^n$ is available (theorem 3.2.30).

Theorem 3.2.9: Continuity and Compactness

The image of a compact space under a continuous map is compact. That is, if $f : X \to Y$ is continuous, and $U \subseteq X$ is compact, then $f(U)$ is compact.

***Proof*:**
Let $\mathcal{A}$ be a cover of $f(U)$ by sets open in Y. Then $\{f^{-1}(V) | V \in \mathcal{A}\}$ is a cover of U by open sets, since f is continuous. By compactness of U finitely many suffice, say $f^{-1}(V_1), \ldots, f^{-1}(V_n)$. Then $V_1, \ldots, V_n$ cover $f(U)$: given $y = f(x)$ with $x \in U$, the point x lies in some $f^{-1}(V_i)$, so $y \in V_i$. Hence every open cover of $f(U)$ has a finite subcover, as we sought to show.

The converse is not necessarily true:

Example 3.2.10: V compact, $f^{-1}(V)$ not Compact
Let

$$f : \mathbb{R} \to \mathbb{R} \qquad f(x) = 0$$

Then $\{0\}$ is compact, while $f^{-1}(0) = \mathbb{R}$, which was shown above not to be compact.

Theorem 3.2.11: Tychonoff's Theorem on finite spaces

If X, and Y are compact, then so is $X \times Y$.

***Proof*:**
A product of two spaces is a product over the two-element index set, so this is the case $|\mathcal{I}| = 2$ of the general Tychonoff theorem theorem 3.2.44, proved at the end of this section.

Example 3.2.12: Maps and Compactness
It was noted earlier that $\pi \colon X \times Y \to X$ is *not* a closed map in general. If Y is compact it *is*.

***Proof*:**
Let $C \subseteq X \times Y$ be closed; the claim is that $X \setminus \pi(C)$ is open. Take $x_0 \notin \pi(C)$, so that no point of $\{x_0\} \times Y$ lies in C, that is, $\{x_0\} \times Y$ is contained in the open set $(X \times Y) \setminus C$.

For each $y \in Y$ the point (x_0, y) has a basic open neighbourhood $U_y \times V_y$ inside $(X \times Y) \setminus C$, by the definition of the product topology. The sets $\{V_y\}_{y \in Y}$ form an open cover of Y, so compactness gives finitely many $V_{y_1}, \ldots, V_{y_n}$ covering Y. Put

$$U = \bigcap_{i=1}^{n} U_{y_i}$$

a finite intersection of open sets containing x_0, hence an open neighbourhood of x_0.

Then $U \times Y$ misses C: given $(u, y) \in U \times Y$, pick i with $y \in V_{y_i}$; since $u \in U \subseteq U_{y_i}$, the point (u, y) lies in $U_{y_i} \times V_{y_i}$, which is disjoint from C. Consequently $U \cap \pi(C) = \emptyset$, so U is a neighbourhood of x_0 inside $X \setminus \pi(C)$. As x_0 was arbitrary, $X \setminus \pi(C)$ is open and $\pi(C)$ is closed, which is what was wanted.

This final theorem is useful for characterising compactness using *closed* sets instead of open sets.

Definition 3.2.13: Finite Intersection Property

Let X be a set and $\mathcal{C}$ a collection of subsets of X. Then $\mathcal{C}$ has the *finite intersection property* (FIP) if for every finite subcollection $\{C_1, ..., C_n\} \subseteq \mathcal{C}$,

$$\bigcap_{i=1}^{n} C_i \neq \emptyset$$

Example 3.2.14: Finite Intersection Property

1. Fixing some $x_0 \in X$ and letting $\mathcal{C}$ be the collection of *all* subsets containing x_0, and so any finite subcollection would also contain x_0.

2. Another would be $\mathcal{C} = \big\{(n, \infty) \big| n \in \mathbb{Z}\big\}$ since any finite subcollection of this set is non-empty because they're all rays.

Proposition 3.2.15: FIP And Compactness

Let X be a topological space. Then X is compact if and only if every collection $\mathcal{C}$ of *closed* subsets of X which has FIP satisfies $\bigcap_{c \in \mathcal{C}} c \neq \emptyset$

***Proof*:**
The two statements are De Morgan duals of each other.

Necessity. Suppose X is compact and let $\mathcal{C}$ be a collection of closed sets with the finite intersection property. If $\bigcap_{C \in \mathcal{C}} C = \emptyset$, then $\{X \setminus C \big| C \in \mathcal{C}\}$ is an open cover of X, so finitely many cover:

$X = \bigcup_{i=1}^{n}(X \setminus C_i)$. Complementing gives $\bigcap_{i=1}^{n} C_i = \emptyset$, contradicting the finite intersection property.

Sufficiency. Suppose every such collection has nonempty intersection, and let $\mathcal{A}$ be an open cover of X with no finite subcover. Put $\mathcal{C} = \{X \setminus U \big| U \in \mathcal{A}\}$, a collection of closed sets. For any finitely many $U_1, \ldots, U_n \in \mathcal{A}$ the union is not all of X, so $\bigcap_i (X \setminus U_i) \neq \emptyset$: the collection $\mathcal{C}$ has the finite intersection property. By hypothesis $\bigcap_{C \in \mathcal{C}} C \neq \emptyset$, which says some point lies outside every member of $\mathcal{A}$, contradicting that $\mathcal{A}$ covers X. So every open cover has a finite subcover, completing the proof.

3.2.3 Compact Hausdorff spaces

Compact Hausdorff spaces are used quite extensively, and so deserve some elaboration on their properties:

Theorem 3.2.16: compact subspaces of Hausdorff Spaces are closed

Suppose X is a Hausdorff space, and $Y \subseteq X$ is compact. Then Y is closed as a subset of X.

***Proof*:**
Let $x_0 \in X \setminus Y$. We will find an open set containing x_0 that's disjoint from Y. Since X is Hausdorff, then $\forall y \in Y$ $\exists$ open U_y, V_y in X such that

1. $y \in Y$, $x_0 \in V_y$
2. $U_y \cap V_y = \emptyset$

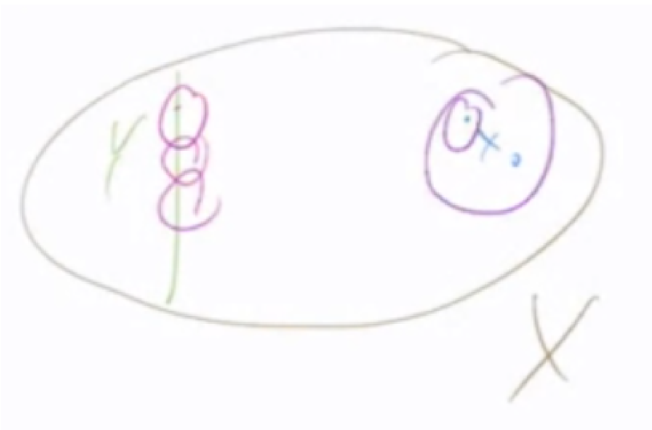

Figure 3.2: subspaceYcompactHausdorff

We can already see what's going on in Y: we're getting an open cover! That is $S = \{U_y \big| y \in Y\}$ covers Y. Therefore, $\exists S' \subseteq S$ such that S' is finite and S' still covers Y. Since $U_y \cap V_y = \emptyset$, and there is now only a *finite* number of U_y, then we can create a new open set which intersects every V_y:

$$\left(\bigcup_{y\in Y'} U_y\right) \cap \left(\bigcap_{y\in Y'} V_y\right) = \emptyset$$

which means that $V_{x_0} := \bigcap_{y\in Y'} V_y$ is disjoint from Y, so V is open and $x_0 \in V$.

From here, we simply repeated the process for every $x_0 \in X \setminus Y$. This will give us a collection set

$$\bigcup_{x\in X\setminus Y} V_x = X \setminus Y$$

meaning that $X \setminus Y$ is the union of open sets, and so is open. Thus, it's complement Y is closed, as we sought to show.

This is a strong statement. In a compact Hausdorff space *every* closed set is compact! So finiteness is a little bit everywhere.

The converse holds by theorem 3.2.7. Without the Hausdorff condition, it is still possible to find spaces in which all closed sets are compact, and all compact sets are closed

Example 3.2.17: Compact Hausdorff Spaces

The trivial topology on any set is compact, and its only closed sets are $\emptyset$ and X. So

3.2.18 Warning All closed sets are compact $\not\Rightarrow$ Compact Hausdorff Space

Lemma 3.2.19: Disjointness of points

If $Y \subseteq X$, Y is compact, X is Hausdorff, $x_0 \in X \setminus Y$. Then there exists open sets U, W such that $x_0 \in U$ and $Y \subseteq W$: an open set separates the point from the compact set.

***Proof*:**

If $Y = \emptyset$ take $U = X$ and $W = \emptyset$. Otherwise, for each $y \in Y$ use the Hausdorff property to choose disjoint open sets $U_y \ni x_0$ and $W_y \ni y$. The family $\{W_y\}_{y\in Y}$ is an open cover of Y, which is compact, so finitely many $W_{y_1}, \ldots, W_{y_n}$ cover Y. Put

$$U = \bigcap_{i=1}^{n} U_{y_i}, \qquad W = \bigcup_{i=1}^{n} W_{y_i}$$

Then U is open as a finite intersection of open sets and contains x_0, and W is open and contains Y. They are disjoint: a point of $U \cap W$ would lie in W_{y_i} for some i and also in U_{y_i}, contradicting $U_{y_i} \cap W_{y_i} = \emptyset$, as we sought to show.

Corollary 3.2.20: making continuous function homeomorphisms

Let $f : X \to Y$ be bijective and continuous. If X is compact, and Y is Hausdorff, then f is a homeomorphism

***Proof*:**
A continuous bijection is a homeomorphism as soon as it is a closed map, since then the image of every closed set is closed, which is exactly the statement that $(f^{-1})^{-1}$ carries closed sets to closed sets, that is, that f^{-1} is continuous. So it suffices to show f is closed.

Let $V \subseteq X$ be closed. Since X is compact, V is compact by theorem 3.2.7. Its image $f(V)$ is then compact by theorem 3.2.9. A compact subspace of a Hausdorff space is closed by theorem 3.2.16, so $f(V)$ is closed in Y.

Thus f is a closed continuous bijection, and therefore a homeomorphism, as claimed.

This fact yields the following result:

Corollary 3.2.21: Incomparability of Compact Hausdorff Spaces

Let $\mathcal{T}, \mathcal{T}' \subseteq P(X)$, where $\mathcal{T}$ is a Compact Hausdorff space

1. if $\mathcal{T}'$ is strictly finer than $\mathcal{T}$, then it is compact, but not Hausdorff
2. if $\mathcal{T}'$ is strictly coarser than $\mathcal{T}$, then it is Hausdorff, but not compact

So for two compact Hausdorff topologies on the same underlying set X, they cannot be comparable.

***Proof*:**

$(\mathcal{T}' \subset \mathcal{T})$ Let $\{U_i\} \subseteq T'$ such that $X = \cup_i U_i$. Since $\mathcal{T}'$ is finer than $\mathcal{T}$, then $\exists x_1, ..., x_n$ such that $X = \cup_{j=1}^{n} U_{x_j}$ still compact.

For the other way, let's take $id : (X, \mathcal{T}) \to (X, \mathcal{T}')$ Note that the identity is a continuous bijection, and so $\mathcal{T}'$ is *nto* Hausdorff, for it it were, then we'd have a continuous bijection from a compact space to a Hausdorff space, meaning the identity is a homeomorphism, meaning $\mathcal{T} = \mathcal{T}'$, a contradiction.

$(\mathcal{T} \subset \mathcal{T}')$ means that $\mathcal{T}'$ is Hausdorff (for easy reasons), then use the identity map again.

Compactness can also be weakened to local compactness, which implies regularity. That is treated with regular spaces in chapter 4.

Theorem 3.2.22: Compact In Metric $\Rightarrow$ Closed And Bounded

if (X, d) is a metric space and $Y \subseteq X$ is compact, then Y is closed and bounded

***Proof*:**
Closed. A metric topology is Hausdorff by lemma 2.6.4, and a compact subspace of a Hausdorff space is closed by theorem 3.2.16.

Bounded. If $Y = \emptyset$ there is nothing to prove, so fix $y_0 \in Y$. The balls $B(y_0, n)$ for $n \in \mathbb{N}$ are open and cover Y, since every point is at some finite distance from y_0. Compactness gives a finite subcover, and the balls are nested, so $Y \subseteq B(y_0, N)$ for the largest index N used. Then any two points of Y are within $2N$ of each other by the triangle inequality, so $\text{diam}(Y) \leq 2N < \infty$, as we sought to show.

The converse is not necessarily true.

Example 3.2.23: Closed and Bounded, but not Compact
Let $X = \mathbb{R}$, and define

$$d(x, y) = \begin{cases} |x-y| & |x-y| < 1 \\ 1 & |x-y| \geq 1 \end{cases}$$

which makes $(\mathbb{R}, d)$ into a metric space (or the discrete metric too). $\mathbb{R}$ is not compact, but $\mathbb{R} \subseteq B_d(0, 2)$, so it is closed and bounded.

One further remark. Being compact Hausdorff is not sufficient for metrizability; the standard counterexample is the ordinal space $\omega_1 + 1$ in the order topology of definition 1.3.13, which is compact introduction another time to explain this:

$\omega_1 + 1$ with the order topology is not metrizable

3.2.4 Compactness of real line

The fact that the any non-finite part of the real line is compact might at first very strange. Surely for an interval $[0, 1]$, there is *some* infinite covering which will not a finite sub-cover. As it turns out, the fact that the "end-points" lie in $[0, 1]$: there is no point of $[0, 1]$ that can be *perpetually approached without reaching* is the key to making it compact.

Theorem 3.2.24: $[0, 1]$ Is Compact (Heine-Borel)

$[0, 1]$ Is Compact

***Proof*:**
Let $X = [0, 1]$, and let S be an open cover for X. We want to show there exists a finite sub-cover for X. Let

$$T := \{t \in [0, 1] | \exists \text{ finite subset of } S \text{ which covers } [0, t]\}$$

We want to show that $1 \in T$. To do so, define $m := \sup(T)$. We'll show hat $m = 1 \in T$.

1. $m > 0$: Since S is a cover, there exists $U \in S$ with $0 \in U$. Since U is open in $\mathbb{R} \cap [0, 1]$, then U is the union of open intervals intersected with $[0, 1]$, so there must exist a $a \in [0, 1]$ such that $[0, a) \subseteq U$, so $\left[0, \frac{a}{2}\right] \subseteq U$. Hence U covers $\left[0, \frac{a}{2}\right]$, so we must at least have $M \geq \frac{a}{2} > 0$. Thus $M > 0$.

2. $m = 1$. Suppose $m < 1$. Note it is not yet known whether $[0, m]$ is compact, m being a supremum, not the maximum. However, we know that $[0, m-\delta]$ *is* compact for appropriate $\delta > 0$. We'll show that with the properties of the Euclidean topology, we can find a compact closed interval $[0, m+\delta]$ for appropriate $\delta > 0$. Since we're in the Euclidean topology, we know that there must exist some $U \in S$ such that $m \in U$. Thus, $\exists \epsilon > 0$ such that $(m-\epsilon, m+\epsilon) \subseteq U$. The fact that this exists is *key*: it is the core property of the Euclidean topology we're using to show compactness. The fact that there always exists this tinnier epsilon band, and that inside it, we'll have $\left[m - \frac{\epsilon}{2}, m + \frac{\epsilon}{2}\right]$, will carry the proof. Since m is the supremum of value such that all closed intervals $[0, m]$ are compact, take the finite sub-cover which covers $\left[0, m - \frac{\epsilon}{2}\right]$. Let $S' \subseteq S$ be the finite sub-cover which covers $\left[0, m - \frac{\epsilon}{2}\right]$. Then define
$$S'' := S' \cup U$$
Then
$$\left[0, m - \frac{\epsilon}{2}\right] \cup U = \left[0, m - \frac{\epsilon}{2}\right] \cup (m-\epsilon, m+\epsilon) \subseteq \bigcup_{V \in S''} V$$
$$\Rightarrow \left[0, m + \frac{\epsilon}{2}\right] \subseteq \bigcup_{V \in S''} V$$
meaning $m + \frac{\epsilon}{2} \in T$, but then m wouldn't be the supermom, a contradiction. Thus, $m = 1$.

3. $m \in T$. Finally, we must show that the supermom is inside T. As before, $\exists U \in S$ such that $m \in U$ $(1 \in U)$, so again by the property of the subspace topology of $[0, 1]$, exists some $\epsilon > 0$ such that $(m-\epsilon, m] \subseteq U$. Since m is the supremum, we know that there exists a finite sub-cover which covers $[0, m - \frac{\epsilon}{2}]$. Let this finite cover be $S' \subseteq S$. Then:
$$S'' = S' \cup \{U\}$$
Then S'' covers $[0, m] = [0, 1]$, completing the proof.

The argument repays a second reading, since the same supremum technique recurs. It is in a sense a novel proof. Out of necessity, With the theorem proved, the topologist's sine curve is worth revisiting to see how this "finiteness" matters the proof of it

Corollary 3.2.25: $[a, b]$ Is Compact

Every closed bounded interval $[a, b] \subseteq \mathbb{R}$ is compact.

***Proof*:**
If $a > b$ the interval is empty and if $a = b$ it is a single point; both are compact, the second because any cover has a one-set subcover. If $a < b$, the map $h(t) = a + t(b-a)$ is a continuous bijection $[0, 1] \to [a, b]$ with continuous inverse $s \mapsto (s-a)/(b-a)$, hence a homeomorphism. Since $[0, 1]$ is compact by theorem 3.2.24, its continuous image $[a, b]$ is compact by theorem 3.2.9, completing the proof.

Corollary 3.2.26: The Unit Cube Is Compact

$[0,1]^n \subseteq \mathbb{R}^n$ is compact for every $n \geq 1$.

***Proof*:**
Induct on n. The case $n = 1$ is theorem 3.2.24. If $[0,1]^{n-1}$ is compact then so is $[0,1]^n = [0,1]^{n-1} \times [0,1]$, being a product of two compact spaces (theorem 3.2.11). The product topology on $[0,1]^n$ agrees with its subspace topology from $\mathbb{R}^n$, since both have the open boxes as a basis, so the result is compact as a subspace of $\mathbb{R}^n$.

Theorem 3.2.27: Closed And Bounded In $\mathbb{R}^n$ Imply Compact

$A \subseteq \mathbb{R}^n$ is compact if and only if A is closed and bounded (for the Euclidean metric)

***Proof*:**

($\Rightarrow$) That it's compact means that implies it's closed and bounded comes from the fact that A is a subspace of the Hausdorff space $\mathbb{R}^n$, so by theorem 3.2.22 it is closed and bounded

($\Leftarrow$) Now we need to show that in $\mathbb{R}^n$, closed and bounded are also *sufficient* conditions for compactness in $\mathbb{R}^n$. This is a direct consequence of what has been shown: a bounded set then there isn't any "rays" when taking the product of $[a, b]$. Since we are closed, A will be the union of:

$$[a_i, b_i]^n$$

And since every $[a_i, b_i]^n$ is compact, and the union of *finitely* many of these will remains compact (notice we're dealing with the finite union of closed sets, so we don't need to worry about the infinite case), thus it will be compact.

3.2.28 Note Closed and bounded implies compact in $\mathbb{R}^n$, and it is tempting to expect this in any the case (like Hausdorff, or metric spaces). This really need not be the case! Take the Euclidean metric on the rational numbers. Closed and bounded is *not* compact. More generally, any non-complete space (that is, spaces where Cauchy sequences need not converge) it fails. Completeness is *not* a topological property, Completeness is *not* a topological property, meaning this implication is not a completely topological one!
Even in cases where there is a complete metric, it's not necessarily the case, though examples of those are more complicated [a].

The condition for the converse being true is unfortunately beyond this course [b]

[a]no infinite-dimensional Banach Spaces as metric spaces have the property that closed and bounded imply compact

[b]A metric space has the "Heine-Borel" property if and only if it is complete, σ-compact, and locally compact. That is not define σ-compactness in this document

Example 3.2.29: Useful Example
The power of this theorem in $\mathbb{R}^n$ should not be understated. For example, S^n is compact: it embeds in $\mathbb{R}^{n+1}$, and there is a closed and bounded set! This is thus a really powerfull tool in the right context.

These theorems make the following theorem a one-liner:

Theorem 3.2.30: Extreme Value Theorem (EVT)

If X is nonempty and compact and $f : X \to \mathbb{R}$ is continuous, then f attains a maximum and a minimum value: there are $c, d \in X$ with $f(c) \leq f(x) \leq f(d)$ for all $x \in X$.

***Proof*:**
The image $f(X)$ is compact by theorem 3.2.9, hence closed and bounded in $\mathbb{R}$ by theorem 3.2.22. It is nonempty, so boundedness gives it a supremum M and an infimum m in $\mathbb{R}$. Every neighbourhood of M meets $f(X)$, by the definition of the supremum as a least upper bound, so $M \in \overline{f(X)} = f(X)$ using theorem 1.4.15 and closedness. Hence $M = f(d)$ for some $d \in X$, and $f(x) \leq M = f(d)$ for all x. The same argument at the infimum produces c with $f(c) = m \leq f(x)$, which completes the proof.

3.2.5 Compactness and metric spaces

Two phenomena arrive together on a compact metric space: a continuity estimate that can be chosen independently of the point, and a scale below which every set is small enough to fit inside a single member of a given cover. Compactness is a *sufficient* condition for both, and it is the one that occurs in practice, but it is worth being explicit that it is not necessary. Give $\mathbb{Z}$ the metric inherited from $\mathbb{R}$. It is not compact, yet every continuous map out of it is uniformly continuous and every open cover has the scale property, both for the trivial reason that distinct points are at distance at least 1. The metric spaces on which every continuous real-valued map is uniformly continuous form a strictly larger class than the compact ones.

Both phenomena rest on a notion that is metric rather than topological, since it compares distances at different points of the space. That is the first hint that neither belongs to metric geometry either: what they require is a structure sitting between a metric and a topology, and section 2.7⋆ identifies it.

Uniform continuity was defined for metric spaces in definition 2.6.15, where proposition 2.6.16 also produced a continuous map that is not uniformly continuous. That counterexample lives on a non-compact domain, and this is forced: on a compact domain the two notions coincide. Before proving it, it is worth isolating the tool that makes such arguments work, since it is used again for the lifting theorems of chapter 6. The statement says that compactness supplies a uniform scale for any open cover.

Lemma 3.2.31: Lebesgue Number Lemma

Let $\mathcal{A}$ be an open cover of a compact metric space (X, d). Then there is a $\delta > 0$, called a *Lebesgue number* for $\mathcal{A}$, such that every subset of X of diameter less than δ is contained in some member of $\mathcal{A}$.

***Proof*:**
If $X \in \mathcal{A}$ any δ will do, so assume otherwise. By compactness choose a finite subcover $A_1, \dots, A_n$ and put $C_i = X \setminus A_i$, a nonempty closed set for each i since $A_i \neq X$. Define

$$\varphi : X \to \mathbb{R}, \qquad \varphi(x) = \frac{1}{n} \sum_{i=1}^{n} d(x, C_i)$$

Each summand is continuous by lemma 2.6.14, so φ is continuous as a finite sum of continuous real-valued functions.

Next, $\varphi(x) > 0$ for every x. Since the A_i cover X, there is an i with $x \in A_i$, and A_i is open, so some ball $B(x, \epsilon)$ lies in A_i and therefore misses C_i. Hence $d(x, C_i) \geq \epsilon > 0$, and the sum of nonnegative terms is positive.

By theorem 3.2.30 the continuous function φ attains a minimum value $\delta > 0$ on the compact space X. This δ is the required Lebesgue number. Let $A \subseteq X$ be nonempty with $\operatorname{diam}(A) < \delta$ and pick $a \in A$; then $A \subseteq B(a, \delta)$. Choose m with $d(a, C_m)$ largest among the n values, so that

$$d(a, C_m) \geq \varphi(a) \geq \delta$$

the first inequality because the largest of n numbers is at least their average. Then $B(a, \delta)$ cannot meet C_m, so

$$A \subseteq B(a, \delta) \subseteq X \setminus C_m = A_m$$

which exhibits the member of $\mathcal{A}$ containing A. The empty set is contained in any A_i, so every set of diameter less than δ is accounted for, as claimed.

With a uniform scale available, the two notions of continuity merge.

Theorem 3.2.32: Uniform Continuity Theorem

Let $f : X \to Y$ be continuous, with X a compact metric space and Y a metric space. Then f is uniformly continuous.

***Proof*:**
Fix $\epsilon > 0$. For each $y \in f(X)$ the ball $B(y, \epsilon/2)$ is open in Y, so its preimage is open in X by continuity, and

$$\mathcal{A} = \left\{ f^{-1}\left(B\left(y, \tfrac{\epsilon}{2}\right)\right) \;\middle|\; y \in f(X) \right\}$$

is an open cover of X. Let δ be a Lebesgue number for $\mathcal{A}$, furnished by lemma 3.2.31.

Now let $x, x' \in X$ with $d_X(x, x') < \delta$. The two-point set $\{x, x'\}$ has diameter $d_X(x, x') < \delta$, so it lies inside a single member of the cover, say $f^{-1}(B(y, \epsilon/2))$. Then both $f(x)$ and $f(x')$ lie in

$B(y, \epsilon/2)$, and the triangle inequality gives

$$d_Y(f(x), f(x')) \leq d_Y(f(x), y) + d_Y(y, f(x')) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

The δ produced depends only on ϵ, not on x, so f is uniformly continuous, as we sought to show.

Combining this with proposition 2.6.16, on a compact metric domain continuity and uniform continuity are the same condition. The one-variable case of this statement is the theorem that a continuous function on a closed bounded interval is uniformly continuous, which is what makes the Riemann integral of a continuous function exist [2].

The metric hypothesis is heavier than the argument needs, and dropping it locates the theorem properly. What the proof used about the domain was compactness and a supply of tolerances, and the second of these is a uniformity in the sense of definition 2.7.2, not a metric.

Theorem 3.2.33: Continuity Is Uniform on a Compact Domain

Let $(X, \mathcal{U})$ be a uniform space whose induced topology is compact, let $(Y, \mathcal{V})$ be a uniform space, and let $f\colon X \to Y$ be continuous. Then f is uniformly continuous.

***Proof*:**
Note first that entourages may always be taken symmetric: for $U \in \mathcal{U}$ the set $U \cap U^{-1}$ is an entourage by conditions (3) and (4) of definition 2.7.2, it is symmetric, and it is contained in U. Note also that $\Delta \subseteq U$ forces $S \subseteq S \circ S$ for any entourage S, so a choice of S with $S \circ S \subseteq U$ automatically has $S \subseteq U$.

Fix $V \in \mathcal{V}$ and choose a symmetric $W \in \mathcal{V}$ with $W \circ W \subseteq V$. For each $x \in X$, continuity of f at x together with lemma 2.7.5 gives an entourage $U_x \in \mathcal{U}$ with

$$f\left(U_x[x]\right) \subseteq W[f(x)]$$

and then a symmetric $S_x \in \mathcal{U}$ with $S_x \circ S_x \subseteq U_x$.

By lemma 2.7.5 each $S_x[x]$ is a neighbourhood of x, so their interiors form an open cover of X. Compactness yields $x_1, \ldots, x_n$ with $X = \bigcup_i \operatorname{int}\left(S_{x_i}[x_i]\right)$. Put

$$S = \bigcap_{i=1}^{n} S_{x_i} \in \mathcal{U}$$

a finite intersection of entourages, hence an entourage by condition (3).

Let $(a, b) \in S$. Choose i with $a \in S_{x_i}[x_i]$, that is $(x_i, a) \in S_{x_i}$. Since also $(a, b) \in S \subseteq S_{x_i}$, composition gives $(x_i, b) \in S_{x_i} \circ S_{x_i} \subseteq U_{x_i}$; and $(x_i, a) \in S_{x_i} \subseteq U_{x_i}$ by the remark above. So both a and b lie in $U_{x_i}[x_i]$, whence

$$f(a),\ f(b) \in W\left[f(x_i)\right]$$

By symmetry of W this gives $(f(a), f(x_i)) \in W$ and $(f(x_i), f(b)) \in W$, so $(f(a), f(b)) \in W \circ W \subseteq V$. Hence $(f \times f)(S) \subseteq V$, and as V was arbitrary, f is uniformly continuous.

The metric statement is the case $\mathcal{U} = \mathcal{U}_d$ of example 2.7.3, and the Lebesgue number lemma is what replaces the entourage bookkeeping when a metric is available. The theorem also explains the remark in box 2.7.6 that compact Hausdorff spaces admit only one compatible uniformity: with continuity and uniform continuity coinciding, the identity map between any two compatible uniformities is uniformly continuous both ways.

3.2.6 Locally Compact

The following is a digression, but a short and worthwhile one.

The coolest thing introduced in this section is the idea of taking a space, and making it a subspace of a *compact* space, and it returns with the Stone-Čech compactification in section 5.4.

Definition 3.2.34: Local Compactness

A space X is said to be *locally compact at* x if there is some compact subspace C of X that contains a neighborhood of x. If X is locally compact at each of its points, X is simply *locally compact.*

Local connectedness has that for *any* neigbhorhood around a point, there is a connected set that is contained in that neigbhorhood. This conditions is thus in a sense *weaker.* It will be satisfied once X is assumed Hausdorff, but there is a reason not to build that in from the start. It leads to introduce the idea of *compactification*

Theorem 3.2.35: Locally Compact in Hausdorff Space

Let X be a Hausdorff space. Then X is locally compact if and only if given x in X, and given neigbhorhood U of x, there is a neighborhood V of x such that $\overline{V}$ is compact and $\overline{V} \subseteq U$

***Proof*:**
Sufficiency. Suppose the stated condition holds. Given $x \in X$, apply it with $U = X$: there is a neighbourhood V of x with $\overline{V}$ compact. Then $\overline{V}$ is a compact subspace containing the neighbourhood V of x, which is local compactness at x in the sense of definition 3.2.34. As x was arbitrary, X is locally compact.

Necessity. Suppose X is locally compact and let U be a neighbourhood of x. Choose a compact $C \subseteq X$ containing a neighbourhood of x, and set $W = U \cap \operatorname{int} C$, an open set with $x \in W \subseteq C$. The set C is compact in a Hausdorff space, hence closed by theorem 3.2.16, so $\overline{W} \subseteq C$; being a closed subset of a compact set, $\overline{W}$ is compact. It remains to shrink W so that its closure lies inside U.

Put $B = \overline{W} \setminus U$, a closed subset of the compact set $\overline{W}$ and hence compact, with $x \notin B$. If $B = \emptyset$ then $V = W$ already works. Otherwise, for each $b \in B$ choose disjoint open sets $U_b \ni x$ and $W_b \ni b$, available because X is Hausdorff. The family $\{W_b\}_{b \in B}$ covers the compact set B, so finitely many $W_{b_1}, \ldots, W_{b_k}$ do. Set

$$V = W \cap U_{b_1} \cap \cdots \cap U_{b_k}$$

which is open and contains x. Since $V \subseteq W$ we have $\overline{V} \subseteq \overline{W}$, so $\overline{V}$ is compact. The set V is disjoint from the open set $W_{b_1} \cup \cdots \cup W_{b_k}$, and a set disjoint from an open set has closure still disjoint from it, so $\overline{V}$ misses that union and in particular misses B. Therefore $\overline{V} \subseteq \overline{W} \setminus B \subseteq U$, as we sought to show.

Theorem 3.2.36: 1-Point Compactification

Let X be a space. Then X is locally compact Hausdorff if and only if there exists a space Y satisfying the following conditions

1. X is a subspace of Y
2. The set $Y - X$ consists of a single point
3. Y is a compact Hausdorff space

If Y and Y' ar two spaces satisfying these conditions, then there is a homeomorphism of Y with Y' that equals the identity map on X

***Proof*:**
Sufficiency. Suppose such a Y exists, and write $Y \setminus X = \{\infty\}$. Then X is Hausdorff as a subspace of the Hausdorff space Y (proposition 2.2.11). Given $x \in X$, use the Hausdorff property of Y to separate x from ∞ by disjoint open $U \ni x$ and $W \ni \infty$. Then $Y \setminus W$ is closed in the compact space Y, hence compact, and it contains U while avoiding ∞, so it is a compact subset of X containing a neighbourhood of x. Thus X is locally compact.

Necessity. Suppose X is locally compact Hausdorff. Let $Y = X \cup \{\infty\}$ with $\infty \notin X$, and declare a subset of Y open if it is either an open subset of X, or of the form $Y \setminus K$ with $K \subseteq X$ compact. This is a topology: the empty set and Y qualify; intersections and unions of the two kinds reduce to the fact that finite unions and arbitrary intersections of compact sets in a Hausdorff space are compact, compact sets being closed by theorem 3.2.16. The subspace topology on X is the original one, since $(Y \setminus K) \cap X = X \setminus K$ is open in X.

Y is compact: an open cover must contain a set of the form $Y \setminus K$ to catch ∞, and the remaining sets cover the compact K, so finitely many suffice. Y is Hausdorff: two points of X are separated inside X, and to separate $x \in X$ from ∞ use local compactness (theorem 3.2.35) to find a neighbourhood V of x with $\overline{V}$ compact, then take V and $Y \setminus \overline{V}$.

Uniqueness. If Y and Y' both satisfy the conditions, define $h\colon Y \to Y'$ to be the identity on X and to send the extra point of Y to that of Y'. It is a bijection, and it carries open sets to open sets: an open set of Y missing ∞ is open in X hence open in Y', and one containing ∞ has compact complement in X, whose image is the complement of a compact set in Y'. The same applies to h^{-1}, so h is a homeomorphism fixing X pointwise, completing the proof.

Definition 3.2.37: One-Point Compactification

IF Y is a compact Hausdorff space and X is a proper subspace of Y whose closure equals Y, then Y is said to be a *compactification* of X. If $Y - X$ is a single point, then Y is the *one-point compactification of* X

Example 3.2.38: One Point Compactification

1. The one point compactification of the real line is homeomorphic with the circle!
2. Similarly, the one-point compactification of $\mathbb{R}$ is homeomorphic to S^2
3. One might ask if the one-point compactification of $\mathbb{C}$ is the same as $\mathbb{R}^2$, since both are homeomorphic, and indeed they are; their algebraic structures differ, which is why they are not usually thought of in this way, and say $\mathbb{C} \cup \{\infty\}$ is the compactification of $\mathbb{C}$, and often call it the *Riemann sphere* or *extended complex plane.*

The rest of these results are trying to match the usual notion of "locality" when a definition of "locally X" is given

Corollary 3.2.39: closed/open subsets of X also locally compact

Let X be a locally compact Hausdorff space; let A be a subspace of X. If A is closed in X or open in X, then A is locally compact

***Proof*:**
Case 1: A closed in X. Let $a \in A$. Since X is locally compact Hausdorff, theorem 3.2.35 applied to the neighbourhood X of a gives a neighbourhood V of a in X with $\overline{V}$ compact. Then $\overline{V} \cap A$ is closed in the compact set $\overline{V}$, hence compact, and it contains the set $V \cap A$, which is a neighbourhood of a in the subspace topology of section 2.1. So A is locally compact at a.

Case 2: A open in X. Let $a \in A$. Now A itself is a neighbourhood of a in X, so theorem 3.2.35 supplies a neighbourhood V of a in X with $\overline{V}$ compact and $\overline{V} \subseteq A$. Since $V \subseteq A$ and A is open, V is also open in A and is therefore a neighbourhood of a in the subspace, while $\overline{V}$ is a compact subset of A containing it. So A is locally compact at a.

In both cases every point of A has a compact subset of A containing one of its neighbourhoods, which is local compactness by definition 3.2.34, completing the proof.

Corollary 3.2.40: Compact Hausdorff space and locally compact

A space X is homeomorphic to an open subspace of a compact Hausdorff space if and only if X is locally compact Hausdorff

***Proof*:**
Sufficiency. Let X be locally compact Hausdorff. If X is compact it is an open subspace of itself, and a compact Hausdorff space is one of the spaces in question. Otherwise theorem 3.2.36 produces a compact Hausdorff $Y = X \cup \{\infty\}$ in which X sits as a subspace with its original topology. The point ∞ is closed in Y, since Y is Hausdorff and lemma 1.5.5 applies, so $X = Y \setminus \{\infty\}$ is open in Y.

Necessity. Suppose X is homeomorphic to an open subspace A of a compact Hausdorff space Y. Then Y is locally compact, being compact, and Hausdorff, so corollary 3.2.39 makes the open subspace A locally compact; and A is Hausdorff by proposition 2.2.11. Both properties transfer along the homeomorphism, so X is locally compact Hausdorff, completing the proof.

3.2.7 The Tychonoff Theorem

The properties proved so far concerned a single compact space or its subspaces. The deepest theorem of the subject concerns *products*: an arbitrary product of compact spaces is compact. For two factors this is theorem 3.2.11; the force of the general statement is that the number of factors may be arbitrary, so that a product of uncountably many compact spaces is still compact. This is what makes compactness the robust substitute for finiteness that it is throughout analysis, and it is the engine behind such results as the existence of Haar measure and the weak-$*$ compactness of the unit ball.

The general theorem cannot be proved from the other axioms of set theory; it is equivalent to the axiom of choice. Choice enters through *Zorn's lemma*, taken here as a foundational principle.

Theorem 3.2.41: Zorn's Lemma

Let $(P, \leq)$ be a nonempty partially ordered set in which every chain, that is, every totally ordered subset, has an upper bound in P. Then P has a maximal element.

Zorn's lemma is equivalent, over the remaining axioms of set theory, to the axiom of choice, and hence not provable from them; it is adopted here without proof. It is the form in which choice is most often invoked in topology, and both results below rest on it.

The path to Tychonoff's theorem runs through a compactness criterion that need only be checked on a subbasis. Recall first that the product topology is generated by the projections.

Lemma 3.2.42: Subbasis of the Product Topology

The product topology on $\prod_{i\in\mathcal{I}} X_i$ (definition 2.2.13) has as a subbasis the collection

$$\mathcal{S} = \left\{\pi_i^{-1}(U) \;\middle|\; i \in \mathcal{I},\ U \text{ open in } X_i\right\}$$

of preimages of open sets under the coordinate projections $\pi_i \colon \prod_{j\in\mathcal{I}} X_j \to X_i$.

***Proof*:**
By theorem 2.2.6 the product topology is the coarsest topology making every projection π_i continuous, so each $\pi_i^{-1}(U)$ is open. Conversely, a basic open set of definition 2.2.13 has the

form $\prod_{i\in\mathcal{I}} U_i$ with $U_i = X_i$ for every i outside a finite set $\{i_1, \ldots, i_n\}$, and

$$\prod_{i\in\mathcal{I}} U_i = \bigcap_{k=1}^{n} \pi_{i_k}^{-1}(U_{i_k})$$

is a finite intersection of members of $\mathcal{S}$. Thus $\mathcal{S}$ generates the product topology, and as each of its members is open, it is a subbasis.

Lemma 3.2.43: Alexander Subbase Lemma

Let X have a subbasis $\mathcal{S}$. If every cover of X by members of $\mathcal{S}$ admits a finite subcover, then X is compact.

***Proof*:**
Suppose, for contradiction, that every subbasic cover has a finite subcover, yet X is not compact. Let P be the collection of all open covers of X that have *no* finite subcover, ordered by inclusion. Since X is not compact, P is nonempty.

Step 1: P has a maximal element. Let $\mathcal{D}$ be a chain in P and put $\mathcal{U}^* = \bigcup \mathcal{D}$, an open cover of X. It has no finite subcover: a finite $F \subseteq \mathcal{U}^*$ has each of its finitely many members lying in some cover of the chain, and as $\mathcal{D}$ is totally ordered there is a single $\mathcal{D}_0 \in \mathcal{D}$ containing all of them, so $F \subseteq \mathcal{D}_0$; were F a cover, $\mathcal{D}_0$ would have a finite subcover, against $\mathcal{D}_0 \in P$. Hence $\mathcal{U}^* \in P$ is an upper bound for $\mathcal{D}$. By theorem 3.2.41, P has a maximal element $\mathcal{M}$.

Step 2: a completion property. If U is open and $U \notin \mathcal{M}$, then $\mathcal{M} \cup \{U\}$ properly contains $\mathcal{M}$, so by maximality it is not in P: it has a finite subcover. That subcover must use U, since otherwise it would lie in $\mathcal{M}$, contradicting $\mathcal{M} \in P$; so there are $V_1, \ldots, V_k \in \mathcal{M}$ with

$$X = U \cup V_1 \cup \cdots \cup V_k \tag{3.1}$$

Step 3: the subbasic members of $\mathcal{M}$ already cover X. Suppose not: some $x \in X$ lies in no member of $\mathcal{M} \cap \mathcal{S}$. Since $\mathcal{M}$ covers X, there is $W \in \mathcal{M}$ with $x \in W$. As $\mathcal{S}$ is a subbasis and W is open, there are $S_1, \ldots, S_n \in \mathcal{S}$ with

$$x \in S_1 \cap \cdots \cap S_n \subseteq W$$

No S_j lies in $\mathcal{M}$: if $S_j \in \mathcal{M}$ then $S_j \in \mathcal{M} \cap \mathcal{S}$ would contain x, against the choice of x. So each $S_j \notin \mathcal{M}$, and (3.1) applies to each, giving $X = S_j \cup C_j$ with C_j a finite union of members of $\mathcal{M}$. Since $\bigcap_j S_j \subseteq W$,

$$X \setminus W \subseteq X \setminus \bigcap_{j=1}^{n} S_j = \bigcup_{j=1}^{n} (X \setminus S_j) \subseteq \bigcup_{j=1}^{n} C_j$$

the last inclusion because $X = S_j \cup C_j$ forces $X \setminus S_j \subseteq C_j$. Therefore $X = W \cup \bigcup_{j=1}^{n} C_j$, a cover of X by W together with finitely many members of $\mathcal{M}$: a finite subcover drawn from $\mathcal{M}$, contradicting $\mathcal{M} \in P$.

Hence $\mathcal{M} \cap \mathcal{S}$ is a cover of X by subbasic sets. By hypothesis it has a finite subcover, which is then a finite subcover contained in $\mathcal{M}$: the final contradiction. Therefore X is compact.

Tychonoff's theorem is now immediate: compactness of the product is tested on its subbasis $\mathcal{S}$, and a subbasic cover is controlled one coordinate at a time.

Theorem 3.2.44: Tychonoff's Theorem

Let $\{X_i\}_{i\in\mathcal{I}}$ be a family of compact spaces. Then the product $\prod_{i\in\mathcal{I}} X_i$, with the product topology, is compact.

***Proof*:**
Write $X = \prod_{i\in\mathcal{I}} X_i$ and let $\mathcal{S} = \left\{\pi_i^{-1}(U) \;\middle|\; i \in \mathcal{I},\ U \text{ open in } X_i\right\}$ be the subbasis of lemma 3.2.42. By the Alexander subbase lemma lemma 3.2.43 it suffices to show that every cover of X by members of $\mathcal{S}$ has a finite subcover.

Let $\mathcal{C} \subseteq \mathcal{S}$ cover X. For each index i collect the open sets used in coordinate i,

$$\mathcal{C}_i = \left\{U \text{ open in } X_i \;\middle|\; \pi_i^{-1}(U) \in \mathcal{C}\right\}$$

Some $\mathcal{C}_i$ covers X_i. Suppose not: for every i there is a point $x_i \in X_i$ lying in no member of $\mathcal{C}_i$. The point $x = (x_i)_{i\in\mathcal{I}}$ lies in X, so it is covered: $x \in \pi_i^{-1}(U)$ for some $\pi_i^{-1}(U) \in \mathcal{C}$. But then $x_i = \pi_i(x) \in U$ with $U \in \mathcal{C}_i$, contradicting the choice of x_i. (This step is where choice, through Zorn's lemma in lemma 3.2.43, does its work: no rule selects the x_i.)

Fix an index i for which $\mathcal{C}_i$ covers X_i. Since X_i is compact, finitely many $U_1, \dots, U_k \in \mathcal{C}_i$ cover X_i. Their preimages $\pi_i^{-1}(U_1), \dots, \pi_i^{-1}(U_k)$ lie in $\mathcal{C}$ and cover X, since

$$\bigcup_{m=1}^{k} \pi_i^{-1}(U_m) = \pi_i^{-1}\left(\bigcup_{m=1}^{k} U_m\right) = \pi_i^{-1}(X_i) = X$$

This is the required finite subcover. By lemma 3.2.43, X is compact.

The theorem contains its finite predecessor theorem 3.2.11 as the case of two factors. Two points about the hypotheses are worth recording. The product topology, not the box topology, is essential: the countable box power of a two-point discrete space is discrete and infinite, hence not compact, even though each factor is compact. And the appeal to choice is not a defect to be engineered away, since Tychonoff's theorem is in fact *equivalent* to the axiom of choice; every proof must use it somewhere.

Tychonoff's theorem also settles the question left open when the limit of an inverse system was constructed in section 2.3. Nothing there ruled out that limit being empty, and for a general system it can be: the coherence conditions may have no simultaneous solution. Compactness rules that out.

Theorem 3.2.45: Limits of Compact Hausdorff Systems

Let $\{X_j, f_{ij}\}$ be an inverse system over a directed set $(J, \leq)$ in which every X_j is compact Hausdorff. Then $\varprojlim_j X_j$ is compact Hausdorff, and if every X_j is nonempty then $\varprojlim_j X_j$ is nonempty.

***Proof*:**
Write $L = \varprojlim_j X_j \subseteq P = \prod_j X_j$. The product P is compact by theorem 3.2.44 and Hausdorff by proposition 2.2.10, and L is Hausdorff as a subspace of a Hausdorff space (proposition 2.2.11).

L is closed in P, hence compact. For each pair $i \le j$ put

$$C_{ij} = \{x \in P \mid f_{ij}(\pi_j(x)) = \pi_i(x)\}$$

so that $L = \bigcap_{i \le j} C_{ij}$. Each C_{ij} is closed: the two maps $f_{ij} \circ \pi_j$ and π_i from P to X_i are continuous, and the set where two continuous maps into a Hausdorff space agree is closed, since its complement is open (if $f_{ij}(\pi_j(x)) \neq \pi_i(x)$, separate the two values by disjoint open sets and intersect their preimages). An arbitrary intersection of closed sets is closed, so L is closed in P and therefore compact by theorem 3.2.7.

Nonemptiness. Assume every $X_j \neq \emptyset$. For $j \in J$ let

$$F_j = \{x \in P \mid f_{ij}(\pi_j(x)) = \pi_i(x) \text{ for all } i \le j\}$$

which is closed by the same argument, being an intersection of sets C_{ij}. It is also nonempty: choose $t \in X_j$, set the i-th coordinate to $f_{ij}(t)$ for each $i \le j$, and choose the remaining coordinates arbitrarily from the nonempty X_k.

The family $\{F_j\}_{j \in J}$ has the finite intersection property. Given $j_1, \dots, j_n$, directedness supplies k with $j_m \le k$ for all m; then $F_k \subseteq F_{j_m}$ for each m, because a point coherent below k is in particular coherent below each j_m by the composition rule $f_{j_m i} \circ f_{ik} = f_{j_m k}$ of definition 2.3.3. So the intersection contains the nonempty F_k.

Since P is compact, proposition 3.2.15 gives $\bigcap_{j \in J} F_j \neq \emptyset$. That intersection is exactly L, so the limit is nonempty, as claimed.

This is the theorem behind every compactness argument for profinite objects. A profinite group is a limit of finite discrete groups, so the theorem makes it compact Hausdorff and totally disconnected, and the nonemptiness half is what guarantees that a system of consistent finite approximations has an actual solution.

Exercise 3.2.1

1. Show that a finite union of compact subspaces is compact, and that the intersection of a nonempty family of compact subspaces of a Hausdorff space is compact.
2. Show that $(0, 1)$ is not compact directly from definition 3.2.2, and that $\mathbb{Q} \cap [0, 1]$ is not compact.
3. Show that a continuous bijection from a compact space to a Hausdorff space is a homeomorphism, and give a continuous bijection that is not a homeomorphism when compactness is dropped.
4. Show that the co-finite topology on any set is compact, and that it is not Hausdorff when the set is infinite. What does this say about the necessity of the Hausdorff hypothesis in theorem 3.2.16?

3.3 Nets and Filters

Sequences are the basic tool of analysis, and three facts about them are used constantly:

1. in a Hausdorff space a sequence has at most one limit;
2. if some sequence in A converges to x, then $x \in \overline{A}$;
3. a continuous map sends convergent sequences to convergent sequences.

Each of these is an implication in one direction only. The converses fail in general spaces: a point of $\overline{A}$ need not be the limit of any sequence from A, and a map preserving convergent sequences need not be continuous. Adding first countability to the space repairs both, but that is a hypothesis, and the alternative is to repair the *notion of convergence* instead so that no hypothesis is needed. There are two standard ways to do it, and they turn out to be equivalent.

3.3.1 Nets

A sequence is a function on $\mathbb{N}$. What the arguments above need is not $\mathbb{N}$ in particular but a domain along which one can always go further, and in which any two positions have a common successor.

Definition 3.3.1: Net

A *net* in X is a function $\eta\colon S \to X$ whose domain is a directed set in the sense of definition 2.3.2. The value $\eta(s)$ is written η_s.

Every sequence is a net, with $S = \mathbb{N}$. The gain is that other directed sets are now allowed, and the useful one is the set of neighbourhoods of a point, ordered by reverse inclusion: it is directed because two neighbourhoods have their intersection below them, and moving along it means shrinking towards the point.

Definition 3.3.2: Convergence of a Net

A net $\eta\colon S \to X$ *converges* to $x \in X$, written $\eta \to x$, if for every neighbourhood N of x there is $t \in S$ such that $\eta_s \in N$ for all $s \geq t$. One says η is *eventually* in N.

With nets, all three implications become equivalences.

Theorem 3.3.3: Nets Detect Closure, Continuity and Separation

Let X and Y be topological spaces.

1. $x \in \overline{A}$ if and only if some net in A converges to x.
2. $f\colon X \to Y$ is continuous if and only if $\eta \to x$ implies $f \circ \eta \to f(x)$ for every net η in X.
3. X is Hausdorff if and only if every net in X converges to at most one point.

***Proof*:**
(1) If a net η in A converges to x, every neighbourhood of x eventually contains $\eta_s \in A$, so every neighbourhood meets A and $x \in \overline{A}$ by theorem 1.4.15. Conversely let $x \in \overline{A}$ and let S be the set of open neighbourhoods of x, directed by reverse inclusion. For each $N \in S$ the intersection $N \cap A$ is nonempty by theorem 1.4.15; choose $\eta_N \in N \cap A$. This is a net in A, and it converges to x: given a neighbourhood M of x, take t to be an open neighbourhood inside M; then $N \geq t$ means $N \subseteq t \subseteq M$, so $\eta_N \in M$.

(2) Suppose f is continuous, $\eta \to x$, and V is a neighbourhood of $f(x)$. Then $f^{-1}(V)$ is a neighbourhood of x, so η is eventually in it and $f \circ \eta$ is eventually in V. Conversely suppose f preserves convergence and let $A \subseteq X$; the claim is $f(\overline{A}) \subseteq \overline{f(A)}$, which gives continuity by theorem 1.4.18 read in reverse, that result being an equivalence for the closure criterion of continuity. Let $x \in \overline{A}$. By (1) there is a net η in A with $\eta \to x$; then $f \circ \eta$ is a net in $f(A)$ converging to $f(x)$, so $f(x) \in \overline{f(A)}$ by (1) again.

(3) If X is Hausdorff and $\eta \to x$ and $\eta \to y$ with $x \neq y$, separate x and y by disjoint neighbourhoods U and V. The net is eventually in each, say beyond t_1 and t_2; directedness gives $t \geq t_1, t_2$, and then $\eta_t \in U \cap V = \emptyset$, a contradiction. Conversely, if X is not Hausdorff there are $x \neq y$ such that every neighbourhood of x meets every neighbourhood of y. Direct the set S of pairs (U, V) of neighbourhoods of x and y by reverse inclusion in both coordinates and choose $\eta_{(U,V)} \in U \cap V$. That net converges to both x and y.

The proof of (1) is the whole point: the directed set was manufactured out of the neighbourhood filter of x, which is available in any space, whereas a sequence would have needed a countable neighbourhood base.

3.3.2 Filters

The second repair keeps the domain and changes the object. A net records where it goes; a filter records the sets it ends up inside. The two carry the same information, and the filter formulation is the one that survives when there are no points to speak of.

Definition 3.3.4: Filter

A *filter* on a set X is a collection $\mathcal{F} \subseteq \mathcal{P}(X)$ such that

1. $X \in \mathcal{F}$ and $\emptyset \notin \mathcal{F}$;
2. if $A, B \in \mathcal{F}$ then $A \cap B \in \mathcal{F}$;
3. if $A \in \mathcal{F}$ and $A \subseteq B \subseteq X$ then $B \in \mathcal{F}$.

A *filter base* is a nonempty collection $\mathcal{B}$ of nonempty sets such that any two members contain a third; the sets containing a member of $\mathcal{B}$ then form a filter, the one *generated by* $\mathcal{B}$.

Example 3.3.5: Filters

1. The *neighbourhood filter* $\mathcal{N}_x$ of a point, consisting of all sets containing an open set containing x.

2. The *tail filter* of a net $\eta\colon S \to X$, generated by the tails $T_t = \{\eta_s \mid s \geq t\}$. These form a base by directedness.

3. The *principal filter* at $A \neq \emptyset$, all sets containing A.

4. The *Fréchet filter* on an infinite X, all sets with finite complement. It has no smallest member, which is what makes it useful.

Definition 3.3.6: Convergence of a Filter

A filter $\mathcal{F}$ on a topological space X *converges* to x, written $\mathcal{F} \to x$, if $\mathcal{N}_x \subseteq \mathcal{F}$, that is, if every neighbourhood of x belongs to $\mathcal{F}$.

Proposition 3.3.7: The Net-Filter Dictionary

Let X be a topological space.

1. A net η converges to x if and only if its tail filter converges to x.

2. For every filter $\mathcal{F}$ there is a net whose tail filter is $\mathcal{F}$.

Consequently every statement about convergence of nets translates into one about convergence of filters and back.

***Proof*:**
(1) The net is eventually in N exactly when some tail $T_t \subseteq N$, which is exactly the condition for N to lie in the filter generated by the tails. So "eventually in every neighbourhood" and "every neighbourhood is in the tail filter" say the same thing.

(2) Let $S = \{(x, A) \mid A \in \mathcal{F},\ x \in A\}$, ordered by $(x, A) \leq (y, B)$ if $B \subseteq A$. This is directed: given (x, A) and (y, B), the set $A \cap B$ lies in $\mathcal{F}$ and is nonempty, so any $(z, A \cap B)$ with $z \in A \cap B$ dominates both. Define $\eta_{(x,A)} = x$. The tail of (x, A) is $\{y \mid y \in B \subseteq A \text{ for some } B \in \mathcal{F}\} = A$, so the tails are exactly the members of $\mathcal{F}$, and the tail filter is $\mathcal{F}$.

The dictionary means the choice between the two is one of convenience. Nets look like sequences and read naturally in analysis; filters are purely set-theoretic, need no auxiliary index set, and have a maximality notion with no clean net analogue.

Definition 3.3.8: Ultrafilter

An *ultrafilter* on X is a filter that is maximal with respect to inclusion among filters on X.

Proposition 3.3.9: Characterization of Ultrafilters

A filter $\mathcal{F}$ on X is an ultrafilter if and only if for every $A \subseteq X$, either $A \in \mathcal{F}$ or $X \setminus A \in \mathcal{F}$. Moreover every filter is contained in an ultrafilter.

***Proof*:**
Suppose $\mathcal{F}$ is an ultrafilter and $A \notin \mathcal{F}$. Then some $B \in \mathcal{F}$ has $B \cap A = \emptyset$: otherwise the sets containing $B \cap A$ for $B \in \mathcal{F}$ would form a filter strictly larger than $\mathcal{F}$ and containing A, contradicting maximality. That B satisfies $B \subseteq X \setminus A$, so $X \setminus A \in \mathcal{F}$ by upward closure.

Conversely suppose the dichotomy holds and let $\mathcal{G} \supseteq \mathcal{F}$ be a filter with some $A \in \mathcal{G} \setminus \mathcal{F}$. Then $X \setminus A \in \mathcal{F} \subseteq \mathcal{G}$, so $\emptyset = A \cap (X \setminus A) \in \mathcal{G}$, contradicting the definition of a filter. So $\mathcal{F}$ is maximal.

For the last claim, order the filters containing $\mathcal{F}$ by inclusion. A chain has its union as an upper bound, which is again a filter: the union is upward closed, omits $\emptyset$, and is closed under pairwise intersection because any two members already lie in a common element of the chain. Zorn's lemma (theorem 3.2.41) supplies a maximal element.

Ultrafilters are what make the filter language pay. Convergence of ultrafilters characterizes compactness, and the characterization is the shortest route to Tychonoff's theorem in the literature.

Theorem 3.3.10: Compactness via Ultrafilters

A space X is compact if and only if every ultrafilter on X converges.

***Proof*:**
Necessity. Let X be compact and $\mathcal{U}$ an ultrafilter, and suppose no point is a limit. Then each $x \in X$ has an open neighbourhood $V_x \notin \mathcal{U}$, hence $X \setminus V_x \in \mathcal{U}$ by proposition 3.3.9. Finitely many $V_{x_1}, \ldots, V_{x_n}$ cover X, so

$$\emptyset = X \setminus \bigcup_i V_{x_i} = \bigcap_i (X \setminus V_{x_i}) \in \mathcal{U}$$

a finite intersection of members of $\mathcal{U}$, which is impossible.

Sufficiency. Suppose every ultrafilter converges and let $\mathcal{C}$ be a family of closed sets with the finite intersection property; by proposition 3.2.15 it suffices to show $\bigcap \mathcal{C} \neq \emptyset$. The finite intersections of members of $\mathcal{C}$ form a filter base, generating a filter $\mathcal{F}$, which extends to an ultrafilter $\mathcal{U}$ by proposition 3.3.9. Let x be a limit of $\mathcal{U}$. For $C \in \mathcal{C}$ and any neighbourhood N of x, both C and N lie in $\mathcal{U}$, so $C \cap N \neq \emptyset$; hence $x \in \overline{C} = C$ by theorem 1.4.15 and closedness. As C was arbitrary, $x \in \bigcap \mathcal{C}$.

The theorem is the reason filters have displaced nets in much of the modern literature. It makes compactness a statement with no quantifier over covers, and from it Tychonoff's theorem follows in a few lines: an ultrafilter on a product pushes forward to an ultrafilter in each coordinate, each of those converges by compactness of the factor, and the resulting point is a limit of the original because convergence in the product topology is convergence coordinatewise. That argument still needs choice, in the form of the extension to an ultrafilter, so it does not evade the issue discussed after theorem 3.2.44; it relocates it.

Filters also outlive the points they were built from. The Stone-Čech compactification of a discrete space, constructed in section 5.4 by embedding in a cube, has an equivalent description as the set of all ultrafilters on that space, the original points corresponding to the principal ones and the remainder to the rest. That the remainder of $\beta\mathbb{N}$ is unreachable by sequences, recorded in example 5.4.6, is the statement that no sequence of principal ultrafilters converges to a non-principal one. The same mechanism underlies Stone duality, ultraproducts in model theory, and the hyperreals.

Exercise 3.3.1

1. Show that a sequence is a net whose tail filter is generated by the tails, and that in a first-countable space nets may be replaced by sequences in theorem 3.3.3(1).
2. Show that the neighbourhood filter $\mathcal{N}_x$ converges to x, and that it is the smallest filter that does.
3. Show that a filter $\mathcal{F}$ converges to x if and only if every net whose tail filter is $\mathcal{F}$ converges to x.
4. Show that an ultrafilter on a finite set is principal, and that the Fréchet filter on an infinite set is contained in no principal ultrafilter.
5. Deduce Tychonoff's theorem from theorem 3.3.10: given an ultrafilter on a product of compact spaces, push it forward to each coordinate and assemble a limit.

3.4 Function Spaces and the Compact-Open Topology

Every construction so far has built new spaces out of the *points* of old ones. There is another supply, at least as important: the continuous maps from one space to another form a set, and that set wants a topology. Convergence of functions is a question analysis asks constantly, and the answer it reaches first, pointwise convergence, is famously too weak, since a pointwise limit of continuous functions need not be continuous. What replaces it is uniform convergence, which is a statement about the behaviour of functions on the whole domain at once. The topological form of that idea is the subject of this section, and it belongs here because the sets it uses are the compact ones.

Throughout, $C(X, Y)$ denotes the set of continuous maps $X \to Y$.

Definition 3.4.1: Compact-Open Topology

For $K \subseteq X$ compact and $U \subseteq Y$ open, put

$$S(K, U) = \{f \in C(X, Y) \mid f(K) \subseteq U\}$$

The *compact-open topology* on $C(X, Y)$ is the topology generated by the collection of all such $S(K, U)$ as a subbasis in the sense of definition 1.3.7.

A basic open set is thus a finite intersection $S(K_1, U_1) \cap \cdots \cap S(K_n, U_n)$: it constrains f on finitely many compact pieces of the domain at once, and says nothing elsewhere. The comparison with pointwise convergence is immediate, since a single point is compact.

Proposition 3.4.2: Compact-Open Refines Pointwise Convergence

The *topology of pointwise convergence* on $C(X, Y)$ is the subspace topology inherited from the product $Y^X = \prod_{x \in X} Y$. The compact-open topology is finer than it, and the two agree when every compact subset of X is finite, in particular when X is discrete.

***Proof*:**
By definition 2.2.13 the product topology on Y^X has as a subbasis the sets $\pi_x^{-1}(U)$ for $x \in X$ and U open in Y, and intersecting with $C(X,Y)$ gives the subbasic sets

$$\{f \in C(X,Y) \mid f(x) \in U\} = S(\{x\},U)$$

Every one-point set is compact, so each of these is compact-open subbasic, and a topology containing a subbasis for another contains that other. Hence the compact-open topology is finer.

If every compact $K \subseteq X$ is finite, then $S(K,U) = \bigcap_{x\in K} S(\{x\},U)$ is a finite intersection of pointwise-subbasic sets, so the compact-open subbasis lies in the pointwise topology and the two coincide. In a discrete space a compact set is finite, since the singletons cover it and a finite subcover must exhaust it.

Proposition 3.4.3: Separation in the Compact-Open Topology

If Y is Hausdorff then $C(X,Y)$ with the compact-open topology is Hausdorff.

***Proof*:**
Let $f \neq g$ in $C(X,Y)$, so $f(x) \neq g(x)$ for some $x \in X$. Choose disjoint open $U \ni f(x)$ and $V \ni g(x)$ in Y. Then $S(\{x\},U)$ and $S(\{x\},V)$ are open, contain f and g respectively, and are disjoint, since a map in both would send x into $U \cap V = \emptyset$. So $C(X,Y)$ is Hausdorff.

The name the definition earns comes from the metric case, where it reproduces uniform convergence exactly.

Theorem 3.4.4: Compact Domain: Compact-Open Is Uniform Convergence

Let X be compact and let (Y,d) be a metric space. Give $C(X,Y)$ the *uniform metric*

$$\rho(f,g) = \sup_{x\in X} \overline{d}\,(f(x),g(x))$$

where $\overline{d} = \min\{d,1\}$ is the bounded metric of theorem 2.6.17. Then the topology induced by ρ is the compact-open topology.

***Proof*:**
That ρ is a metric is inherited from $\overline{d}$ pointwise, the supremum being finite because $\overline{d} \leq 1$.

The uniform topology is finer. Let $f \in S(K,U)$ with $K \subseteq X$ compact and $U \subseteq Y$ open. For each $x \in K$ the point $f(x)$ lies in the open set U, so some ball $B_{\overline{d}}(f(x),\epsilon_x) \subseteq U$. The sets $f^{-1}\left(B_{\overline{d}}(f(x),\epsilon_x/2)\right)$ form an open cover of the compact K, so finitely many suffice, say indexed by $x_1,\dots,x_n$; let $\epsilon = \frac{1}{2}\min_i \epsilon_{x_i} > 0$. If $\rho(f,g) < \epsilon$ and $y \in K$, pick i with $\overline{d}(f(y),f(x_i)) < \epsilon_{x_i}/2$; then

$$\overline{d}\,(g(y),f(x_i)) \leq \overline{d}\,(g(y),f(y)) + \overline{d}\,(f(y),f(x_i)) < \epsilon + \frac{\epsilon_{x_i}}{2} \leq \epsilon_{x_i}$$

so $g(y) \in B_{\overline{d}}(f(x_i),\epsilon_{x_i}) \subseteq U$. Hence the ρ-ball of radius ϵ about f lies in $S(K,U)$.

The compact-open topology is finer. Let $f \in C(X,Y)$ and $\epsilon > 0$; the aim is a compact-open neighbourhood of f inside the ρ-ball of radius ϵ. The sets $f^{-1}\left(B_{\overline{d}}(f(x), \epsilon/4)\right)$ for $x \in X$ are open and cover X, so compactness yields finitely many points $x_1, \ldots, x_n$ whose preimages

$$W_i = f^{-1}\left(B_{\overline{d}}\left(f(x_i), \tfrac{\epsilon}{4}\right)\right)$$

cover X. Each $\overline{W_i}$ is closed in the compact space X, hence compact by theorem 3.2.7, and by theorem 1.4.18,

$$f\left(\overline{W_i}\right) \subseteq \overline{f(W_i)} \subseteq \overline{B_{\overline{d}}\left(f(x_i), \tfrac{\epsilon}{4}\right)} \subseteq B_{\overline{d}}\left(f(x_i), \tfrac{\epsilon}{3}\right)$$

the last inclusion because a point in the closure of a ball of radius $\epsilon/4$ is at distance at most $\epsilon/4 < \epsilon/3$ from its centre. Put

$$N = \bigcap_{i=1}^{n} S\left(\overline{W_i},\ B_{\overline{d}}\left(f(x_i), \tfrac{\epsilon}{3}\right)\right)$$

an open set containing f by the display above. If $g \in N$ and $y \in X$, pick i with $y \in W_i \subseteq \overline{W_i}$; then $\overline{d}(g(y), f(x_i)) < \epsilon/3$ and $\overline{d}(f(y), f(x_i)) < \epsilon/4$, so $\overline{d}(f(y), g(y)) < \epsilon/3 + \epsilon/4$. Taking the supremum over y,

$$\rho(f, g) \leq \frac{\epsilon}{3} + \frac{\epsilon}{4} = \frac{7\epsilon}{12} < \epsilon$$

so N lies inside the ρ-ball of radius ϵ about f.

Each topology refines the other, so they are equal.

The hypothesis that X be compact is what makes "uniformly on X" the same as "uniformly on compact sets". Dropping it, the compact-open topology becomes the topology of uniform convergence on compact subsets, which is the right notion for spaces of holomorphic functions and the setting in which normal families and Montel's theorem are stated [8].

One property separates the compact-open topology from its competitors, and it is the reason it is the standard choice: it makes evaluation continuous.

Theorem 3.4.5: Continuity of Evaluation

Let X be locally compact Hausdorff and give $C(X, Y)$ the compact-open topology. Then the evaluation map

$$e\colon C(X, Y) \times X \to Y, \qquad e(f, x) = f(x)$$

is continuous.

***Proof*:**
Fix (f, x) and an open $W \ni f(x)$ in Y. Then $f^{-1}(W)$ is open in X and contains x. Since X is locally compact Hausdorff, theorem 3.2.35 supplies an open $V \ni x$ with $\overline{V}$ compact and $\overline{V} \subseteq f^{-1}(W)$.

Consider $S(\overline{V}, W) \times V$, an open set of the product containing (f, x): it contains f because $f(\overline{V}) \subseteq W$, and it contains x because $x \in V$. For any (g, y) in it, $y \in V \subseteq \overline{V}$ and $g(\overline{V}) \subseteq W$, so $e(g, y) = g(y) \in W$. Hence e maps this neighbourhood into W, and e is continuous at (f, x). The point was arbitrary, so e is continuous.

Evaluation being continuous is what licenses treating a family of maps as a single map of two variables, and the exchange runs in both directions.

Theorem 3.4.6: The Exponential Law

Let X be locally compact Hausdorff and let Z be any space. A map $F\colon Z \times X \to Y$ is continuous if and only if the associated map

$$\widehat{F}\colon Z \to C(X,Y), \qquad \widehat{F}(z)(x) = F(z,x)$$

is well defined and continuous, where $C(X,Y)$ carries the compact-open topology.

***Proof*:**
From $\widehat{F}$ to F. Suppose $\widehat{F}$ is continuous. Then F is the composite

$$Z \times X \xrightarrow{\widehat{F}\times \mathrm{id}} C(X,Y) \times X \xrightarrow{e} Y$$

The first map is continuous by theorem 2.2.15, since each coordinate is, and the second is continuous by theorem 3.4.5. A composite of continuous maps is continuous.

From F to $\widehat{F}$. Suppose F is continuous. For fixed z the map $x \mapsto F(z,x)$ is continuous, being the composite of $x \mapsto (z,x)$ with F, so $\widehat{F}(z) \in C(X,Y)$ and $\widehat{F}$ is well defined. For continuity it suffices to check preimages of subbasic sets. Let $S(K,U)$ be subbasic and let $z_0 \in \widehat{F}^{-1}(S(K,U))$, so $F(\{z_0\} \times K) \subseteq U$. For each $x \in K$ continuity of F at (z_0,x) gives open sets $A_x \ni z_0$ and $B_x \ni x$ with $F(A_x \times B_x) \subseteq U$. The B_x cover the compact K, so finitely many $B_{x_1},\dots,B_{x_n}$ do; put $A = \bigcap_{i=1}^n A_{x_i}$, an open set containing z_0. For $z \in A$ and $y \in K$, choose i with $y \in B_{x_i}$; then $(z,y) \in A_{x_i} \times B_{x_i}$, so $F(z,y) \in U$. Hence $\widehat{F}(A) \subseteq S(K,U)$, and $\widehat{F}^{-1}(S(K,U))$ is open.

The law is what makes homotopy theory possible in the form used in chapter 6: a homotopy $F\colon X \times I \to Y$ and a path in the function space $C(X,Y)$ are the same datum, so deforming a map and moving through a space of maps are two descriptions of one thing. It is also the reason that “the space of loops in X” is a space rather than a figure of speech, and the fundamental group is then the set of its path components.

Local compactness of one factor gives more than continuous evaluation. It also forces a product to respect quotients, which is what makes it valid to build a space by gluing and then multiply the result by an interval.

Theorem 3.4.7: Products of Quotient Maps

Let $p\colon X \to X'$ be a quotient map and let Z be locally compact Hausdorff. Then

$$p \times \mathrm{id}_Z\colon X \times Z \to X' \times Z$$

is a quotient map.

***Proof*:**
The map is a continuous surjection, so by the characterization in definition 2.4.1 it is enough to show that it carries saturated open sets to open sets. Fix $W \subseteq X \times Z$ open and saturated for $p \times \mathrm{id}_Z$, and let $(p(x_0), z_0)$ be a point of $(p \times \mathrm{id}_Z)(W)$, chosen with $(x_0, z_0) \in W$.

Step 1: A compact tube through (x_0, z_0). By corollary 2.2.2 there are open sets $A \ni x_0$ and $B \ni z_0$ with $A \times B \subseteq W$. Since Z is locally compact Hausdorff, theorem 3.2.35 supplies an open $V \ni z_0$ with $\overline{V}$ compact and $\overline{V} \subseteq B$. Then $\{x_0\} \times \overline{V} \subseteq A \times B \subseteq W$.

Step 2: The set of points whose tube lies in W is open. Put

$$A_0 = \{x \in X \mid \{x\} \times \overline{V} \subseteq W\}$$

so that $x_0 \in A_0$ by Step 1. Let $x \in A_0$. For each $z \in \overline{V}$ the point (x, z) lies in the open set W, so corollary 2.2.2 gives open sets $A_z \ni x$ and $B_z \ni z$ with $A_z \times B_z \subseteq W$. The sets B_z cover the compact $\overline{V}$, so finitely many $B_{z_1}, \dots, B_{z_n}$ already do; put $A' = \bigcap_{i=1}^n A_{z_i}$, an open set containing x. Given $a \in A'$ and $z \in \overline{V}$, choose i with $z \in B_{z_i}$; then $(a, z) \in A_{z_i} \times B_{z_i} \subseteq W$. So $A' \times \overline{V} \subseteq W$, whence $A' \subseteq A_0$, and A_0 is open.

Step 3: A_0 is saturated for p. Let $x_1 \in A_0$ and let x_2 satisfy $p(x_2) = p(x_1)$. For each $z \in \overline{V}$ the point (x_1, z) lies in W, so $(p(x_1), z) \in (p \times \mathrm{id}_Z)(W)$; since $p(x_2) = p(x_1)$, the point (x_2, z) has the same image, and W being saturated forces $(x_2, z) \in W$. Hence $\{x_2\} \times \overline{V} \subseteq W$, that is $x_2 \in A_0$.

Step 4: The image is a neighbourhood of the chosen point. By Steps 2 and 3 the set A_0 is open and saturated for p, so $p(A_0)$ is open in X' because p is a quotient map. It contains $p(x_0)$, and V is an open neighbourhood of z_0. Finally $p(A_0) \times V \subseteq (p \times \mathrm{id}_Z)(W)$: given $x \in A_0$ and $z \in V \subseteq \overline{V}$ one has $(x, z) \in W$, so $(p(x), z)$ lies in the image. The point $(p(x_0), z_0)$ of the image therefore has an open neighbourhood inside it, and since that point was arbitrary the image is open, completing the proof.

The hypothesis on Z enters exactly once, at Step 1, where it produces a *compact* tube $\{x_0\} \times \overline{V}$; compactness is then what lets Step 2 widen the tube using finitely many rectangles. Without it Step 2 has no finite subcover to take, and the conclusion fails, so the hypothesis is carried rather than assumed away. Dropping it altogether breaks the statement: a product of two quotient maps need not be a quotient map, and repairing that is one of the reasons for restricting to a smaller category of spaces, as [12] sets out. The theorem is the reason a homotopy descends through a gluing: given a quotient $p \colon X \to X'$ and a homotopy $F \colon X \times I \to Y$ constant on the fibres of $p \times \mathrm{id}_I$, the induced map $X' \times I \to Y$ is continuous by theorem 2.4.8 applied to the quotient map $p \times \mathrm{id}_I$, the interval being compact Hausdorff and hence locally compact.

One theorem about $C(X, Y)$ is conspicuously absent. The Arzelà-Ascoli theorem, which characterizes the compact subsets of $C(X, Y)$ for X compact and Y metric in terms of equicontinuity and pointwise boundedness, is proved in [1] and cited rather than reproved here; the compact-open topology is the topology in whose terms it is stated.

Exercise 3.4.1

1. Show that $C(\{*\}, Y)$ is homeomorphic to Y, and that $C(X, Y)$ is a single point when Y is.
2. Show that if Y is Hausdorff then the set of constant maps is closed in $C(X, Y)$ for the compact-open topology.

3. Let X be compact Hausdorff, $K \subseteq X$ compact and $U \subseteq Y$ open. Show that every $f \in S(K, U)$ admits an open V with $K \subseteq V \subseteq \overline{V} \subseteq f^{-1}(U)$, and deduce that the sets $S(\overline{V}, U)$ with V open already generate the compact-open topology.

4. Show that if X is discrete then $C(X, Y)$ with the compact-open topology is the product Y^X with the product topology.

5. Show that evaluation can fail to be continuous when X is not locally compact, so the hypothesis of theorem 3.4.5 is not removable. Take $X = \mathbb{Q}$, $Y = \mathbb{R}$, and work at the point $(\mathbf{0}, q_0)$, where $\mathbf{0}$ is the zero function and $q_0 \in \mathbb{Q}$.

4

Separation Axioms

The Hausdorff condition removed many pathologies, but it still admits edge cases at odds with Euclidean intuition. A stronger condition, *normal* (or T_4), restores more of that intuition. This chapter climbs from the Hausdorff condition to normality, collecting the results that strengthening gives, and closes with Urysohn's lemma and its consequences.

4.1 Regularity and Normality

Definition 4.1.1: Regular

Suppose X is a topological space in which one-point sets are closed. Then X is *regular* if given $x \in X$, and a closed set B in X such that $x \notin B$, then there exists disjoint open sets U, W such that $x \in U$ $B \subseteq W$.

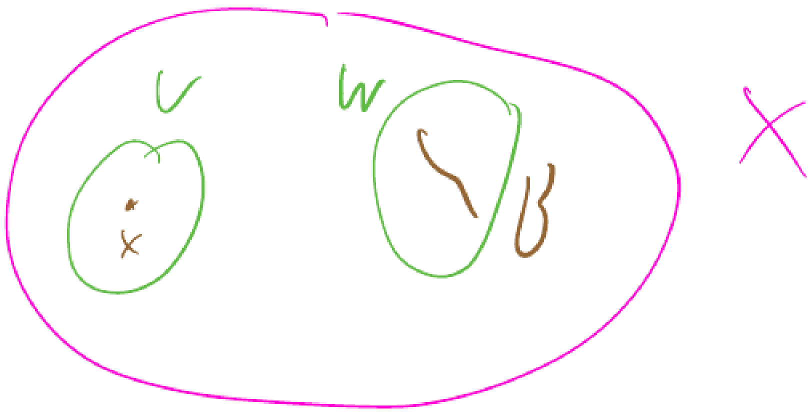

Figure 4.1: Regularity: a point and a disjoint closed set separated by open sets.

4.1.2 Note Note the standing assumption that one-point sets are closed. The converse implication, though it might seem intuitive enough, is not necessarily true (perhaps the trivial topology is a "pathological" example)

Example 4.1.3: Hausdorff but Not Regular Space

Here's a Hausdorff space that is not regular. One example is the space $\mathbb{R}_K$ constructed by Munkres. The underlying set is the reals, and the basis is chosen as the usual open intervals, along with all sets of the form $(a, b) \setminus K$ where $K = \{1/n | n \in \mathbb{N}\}$. This topology is finer than the usual topology on the reals, and the reals form a Hausdorff space under the usual topology, so $\mathbb{R}_K$ is also Hausdorff (passing to a finer topology preserves Hausdorffness).

However, $\mathbb{R}_K$ is not a regular space. For instance, the closed subset K in this space (closed because its complement is open by construction in this topology) and the point 0 cannot be separated by disjoint open subsets.

Definition 4.1.4: Normal

X is *normal* if for disjoint closed subsets $A, B \subseteq X$ there exist disjoint open subsets U, W such that $A \subseteq U$ and $B \subseteq W$

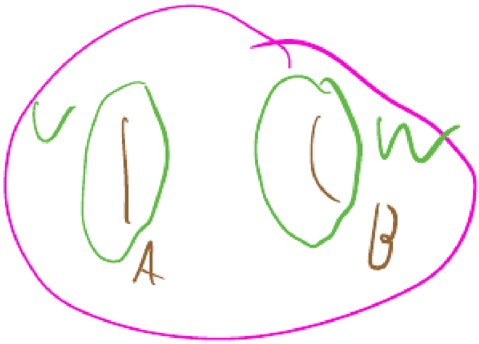

Figure 4.2: Normality: two disjoint closed sets separated by open sets.

The picture should not suggest a rise in "dimension" from a point to a line; the move is from points to *closed sets*. Nothing is dimensionally "higher" than an open or closed set, the whole space aside, so the visual is a little misleading.

With closed one-point sets, normality is the stronger condition: it separates two closed sets, of which separating a point from a closed set is the special case where one of them is a singleton. The converse fails, so the implication is strict: example 4.1.9 below is regular, being a product of regular spaces by theorem 4.1.8, and is not normal. Without closed points the two conditions are not comparable at all, as the two-point example after theorem 4.1.8 shows.

4.1.5 Chain of Implications Given that one-point sets are closed:

$$\text{Normal} \Rightarrow \text{Regular} \Rightarrow \text{Hausdorff}$$

Lemma 4.1.6: equivalent formulations ("zoom-in" axiom)

Let X be a topological space such that one-point sets are closed then

1. X is regular if and only if given $x \in X$, and open U containing x, $\exists$ open set V such that $x \in V$, and $\overline{V} \subseteq U$
2. X is normal if and only if given a closed set $A \subseteq X$ and open set U containing A, then $\exists$ open set V such that $A \subseteq V$ and $\overline{V} \subseteq U$

***Proof*:**
Both parts follow the same pattern: the separating open set around a closed set is traded for the complement of the other separating set, which is closed and therefore contains the closure.

1. *Necessity.* Suppose X is regular, let U be open with $x \in U$, and put $B = X \setminus U$, which is closed with $x \notin B$. Regularity supplies disjoint open sets $V \ni x$ and $W \supseteq B$. Since $V \cap W = \emptyset$ we have $V \subseteq X \setminus W$, and $X \setminus W$ is closed, so $\overline{V} \subseteq X \setminus W$ by proposition 1.4.12.

As $B \subseteq W$, this gives $\overline{V} \subseteq X \setminus B = U$.

Sufficiency. Suppose the stated condition holds, let B be closed and $x \notin B$. Then $U = X \setminus B$ is open and contains x, so there is an open V with $x \in V$ and $\overline{V} \subseteq U$. The sets V and $X \setminus \overline{V}$ are open, disjoint, and contain x and B respectively, so X is regular.

2. The argument is the same with the closed set A in place of the point x. For necessity, put $B = X \setminus U$, disjoint from A and closed; normality gives disjoint open $V \supseteq A$ and $W \supseteq B$, and as above $\overline{V} \subseteq X \setminus W \subseteq U$. For sufficiency, given disjoint closed A and B, apply the condition to A and the open set $U = X \setminus B$ to obtain V, and then V and $X \setminus \overline{V}$ separate A from B.

Both equivalences hold, as we sought to show.

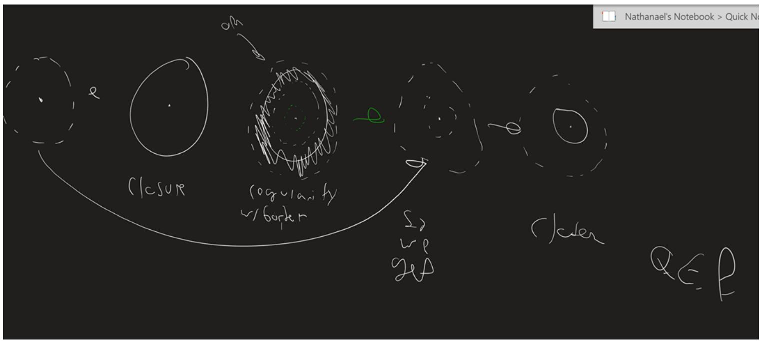

Figure 4.3: Regularity restated: every neighbourhood of x contains one whose closure still lies inside.

Note that regular spaces behave nicely under subspaces and product spaces

Theorem 4.1.7: subspaces of Regular Spaces

A subspace of a regular space is regular.

***Proof*:**

Let X be regular and $Y \subseteq X$ a subspace. One-point sets are closed in Y, since $\{y\} = \{y\} \cap Y$ is the trace of a closed set of X. Let $y \in Y$ and let B be closed in Y with $y \notin B$. Then $B = C \cap Y$ for some closed $C \subseteq X$, and $y \notin C$. Regularity of X supplies disjoint open $U \ni y$ and $W \supseteq C$ in X. Their traces $U \cap Y$ and $W \cap Y$ are open in Y, disjoint, and contain y and B respectively, completing the proof.

Theorem 4.1.8: Product of Regular Spaces

A Product of regular spaces is regular.

***Proof*:**
Let $\{X_j\}_{j\in J}$ be regular and let $X = \prod_j X_j$. One-point sets are closed in X, since $\{x\} = \bigcap_j \pi_j^{-1}(\{x_j\})$ is an intersection of closed sets.

It suffices to show that for $x \in X$ and a basic open $U \ni x$ there is an open V with $x \in V \subseteq \overline{V} \subseteq U$; this reformulation of regularity is the one supplied by the closure form of the axiom. Write $U = \prod_j U_j$ with U_j open and $U_j = X_j$ for j outside a finite set F. For $j \in F$ regularity of X_j gives an open V_j with $x_j \in V_j \subseteq \overline{V_j} \subseteq U_j$; for $j \notin F$ put $V_j = X_j$. Set $V = \prod_j V_j$, an open set containing x.

By proposition 2.2.17 the closure of a product is the product of the closures, so $\overline{V} = \prod_j \overline{V_j} \subseteq \prod_j U_j = U$. Hence X is regular, as we sought to show.

However, Though normal spaces have a stronger condition, they do not behave as nicely!

Example 4.1.9: Sorgenfrey Plane
For example, take the Sorgenfrey plane:

$$\beta_{\mathbb{R}_l^2} = \left\{[a,b) \times [c,d) \middle| a < b, c < d, a, b, c, d \in \mathbb{R}\right\}$$

This space is not normal.

***Proof*:**
Suppose $\mathbb{R}_l^2$ were normal. Consider the anti-diagonal

$$L = \{(x, -x) \mid x \in \mathbb{R}\}$$

Every basic set $[a,b) \times [c,d)$ meets L in at most one point, since $(x, -x)$ lies in it exactly when $a \leq x < b$ and $c \leq -x < d$, and taking the basic set $[x, x+1) \times [-x, -x+1)$ shows each point of L is isolated in the subspace L. So L is discrete. It is also closed: a point (u, v) with $v \neq -u$ has a basic neighbourhood missing L, obtained by shrinking both factors so that the sum of coordinates stays on one side of 0.

Consequently every subset $A \subseteq L$ is closed in $\mathbb{R}_l^2$, being closed in the closed discrete subspace L. Fix $A \subseteq L$. The sets A and $L \setminus A$ are disjoint closed sets, so normality provides disjoint open $U_A \supseteq A$ and $V_A \supseteq L \setminus A$.

The set $D = \mathbb{Q} \times \mathbb{Q}$ is dense in $\mathbb{R}_l^2$, since every basic set $[a,b) \times [c,d)$ with $a < b$ and $c < d$ contains a point with both coordinates rational. Consider the assignment $A \mapsto U_A \cap D$. It is injective: suppose $A \neq B$ and take, without loss of generality, $x \in A \setminus B$. Then $x \in U_A$ and $x \in L \setminus B \subseteq V_B$, so $W = U_A \cap V_B$ is a nonempty open set. By density there is $d \in W \cap D$. If $U_A \cap D = U_B \cap D$ then $d \in U_B$, while also $d \in V_B$, contradicting $U_B \cap V_B = \emptyset$. So the two traces differ.

Injectivity gives $|\mathcal{P}(L)| \leq |\mathcal{P}(D)|$, that is, $2^{|\mathbb{R}|} \leq 2^{\aleph_0} = |\mathbb{R}|$. Cantor's theorem forbids this. Hence $\mathbb{R}_l^2$ is not normal, as claimed.

Note that Since $\mathbb{R}_l$ is regular, then $(\mathbb{R}_l)^2$ is also regular. Normal spaces are in a sense worse than regular spaces because they lack this condition. Adding further restrictions to a space leaves a concept) then there is less flexibility given because more conditions need to be satisfied in order for things to work.

The converse does hold: if the product space is normal, then so is each factor.

Proposition 4.1.10: $X \times Y$ Normal Implies X, Y Normal

Let X and Y be nonempty spaces. If $X \times Y$ is normal, then so are X and Y.

***Proof*:**
If A, A' are disjoint closed subsets of X, then $A \times Y$, $A' \times Y$ are disjoint closed of $X \times Y$. Therefore, we can use normality and find O, O' open disjoint neighbourhoods. Write $O = \bigcup_i U_i \times V_i$, and $O' = \bigcup_j U'_j \times V'_j$ where eac hU_i, U'_j is open in X and V_i, V'_j is open in Y. Pick $y \in Y$. Then

$$\bigcup \{U_i | y \in V_i\} \qquad \bigcup \{U'_j | y \in V'_j\}$$

are disjoint open neighbourhoods of A and A'

One space is "vacuously normal" while being neither regular nor Hausdorff, which is worth care. Let $S = \{0, 1\}$ be the underlying set, with open sets

$$\{\emptyset, \{1\}, \{0, 1\}\}$$

Note that the closed sets are:

$$\{\emptyset, \{0\}, \{0, 1\}\}$$

This space is not Hausdorff: the point 1 is not closed, since its complement $\{0\}$ is not open, so 0 and 1 cannot be separated. Normality, however, holds vacuously.

Exercise 4.1.1

1. Show that a regular space is Hausdorff, and that a normal space whose one-point sets are closed is regular. Why does definition 4.1.1 build the hypothesis on points into the definition while definition 4.1.4 does not?

2. Show that in the co-finite topology on an infinite set any two nonempty open sets meet, and deduce that the space is T_1 but neither regular nor normal. Why does this not contradict the previous part?

3. Show that regularity is inherited by subspaces and preserved by products, using theorem 4.1.7 and theorem 4.1.8, and identify precisely where the argument fails for normality.

4. Use lemma 4.1.6 to show that in a regular space every point has a neighbourhood base of closed sets.

5. Show that a normal space in which every point is closed satisfies: any two disjoint closed sets A, B admit open $U \supseteq A$, $V \supseteq B$ with $\overline{U} \cap \overline{V} = \emptyset$.

4.2 Sufficient Conditions for Normality

Though regular spaces act more nicely when being manipulated, normal spaces on themselves are strong enough to have lots of strong equivalent implications.

Lemma 4.2.1: Compact Hausdorff Spaces Are Normal

Every compact Hausdorff space is normal.

***Proof*:**
Let X be compact Hausdorff and let A, B be disjoint closed sets. One-point sets are closed, since X is Hausdorff. Being closed in a compact space, A and B are compact.

For each $a \in A$, lemma 3.2.19 applied to the point a and the compact set B supplies disjoint open $U_a \ni a$ and $W_a \supseteq B$. The family $\{U_a\}_{a \in A}$ covers the compact set A, so finitely many $U_{a_1}, \ldots, U_{a_n}$ do. Put

$$U = \bigcup_{i=1}^{n} U_{a_i}, \qquad W = \bigcap_{i=1}^{n} W_{a_i}$$

Then U is open and contains A, and W is open as a finite intersection and contains B. They are disjoint: a point of $U \cap W$ lies in some U_{a_i} and in every W_{a_j}, in particular in W_{a_i}, contradicting $U_{a_i} \cap W_{a_i} = \emptyset$. So X is normal, as we sought to show.

Weakening compactness to local compactness still yields regularity:

Lemma 4.2.2: Locally Compact Hausdorff Spaces Are Regular

Every locally compact Hausdorff space is regular.

***Proof*:**
Let X be locally compact Hausdorff. One-point sets are closed in a Hausdorff space, so the standing hypothesis of definition 4.1.1 is met. Let $B \subseteq X$ be closed with $x \notin B$. Then $X \setminus B$ is an open neighbourhood of x, so by theorem 3.2.35 there is a neighbourhood V of x with $\overline{V}$ compact and $\overline{V} \subseteq X \setminus B$. The sets V and $X \setminus \overline{V}$ are open, disjoint, and contain x and B respectively, completing the proof.

Theorem 4.2.3: Second-Countable Regular Implies Normal

Every second-countable regular space is normal

4.2.4 Note The converse fails: a normal space need not be second-countable. The Sorgenfrey line of example 4.2.6 is normal, yet every basis for it is uncountable.

***Proof*:**
Let X be regular with countable basis $\mathcal{B}$, and let A and B be disjoint closed subsets. One-point sets are closed by the standing hypothesis of definition 4.1.1.

Step 1: countable covers with controlled closures. Let $a \in A$. Since $X \setminus B$ is open and contains a, regularity gives an open W with $a \in W$ and $\overline{W} \subseteq X \setminus B$, and then a basis element $U \in \mathcal{B}$ with $a \in U \subseteq W$, so that $\overline{U} \subseteq \overline{W}$ misses B. Collecting these over all $a \in A$ and discarding repetitions leaves a countable family, since $\mathcal{B}$ is countable; enumerate it $U_1, U_2, \ldots$. It covers A and satisfies $\overline{U_n} \cap B = \emptyset$ for every n. Symmetrically choose $V_1, V_2, \ldots$ covering B with $\overline{V_n} \cap A = \emptyset$.

Step 2: the correction. The unions $\bigcup U_n$ and $\bigcup V_n$ need not be disjoint, so trim them against each other. Put

$$U_n' = U_n \setminus \bigcup_{k \leq n} \overline{V_k}, \qquad V_n' = V_n \setminus \bigcup_{k \leq n} \overline{U_k}$$

Each is open, being an open set minus a finite union of closed sets, and we set $U' = \bigcup_n U_n'$ and $V' = \bigcup_n V_n'$.

Step 3: they still cover. If $a \in A$ then $a \in U_n$ for some n, and $a \notin \overline{V_k}$ for every k because $\overline{V_k} \cap A = \emptyset$; hence $a \in U_n' \subseteq U'$. So $A \subseteq U'$, and symmetrically $B \subseteq V'$.

Step 4: they are disjoint. Suppose $x \in U' \cap V'$, say $x \in U_j' \cap V_k'$. If $j \leq k$ then $x \in U_j' \subseteq U_j \subseteq \overline{U_j}$, while $x \in V_k'$ excludes x from $\overline{U_k}$ for every index up to k, in particular from $\overline{U_j}$, a contradiction. If $k < j$ the symmetric argument applies with the roles exchanged. So $U' \cap V' = \emptyset$, and X is normal, as we sought to show.

Proposition 4.2.5: Metrizable Spaces Are Normal

Every metrizable space is normal.

***Proof*:**
Let (X, d) be a metric space and let A, B be disjoint closed subsets. One-point sets are closed in a metric space, so the standing hypothesis for normality is met. For $a \in A$, the complement of B is open and contains a, so there is $\epsilon_a > 0$ with $B(a, \epsilon_a) \cap B = \emptyset$; choose such an ϵ_a for each $a \in A$, and symmetrically $\epsilon_b > 0$ for each $b \in B$ with $B(b, \epsilon_b) \cap A = \emptyset$. Put

$$U = \bigcup_{a \in A} B\left(a, \tfrac{\epsilon_a}{2}\right), \qquad V = \bigcup_{b \in B} B\left(b, \tfrac{\epsilon_b}{2}\right)$$

Both are open, being unions of balls, and $A \subseteq U$, $B \subseteq V$.

They are disjoint. Suppose $z \in U \cap V$, so $d(z, a) < \epsilon_a/2$ and $d(z, b) < \epsilon_b/2$ for some $a \in A$, $b \in B$. The triangle inequality gives

$$d(a, b) \leq d(a, z) + d(z, b) < \frac{\epsilon_a}{2} + \frac{\epsilon_b}{2} \leq \max\{\epsilon_a, \epsilon_b\}$$

If the maximum is ϵ_a this says $b \in B(a, \epsilon_a)$, contradicting $B(a, \epsilon_a) \cap B = \emptyset$; if it is ϵ_b it says $a \in B(b, \epsilon_b)$, contradicting the symmetric choice. So no such z exists, and X is normal, as we sought to show.

However, normal does not imply metrizable. The Sorgenfrey line settles both this and the converse question left open above.

Example 4.2.6: Normal, but not Metrizable, Spaces
The Sorgenfrey line $\mathbb{R}_l$, whose basis is $\{[a, b) \mid a < b\}$, is Hausdorff and normal, hence regular, but neither second-countable nor metrizable.

***Proof*:**
Hausdorff, and one-point sets closed. The topology $\mathbb{R}_l$ is finer than the standard topology, since $(a, b) = \bigcup_n [a + \frac{1}{n}, b)$, so the standard separating intervals remain open and $\mathbb{R}_l$ is Hausdorff by example 1.5.2(5). One-point sets are therefore closed by lemma 1.5.5, which is the standing hypothesis definition 4.1.1 requires, so normality below will also give regularity.

Normality. Let $A, B \subseteq \mathbb{R}_l$ be disjoint closed sets. For each $a \in A$ the complement of B is an open set containing a, so there is a basic set $[a, x_a)$ disjoint from B; symmetrically each $b \in B$ has $[b, y_b)$ disjoint from A. Put

$$U = \bigcup_{a \in A} [a, x_a), \qquad V = \bigcup_{b \in B} [b, y_b)$$

which are open and contain A and B respectively. They are disjoint: suppose $z \in [a, x_a) \cap [b, y_b)$. If $a \leq b$ then $a \leq b \leq z < x_a$, so $b \in [a, x_a)$, contradicting $[a, x_a) \cap B = \emptyset$. If $b < a$ then $a \in [b, y_b)$, contradicting the symmetric choice. So no such z exists and $\mathbb{R}_l$ is normal.

Not second-countable. Let $\mathcal{B}$ be any basis. For each $x \in \mathbb{R}$ the set $[x, x + 1)$ is open, so there is $B_x \in \mathcal{B}$ with $x \in B_x \subseteq [x, x + 1)$. Then $\inf B_x = x$, because B_x contains x and is contained in $[x, x + 1)$. The assignment $x \mapsto B_x$ is therefore injective, so $\mathcal{B}$ is uncountable.

Not metrizable. The set $\mathbb{Q}$ is dense in $\mathbb{R}_l$, since every $[a, b)$ with $a < b$ contains a rational, so $\mathbb{R}_l$ is separable. A separable metrizable space is second-countable by proposition 1.6.7. Since $\mathbb{R}_l$ is separable but not second-countable, no metric induces its topology.

Exercise 4.2.1

1. Show that every compact Hausdorff space is normal, and that the converse fails.
2. Show that a closed subspace of a normal space is normal, and give an example showing an arbitrary subspace need not be.
3. Show that the Sorgenfrey line $\mathbb{R}_l$ is normal but $\mathbb{R}_l \times \mathbb{R}_l$ is not, and explain what this says about theorem 4.1.8.
4. Show that a second-countable normal space whose one-point sets are closed is metrizable, using theorem 4.4.1 (proved two sections below). Why is the hypothesis on points needed?
5. Show that a metrizable space is normal without invoking Urysohn's lemma, and identify which step of proposition 4.2.5 uses the triangle inequality.

4.3 Urysohn Lemma

Urysohn lemma is one of the larger topological insights, and has a great multitude of application in other very large and important theorems (hence why it's called a lemma, it's like the key step for many important theorems).

Theorem 4.3.1: Urysohn

Let X be a normal space and let $A, B \subseteq X$ be disjoint closed sets. Then there exists a continuous function $f\colon X \to [0,1]$ with $A \subseteq f^{-1}(0)$ and $B \subseteq f^{-1}(1)$.

***Proof*:**
The construction rests on a rewording of normality: whenever a closed set sits inside an open set, an open set can be interposed with its closure still inside. Precisely, if C is closed, U is open, and $C \subseteq U$, then there is an open V with

$$C \subseteq V \subseteq \overline{V} \subseteq U$$

Indeed C and $X \setminus U$ are disjoint closed sets, so normality (definition 4.1.4) gives disjoint open sets $V \supseteq C$ and $W \supseteq X \setminus U$; then $V \subseteq X \setminus W$, and $X \setminus W$ is closed, so $\overline{V} \subseteq X \setminus W \subseteq U$.

Step 1: a dyadic scale of open sets. Let $D = \{k/2^n \mid n \geq 0,\ 0 \leq k \leq 2^n\}$ be the dyadic rationals in $[0,1]$. We build open sets U_q, one for each $q \in D$, satisfying

$$p < q \implies \overline{U_p} \subseteq U_q \tag{4.1}$$

with $A \subseteq U_0$ and $U_1 = X \setminus B$. Begin with $U_1 = X \setminus B$, open and containing A (since $A \cap B = \emptyset$); as A is closed and $A \subseteq U_1$, the interposition gives an open U_0 with $A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1$. Now induct on the level n: given U_q for all dyadics q of denominator dividing 2^{n-1} satisfying (4.1), each new dyadic $q = (2j+1)/2^n$ lies between its already-defined neighbours $p = 2j/2^n$ and $r = (2j+2)/2^n$, for which $\overline{U_p} \subseteq U_r$; the interposition applied to the closed set $\overline{U_p}$ inside the open set U_r yields an open U_q with $\overline{U_p} \subseteq U_q \subseteq \overline{U_q} \subseteq U_r$. This gives (4.1) for q against its immediate neighbours; for a non-adjacent older dyadic t it follows by transitivity through the previous level, chaining $\overline{U_s} \subseteq U_t$ with $U_t \subseteq \overline{U_t}$. Thus (4.1) holds at level n, completing the construction.

Step 2: the function. Define $f\colon X \to [0,1]$ by

$$f(x) = \inf\{q \in D \mid x \in U_q\}, \qquad \text{with } f(x) = 1 \text{ if } x \in U_q \text{ for no } q$$

If $x \in A$ then $x \in U_0$, so $f(x) = 0$; hence $A \subseteq f^{-1}(0)$. If $x \in B$ then $x \notin U_1 = X \setminus B$, and by (4.1) also $x \notin U_q$ for every $q < 1$ (as $\overline{U_q} \subseteq U_1$), so x lies in no U_q and $f(x) = 1$; hence $B \subseteq f^{-1}(1)$.

Step 3: continuity. It suffices to check that preimages of the subbasic sets $[0,a)$ and $(a,1]$ of $[0,1]$ are open. For the first,

$$f^{-1}\big([0,a)\big) = \{x \mid f(x) < a\} = \bigcup_{q<a} U_q$$

which is open, since $f(x) < a$ holds exactly when some dyadic $q < a$ has $x \in U_q$. For the second,

$$f^{-1}\big((a,1]\big) = \{x \mid f(x) > a\} = \bigcup_{q>a} \big(X \setminus \overline{U_q}\big)$$

also open. Indeed, if $x \notin \overline{U_q}$ for some $q > a$ then $x \notin U_p$ for all $p \leq q$ by (4.1), so $f(x) \geq q > a$; conversely if $f(x) > a$, pick dyadics $q < r$ with $a < q < r < f(x)$, so $x \notin U_r$, and $\overline{U_q} \subseteq U_r$ then forces $x \notin \overline{U_q}$ with $q > a$. Since every open subset of $[0,1]$ is a union of such subbasic sets, f is continuous.

4.3.2 Categorical Role and Universality of $[0,1]$ The codomain $[0,1]$ in theorem 4.3.1 acts as a universal separating object for normal spaces.

Let $S = \{0,1\}$ be the *Sierpiński space* equipped with open sets $\{\emptyset, \{1\}, \{0,1\}\}$. For any space X, continuous maps $g\colon X \to S$ are in bijection with open sets $U \subseteq X$ via $U = g^{-1}(\{1\})$, making S a subobject classifier for open sets in **Top**. Normality (definition 4.1.4) separates disjoint closed sets $A, B \subseteq X$ via indicator maps into S.

The proof of theorem 4.3.1 replaces S with its continuum completion $[0,1]$ by constructing a nested family of open sets $\{U_q\}_{q \in D}$ indexed by the dyadic rationals $D \subset [0,1]$. The function $f(x) = \inf\{q \in D \,|\, x \in U_q\}$ integrates these open classifications into a continuous map into $[0,1]$.

An object Q in a category $\mathcal{C}$ is a *cogenerator* if for any distinct morphisms $g_1, g_2\colon X \to Y$ in $\mathcal{C}$, there exists $h\colon Y \to Q$ such that $h \circ g_1 \neq h \circ g_2$. Theorem 4.3.1 establishes $[0,1]$ as a cogenerator for the category of normal T_1 spaces. Consequently, evaluation maps into powers $[0,1]^J$ yield topological embeddings (theorem 5.4.3).

The lemma turns a separation hypothesis into a continuous function, and having a function opens the door to arguments that no purely set-theoretic manipulation could reach. Here is a first instance: a bound on the *size* of a space, extracted from separation and connectedness alone.

Example 4.3.3: A Cardinality Bound From Urysohn
Let X be normal and connected, with one-point sets closed. If X has more than one point, then X is uncountable.

***Proof*:**
Take distinct $x, y \in X$. The sets $\{x\}$ and $\{y\}$ are closed by hypothesis and disjoint, so theorem 4.3.1 supplies a continuous $f : X \to [0,1]$ with $f(x) = 0$ and $f(y) = 1$.

The image $f(X)$ contains both 0 and 1, so theorem 3.1.23 places every value between them in $f(X)$ as well. Combined with $f(X) \subseteq [0,1]$ this gives $f(X) = [0,1]$.

The interval $[0,1]$ is uncountable. A function defined on a countable set has countable image, so X cannot be countable, which is the claim.

Exercise 4.3.1

1. Show that Urysohn's lemma gives no control over $f^{-1}(0)$ beyond $A \subseteq f^{-1}(0)$: exhibit a normal space, disjoint closed A, B, and a Urysohn function with $f^{-1}(0) \supsetneq A$.
2. Deduce from theorem 4.3.1 that in a normal space, two disjoint closed sets are separated by open sets whose closures are disjoint.

3. Show that a space in which every pair of disjoint closed sets admits a Urysohn function is normal, so the converse of theorem 4.3.1 holds.
4. Show that Urysohn's lemma fails without normality by exhibiting a Hausdorff space, two disjoint closed sets, and an argument that no continuous $f\colon X \to [0,1]$ takes the value 0 on one and 1 on the other. (Naming the space and the obstruction suffices.)
5. Use theorem 4.3.1 to show that a normal space with closed points and more than one point admits a non-constant continuous map to $[0,1]$, and deduce example 4.3.3 again.

4.4 Metrization Theorem

Urysohn's lemma has two immediate and substantial consequences:

Theorem 4.4.1: Urysohn Metrization Theorem

If X is regular and second-countable, then X is metrizable

***Proof*:**
Throughout, X is regular with a countable basis $\mathcal{B} = \{B_1, B_2, \dots\}$, so it is normal by theorem 4.2.3 and one-point sets are closed. The plan is to embed X in the cube $[0,1]^{\mathbb{Z}_+}$ and metrize the cube directly.

Step 1: a countable separating family of functions. Let P be the set of pairs (n,m) with $\overline{B_n} \subseteq B_m$; it is countable. For $(n,m) \in P$ the sets $\overline{B_n}$ and $X \setminus B_m$ are disjoint and closed, so Urysohn's lemma (theorem 4.3.1) supplies a continuous $g_{n,m}\colon X \to [0,1]$ equal to 1 on $\overline{B_n}$ and 0 on $X \setminus B_m$. Re-index this countable family as $f_1, f_2, \dots$.

The family separates points from closed sets: given $x \in X$ and an open $U \ni x$, choose B_m with $x \in B_m \subseteq U$, use regularity to find an open $W \ni x$ with $\overline{W} \subseteq B_m$, and then B_n with $x \in B_n \subseteq W$. Then $\overline{B_n} \subseteq \overline{W} \subseteq B_m$, so $(n,m) \in P$, and the corresponding function takes the value 1 at x and vanishes off $B_m \subseteq U$.

Step 2: the embedding. Define $F\colon X \to [0,1]^{\mathbb{Z}_+}$ by $F(x) = (f_1(x), f_2(x), \dots)$. Each coordinate is continuous, so F is continuous by theorem 2.2.15. It is injective: if $x \neq y$ then $X \setminus \{y\}$ is open and contains x, so some f_k has $f_k(x) = 1$ and $f_k(y) = 0$.

F is open onto its image. Let $U \subseteq X$ be open and $x_0 \in U$. Choose k with $f_k(x_0) = 1$ and $f_k = 0$ off U, and put $W = \pi_k^{-1}\big((0,1]\big) \cap F(X)$, open in $F(X)$ and containing $F(x_0)$. If $F(x) \in W$ then $f_k(x) > 0$, forcing $x \in U$, so $W \subseteq F(U)$. Hence F is a homeomorphism onto its image.

Step 3: the cube is metrizable. On $[0,1]^{\mathbb{Z}_+}$ set

$$D(x,y) = \sup_{n \in \mathbb{Z}_+} \frac{|x_n - y_n|}{n}$$

which is finite and bounded by 1. Symmetry and the triangle inequality pass through the supremum, and $D(x,y) = 0$ forces $x = y$, so D is a metric.

Its topology is the product topology. This is the case $X_n = [0,1]$, $d_n = |\cdot|$ of the computation in theorem 2.6.21, whose two inclusions apply verbatim.

Step 4: conclusion. A subspace of a metric space carries the topology of the restricted metric, so $F(X)$ is metrizable; X is homeomorphic to it by Step 2, hence metrizable in the sense of definition 2.6.5, completing the proof.

Exercise 4.4.1

1. Show that the hypotheses of theorem 4.4.1 cannot be weakened to regular alone, nor to second-countable alone.
2. Show that a compact Hausdorff second-countable space is metrizable.
3. Show that $\mathbb{R}_l$ is regular, separable and first-countable but not metrizable, and identify which hypothesis of the metrization theorem fails.
4. Show that for a metrizable space, second-countable, separable and Lindelöf are equivalent.
5. Show that the cube $[0,1]^J$ is metrizable when J is countable and not metrizable when J is uncountable.

4.5 Tietze Extension Theorem

Theorem 4.5.1: Tietze Extension Theorem

Let X be a normal space and let $A \subseteq X$ be a closed subset. let $f : A \to [0,1]$ be a continuous function. Then there exists a continuous function $F : X \to [0,1]$ such that $F|_A = f$.

The same holds with $[0,1]$ replaced by $\mathbb{R}$.

***Proof*:**
It is convenient to work with $[-1,1]$, which is homeomorphic to $[0,1]$ by an affine map, so a statement for one transfers to the other.

Step 1: the approximation lemma. Let $r > 0$ and $g\colon A \to [-r,r]$ be continuous. The sets $C = g^{-1}\big([-r,-r/3]\big)$ and $D = g^{-1}\big([r/3,r]\big)$ are closed in A, hence closed in X because A is closed in X, and they are disjoint. If both are nonempty, Urysohn's lemma (theorem 4.3.1) gives a continuous $h\colon X \to [-r/3,r/3]$ with $h = -r/3$ on C and $h = r/3$ on D; if one is empty, take h the appropriate constant. In every case $|g(a) - h(a)| \le 2r/3$ for all $a \in A$: on C both values lie in $[-r,-r/3]$, on D both lie in $[r/3,r]$, and elsewhere $|g(a)| < r/3$ while $|h(a)| \le r/3$.

Step 2: iteration. Set $g_1 = f$ viewed in $[-1,1]$. Given g_k with $|g_k| \le (2/3)^{k-1}$ on A, Step 1 with $r = (2/3)^{k-1}$ produces $h_k\colon X \to \mathbb{R}$ with $|h_k| \le \frac{1}{3}(2/3)^{k-1}$ and $|g_k - h_k| \le (2/3)^k$ on A; put $g_{k+1} = g_k - h_k|_A$.

Step 3: the sum. Let $F = \sum_{k\ge1} h_k$. The series is dominated by $\sum_k \frac{1}{3}(2/3)^{k-1} = 1$, so it converges uniformly on X; a uniform limit of continuous real functions is continuous, and F takes values in $[-1,1]$. On A the partial sums telescope to $f - g_{N+1}$ with $|g_{N+1}| \le (2/3)^N \to 0$, so $F|_A = f$. Rescaling to $[0, 1]$ gives the bounded case.

Step 4: the real-valued case. For continuous $f : A \to \mathbb{R}$, compose with a homeomorphism $\mathbb{R} \to (-1, 1)$ to get f', and extend it by Step 3 to $F' : X \to [-1, 1]$. The extension may attain ± 1, so let $E = (F')^{-1}(\{-1, 1\})$, closed and disjoint from A. Urysohn's lemma gives continuous $\varphi : X \to [0, 1]$ with $\varphi = 1$ on A and $\varphi = 0$ on E. Then $\varphi F'$ is continuous, agrees with f' on A, and takes values in $(-1, 1)$, being 0 on E and of modulus < 1 elsewhere. Composing back with $(-1, 1) \to \mathbb{R}$ extends f, completing the proof.

Exercise 4.5.1

1. Show that theorem 4.5.1 fails for the domain $\mathbb{R}$ and the subspace $A = \mathbb{Q}$, and identify the hypothesis that is violated.
2. Deduce Urysohn's lemma from the Tietze extension theorem.
3. Show that the extension in theorem 4.5.1 is not unique, and describe the set of all extensions when A is a single point of $\mathbb{R}$.
4. In the passage from the bounded case to the real-valued case, a Urysohn function is used to push the extension off the values ± 1. Show that this step fails if A is not closed, by identifying which hypothesis of theorem 4.3.1 is lost.
5. Show that Tietze extension fails for maps into S^1: exhibit a normal X, a closed A, and a continuous $A \to S^1$ with no continuous extension. (You may quote the no-retraction theorem, proved later as theorem 6.6.4.)

5

Paracompactness, Partitions of Unity, and Compactifications

The separation axioms of the previous chapter answered a question about existence: given a normal space, a closed set, and a disjoint closed set, there is a continuous function separating them. Urysohn's lemma produces that function one pair at a time, and for many purposes one function is not enough. Building a metric, or a measure, or a vector field on a space means assembling something globally out of pieces defined only on the members of an open cover, and the assembly needs a family of functions adapted to that cover: one bump per open set, each vanishing outside it, the whole family summing to one.

Such a family cannot exist on every space. The obstruction is not separation but size: an open cover can be so unruly that no point has a neighbourhood meeting only finitely many of its members, and then the sum defining the assembly need not converge. Paracompactness is the hypothesis that removes the obstruction, and this chapter develops it. The first two sections define paracompactness and identify the spaces that have it, including the metrizable ones and the second-countable locally compact Hausdorff ones. The third section proves the theorem the chapter exists for: a Hausdorff space is paracompact exactly when every open cover admits a subordinate partition of unity. The last turns from gluing to embedding, using the completely regular spaces to construct the Stone-Čech compactification, the largest compactification a space admits. The payoff outside this book is immediate: a space that looks locally like $\mathbb{R}^n$ and is Hausdorff and second countable falls under the criterion of the second section, so it is paracompact and carries partitions of unity. That is what allows the companion volume on differential topology, *Differential Topology* [9], to differentiate.

5.1 Locally Finite Families and Paracompactness

Two pieces of vocabulary are needed. The first is a smallness condition on a family of subsets: no point of the space may see infinitely many members of the family at once. The second is a loosening of "subcover" that makes the first condition attainable. Paracompactness is the conjunction of the two, and this section defines it, exhibits the first spaces that have it together with one Hausdorff space that does not, and then extracts what the hypothesis delivers in the way of separation: a paracompact Hausdorff space is normal.

A family of subsets can be infinite and still be manageable, provided its infinitude is spread out. What makes an infinite family unusable for assembly is a point at which infinitely many members pile up, because any construction that adds one contribution per member of the family then has infinitely many contributions to add at that point. The following condition forbids exactly that accumulation, and forbids it at every point simultaneously.

Definition 5.1.1: Locally Finite Family

Let X be a topological space and let $\mathcal{A} = \{A_\alpha\}_{\alpha \in J}$ be an indexed family of subsets of X. The family $\mathcal{A}$ is *locally finite* if every point of X has an open neighbourhood N such that $N \cap A_\alpha \neq \emptyset$ for only finitely many indices α. It is *point-finite* if every point of X lies in A_α for only finitely many indices α.

The family $\mathcal{A}$ is σ*-locally finite* if it can be written as a countable union

$$\mathcal{A} = \bigcup_{n \in \mathbb{N}} \mathcal{A}_n$$

in which each subfamily $\mathcal{A}_n$ is locally finite.

Demanding an *open* neighbourhood costs nothing: a neighbourhood of x meeting only finitely many members contains its own interior, which is an open neighbourhood of x meeting no more of them. Two consequences are immediate from the shape of the definition. A locally finite family is point-finite, since the neighbourhood at x already meets only finitely many members. And any subfamily of a locally finite family is locally finite, since the same neighbourhoods work; this is used constantly below and will not be remarked on again.

Local finiteness is a real restriction, not a formality. Cover the interval $(0, 1)$ by the sets

$$U_n = \left(\tfrac{1}{n}, 1\right), \qquad n \geq 2$$

This is an open cover: given $x \in (0, 1)$, any n with $1/n < x$ puts x in U_n. But the family $\{U_n\}_{n \geq 2}$ fails local finiteness at every single point, and fails it in the strongest way. Fix $x \in (0, 1)$ and choose n_0 with $1/n_0 < x$. Then $x \in U_n$ for every $n \geq n_0$, so *every* neighbourhood of x meets infinitely many members of the family. The family is not even point-finite. The failure is caused by the way the sets stack: each U_n is contained in all its successors, so they accumulate on the whole of $(0, 1)$ rather than spreading out. It is, however, σ-locally finite for a trivial reason: it is the countable union of the one-member subfamilies $\{U_n\}$, each of which is locally finite. The σ condition is correspondingly weaker, and it is the form in which a refinement is first obtained for metric spaces in section 5.2.

The repair is to slide the sets apart. Put

$$V_n = \left(\tfrac{1}{n+1}, \tfrac{1}{n-1}\right), \qquad n \geq 2$$

so that $V_2 = (1/3, 1)$, $V_3 = (1/4, 1/2)$, $V_4 = (1/5, 1/3)$, and so on. These sets still cover $(0,1)$: if $x > 1/3$ then $x \in V_2$, and otherwise there is a unique $m \geq 3$ with $1/(m+1) < x \leq 1/m$, and then $x \in V_m$ because $1/m < 1/(m-1)$. Each V_n sits inside a member of the original family, namely $V_n \subseteq U_{n+1}$. And the new family is locally finite: given $x \in (0,1)$, choose n_0 with $1/n_0 < x$, and observe that the open neighbourhood $(1/n_0, 1)$ of x meets V_n only when $1/(n-1) > 1/n_0$, that is only when $n \leq n_0$. The two families cover the same interval; only the second does so without piling up.

What the arguments of this chapter use is that closure commutes with the union of a locally finite family. For an arbitrary family one has only the containment $\bigcup_\alpha \overline{A_\alpha} \subseteq \overline{\bigcup_\alpha A_\alpha}$, and it is usually strict: the singletons $\{q\}$ for $q \in \mathbb{Q}$ have union $\mathbb{Q}$ with closure $\mathbb{R}$, while the union of their closures is $\mathbb{Q}$ again. Local finiteness removes the gap.

Let $\mathcal{A} = \{A_\alpha\}_{\alpha \in J}$ be locally finite in X. Then

$$\overline{\bigcup_{\alpha \in J} A_\alpha} = \bigcup_{\alpha \in J} \overline{A_\alpha} \tag{5.1}$$

The containment from right to left holds for any family, since each A_α lies in the union and closure preserves inclusion. For the other containment, let x lie in the closure of the union and let N be an open neighbourhood of x meeting only $A_{\alpha_1}, \dots, A_{\alpha_k}$. If x belonged to none of $\overline{A_{\alpha_1}}, \dots, \overline{A_{\alpha_k}}$, then for each i there would be an open $O_i \ni x$ with $O_i \cap A_{\alpha_i} = \emptyset$, and the open set

$$O = N \cap O_1 \cap \cdots \cap O_k$$

would be a neighbourhood of x meeting no member of $\mathcal{A}$ at all: it meets none of the A_{α_i} by construction, and none of the remaining members because it lies in N. That contradicts $x \in \overline{\bigcup_\alpha A_\alpha}$. Hence $x \in \overline{A_{\alpha_i}}$ for some i, which is (5.1). Taking each A_α closed gives the corollary that a locally finite family of closed sets has closed union, a statement that is false without local finiteness for the same reason the singleton example above fails.

The $(1/n, 1)$ cover of $(0,1)$ has no locally finite subfamily that still covers. A finite subfamily has union $(1/n, 1)$, where n is the largest index occurring in it, and so misses the piece $(0, 1/n]$; a covering subfamily is therefore infinite, and the computation above applies to any infinite subfamily just as it did to the whole. Passing to a subcover is too rigid an operation to produce local finiteness. The right operation was already performed on that example when the U_n were replaced by the V_n: the new sets are not members of the old family, only subsets of members.

Definition 5.1.2: Refinement of a Cover

Let $\mathcal{A}$ be a cover of a space X (definition 3.2.1). A family $\mathcal{B}$ of subsets of X *refines* $\mathcal{A}$, and is called a *refinement* of $\mathcal{A}$, if

1. $\mathcal{B}$ covers X, and
2. every $B \in \mathcal{B}$ is contained in some $A \in \mathcal{A}$.

If every member of $\mathcal{B}$ is open, $\mathcal{B}$ is an *open refinement* of $\mathcal{A}$; if every member is closed, a *closed refinement*.

The index sets of $\mathcal{A}$ and $\mathcal{B}$ are unrelated, and no map from $\mathcal{B}$ to $\mathcal{A}$ is part of the data. For each B there is at least one A containing it, but there may be many, and nothing in the definition records a choice; when an argument needs such a choice it must make one, and does so explicitly. A refinement may also be far larger than the cover it refines: the family of all one-point subsets of X refines every cover of X. A subfamily of $\mathcal{A}$ that still covers X, a subcover of $\mathcal{A}$, is a refinement of it, since each of its members is a member of $\mathcal{A}$ and so is contained in one, but the previous paragraph shows the converse fails. Finally, refinement is transitive: if $\mathcal{C}$ refines $\mathcal{B}$ and $\mathcal{B}$ refines $\mathcal{A}$, then each $C \in \mathcal{C}$ lies in some $B \in \mathcal{B}$, which lies in some $A \in \mathcal{A}$.

Compactness asks that every open cover be replaceable by a *finite* one lying under it. Weakening "finite" to "locally finite" and "subcover" to "refinement" gives the hypothesis this chapter is built on.

Definition 5.1.3: Paracompact Space

A topological space X is *paracompact* if every open cover of X admits a locally finite open refinement.

By definition 5.1.2 the refinement is again a cover of X; the requirement is not that some subfamily be locally finite but that the whole refining family be. Note what the definition does *not* say. It carries no separation axiom: a paracompact space here is not assumed Hausdorff, and every result below that needs the Hausdorff property states it as a hypothesis. Bourbaki, and much of the literature that follows it, builds Hausdorffness into the word; the convention is harmless in that setting, but it hides which arguments depend on separation, and the two theorems ending this section depend on it at every step.

Compactness implies the definition at once. Two further classes of spaces satisfy it by arguments of quite different shapes, and the last of the three is the model for the general theorems of section 5.2.

Example 5.1.4: Compact Spaces, Discrete Spaces, and Euclidean Space

1. Every compact space is paracompact. Let $\mathcal{A}$ be an open cover of a compact X. By definition 3.2.2 there is a finite subcover $\mathcal{A}' \subseteq \mathcal{A}$, which is an open refinement of $\mathcal{A}$ by the remark following definition 5.1.2. It is locally finite because X itself is an open neighbourhood of each of its points meeting only the finitely many members of $\mathcal{A}'$.

2. Every discrete space is paracompact. Let X carry the topology in which every subset is open, and let $\mathcal{A}$ be any open cover. The family $\mathcal{B} = \{\, \{x\} \,|\, x \in X \,\}$ consists of open sets and covers X; each $\{x\}$ lies in whichever member of $\mathcal{A}$ contains x, so $\mathcal{B}$ refines $\mathcal{A}$. It is locally finite because $\{x\}$ is an open neighbourhood of x meeting exactly one member of $\mathcal{B}$. Taking X uncountable gives a Hausdorff paracompact space that is neither compact nor second countable, so the weakening in definition 5.1.3 is not vacuous.

3. $\mathbb{R}^n$ is paracompact. Write $\|x\|$ for the Euclidean norm and set

$$B_m = \{x \in \mathbb{R}^n \mid \|x\| < m\} \quad (m \geq 1), \qquad B_m = \emptyset \quad (m \leq 0)$$

The annuli

$$K_m = \overline{B_m} \setminus B_{m-1} = \{x \in \mathbb{R}^n \mid m - 1 \leq \|x\| \leq m\}, \qquad m \geq 1$$

are closed and bounded, hence compact by the Heine-Borel characterization of the compact subsets of $\mathbb{R}^n$, and they cover $\mathbb{R}^n$: a point with $\|x\| = t$ lies in K_1 if $t \leq 1$ and in K_m for m

the least integer $\geq t$ otherwise. Around each annulus put the open collar

$$W_m = B_{m+1} \setminus \overline{B_{m-2}}, \qquad m \geq 1$$

so that $W_1 = B_2$ and $W_2 = B_3$, while $W_m = \{x \in \mathbb{R}^n \mid m-2 < \|x\| < m+1\}$ for $m \geq 3$. In every case $K_m \subseteq W_m$: for $m \geq 3$ because $m-1 > m-2$ and $m < m+1$, and for $m = 1, 2$ because $K_m \subseteq \overline{B_m} \subseteq B_{m+1}$.

Now let $\mathcal{A} = \{U_\alpha\}_{\alpha \in J}$ be an open cover of $\mathbb{R}^n$. For fixed m the sets $U_\alpha \cap W_m$ are open and cover the compact set K_m, so finitely many of them do; let $F_m \subseteq J$ be a finite index set with $K_m \subseteq \bigcup_{\alpha \in F_m}(U_\alpha \cap W_m)$. Put

$$\mathcal{V} = \{U_\alpha \cap W_m \mid m \geq 1,\ \alpha \in F_m\}$$

Every member of $\mathcal{V}$ is open and contained in some U_α, so $\mathcal{V}$ refines $\mathcal{A}$, and $\mathcal{V}$ covers $\mathbb{R}^n$ because every point lies in some K_m and hence in a member drawn from the m-th finite family.

Local finiteness is what the collars are for. Given $x \in \mathbb{R}^n$, choose an integer $m > \|x\|$, so that B_m is an open neighbourhood of x. Every point of B_m has norm less than m, while for $k \geq 3$ every point of W_k has norm exceeding $k-2$; hence $B_m \cap W_k = \emptyset$ whenever $k \geq m+2$, all such k being at least 3 because $m \geq 1$. Thus B_m meets only those members of $\mathcal{V}$ built from $W_1, \ldots, W_{m+1}$, and the collar W_k contributes at most $|F_k|$ of them. The neighbourhood B_m therefore meets at most $|F_1| + \cdots + |F_{m+1}|$ members of $\mathcal{V}$, a finite number.

The first two items are cheap, and the second shows how far paracompactness is from compactness: an uncountable discrete space is as far from compact as a space can be. The mechanism behind the third item is the general one. $\mathbb{R}^n$ was exhausted by a sequence of compact pieces K_m, each piece was handled by compactness alone, and the collars W_m kept the resulting finite families from interfering with each other beyond a bounded range. Exhaustion by compact pieces plus a device for keeping the pieces apart is the standard route to paracompactness, and section 5.2 runs it in general.

Since $\mathbb{R}^n$ is Hausdorff, locally compact, and second countable, one might guess that Hausdorffness is already enough. It is not, and the standard counterexample is a space too long to admit any countable exhaustion at all. It is assembled from the smallest uncountable well-ordering.

Example 5.1.5: The Minimal Uncountable Well-Ordered Set

Call a well-ordered set $(\Omega, <)$ *minimal uncountable* if Ω is uncountable while every proper initial segment

$$S_\alpha = \{x \in \Omega \mid x < \alpha\}, \qquad \alpha \in \Omega$$

is countable. Such a set exists. By the well-ordering theorem, a form of the axiom of choice equivalent to theorem 3.2.41, some uncountable set W admits a well-ordering. If every proper initial segment of W is countable, take $\Omega = W$. Otherwise the set of $\alpha \in W$ whose initial segment is uncountable is nonempty, so it has a least element α_0, and $\Omega = S_{\alpha_0}$ is uncountable with every proper initial segment countable by minimality of α_0. Write 0 for the least element of Ω.

The property that drives everything is that Ω cannot be reached from below by countably many steps:

$$\text{every countable } S \subseteq \Omega \text{ has a strict upper bound in } \Omega \tag{5.2}$$

For if no $\beta \in \Omega$ satisfied $s < \beta$ for all $s \in S$, then every $x \in \Omega$ would satisfy $x \leq s$ for some $s \in S$, giving $\Omega = \bigcup_{s \in S}(S_s \cup \{s\})$, a countable union of countable sets and hence countable, which it is not.

Give Ω the order topology of definition 1.3.13, which is generated equally well by the subbasis consisting of the initial segments S_α and the final segments $T_\alpha = \{x \in \Omega \mid x > \alpha\}$. This space is Hausdorff: given $\alpha < \beta$, if some γ satisfies $\alpha < \gamma < \beta$ then S_γ and T_γ are disjoint open sets containing α and β, while if no such γ exists then S_β and T_α are disjoint and contain α and β respectively.

Now consider the open cover $\mathcal{U} = \{S_\alpha\}_{\alpha \in \Omega}$, which covers by (5.2) applied to the one-point set $\{x\}$. Suppose, for contradiction, that $\mathcal{V}$ is a locally finite open refinement of $\mathcal{U}$. Each $V \in \mathcal{V}$ lies in some member of $\mathcal{U}$; fix such a bound and write $V \subseteq S_{\alpha_V}$.

Build a strictly increasing sequence in Ω as follows. Set $\beta_0 = 0$. Given β_n, local finiteness supplies an open $N_n \ni \beta_n$ meeting only finitely many members of $\mathcal{V}$, say $V_1^n, \ldots, V_{k_n}^n$. The finite set $\{\beta_n, \alpha_{V_1^n}, \ldots, \alpha_{V_{k_n}^n}\}$ is countable, so by (5.2) it has a strict upper bound; let β_{n+1} be one. Then $\beta_{n+1} > \beta_n$ and $\beta_{n+1} > \alpha_{V_i^n}$ for every $i \leq k_n$.

The set $\{\beta_n \big| n \in \mathbb{N}\}$ is countable, so it has strict upper bounds, and the set of them is well-ordered; let β be its least element. Then $\beta_n < \beta$ for every n, and no element below β is a strict upper bound, so for every $\gamma < \beta$ there is an n with $\gamma < \beta_n < \beta$.

Since $\mathcal{V}$ covers Ω, some $V \in \mathcal{V}$ contains β. Finite intersections of subbasic sets form a basis, so a set of one of the forms Ω, S_δ, T_η, $S_\delta \cap T_\eta$ contains β and lies inside V. In each case there is a $\gamma < \beta$ with $\{x \in \Omega \mid \gamma < x \leq \beta\} \subseteq V$: take $\gamma = \eta$ when a final segment occurs, which satisfies $\eta < \beta$ because $\beta \in T_\eta$, and $\gamma = 0$ otherwise, which is below β because $\beta > \beta_0 = 0$. In the two cases involving S_δ the containment holds because $x \leq \beta < \delta$. Choose n with $\gamma < \beta_n < \beta$. Then $\beta_n \in V$, and also $\beta_n \in N_n$, so V meets N_n and is therefore one of $V_1^n, \ldots, V_{k_n}^n$. Consequently $\alpha_V < \beta_{n+1} < \beta$, and $V \subseteq S_{\alpha_V}$ forces $\beta \notin V$, contradicting $\beta \in V$. No locally finite open refinement of $\mathcal{U}$ exists, so Ω is a Hausdorff space that is not paracompact.

The obstruction is not a shortage of open sets, and not a failure of separation: the space is Hausdorff. The cover $\mathcal{U}$ climbs Ω from below, and any refinement of it consists of bounded sets; the argument shows that a locally finite family of bounded sets can only climb countably far, since each step past a point is a step past the finitely many members visible near that point, and countably many such steps stop short of Ω by (5.2). Compare the $\mathbb{R}^n$ computation, where the whole proof was the countable exhaustion $\mathbb{R}^n = \bigcup_m K_m$: here no countable family of bounded sets exhausts the space, and paracompactness fails for exactly that reason.

Insert a copy of the interval $[0, 1)$ between each element of Ω and its successor and order the result lexicographically: on

$$Y = \Omega \times [0, 1)$$

declare $(\alpha, s) < (\beta, t)$ if $\alpha < \beta$, or if $\alpha = \beta$ and $s < t$. This is a linear order whose smallest element is the pair $(0, 0)$, and Y carries its order topology, generated as before by the initial segments $\{y \in Y \mid y < s\}$ and the final segments $\{y \in Y \mid y > s\}$. Deleting the smallest element gives the *long line*

$$L = Y \setminus \{(0, 0)\}$$

with the subspace topology, in which $\alpha \in \Omega$ is identified with $(\alpha, 0)$.[1] Since $(0, 0)$ is the smallest element of Y, the traces on L of the subbasic sets of Y are the initial and final segments of L together with $\emptyset$ and L itself, so the subspace topology on L is the order topology of the inherited order.

The deletion matters, because the neighbourhoods of the smallest element of Y are half-open. The sets

$$P_u = \{y \in Y \mid y < (0, u)\} = \{0\} \times [0, u), \qquad 0 < u < 1$$

form a neighbourhood base at $(0, 0)$. An open set containing $(0, 0)$ contains a finite intersection of subbasic sets containing $(0, 0)$; no final segment contains the smallest element, so that intersection is either Y itself, which contains every P_u, or an initial segment $\{y \in Y \mid y < s\}$ with $s > (0, 0)$, which contains P_u as soon as $(0, u) \le s$, and such a u exists precisely because s exceeds $(0, 0)$. Each P_u is carried homeomorphically onto $[0, u) \subseteq \mathbb{R}$ by $\theta\colon (0, t) \mapsto t$: the map is an order isomorphism, and the traces on P_u of the subbasic sets of Y are $\emptyset$, P_u, and the sets $\theta^{-1}\big([0, v)\big)$ and $\theta^{-1}\big(\{t \in \mathbb{R} \mid v < t < u\}\big)$ for $0 \le v < u$, whose images generate the Euclidean topology on $[0, u)$.

Suppose now that h were a homeomorphism of an open $U \ni (0, 0)$ onto an open interval $I \subseteq \mathbb{R}$, and put $c = h((0, 0))$. Choose u with $P_u \subseteq U$. Then $h(P_u)$ is open and contains c, so it contains an open interval I' around c, and $g = \theta \circ h^{-1}$ carries I' homeomorphically onto a set $G \subseteq [0, u)$ which is open in $[0, u)$, contains $g(c) = 0$, and therefore contains $[0, \epsilon)$ for some $\epsilon > 0$. Being a continuous image of the connected set I', the set G is connected (proposition 3.1.17), and a connected subset of $\mathbb{R}$ is convex: if $p < w < q$ with $p, q \in G$ and $w \notin G$, then $\{x \in G \mid x < w\}$ and $\{x \in G \mid x > w\}$ would be nonempty disjoint open subsets of G covering it. Let $r = \sup G$, which is at most u and at least ϵ. Convexity gives $[0, r) \subseteq G$, while $r \notin G$, since $r \in G$ would force $r < u$ and then openness of G in $[0, u)$ would supply points of G above r. So $G = [0, r)$, and $G \setminus \{0\}$ is the open interval from 0 to r, which is connected (corollary 3.1.21). But g carries $I' \setminus \{c\}$ homeomorphically onto $G \setminus \{0\}$, and $I' \setminus \{c\}$ is disconnected, being split by the two open pieces of I' on either side of c. So $(0, 0)$ has no neighbourhood homeomorphic to an open interval of $\mathbb{R}$, and its removal is what makes it possible for every point to have one. That every point of L does have one is a transfinite induction along Ω, and it is carried out where the long line is put to work as a counterexample, in *Differential Topology* [9].

What is needed here is only the covering behaviour of L, and there the work done for Ω largely transfers. First, L is Hausdorff: the separation of two points carried out in example 5.1.5 used nothing about Ω beyond its being linearly ordered and carrying the order topology, so it applies to L word for word. Second, L is not paracompact. Cover it by the initial segments

$$Q_\alpha = \{y \in L \mid y < (\alpha, 0)\}, \qquad \alpha \in \Omega$$

which are open (that for $\alpha = 0$ being empty) and do cover, since $(\gamma, t) < (\alpha, 0)$ whenever $\alpha > \gamma$, and such an α exists by (5.2). Suppose $\mathcal{V}$ were a locally finite open refinement, and fix for each $V \in \mathcal{V}$ an $\alpha_V \in \Omega$ with $V \subseteq Q_{\alpha_V}$. Two steps of the argument of example 5.1.5 are then not transcriptions, because L is not well-ordered: the set $\{(0, t) \mid 0 < t < 1\}$ of points of L with first coordinate 0 has no least element.

The first change is in the recursion, whose bounds α_V live in Ω while the points being constructed live in L; the two are kept apart by running the recursion in the first coordinate. Set $\beta_0 = (0, 1/2)$, a

[1] The name follows Munkres. Elsewhere L is called the long ray and "long line" is kept for the space obtained by gluing two copies of $\Omega \times [0, 1)$ at their smallest elements; both properties recorded here hold for that space as well.

point of L because only $(0,0)$ was removed, and write $\delta_n \in \Omega$ for the first coordinate of β_n, so that $\delta_0 = 0$. Given β_n, local finiteness supplies an open $N_n \ni \beta_n$ meeting only $V_1^n, \dots, V_{k_n}^n$, and (5.2) supplies a strict upper bound $\delta_{n+1} \in \Omega$ for the finite set $\{\delta_n, \alpha_{V_1^n}, \dots, \alpha_{V_{k_n}^n}\}$; put $\beta_{n+1} = (\delta_{n+1}, 0)$. Then $\delta_{n+1} > \delta_n$ forces $\beta_{n+1} > \beta_n$, and $\delta_{n+1} > \alpha_{V_i^n}$ for every $i \le k_n$.

The second change is in the passage to a limit. In Ω that limit was the least strict upper bound of the β_n, and its existence came from the strict upper bounds forming a subset of a well-ordered set, which is exactly what fails in L. So the limit too is formed in the first coordinate: the countable set $\{\delta_n \big| n \in \mathbb{N}\}$ has strict upper bounds by (5.2), those bounds form a subset of the well-ordered Ω, and γ is taken to be its least element. Put $\beta = (\gamma, 0)$, a point of L because $\gamma > \delta_0 = 0$. Then $\beta_n < \beta$ for every n, and every $y \in L$ with $y < \beta$ satisfies $y < \beta_m < \beta$ for some m: writing $y = (\zeta, t)$, the inequality $y < (\gamma, 0)$ forces $\zeta < \gamma$, so ζ is not a strict upper bound of the δ_n by minimality of γ, whence $\delta_n \ge \zeta$ for some n and therefore $\delta_{n+1} > \zeta$, giving $y < \beta_{n+1} < \beta$.

From here the original argument transcribes. Some $V \in \mathcal{V}$ contains β, and a basic open set containing β and lying inside V has one of the forms L, $\{y \in L \mid y < s\}$, $\{y \in L \mid y > s'\}$, or the intersection of the last two, so in each case there is a $y_0 < \beta$ with $\{y \in L \mid y_0 < y \le \beta\} \subseteq V$: take $y_0 = s'$ when a final segment occurs, which lies below β because β lies in that segment, and $y_0 = \beta_0$ otherwise, the containment in the cases involving an initial segment holding because $y \le \beta < s$. Choose m with $y_0 < \beta_m < \beta$. Then β_m lies in V and in N_m, so V meets N_m and is one of $V_1^m, \dots, V_{k_m}^m$, whence $\alpha_V < \delta_{m+1} < \gamma$. But then $\beta = (\gamma, 0)$ exceeds $(\alpha_V, 0)$, so $\beta \notin Q_{\alpha_V}$ and $V \subseteq Q_{\alpha_V}$ forces $\beta \notin V$, contradicting $\beta \in V$. So L is a Hausdorff space that is not paracompact, which is the reason local Euclidean behaviour on its own cannot serve as the definition of a manifold.

Paracompactness was defined purely by a covering condition, with no mention of separating open sets. In the presence of the Hausdorff axiom it manufactures them, and it does so by the argument compactness uses in the same situation, with the finite subcover replaced by a locally finite refinement and finiteness of the union replaced by (5.1).

Theorem 5.1.6: Paracompact Hausdorff Spaces Are Regular

Every paracompact Hausdorff space is regular.

***Proof*:**
Let X be paracompact and Hausdorff.

Step 1: One-point sets are closed. Fix $x \in X$. For each $y \neq x$ the Hausdorff property supplies an open $W_y \ni y$ with $x \notin W_y$, and then $X \setminus \{x\} = \bigcup_{y \neq x} W_y$ is open. Hence the standing hypothesis of definition 4.1.1 is met, and it remains to separate a point from a closed set not containing it.

Step 2: A cover adapted to the point. Fix $a \in X$ and a closed $B \subseteq X$ with $a \notin B$. For each $b \in B$ the Hausdorff property gives disjoint open sets $U_b \ni b$ and $N_b \ni a$. Since N_b is an open neighbourhood of a disjoint from U_b, we have $a \notin \overline{U_b}$. The family

$$\mathcal{A} = \{U_b\}_{b \in B} \cup \{X \setminus B\}$$

is an open cover of X: a point of B lies in its own U_b, and a point outside B lies in $X \setminus B$, which is open because B is closed.

Step 3: Refining. By paracompactness choose a locally finite open refinement $\mathcal{V}$ of $\mathcal{A}$, and set

$$\mathcal{V}' = \{V \in \mathcal{V} \mid V \cap B \neq \emptyset\}, \qquad W = \bigcup_{V \in \mathcal{V}'} V$$

Then W is open and $B \subseteq W$: each $b \in B$ lies in some $V \in \mathcal{V}$, and that V meets B, so $V \in \mathcal{V}'$. Each $V \in \mathcal{V}'$ is contained in some member of $\mathcal{A}$, and it cannot be contained in $X \setminus B$ because it meets B; so $V \subseteq U_b$ for some $b \in B$, whence $\overline{V} \subseteq \overline{U_b}$ and $a \notin \overline{V}$.

Step 4: Closing off the union. The family $\mathcal{V}'$ is a subfamily of a locally finite family, hence locally finite, so (5.1) gives

$$\overline{W} = \bigcup_{V \in \mathcal{V}'} \overline{V}$$

By Step 3 the point a lies in none of the sets on the right, so $a \notin \overline{W}$. Therefore $X \setminus \overline{W}$ is an open set containing a, and it is disjoint from W, which is an open set containing B. These separate a from B, as we sought to show.

The Hausdorff property separates a from each point of B one at a time, producing infinitely many open sets U_b whose union contains B but whose closure may contain a all the same. What the refinement gives is a family whose union has a computable closure: local finiteness converts the pointwise information $a \notin \overline{U_b}$ into the global statement $a \notin \overline{W}$. In the compact case the same conversion is performed by finiteness, since a finite union of closures is the closure of the union without any hypothesis. Paracompactness supplies enough finiteness to justify that step.

The Hausdorff hypothesis cannot be dropped. The two-point space with the indiscrete topology is compact, hence paracompact by example 5.1.4, yet its one-point sets are not closed, so it does not even satisfy the standing hypothesis of definition 4.1.1.

Regularity separates a point from a closed set. Normality (definition 4.1.4) separates two closed sets, and it is normality, not regularity, that Urysohn's lemma (theorem 4.3.1) requires. The step from one to the other is the same argument run again, with the point a replaced by a closed set A; what justifies the replacement is exactly the regularity just proved, which is what supplies open sets around b whose closures miss all of A rather than a single point.

Theorem 5.1.7: Paracompact Hausdorff Spaces Are Normal

Every paracompact Hausdorff space is normal.

***Proof*:**
Let X be paracompact and Hausdorff, and let $A, B \subseteq X$ be disjoint closed sets.

Step 1: A cover adapted to the closed set A. Fix $b \in B$. Since A is closed and $b \notin A$, the regularity established in theorem 5.1.6 gives disjoint open sets $U_b \ni b$ and $N_b \supseteq A$. Because N_b is open, contains A, and is disjoint from U_b, no point of A is in the closure of U_b:

$$\overline{U_b} \cap A = \emptyset$$

As in theorem 5.1.6, the family $\mathcal{A} = \{U_b\}_{b \in B} \cup \{X \setminus B\}$ is an open cover of X.

Step 2: Refining. Choose a locally finite open refinement $\mathcal{V}$ of $\mathcal{A}$ and set

$$\mathcal{V}' = \{V \in \mathcal{V} \mid V \cap B \neq \emptyset\}, \qquad W = \bigcup_{V \in \mathcal{V}'} V$$

The set W is open and contains B, by the argument of Step 3 of theorem 5.1.6, which used nothing about a. Each $V \in \mathcal{V}'$ meets B, so it is not contained in $X \setminus B$ and therefore lies in some U_b; hence $\overline{V} \cap A \subseteq \overline{U_b} \cap A = \emptyset$.

Step 3: Closing off the union. The family $\mathcal{V}'$ is locally finite, so (5.1) applies and

$$\overline{W} \cap A = \Big(\bigcup_{V \in \mathcal{V}'} \overline{V} \Big) \cap A = \emptyset$$

Thus $X \setminus \overline{W}$ is an open set containing A, and it is disjoint from the open set $W \supseteq B$. Disjoint open sets therefore separate A from B, completing the proof.

The theorem also supplies the cheapest way to certify that a space fails to be paracompact, since it may be read in contrapositive: a Hausdorff space that is not normal cannot be paracompact. The book already carries such a space.

Example 5.1.8: The Sorgenfrey Plane Is Not Paracompact

The Sorgenfrey plane $\mathbb{R}_l^2$ of example 4.1.9, the product of two copies of the line with the half-open intervals $[a, b)$ as basis, is Hausdorff, being a product of Hausdorff spaces by proposition 2.2.10, and is not normal. By theorem 5.1.7 it is therefore not paracompact.

Two spaces have now been separated from paracompactness by different routes. For Ω the obstruction was exhibited directly, by producing a cover no locally finite family refines; for $\mathbb{R}_l^2$ it is inherited from a separation failure, at the cost of citing a harder theorem. The second route is shorter whenever normality is already known to fail, and the first is the only one available when the space is normal, as L is.

The two theorems together say that a covering hypothesis, imposed on a space that is only assumed Hausdorff, forces regularity and normality, the two strongest separation axioms of chapter 4. Both classical sources of normality are recovered as instances. Compact Hausdorff spaces are paracompact by example 5.1.4, so lemma 4.2.1 is the case in which the refinement may be taken finite. Metrizable spaces will be shown paracompact by A. H. Stone's theorem, at which point proposition 4.2.5 becomes another instance of the same argument rather than a separate metric computation.

Normality supplies a stock of functions. By theorem 4.3.1, disjoint closed sets in a normal space are separated by a continuous $f\colon X \to [0, 1]$ taking the value 0 on one and 1 on the other, so a paracompact Hausdorff space carries such a function for every pair of disjoint closed sets it contains.

Exercise 5.1.1

1. Show that a locally finite family is closure-preserving: $\overline{\bigcup A_\alpha} = \bigcup \overline{A_\alpha}$, and that this fails for an arbitrary family.

2. Show that every compact space is paracompact, and that a discrete space is paracompact.

3. Show that a closed subspace of a paracompact space is paracompact, and explain why the argument does not extend to arbitrary subspaces.

4. Show that a refinement of a refinement is a refinement, and that a paracompact space admits, for every open cover $\{U_\alpha \mid \alpha \in A\}$, a locally finite open refinement indexed by the *same* set A, with the α-th member contained in U_α.

5. Show that Ω of example 5.1.5 is not paracompact but is countably compact, so paracompactness is not implied by limit-point conditions.

5.2 Metrizable and Locally Compact Spaces are Paracompact

Two theorems do the work of this section. The first, due to A. H. Stone, is that every metrizable space is paracompact, which delivers the whole metric world of section 2.6 at one stroke. The second is that a second-countable locally compact Hausdorff space is paracompact, which delivers $\mathbb{R}^n$, its open subsets, and everything locally modelled on them without any metric being named. The two proofs have no machinery in common: the first cuts the members of a cover down by a scale of radii, the second exhausts the space by compact sets. The section closes by reconciling them for locally compact Hausdorff spaces that are locally connected and locally second countable, where paracompactness, metrizability, and a countability condition on the connected components are a single condition.

Producing a locally finite open cover refining a given open cover (definitions 3.2.1 and 5.1.2) directly is awkward, because local finiteness is a constraint that every point of the space must certify at once. What is often easy to produce instead is a refinement that is locally finite *in layers*: a σ-locally finite family in the sense of definition 5.1.1, a countable union of subfamilies each of which is locally finite on its own, while the whole need not be. The stacked cover of $(0,1)$ by the intervals $(1/n, 1)$ for $n \geq 2$, in section 5.1, was one. The layers can then be pared against one another, each layer deleting what the earlier layers already covered. The following lemma carries out that paring.

Lemma 5.2.1: σ-Locally Finite Refinements Suffice

Let X be a regular space (definition 4.1.1) in which every open cover admits a σ-locally finite open cover refining it. Then X is paracompact (definition 5.1.3).

***Proof*:**
Two facts about a locally finite family $\mathcal{C}$ are used repeatedly. First, $\left\{\overline{C} \;\middle|\; C \in \mathcal{C}\right\}$ is again locally finite: an open set meets $\overline{C}$ only if it meets C, so a neighbourhood meeting only finitely many members of $\mathcal{C}$ meets only finitely many of their closures. Second, if the members of $\mathcal{C}$ are closed, then the union of any subfamily $\mathcal{C}' \subseteq \mathcal{C}$ is closed: given $x \notin \bigcup \mathcal{C}'$, choose an open $W \ni x$ meeting only $C_1, \ldots, C_r$ of the members of $\mathcal{C}'$; then $W \setminus (C_1 \cup \cdots \cup C_r)$ is an open neighbourhood of x disjoint from $\bigcup \mathcal{C}'$.

Step 1: every open cover has a locally finite refinement, by sets not necessarily open. Let $\mathcal{U}$ be an open cover and let $\mathcal{V} = \bigcup_{n \geq 1} \mathcal{V}_n$ be a σ-locally finite open cover refining it. Write $G_n = \bigcup \mathcal{V}_n$ and, for $V \in \mathcal{V}_n$, set

$$S_V = V \setminus \bigcup_{i<n} G_i$$

Each S_V is contained in V, hence in a member of $\mathcal{U}$, so the collection of all S_V refines $\mathcal{U}$. It covers X: given x, let n be the least index with $x \in G_n$ and pick $V \in \mathcal{V}_n$ with $x \in V$; then x lies in no G_i with $i < n$, so $x \in S_V$. For local finiteness, keep this n and this V. If $m > n$ and $W \in \mathcal{V}_m$, then S_W is disjoint from G_n and hence from V. For each $i \leq n$ choose an open $W_i \ni x$ meeting only finitely many members of $\mathcal{V}_i$; since $S_W \subseteq W$ for every W, the open neighbourhood

$$V \cap W_1 \cap \cdots \cap W_n$$

of x meets only finitely many of the sets S_W.

Step 2: every open cover has a locally finite closed refinement. Let $\mathcal{U}$ be an open cover and let

$\mathcal{P}$ be the collection of all open sets whose closure is contained in some member of $\mathcal{U}$. Then $\mathcal{P}$ covers X: given x, choose $U \in \mathcal{U}$ with $x \in U$; the set $X \setminus U$ is closed and misses x, so regularity provides disjoint open sets $P \ni x$ and $Q \supseteq X \setminus U$, and then $P \subseteq X \setminus Q$ with $X \setminus Q$ closed, whence $\overline{P} \subseteq X \setminus Q \subseteq U$. Being an open cover, $\mathcal{P}$ has a locally finite refinement $\mathcal{C}$ by Step 1. The closures $\{\overline{C} \mid C \in \mathcal{C}\}$ form a locally finite family of closed sets covering X, and if $C \subseteq P \in \mathcal{P}$ with $\overline{P} \subseteq U \in \mathcal{U}$, then $\overline{C} \subseteq U$. So the closures refine $\mathcal{U}$.

Step 3: every open cover has a locally finite open refinement. Let $\mathcal{U}$ be an open cover and let $\mathcal{D}$ be a locally finite closed cover refining it, as furnished by Step 2. Let $\mathcal{B}$ be the collection of all open sets meeting only finitely many members of $\mathcal{D}$; local finiteness of $\mathcal{D}$ says exactly that $\mathcal{B}$ covers X, so Step 2 applies to $\mathcal{B}$ and yields a locally finite closed cover $\mathcal{E}$ refining it. For each $D \in \mathcal{D}$ fix $U_D \in \mathcal{U}$ with $D \subseteq U_D$, and put

$$F(D) = U_D \setminus \bigcup \{E \in \mathcal{E} \mid E \cap D = \emptyset\}$$

The subtracted set is a union of a subfamily of the locally finite closed family $\mathcal{E}$, hence closed, so $F(D)$ is open; and $F(D) \subseteq U_D$, so these sets refine $\mathcal{U}$. They cover X, because $D \subseteq F(D)$ for every D: a point of D lies in U_D and in no member of $\mathcal{E}$ disjoint from D.

It remains to see that $\{F(D) \mid D \in \mathcal{D}\}$ is locally finite. Let $x \in X$ and choose an open $W \ni x$ meeting only the members $E_1, \ldots, E_k$ of $\mathcal{E}$. Since $\mathcal{E}$ covers X, every point of W lies in one of $E_1, \ldots, E_k$, so $W \subseteq E_1 \cup \cdots \cup E_k$. If $F(D)$ meets W then $F(D)$ meets some E_j, and by the construction of $F(D)$ this forces $E_j \cap D \neq \emptyset$. Each E_j lies in some $B_j \in \mathcal{B}$, and B_j meets only finitely many members of $\mathcal{D}$; so for each j there are only finitely many D with $E_j \cap D \neq \emptyset$, and therefore only finitely many D with $F(D) \cap W \neq \emptyset$. Every open cover of X thus has a locally finite open cover refining it, which is to say X is paracompact, as we sought to show.

The lemma converts a global requirement into a layered one, and layers are easy to come by: a countable cover is σ-locally finite for trivial reasons, each layer being a single set. Regularity is the hypothesis that drives the conversion, and it enters at exactly one point, Step 2, where a cover must be replaced by one whose members have small closures. The lemma will be applied to metric spaces, where regularity is a direct computation with the metric.

The first of the two main theorems is A. H. Stone's, from 1948. The construction below, which well-orders the index set of the cover and cuts each member down by a dyadic scale of radii, follows the short proof M. E. Rudin published in 1969. The well-ordering breaks the symmetry among the members of the cover by declaring one of them responsible for each point, and the radii then keep the pieces assigned to distinct members a definite distance apart.

Theorem 5.2.2: Stone's Theorem: Metrizable Spaces are Paracompact

Every metrizable space (definition 2.6.5) is paracompact.

***Proof*:**
Let X be metrizable and fix a metric d inducing its topology (section 2.6); write $B(x, r) = \{y \in X \mid d(x, y) < r\}$.

First, X is regular, and the metric proves this directly. One-point sets are closed, since for $y \neq x$ the ball $B(y, d(x, y))$ misses x, so that $X \setminus \{x\}$ is a union of open balls. Now let B be a closed set and $x \notin B$; the case $B = \emptyset$ is trivial, so take B nonempty. Since $X \setminus B$ is open there is an

$\varepsilon > 0$ with $B(x, \varepsilon) \cap B = \emptyset$, so that $r = \inf\{d(x, b) \mid b \in B\}$ satisfies $r \geq \varepsilon > 0$. The sets

$$U = B(x, r/3), \qquad W = \bigcup_{b \in B} B(b, r/3)$$

are open and contain x and B respectively, and they are disjoint: a point $w \in U \cap W$ would lie in $B(b, r/3)$ for some $b \in B$, whence $d(x, b) \leq d(x, w) + d(w, b) < 2r/3 < r$, contradicting the definition of r. So disjoint open sets separate x from B. By lemma 5.2.1 it therefore suffices to produce, for an arbitrary open cover, a σ-locally finite open cover refining it.

Let $\{C_\alpha\}_{\alpha \in A}$ be an open cover of X and fix a well-ordering of the index set A; this is the axiom of choice again, in the form equivalent to Zorn's lemma (theorem 3.2.41). For $\alpha \in A$ and $n \geq 1$, call a point $x \in X$ an (α, n)*-centre* if

1. α is the least index with $x \in C_\alpha$, and
2. $B(x, 3 \cdot 2^{-n}) \subseteq C_\alpha$.

Define

$$D_{\alpha,n} = \bigcup \{B(x, 2^{-n}) \mid x \text{ is an } (\alpha, n)\text{-centre}\}$$

Step 1: the sets $D_{\alpha,n}$ are open and refine the cover. Each is a union of open balls, hence open. If x is an (α, n)-centre then $B(x, 2^{-n}) \subseteq B(x, 3 \cdot 2^{-n}) \subseteq C_\alpha$, so $D_{\alpha,n} \subseteq C_\alpha$.

Step 2: the sets $D_{\alpha,n}$ cover X. Let $x \in X$. The well-ordering gives a least α with $x \in C_\alpha$, and since C_α is open there is an $n \geq 1$ with $B(x, 3 \cdot 2^{-n}) \subseteq C_\alpha$. Then x is an (α, n)-centre, so $x \in B(x, 2^{-n}) \subseteq D_{\alpha,n}$.

Step 3: for each fixed n, the family $\{D_{\alpha,n}\}_{\alpha \in A}$ is locally finite. The claim is the quantitative statement that any two points taken from distinct members of this family are more than 2^{-n} apart. Let $\beta \neq \gamma$, say $\beta < \gamma$ in the well-ordering, and let $p \in D_{\beta,n}$ and $q \in D_{\gamma,n}$. Choose a (β, n)-centre y with $p \in B(y, 2^{-n})$ and a (γ, n)-centre z with $q \in B(z, 2^{-n})$. Condition (2) for y gives $B(y, 3 \cdot 2^{-n}) \subseteq C_\beta$, while condition (1) for z says that γ is the least index with $z \in C_\gamma$; as $\beta < \gamma$, this forces $z \notin C_\beta$ and hence $d(y, z) \geq 3 \cdot 2^{-n}$. The triangle inequality now gives

$$d(p, q) \geq d(y, z) - d(y, p) - d(z, q) > 3 \cdot 2^{-n} - 2^{-n} - 2^{-n} = 2^{-n}$$

Consequently the ball $B(w, 2^{-n-1})$ about any point w meets at most one member of $\{D_{\alpha,n}\}_{\alpha \in A}$, since two points of that ball are at distance less than 2^{-n}. A family for which every point has a neighbourhood meeting at most one member is in particular locally finite.

The collection $\{D_{\alpha,n} \mid \alpha \in A,\ n \geq 1\}$ is therefore an open cover refining $\{C_\alpha\}_{\alpha \in A}$ and σ-locally finite, the layer indexed by n being $\{D_{\alpha,n}\}_{\alpha \in A}$. By lemma 5.2.1, X is paracompact, completing the proof.

Every metric space is thus paracompact: $\mathbb{R}^n$ and each of its subspaces, every normed space, every discrete space. Separation is not what stands in the way of building locally finite refinements, the only separation used above being the regularity computed on the spot from the metric; what the proof needed on top of that was a device for choosing, once and for all, which member of the cover owns a given point. That is what the well-ordering supplies, and it is where the axiom of choice enters. Because normality was nowhere an input, the theorem supplies a second route to proposition 4.2.5: a metrizable space is Hausdorff and, by the above, paracompact, hence normal by theorem 5.1.7.

The implication runs one way only. A paracompact Hausdorff space need not be metrizable, and the cheapest witnesses are compact: a compact space is paracompact, a finite subcover of a cover being in particular a locally finite refinement of it. Take J uncountable and consider the cube $[0,1]^J$ with the product topology (definition 2.2.13), compact by Tychonoff's theorem (theorem 3.2.44) and Hausdorff. It is not metrizable, because in a metric space every one-point set is a countable intersection of open sets, $\{x\} = \bigcap_{n\geq 1} B(x, 1/n)$, whereas in $[0,1]^J$ no one-point set is. Indeed, suppose $\{x\} = \bigcap_{n\geq 1} W_n$ with each W_n open. Each W_n contains a basic open set around x which constrains only finitely many coordinates, say those in a finite set F_n; since J is uncountable there is a coordinate $j \notin \bigcup_n F_n$, and the point y that agrees with x off j and differs from x at j lies in every W_n yet is not x.

The second theorem drops the metric and asks only for countability and local compactness. Its proof is a different mechanism entirely: instead of separating pieces by distance, it exhausts the space by compact sets that grow slowly enough for a neighbourhood of any point to see only finitely many stages of the growth. The lemma above plays no part in it.

Theorem 5.2.3: Second-Countable Locally Compact Hausdorff Spaces are Paracompact

Let X be a second-countable (definition 1.6.1) locally compact (definition 3.2.34) Hausdorff space. Then X is paracompact.

***Proof*:**
Assume $X \neq \emptyset$; the empty space is paracompact, every open cover of it being locally finite already.

Step 1: X has a compact exhaustion. The claim is that there are compact sets $K_1, K_2, \ldots$ with

$$K_n \subseteq \mathrm{int} K_{n+1} \quad \text{for all } n, \qquad \bigcup_{n\geq 1} K_n = X$$

Let $\mathcal{B}$ be a countable basis for X and let $\mathcal{B}'$ consist of those members of $\mathcal{B}$ whose closure is compact. Then $\mathcal{B}'$ still covers X: given x, theorem 3.2.35 applied to x and the neighbourhood X produces a neighbourhood V of x with $\overline{V}$ compact, and choosing $B \in \mathcal{B}$ with $x \in B \subseteq \mathrm{int} V$ gives $\overline{B} \subseteq \overline{V}$, so that $\overline{B}$ is a closed subset of a compact space and hence compact by theorem 3.2.7. In particular $\mathcal{B}'$ is nonempty, since X is; enumerate $\mathcal{B}' = \{W_1, W_2, \ldots\}$, repeating a set if the list is finite.

Put $K_1 = \overline{W_1}$. Given the compact set K_n, the sets W_i cover it, so finitely many suffice; choose an index $j_n \geq n$ with $K_n \subseteq W_1 \cup \cdots \cup W_{j_n}$ and set $K_{n+1} = \overline{W_1} \cup \cdots \cup \overline{W_{j_n}}$, a finite union of compact sets and hence compact. Since $W_1 \cup \cdots \cup W_{j_n}$ is an open set contained in K_{n+1}, it lies in $\mathrm{int} K_{n+1}$, so $K_n \subseteq \mathrm{int} K_{n+1}$. And $j_n \geq n$ gives $W_n \subseteq K_{n+1}$ for every n, so the K_n exhaust X.

Step 2: an exhaustion produces locally finite refinements. Set $K_{-1} = K_0 = \emptyset$ and, for $n \geq 1$,

$$A_n = K_n \setminus \mathrm{int} K_{n-1}, \qquad O_n = \mathrm{int} K_{n+1} \setminus K_{n-2}$$

Each A_n is closed in the compact set K_n, hence compact by theorem 3.2.7. Each O_n is open, since K_{n-2} is compact in a Hausdorff space and therefore closed by theorem 3.2.16. Moreover $A_n \subseteq O_n$: on one hand $A_n \subseteq K_n \subseteq \mathrm{int} K_{n+1}$, and on the other $K_{n-2} \subseteq \mathrm{int} K_{n-1}$ is deleted from K_n in forming A_n. The A_n cover X, for if $x \in X$ and n is least with $x \in K_n$, then $x \notin K_{n-1} \supseteq \mathrm{int} K_{n-1}$ and so $x \in A_n$.

Now let $\mathcal{U}$ be an open cover of X. Fix n. For each $x \in A_n$ choose $U \in \mathcal{U}$ containing x; then $U \cap O_n$ is an open neighbourhood of x contained in a member of $\mathcal{U}$. These sets cover the compact set A_n, so finitely many of them do; let $\mathcal{V}_n$ be such a finite family and put $\mathcal{V} = \bigcup_{n \geq 1} \mathcal{V}_n$. Every member of $\mathcal{V}$ lies in a member of $\mathcal{U}$, and $\mathcal{V}$ covers X because the A_n do.

Finally $\mathcal{V}$ is locally finite. Given $x \in X$, choose m with $x \in K_m$; then $x \in \mathrm{int}K_{m+1}$, an open neighbourhood of x. If $n \geq m+3$ then $K_{m+1} \subseteq K_{n-2}$, so $\mathrm{int}K_{m+1}$ is disjoint from O_n and hence from every member of $\mathcal{V}_n$. The neighbourhood $\mathrm{int}K_{m+1}$ therefore meets only members of $\mathcal{V}_1, \ldots, \mathcal{V}_{m+2}$, finitely many sets in all. So $\mathcal{U}$ has a locally finite open cover refining it, completing the proof.

Step 1 is where second countability was used, and it was used for one thing only: a countable family of compact sets exhausting the space. Call a space *σ-compact* if it is the union of countably many compact subspaces. Step 2 used nothing about X beyond local compactness, the Hausdorff property, and the existence of the exhaustion, so it proves more than was asked: a locally compact Hausdorff space with a compact exhaustion is paracompact. Every σ-compact locally compact Hausdorff space has one. Indeed, write $X = \bigcup_{m \geq 1} L_m$ with each L_m compact, take $K_1 = \emptyset$, and given K_n cover the compact set $K_n \cup L_n$ by finitely many open sets with compact closure (theorem 3.2.35) and let K_{n+1} be the union of their closures; then $K_n \subseteq \mathrm{int}K_{n+1}$, and $L_n \subseteq K_{n+1}$ for every n, so the K_n exhaust X. In $\mathbb{R}^n$ the exhaustion can be taken to be K_m, the closed ball of radius m, and the sets O_m are then the open annuli; this is worked out in example 5.2.5.

The second theorem is not more general than the first. A second-countable locally compact Hausdorff space is regular by lemma 4.2.2 and hence metrizable by the Urysohn metrization theorem of chapter 4, so its hypotheses cut out a subclass of the metrizable spaces, and a proper one: a discrete space of uncountable cardinality is metrizable but not second countable. What the second theorem contributes is not reach but the exhaustion, which builds the refinement stage by stage in place of an appeal to the well-ordering theorem, and hypotheses that can be checked without producing a metric. In the locally compact case the two theorems can be reconciled into a single equivalence. Two hypotheses are needed beyond local compactness and the Hausdorff property, and both hold in any space locally modelled on $\mathbb{R}^n$: local connectedness, so that the connected components are open and the space is the disjoint assembly of them, and local second countability, so that a countable exhaustion of a component forces a countable basis on it. Without the second the equivalence fails, and $[0,1]^J$ is again the witness.

Corollary 5.2.4: Paracompactness for Locally Compact Spaces

Let X be a locally compact Hausdorff space which is locally connected, in the sense that every point has a neighbourhood base of connected open sets (section 3.1), and in which every point has a second-countable open neighbourhood. Then the following are equivalent:

1. X is paracompact;
2. every connected component of X is σ-compact;
3. every connected component of X is second countable;
4. X is metrizable.

***Proof*:**
Begin with three observations about the components of X. Each component C is open: given $x \in C$, local connectedness supplies a connected open V with $x \in V$, and a connected set meeting a component is contained in it, so $x \in V \subseteq C$. Since the components partition X and each is open, each is also closed, its complement being the union of the others. Finally, C is again locally compact and Hausdorff: it is open in X, so corollary 3.2.39 applies, and the Hausdorff property passes to subspaces at once. The four statements are then linked in the cycle (1) ⇒ (2) ⇒ (3) ⇒ (4) ⇒ (1).

1. (1) *implies* (2). Let C be a component and let $\mathcal{A}$ be a cover of C by sets open in C, hence open in X as C is open. Adding the open set $X \setminus C$ gives an open cover of X; let $\mathcal{W}$ be a locally finite open cover of X refining it. Discarding the members disjoint from C and intersecting the rest with C produces a locally finite open cover of C refining $\mathcal{A}$, since a member of $\mathcal{W}$ that meets C cannot be contained in $X \setminus C$ and so lies in a member of $\mathcal{A}$. Thus C is paracompact.

   Now use paracompactness of C to exhaust it. By theorem 3.2.35 applied to a point of C and to the neighbourhood C itself, every point of C has a neighbourhood with compact closure, and passing to its interior gives an open one; those interiors form an open cover of C, so let $\mathcal{V}$ be a locally finite open cover of C refining it. Each $V \in \mathcal{V}$ has $\overline{V}$ compact, being a closed subset of a compact closure (theorem 3.2.7). A component is nonempty, so $\mathcal{V}$ has a nonempty member; fix one, call it V_0, and define

   $$G_1 = V_0, \qquad G_{k+1} = \bigcup \{ V \in \mathcal{V} \mid V \cap \overline{G_k} \neq \emptyset \}$$

   Each $\overline{G_k}$ is compact, by induction: $\overline{G_1}$ is compact, and if $\overline{G_k}$ is compact then, covering it by finitely many neighbourhoods each meeting only finitely many members of $\mathcal{V}$, it meets only finitely many members $V_1, \ldots, V_r$; so $G_{k+1} = V_1 \cup \cdots \cup V_r$ and $\overline{G_{k+1}} \subseteq \overline{V_1} \cup \cdots \cup \overline{V_r}$ is a closed subset of a compact set. Also $\overline{G_k} \subseteq G_{k+1}$, since every point of $\overline{G_k}$ lies in some member of $\mathcal{V}$ and that member meets $\overline{G_k}$. Hence $G = \bigcup_{k \geq 1} G_k$ equals $\bigcup_{k \geq 1} \overline{G_k}$ and is σ-compact. It is open, being a union of open sets, and it is closed: if $x \in \overline{G}$, pick $V \in \mathcal{V}$ with $x \in V$; then V meets G, hence meets some $\overline{G_k}$, hence $V \subseteq G_{k+1} \subseteq G$ and $x \in G$. As C is connected and G is a nonempty clopen subset, $G = C$, and C is σ-compact.

2. (2) *implies* (3). Let C be a component, say $C = \bigcup_{m \geq 1} L_m$ with each L_m compact. Every $x \in C$ has a second-countable open neighbourhood N in X, and $N \cap C$ is then an open subset of X containing x and second countable, the traces of a countable basis of N forming a countable basis of $N \cap C$. Each compact L_m is covered by finitely many such sets, so C is the union of countably many open second-countable subspaces $P_1, P_2, \ldots$. Collecting a countable basis of each P_i gives a countable family of sets open in C, and it is a basis: if $x \in U$ with U open in C, then $x \in P_i$ for some i, and $U \cap P_i$ is open in P_i, so some basic set of P_i contains x and lies in U. Hence C is second countable.

3. (3) *implies* (4). Each component C is locally compact Hausdorff, hence regular by lemma 4.2.2, and second countable by hypothesis; the Urysohn metrization theorem of chapter 4 makes it metrizable. Choose for each component C_i a metric d_i inducing its topology and replace d_i by $\min(d_i, 1)$, which induces the same topology because balls of radius less than 1 are unchanged. Define d on X by $d(x, y) = d_i(x, y)$ when x and y lie in the same component C_i, and $d(x, y) = 1$ otherwise. This is a metric: symmetry and the vanishing condition

are clear, and for the triangle inequality note that all values lie in $[0,1]$, so the only case needing comment is when y lies in a different component from x or from z, and then $d(x,y)+d(y,z) \geq 1 \geq d(x,z)$. For $r<1$ the d-ball $B(x,r)$ is the d_i-ball of the same radius in the component of x, so a set is d-open exactly when its trace on each component is open in that component, which, the components being open and partitioning X, is exactly openness in X. Hence d induces the topology of X and X is metrizable.

4. (4) *implies* (1). This is theorem 5.2.2.

The cycle is closed, completing the proof.

The equivalence localizes paracompactness into a countability condition, one component at a time, and that is what makes it usable: to check that a space of this kind is paracompact, exhibit a countable basis or a countable family of compact sets on each component and stop. The hypotheses cannot simply be dropped. Local second countability fails for $[0,1]^J$ with J uncountable, and that space satisfies (1), being compact, and (2), its components being closed in a compact space and hence compact, yet it is not metrizable. Local connectedness is what makes the componentwise conditions say anything at all, since without open components the decomposition in the first implication is unavailable; outright connectedness would serve as well in its place, a connected space being its own only component. There is no such substitute for local second countability, since $[0,1]^J$ is itself connected and locally connected. Nor do the hypotheses on their own deliver the four conditions: the long line L constructed in section 5.1 is connected, locally connected, locally compact and Hausdorff, and each of its points has a second-countable neighbourhood, so the corollary applies to it; since it is not paracompact, it satisfies none of the four.

The concrete instance behind all of this is Euclidean space and the open sets inside it, where both routes to paracompactness are available and both can be made explicit.

Example 5.2.5: Euclidean Space and Its Open Subsets

1. $\mathbb{R}^n$ is metrizable, so theorem 5.2.2 already makes it paracompact. The second route is more informative. The balls with rational centres and rational radii form a countable basis, so $\mathbb{R}^n$ is second countable; the closed balls are closed and bounded, hence compact by the Heine-Borel theorem, so $\mathbb{R}^n$ is locally compact; and it is Hausdorff. Thus theorem 5.2.3 applies, and its proof can be followed with $K_m = \{x \in \mathbb{R}^n \mid \|x\| \leq m\}$ as the exhaustion, for which $\mathrm{int}K_{m+1}$ is the open ball of radius $m+1$ and $K_m \subseteq \mathrm{int}K_{m+1}$. The compact shells and the open annuli of Step 2 are

$$A_m = \{x \mid m-1 \leq \|x\| \leq m\}\,, \qquad O_m = \{x \mid m-2 < \|x\| < m+1\}$$

   the second formula holding for $m \neq 2$; at $m=2$ the convention $K_{-1} = K_0 = \emptyset$ of the proof gives $O_2 = \mathrm{int}K_3 \setminus K_0 = \{x \mid \|x\| < 3\}$, the whole open ball rather than the punctured one, while at $m=1$ the displayed set $\{x \mid -1 < \|x\| < 2\}$ agrees with O_1, being the whole open ball of radius 2. These are exactly the compact annuli and the open collars arranged by hand in item 3 of example 5.1.4, edge cases included; what the theorem adds is that they are not a device invented for $\mathbb{R}^n$ but the output of Step 1. So the recipe reads: given any open cover of $\mathbb{R}^n$, cover the shell A_m by finitely many sets of the cover cut down to the annulus O_m, and take all of these over all m. A point of norm r lies in the open ball of radius $m+1$ for any integer $m \geq r$, and that ball misses every annulus O_k with $k \geq m+3$, so it meets only finitely many members of the refinement.

2. Let $U \subseteq \mathbb{R}^n$ be open and nonempty, with $U \neq \mathbb{R}^n$. Then U is a metric space in the induced metric, so it is paracompact by theorem 5.2.2; it is also open in $\mathbb{R}^n$, hence locally compact by corollary 3.2.39, and second countable, the traces of a countable basis of $\mathbb{R}^n$ forming a countable basis. Its exhaustion can be written down:

$$K_m = \{x \in U \mid \|x\| \leq m \text{ and } d(x, \mathbb{R}^n \setminus U) \geq 1/m\}$$

   Each K_m is closed in $\mathbb{R}^n$ (both conditions are closed conditions, the distance to a fixed set being continuous) and bounded, hence compact by the Heine-Borel theorem. If $x \in K_m$ and $\|y - x\| < \min\{1, 1/m - 1/(m+1)\}$, then $\|y\| < m+1$ and $d(y, \mathbb{R}^n \setminus U) \geq d(x, \mathbb{R}^n \setminus U) - \|x - y\| > 1/(m+1)$, so $y \in K_{m+1}$; thus $K_m \subseteq \mathrm{int} K_{m+1}$. And every $x \in U$ lies in some K_m, because $\mathbb{R}^n \setminus U$ is closed and misses x, so $d(x, \mathbb{R}^n \setminus U) > 0$.

3. An open $U \subseteq \mathbb{R}^n$ satisfies every hypothesis of corollary 5.2.4: it is locally compact and Hausdorff, it is locally connected because the balls contained in it form a neighbourhood base of connected sets, and it is locally second countable. All four conditions of the corollary hold for it, and they can be read off directly. Its components are open connected subsets of $\mathbb{R}^n$; each is second countable, and each is σ-compact by the exhaustion of item (2) applied to it when it is a proper subset of $\mathbb{R}^n$ and by that of item (1) when $U = \mathbb{R}^n$; U is metrizable; and U is paracompact. The corollary says these four facts are one fact, so for open subsets of $\mathbb{R}^n$ any one of them may be checked in place of the others.

Exercise 5.2.1

1. Show that $\mathbb{R}^n$ is paracompact in two ways: from theorem 5.2.2 and from theorem 5.2.3.
2. Show that a σ-compact locally compact Hausdorff space is paracompact, and that $\mathbb{R}^\omega$ in the product topology is paracompact but not locally compact.
3. Show that the Sorgenfrey line is paracompact, and reconcile this with example 5.1.8.
4. Show that a regular Lindelöf space is normal, and note that this gives a second route to normality of $\mathbb{R}_l$.
5. Show that paracompactness is not productive, using example 5.1.8.

The exercise closes the section on a warning: a product of two paracompact spaces need not be paracompact, the Sorgenfrey line against itself being the standard witness. That failure is a failure of one factor to be small, and compactness repairs it. The theorem below records the repair in the form the rest of the series needs, where the product is only local: a map that looks like a projection onto a compact factor near each point of the base carries paracompactness upward. The fiber bundles of *Algebraic Topology* [7] and *Vector Bundles and K-Theory* [11], whose defining property is exactly the local product form below, are the intended consumers, and the paracompactness of their total spaces is what this supplies.

Theorem 5.2.6: Locally Trivial Maps with Compact Fiber Preserve Paracompactness

Let F be a compact space and let $p\colon E \to B$ be a continuous surjection that is *locally trivial with fiber* F: every $b \in B$ has an open neighbourhood U and a homeomorphism $h\colon p^{-1}(U) \to U \times F$ satisfying $\pi_U \circ h = p$ on $p^{-1}(U)$, where π_U denotes the projection to U. If B is paracompact, then E is paracompact. If moreover B and F are Hausdorff, then E is Hausdorff, so that E is paracompact Hausdorff and therefore normal by theorem 5.1.7.

***Proof*:**
Let $\mathcal{V}$ be an open cover of E. A locally finite open refinement of it is produced by refining downstairs and cutting the result against finitely many members of $\mathcal{V}$ over each piece.

Step 1: Over a small enough neighbourhood, finitely many members of $\mathcal{V}$ suffice. Fix $b \in B$ and a trivializing neighbourhood N of b with homeomorphism $h\colon p^{-1}(N) \to N \times F$ over N. The fiber $p^{-1}(b)$ is carried by h onto $\{b\} \times F$, which is compact, so finitely many members $V_1, \dots, V_m$ of $\mathcal{V}$ cover $p^{-1}(b)$. The image

$$W = h\big((V_1 \cup \dots \cup V_m) \cap p^{-1}(N)\big)$$

is open in $N \times F$ and contains $\{b\} \times F$. Since F is compact, the projection $\pi_N\colon N \times F \to N$ is a closed map (example 3.2.12), so

$$U_b = N \setminus \pi_N\big((N \times F) \setminus W\big)$$

is open. It contains b: were $b \in \pi_N((N \times F) \setminus W)$ there would be $y \in F$ with $(b, y) \notin W$, contradicting $\{b\} \times F \subseteq W$. And $U_b \times F \subseteq W$, since $u \in U_b$ with $(u, y) \notin W$ for some y would put u in the removed image. Pulling back through h, which carries $p^{-1}(U_b)$ onto $U_b \times F$,

$$p^{-1}(U_b) \subseteq V_1 \cup \dots \cup V_m$$

This is the tube lemma, and compactness of F is what example 3.2.12 needs.

Step 2: Refine downstairs. The sets U_b for $b \in B$ form an open cover of B. Since B is paracompact, it admits a locally finite open refinement $\{W_\gamma\}_{\gamma \in \Gamma}$ covering B, and for each γ there is a point $b(\gamma)$ with $W_\gamma \subseteq U_{b(\gamma)}$.

Step 3: Assemble upstairs. For each γ write $V_1^\gamma, \dots, V_{m_\gamma}^\gamma$ for the finitely many members of $\mathcal{V}$ that Step 1 produced at $b(\gamma)$, and set

$$O_i^\gamma = p^{-1}(W_\gamma) \cap V_i^\gamma, \qquad 1 \le i \le m_\gamma$$

Each O_i^γ is open and contained in the member V_i^γ of $\mathcal{V}$, so the family $\{O_i^\gamma\}$ refines $\mathcal{V}$. It covers E: a point $e \in E$ has $p(e) \in W_\gamma$ for some γ, and Step 1 gives $e \in p^{-1}(W_\gamma) \subseteq p^{-1}(U_{b(\gamma)}) \subseteq V_1^\gamma \cup \dots \cup V_{m_\gamma}^\gamma$, so e lies in some O_i^γ.

Step 4: Local finiteness. Let $e \in E$. Local finiteness of $\{W_\gamma\}$ gives an open $S \ni p(e)$ meeting W_γ for only finitely many γ. Then $p^{-1}(S)$ is an open neighbourhood of e, and

$$p^{-1}(S) \cap O_i^\gamma \subseteq p^{-1}(S \cap W_\gamma)$$

which is empty whenever S misses W_γ. So $p^{-1}(S)$ meets O_i^γ for only those finitely many γ, and for each of them only for the finitely many indices $i \le m_\gamma$. Hence $p^{-1}(S)$ meets only finitely many members of the family, which is therefore locally finite. The cover $\mathcal{V}$ has a locally finite open refinement, so E is paracompact.

Step 5: The Hausdorff clause. Suppose B and F are Hausdorff, and let $e \neq e'$ in E. If $p(e) \neq p(e')$, separate them in B and take preimages. If $p(e) = p(e') = b$, both lie in $p^{-1}(N)$ for a trivializing $N \ni b$, and h carries them to distinct points of $N \times F$ with the same first coordinate, hence to points differing in the F coordinate; separating those in the Hausdorff F and pulling back through the homeomorphism h separates e from e'. So E is Hausdorff, and being paracompact it is normal by theorem 5.1.7, completing the proof.

Compactness of the fiber carries the argument, and the Sorgenfrey square shows it cannot be dropped: the projection $\mathbb{R}_l \times \mathbb{R}_l \to \mathbb{R}_l$ is globally trivial with fiber $\mathbb{R}_l$ over a paracompact base, yet its total space is not paracompact (example 5.1.8). The place compactness enters is Step 1, where one fiber is covered by finitely many members of $\mathcal{V}$ and that covering then spreads across a whole tube; with a noncompact fiber no finite subfamily need exist, and the refinement obtained downstairs has nothing to be cut against. No separation hypothesis is used anywhere in the proof, so the theorem holds for the bare covering definition of definition 5.1.3.

5.3 Partitions of Unity

A paracompact Hausdorff space is normal (theorem 5.1.7), so Urysohn's lemma (theorem 4.3.1) is available on it. What is still missing is slack. Urysohn's lemma needs a closed set sitting inside an open one, and an open cover supplies no closed set at all. The device that manufactures one, for every member of the cover simultaneously and without destroying the covering property, is the shrinking lemma; the rest of the section is built on top of it.

Shrinking a single member of a cover is immediate. If $\{U_\alpha\}_{\alpha\in A}$ covers X, then the set $X \setminus \bigcup_{\gamma\neq\alpha} U_\gamma$ is closed and contained in U_α, so normality interposes an open set whose closure lies in U_α, and replacing U_α by that set leaves a cover. The difficulty is doing this at every index at once: each replacement discards points from the union, and after infinitely many replacements there is no reason for the union to still be X. The resolution is to well-order the index set and shrink one member at a time, arranging at every stage that the members already shrunk, together with those not yet touched, still cover. Point-finiteness is what makes the arrangement survive: a point lies in only finitely many members of the cover, so only finitely many stages can take it away, and the last such stage is forced to give it back.

Lemma 5.3.1: Shrinking Lemma

Let X be a normal space and let $\{U_\alpha\}_{\alpha\in A}$ be a locally finite open cover of X. Then there is an open cover $\{V_\alpha\}_{\alpha\in A}$ of X, indexed by the same set, such that

$$\overline{V_\alpha} \subseteq U_\alpha \qquad \text{for every } \alpha \in A$$

***Proof*:**
Step 1: a rewording of normality. If C is closed, U is open, and $C \subseteq U$, then there is an open set V with

$$C \subseteq V \subseteq \overline{V} \subseteq U$$

Indeed C and $X \setminus U$ are disjoint closed sets, so normality (definition 4.1.4) supplies disjoint open sets $V \supseteq C$ and $W \supseteq X \setminus U$. Then $V \subseteq X \setminus W$, and $X \setminus W$ is closed, so $\overline{V} \subseteq X \setminus W \subseteq U$. This is the same rewording that opens the proof of theorem 4.3.1.

Step 2: point-finiteness. The cover is locally finite, so for each $x \in X$ the set

$$F(x) = \{\alpha \in A \mid x \in U_\alpha\}$$

is finite: a neighbourhood of x meeting only finitely many U_α meets U_α whenever $x \in U_\alpha$. It is nonempty because $\{U_\alpha\}_{\alpha \in A}$ covers X. Once A is well-ordered, $F(x)$ therefore has a largest element. Nothing beyond this consequence of local finiteness is used below.

Step 3: the recursion. Well-order the index set A; this is the well-ordering theorem, a form of the axiom of choice equivalent to Zorn's lemma (theorem 3.2.41). Open sets V_α will be constructed by transfinite recursion so that $\overline{V_\alpha} \subseteq U_\alpha$ and so that, for every $\beta \in A$,

$$X = \Big(\bigcup_{\alpha \le \beta} V_\alpha\Big) \cup \Big(\bigcup_{\alpha > \beta} U_\alpha\Big) \tag{5.3}$$

Fix $\beta \in A$ and suppose V_α has been defined for every $\alpha < \beta$, with $\overline{V_\alpha} \subseteq U_\alpha$ and with (5.3) holding at every stage $\gamma < \beta$. We first check that the family

$$\{V_\alpha \mid \alpha < \beta\} \cup \{U_\alpha \mid \alpha \ge \beta\}$$

covers X. Let $x \in X$. If some $\alpha \in F(x)$ satisfies $\alpha \ge \beta$, then x lies in the member U_α of the family and there is nothing to prove. Otherwise every element of $F(x)$ is smaller than β; put $\gamma = \max F(x)$, so $\gamma < \beta$. Condition (5.3) at stage γ places x either in some V_α with $\alpha \le \gamma$ or in some U_α with $\alpha > \gamma$. The second alternative is impossible, since $\alpha > \gamma = \max F(x)$ forces $\alpha \notin F(x)$, that is, $x \notin U_\alpha$. Hence $x \in V_\alpha$ for some $\alpha \le \gamma < \beta$, and the family covers X.

Now set

$$C_\beta = X \setminus \Big(\bigcup_{\alpha < \beta} V_\alpha \cup \bigcup_{\alpha > \beta} U_\alpha\Big)$$

The set subtracted is a union of open sets, so C_β is closed. By the covering property just established, every point of C_β lies in U_α for some $\alpha \ge \beta$, and the only such index available is β itself; hence $C_\beta \subseteq U_\beta$. Step 1 applied to the pair $C_\beta \subseteq U_\beta$ produces an open set V_β with

$$C_\beta \subseteq V_\beta \subseteq \overline{V_\beta} \subseteq U_\beta$$

and we take this as the value of the recursion at β (one such V_β is selected at each stage). Condition (5.3) then holds at stage β: a point lying in no V_α with $\alpha < \beta$ and in no U_α with $\alpha > \beta$ belongs to C_β, hence to V_β. Transfinite recursion therefore defines V_α for every $\alpha \in A$, with $\overline{V_\alpha} \subseteq U_\alpha$ throughout and with (5.3) valid at every stage.

Step 4: the shrunken family covers. Let $x \in X$ and put $\gamma = \max F(x)$, which exists by Step 2. Condition (5.3) at stage γ places x in some V_α with $\alpha \le \gamma$ or in some U_α with $\alpha > \gamma$; as in Step 3 the latter contradicts the maximality of γ in $F(x)$. So $x \in V_\alpha$ for some α, and $\{V_\alpha\}_{\alpha \in A}$ is an open cover of X with $\overline{V_\alpha} \subseteq U_\alpha$, as we sought to show.

Three features of the lemma outlive the argument that produced them. First, the index set is preserved. A refinement is free to change indices (definition 5.1.2), and that freedom is fatal here: the partition of unity to be built must attach one function to each U_α, so the sets it is built from have to be labelled by A as well. Individual V_α may well be empty, and this costs nothing. Second, the shrunken cover is again locally finite, since $V_\alpha \subseteq U_\alpha$ and a neighbourhood meeting V_α meets U_α; the lemma can therefore be applied to its own output. Third, the proof used local finiteness only through Step 2, that is, only through point-finiteness. The shrinking lemma is true for point-finite open covers of a normal space, and local finiteness is assumed here because that is the form paracompactness delivers.

The object the shrinking lemma is being sharpened for can now be named. What an open cover $\{U_\alpha\}_{\alpha\in A}$ fails to provide is any way of weighting its members against each other: a point lying in three of them is described three times over, and nothing in the cover says how to reconcile the three descriptions. A family of continuous functions φ_α, one attached to each U_α, vanishing outside it and summing to 1 at every point, supplies exactly that arbitration. The value $\varphi_\alpha(x)$ is the share of x assigned to U_α, and the constant function 1 is decomposed into contributions indexed by the cover. The definition records the two requirements, and is stated for its own sake first, without reference to a cover.

Definition 5.3.2: Partition of Unity

Let X be a topological space. The *support* of a continuous function $f\colon X \to \mathbb{R}$ is the closed set

$$\operatorname{supp} f = \overline{\{x \in X \mid f(x) \neq 0\}}$$

A *partition of unity* on X is a family $\{\varphi_\alpha\}_{\alpha\in A}$ of continuous functions $\varphi_\alpha\colon X \to [0,1]$ such that

1. the family of supports $\{\operatorname{supp}\varphi_\alpha\}_{\alpha\in A}$ is locally finite (definition 5.1.1), and
2. $\sum_{\alpha\in A} \varphi_\alpha(x) = 1$ for every $x \in X$.

Condition (1) is what makes condition (2) meaningful. The index set A may be uncountable, so a sum over it carries no meaning on its own; but local finiteness of the supports gives each $x \in X$ a neighbourhood N meeting only finitely many of them, and φ_α vanishes identically on N for every α whose support misses N. The sum in (2) is thus a finite sum at each point, with no question of convergence, and the finitely many surviving terms are the same ones throughout N. That uniformity is what carries continuity: on N the function $\sum_\alpha \varphi_\alpha$ is a finite sum of continuous functions, hence continuous on N, and continuity is a local property, so the sum is continuous on X. Every sum over A written in this section is finite in this local sense; no limiting process occurs anywhere.

The support is the *closure* of the set where φ_α is nonzero, not that set itself. The distinction is the entire content of the next definition.

Definition 5.3.3: Subordinate to a Cover

Let $\{U_\alpha\}_{\alpha\in A}$ be an open cover of a space X. A partition of unity $\{\varphi_\alpha\}_{\alpha\in A}$ on X, indexed by the same set, is *subordinate* to the cover if

$$\operatorname{supp}\varphi_\alpha \subseteq U_\alpha \qquad \text{for every } \alpha \in A$$

Demanding $\operatorname{supp}\varphi_\alpha \subseteq U_\alpha$ rather than the weaker $\{x \mid \varphi_\alpha(x) \neq 0\} \subseteq U_\alpha$ is what makes the notion useful: the stronger containment leaves an open collar $X \setminus \operatorname{supp}\varphi_\alpha$ around the complement of U_α on which φ_α vanishes identically, and that collar is exactly the room needed to extend a function defined only on U_α by zero without breaking continuity. A weaker convention asks only that each support lie in *some* member of the cover, with no matching of indices; the indexed version defined here implies that one, and it is the version the theorem below produces.

Both conditions are visible on the line with two functions. Cover $\mathbb{R}$ by $U_1 = (-\infty, 1)$ and $U_2 = (-1, \infty)$, and set

$$\varphi_1(x) = \begin{cases} 1 & x \leq 0 \\ 1 - 2x & 0 \leq x \leq \frac{1}{2} \\ 0 & x \geq \frac{1}{2} \end{cases} \qquad\qquad \varphi_2 = 1 - \varphi_1$$

Both are continuous, being piecewise linear with matching values at the break points, and both take values in $[0, 1]$. Their sum is 1 by construction. The supports are $\operatorname{supp}\varphi_1 = (-\infty, \frac{1}{2}]$ and $\operatorname{supp}\varphi_2 = [0, \infty)$, which sit inside U_1 and U_2 respectively with room to spare, and a family of two sets is locally finite. Had the ramp been stretched to run from 0 to 1, the support of φ_1 would have been $(-\infty, 1]$, which is not contained in U_1: the partition of unity would have survived, but subordination would have failed.

The general construction assembles such ramps over an arbitrary open cover. Paracompactness enters twice, once to replace the cover by a locally finite one indexed by the same set, and once through normality, which is what Urysohn's lemma needs.

Theorem 5.3.4: Existence of Subordinate Partitions of Unity

Let X be a paracompact Hausdorff space and let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of X. Then there is a partition of unity on X subordinate to $\{U_\alpha\}_{\alpha \in A}$.

***Proof*:**
Since X is paracompact and Hausdorff it is normal (theorem 5.1.7), so both lemma 5.3.1 and theorem 4.3.1 are available; this is the only use made of the Hausdorff hypothesis.

Step 1: a locally finite cover indexed by A. Paracompactness gives a locally finite open refinement $\mathcal{S}$ of $\{U_\alpha\}_{\alpha \in A}$. A refinement need not be indexed by A, so choose for each $S \in \mathcal{S}$ an index $r(S) \in A$ with $S \subseteq U_{r(S)}$, and set

$$U'_\alpha = \bigcup \{S \in \mathcal{S} \mid r(S) = \alpha\}$$

Each U'_α is open and contained in U_α, possibly empty. The family $\{U'_\alpha\}_{\alpha \in A}$ covers X, since every $S \in \mathcal{S}$ is contained in $U'_{r(S)}$ and $\mathcal{S}$ covers X. It is locally finite: given $x \in X$, choose a neighbourhood N meeting only the members $S_1, \ldots, S_n$ of $\mathcal{S}$; if N meets U'_α then N meets some S with $r(S) = \alpha$, so α lies in the finite set $\{r(S_1), \ldots, r(S_n)\}$. Thus $\{U'_\alpha\}_{\alpha \in A}$ is a locally finite open cover, indexed by A, refining the original cover index by index.

Step 2: two shrinkings. Apply lemma 5.3.1 to $\{U'_\alpha\}_{\alpha \in A}$ to obtain an open cover $\{V_\alpha\}_{\alpha \in A}$ with $\overline{V_\alpha} \subseteq U'_\alpha$. Since $V_\alpha \subseteq U'_\alpha$, this cover is again locally finite, so the lemma applies to it in turn and yields an open cover $\{W_\alpha\}_{\alpha\in A}$ with $\overline{W_\alpha} \subseteq V_\alpha$. The chain now reads

$$\overline{W_\alpha} \subseteq V_\alpha \subseteq \overline{V_\alpha} \subseteq U'_\alpha \subseteq U_\alpha$$

for every $\alpha \in A$.

Step 3: the Urysohn functions. Fix $\alpha \in A$. The sets $\overline{W_\alpha}$ and $X \setminus V_\alpha$ are closed and disjoint, so theorem 4.3.1 provides a continuous $g_\alpha \colon X \to [0,1]$ with

$$g_\alpha = 0 \text{ on } X \setminus V_\alpha, \qquad g_\alpha = 1 \text{ on } \overline{W_\alpha}$$

(If either closed set is empty the constant function 0 or 1 serves.) Since g_α vanishes off V_α, its nonvanishing set lies in V_α and therefore

$$\mathrm{supp} g_\alpha \subseteq \overline{V_\alpha} \subseteq U'_\alpha \subseteq U_\alpha$$

This is where the second shrinking earns its place. Had g_α been produced from the single chain $\overline{V_\alpha} \subseteq U'_\alpha$, it would vanish only off U'_α, and the containment available for its support would be $\mathrm{supp} g_\alpha \subseteq \overline{U'_\alpha}$, a set that need not lie inside U_α. The extra layer supplies a set whose closure is interior to the next one, and the support is trapped there. Because $\mathrm{supp} g_\alpha \subseteq U'_\alpha$ and $\{U'_\alpha\}_{\alpha\in A}$ is locally finite, the family of supports $\{\mathrm{supp} g_\alpha\}_{\alpha\in A}$ is locally finite as well.

Step 4: normalizing. Define

$$g(x) = \sum_{\alpha\in A} g_\alpha(x)$$

Each $x \in X$ has a neighbourhood N meeting only finitely many of the sets $\mathrm{supp} g_\alpha$, and every g_α whose support misses N vanishes identically on N; so on N the sum is a finite sum of continuous functions, and g is continuous on N. Continuity being local, g is continuous on X. Moreover $g(x) \geq 1$ for every x: the family $\{W_\alpha\}_{\alpha\in A}$ covers X, so $x \in W_\alpha$ for some α, and then $g_\alpha(x) = 1$ while every other term is nonnegative. In particular g never vanishes, and

$$\varphi_\alpha = \frac{g_\alpha}{g}$$

is continuous with values in $[0,1]$.

It remains to verify the two conditions of definition 5.3.2 and subordination. Since g is nowhere zero, $\varphi_\alpha(x) \neq 0$ exactly when $g_\alpha(x) \neq 0$, so $\mathrm{supp}\varphi_\alpha = \mathrm{supp} g_\alpha$; the supports are therefore locally finite and satisfy $\mathrm{supp}\varphi_\alpha \subseteq U_\alpha$. Finally, for each $x \in X$ the sum $\sum_\alpha \varphi_\alpha(x)$ has the same finitely many nonzero terms as $\sum_\alpha g_\alpha(x)$, and

$$\sum_{\alpha\in A} \varphi_\alpha(x) = \frac{1}{g(x)} \sum_{\alpha\in A} g_\alpha(x) = \frac{g(x)}{g(x)} = 1$$

So $\{\varphi_\alpha\}_{\alpha\in A}$ is a partition of unity subordinate to $\{U_\alpha\}_{\alpha\in A}$, completing the proof.

The theorem converts local data into global data, and every application uses the mechanism in the same form. Suppose a continuous function $f_\alpha \colon U_\alpha \to \mathbb{R}$ has been chosen on each member of the cover, with no compatibility assumed on the overlaps. The product $\varphi_\alpha f_\alpha$, defined on U_α and extended by 0 on $X \setminus \mathrm{supp}\varphi_\alpha$, is a well-defined continuous function on all of X: the two open sets U_α and $X \setminus \mathrm{supp}\varphi_\alpha$ cover X because $\mathrm{supp}\varphi_\alpha \subseteq U_\alpha$, the two formulas agree on the overlap (where $\varphi_\alpha = 0$), and continuity of a function defined by cases on an open cover is the pasting lemma. Local finiteness then makes

$$f = \sum_{\alpha \in A} \varphi_\alpha f_\alpha$$

a locally finite sum, hence a continuous function on X, whose value at each point is a convex combination of the values of those f_α that are defined there. Choices made independently on the pieces are averaged into one global object, with the weights φ_α recording how much each piece contributes. Example 5.3.6 runs the formula on the line, gluing the constant choices $f_n \equiv n$ over the cover $U_n = (n-1, n+1)$ of $\mathbb{R}$ into a single continuous function that equals n near each integer and $n + \frac{1}{2}$ in the middle of each overlap. Later uses replace the numbers f_n by any local data that can be added and scaled, such as continuous approximations to a target function chosen separately on each U_α and averaged into one global approximation; the formula itself does not change.

The implication also runs backwards, and for a cheaper reason than the one that made it run forwards. A subordinate partition of unity carries a locally finite open refinement inside it, ready to be read off: the sets on which the individual φ_α fail to vanish.

Corollary 5.3.5: Paracompactness from Partitions of Unity

Let X be a Hausdorff space. Then X is paracompact if and only if every open cover of X admits a subordinate partition of unity.

***Proof*:**
The forward implication is theorem 5.3.4. For the converse, let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of X and let $\{\varphi_\alpha\}_{\alpha \in A}$ be a partition of unity subordinate to it. Put

$$V_\alpha = \{x \in X \mid \varphi_\alpha(x) > 0\}$$

Each V_α is open, being the preimage of $(0, 1]$, which is open in the codomain $[0, 1]$, under the continuous map φ_α, and

$$V_\alpha \subseteq \mathrm{supp}\varphi_\alpha \subseteq U_\alpha$$

so $\{V_\alpha\}_{\alpha \in A}$ refines $\{U_\alpha\}_{\alpha \in A}$ in the sense of definition 5.1.2. It covers X: for $x \in X$ the sum $\sum_\alpha \varphi_\alpha(x)$ equals 1, so at least one term is nonzero, and $x \in V_\alpha$ for that index. It is locally finite: a neighbourhood of x meeting only finitely many of the sets $\mathrm{supp}\varphi_\alpha$ meets only finitely many of the smaller sets V_α. Every open cover of X thus has a locally finite open refinement, which is definition 5.1.3, completing the proof.

The corollary replaces a condition on covers by a condition on functions. Paracompactness was defined by a combinatorial requirement, that arbitrary covers can be tamed, and the equivalence says the requirement is met exactly when the space carries enough continuous functions to distribute the constant 1 over any prescribed open cover. The converse half used no separation hypothesis at all, since paracompactness as defined in definition 5.1.3 does not include the Hausdorff property; Hausdorff appears in the statement only because the forward half needs normality, and normality is what makes the functions exist.

The construction behind the forward half is explicit, and on the line every step of it can be run by hand with the resulting functions written down in closed form.

Example 5.3.6: A Partition of Unity on the Line

The line is Hausdorff and paracompact (example 5.2.5), so theorem 5.3.4 applies to every open cover of it. Take the cover by unit-radius intervals about the integers,

$$U_n = (n-1, n+1), \qquad n \in \mathbb{Z}$$

which is already locally finite and already indexed by $\mathbb{Z}$, so Step 1 of the proof has nothing to do. The remaining steps can be carried out explicitly.

The two shrinkings. Take $V_n = \left(n - \frac{7}{8}, n + \frac{7}{8}\right)$ and $W_n = \left(n - \frac{3}{4}, n + \frac{3}{4}\right)$. Both are covers of $\mathbb{R}$: consecutive members overlap, since $(n+1) - \frac{7}{8} = n + \frac{1}{8} < n + \frac{7}{8}$ and $(n+1) - \frac{3}{4} = n + \frac{1}{4} < n + \frac{3}{4}$. The closures satisfy $\overline{V_n} = \left[n - \frac{7}{8}, n + \frac{7}{8}\right] \subseteq U_n$ and $\overline{W_n} = \left[n - \frac{3}{4}, n + \frac{3}{4}\right] \subseteq V_n$.

The Urysohn functions. On $\mathbb{R}$ no dyadic construction is needed; a piecewise linear formula does the work. Set

$$g_n(x) = \min\bigl\{1, \max\{0,\ 7 - 8|x-n|\}\bigr\}$$

which is continuous as a composite of continuous functions. If $|x-n| \le \frac{3}{4}$ then $7 - 8|x-n| \ge 1$ and $g_n(x) = 1$, so $g_n = 1$ on $\overline{W_n}$; if $|x-n| \ge \frac{7}{8}$ then $7 - 8|x-n| \le 0$ and $g_n(x) = 0$, so $g_n = 0$ off V_n. Its nonvanishing set is the open interval V_n, so $\operatorname{supp} g_n = \left[n - \frac{7}{8}, n + \frac{7}{8}\right] \subseteq U_n$.

The sum. If $x \in [n, n+1]$ then $|x - m| < \frac{7}{8}$ forces $m \in \{n, n+1\}$, so only two terms of $g = \sum_m g_m$ are nonzero there. Translating so that $n = 0$ and reading off the two formulas on $[0,1]$,

$$g_0(x) = \begin{cases} 1 & 0 \le x \le \frac{3}{4} \\ 7 - 8x & \frac{3}{4} \le x \le \frac{7}{8} \\ 0 & \frac{7}{8} \le x \le 1 \end{cases} \qquad g_1(x) = \begin{cases} 0 & 0 \le x \le \frac{1}{8} \\ 8x - 1 & \frac{1}{8} \le x \le \frac{1}{4} \\ 1 & \frac{1}{4} \le x \le 1 \end{cases}$$

so that on $[0,1]$

$$g(x) = \begin{cases} 1 & 0 \le x \le \frac{1}{8} \\ 8x & \frac{1}{8} \le x \le \frac{1}{4} \\ 2 & \frac{1}{4} \le x \le \frac{3}{4} \\ 8 - 8x & \frac{3}{4} \le x \le \frac{7}{8} \\ 1 & \frac{7}{8} \le x \le 1 \end{cases}$$

The values of g lie between 1 and 2, in agreement with the bound $g \ge 1$ established in the proof.

The partition. Dividing, $\varphi_n = g_n/g$, and on $[0,1]$ this reads

$$\varphi_0(x) = \begin{cases} 1 & 0 \le x \le \frac{1}{8} \\ \dfrac{1}{8x} & \frac{1}{8} \le x \le \frac{1}{4} \\ \dfrac{1}{2} & \frac{1}{4} \le x \le \frac{3}{4} \\ \dfrac{7-8x}{8-8x} & \frac{3}{4} \le x \le \frac{7}{8} \\ 0 & \frac{7}{8} \le x \le 1 \end{cases} \qquad \varphi_1 = 1 - \varphi_0 \text{ on } [0,1]$$

with $\varphi_n(x) = \varphi_0(x-n)$ in general. The pieces match at the four break points displayed, where the formulas give 1, $\frac{1}{2}$, $\frac{1}{2}$ and 0 respectively, so φ_0 is continuous. Each g_m is even about m, and reindexing $m \mapsto -m$ leaves g unchanged, so g is even and hence so is φ_0: its graph on $[-1,0]$ is the mirror image of the one displayed, and it vanishes outside $\left[-\frac{7}{8}, \frac{7}{8}\right]$. The supports are $\operatorname{supp}\varphi_n = \operatorname{supp} g_n = \left[n - \frac{7}{8}, n + \frac{7}{8}\right] \subseteq U_n$, and any interval of length 1 meets at most three of them, so the family is locally finite. Since only φ_n and φ_{n+1} are nonzero on $[n, n+1]$ and they sum to $(g_n + g_{n+1})/g = 1$ there, the family sums to 1 everywhere. It is a partition of unity subordinate to $\{U_n\}_{n\in\mathbb{Z}}$.

Gluing local data. These weights can now be used on local functions. Take $f_n \colon U_n \to \mathbb{R}$ to be the constant $f_n \equiv n$, so that overlapping pieces disagree everywhere, and form $f = \sum_n \varphi_n f_n$. On $\left[n - \frac{1}{8}, n + \frac{1}{8}\right]$ the formulas above give $g = 1$ and $\varphi_n = 1$, the neighbouring $\varphi_{n\pm1}$ vanishing there, so $f = n$: only the nth piece is in play. On $\left[n + \frac{1}{4}, n + \frac{3}{4}\right]$ they give $g = 2$ and $\varphi_n = \varphi_{n+1} = \frac{1}{2}$, so $f = \frac{1}{2}n + \frac{1}{2}(n+1) = n + \frac{1}{2}$: the two overlapping pieces are weighted equally. On the intervals between, f climbs continuously from one plateau to the next. Constants chosen independently on the members of the cover have been averaged into a single continuous function on $\mathbb{R}$, one that agrees with the nth choice on a neighbourhood of n.

The computation exposes what subordination costs. The tent functions $t_n(x) = \max\{0,\ 1 - |x-n|\}$ already sum to 1 on $\mathbb{R}$, by the same two-term argument, and they are far simpler than the φ_n above; but $\operatorname{supp} t_n = [n-1, n+1]$, which is not contained in $U_n = (n-1, n+1)$. The tents form a partition of unity that is not subordinate to this cover, and the two shrinkings are exactly the price of pulling the supports inside. Normalization is the other visible cost: dividing by g turns the piecewise linear g_0 into a function with two plateaus at height $\frac{1}{2}$ and four hyperbolic arcs, so the members of a partition of unity produced by the theorem are rarely as tidy as the sets they are attached to. Each φ_n here is continuous but fails to be differentiable at the eight points where its pieces meet, namely $n \pm \frac{1}{8}$, $n \pm \frac{1}{4}$, $n \pm \frac{3}{4}$ and $n \pm \frac{7}{8}$. Sharpening the conclusion to C^∞ functions is a different construction, resting on the flat bump function of one real variable rather than on Urysohn's lemma; it is carried out for $\mathbb{R}^n$ in *Multivariable Calculus* [4] and for manifolds in the series' volume on differential topology, and nothing in this chapter asserts it.

Exercise 5.3.1

1. Show that a partition of unity subordinate to a cover of a compact *Hausdorff* space may be taken finite, and construct one for the cover of S^1 by two overlapping arcs.
2. Show that if $\{\varphi_\alpha\}$ is a partition of unity then at most countably many φ_α are nonzero at any given point, and that the sum defining 1 is therefore a genuine sum.

3. Use theorem 5.3.4 to show that on a paracompact Hausdorff space, any two disjoint closed sets admit a Urysohn function, recovering theorem 4.3.1 for such spaces.

4. Show that a partition of unity subordinate to $\{U_\alpha\}$ patches locally defined continuous functions $f_\alpha\colon U_\alpha \to \mathbb{R}$ into a global continuous function, and that the result depends on the partition.

5. Exhibit continuous $\psi_n \geq 0$ on $\mathbb{R}$ summing to 1 pointwise whose supports are not locally finite, so that condition (2) of definition 5.3.2 does not imply condition (1).

5.4 Complete Regularity and the Stone-Čech Compactification

The partitions of unity of section 5.3 spend Urysohn functions on an open cover. This section spends them on the points of the space instead. A space whose one-point sets are closed and which carries enough continuous functions into $[0, 1]$ to separate each point from each closed set missing it can be mapped into a cube $[0, 1]^J$ by using all of those functions at once as coordinates, and that map is a homeomorphism onto its image. Two questions organize the section: which spaces admit such a supply of functions, and what the embedding gives once it exists.

What it gives is a compactification: the cube is a product of compact spaces, hence compact by theorem 3.2.44, so the closure of the image of X is a compact Hausdorff space with a dense copy of X inside it. This is the Stone-Čech compactification.

What theorem 4.3.1 supplies is more than the embedding needs. The embedding needs only the case where one of the two disjoint closed sets is a single point, and it needs the function rather than the pair of open sets. Weakening the hypothesis to exactly what is used gives an axiom that, unlike normality, survives the two constructions the embedding runs on: passage to a subspace and passage to a product.

Definition 5.4.1: Completely Regular Space

Let X be a topological space in which one-point sets are closed, as in definition 4.1.1. Then X is *completely regular* if for every point $x \in X$ and every closed set $A \subseteq X$ with $x \notin A$ there is a continuous function

$$f\colon X \to [0, 1]$$

with $f(x) = 0$ and $f(a) = 1$ for every $a \in A$. Such an f is a *Urysohn function* for the pair (x, A). A completely regular space is also called a *Tychonoff space*.

The axiom slots into the tower of chapter 4 between regularity and normality, and both comparisons are immediate. A completely regular space is regular in the sense of definition 4.1.1: given $x \notin A$ with A closed, take a Urysohn function f for (x, A) and set $U = f^{-1}\big([0, \tfrac{1}{2})\big)$ and $V = f^{-1}\big((\tfrac{1}{2}, 1]\big)$. These are open, being preimages of open subsets of $[0, 1]$ under a continuous map; they are disjoint, because $[0, \tfrac{1}{2})$ and $(\tfrac{1}{2}, 1]$ are; and $x \in U$ while $A \subseteq V$. In the other direction, a normal space (definition 4.1.4) with closed one-point sets is completely regular: the sets $\{x\}$ and A are disjoint closed sets, so theorem 4.3.1 supplies a continuous $f\colon X \to [0, 1]$ with $\{x\} \subseteq f^{-1}(0)$ and $A \subseteq f^{-1}(1)$, which is precisely a Urysohn function for the pair. The chain therefore reads

$$\text{normal} \implies \text{completely regular} \implies \text{regular} \implies \text{Hausdorff}$$

with one-point sets closed throughout. Complete regularity is also the axiom that chapter 7 will name, further ahead, when it records that a Hausdorff topological group separates a point from a disjoint closed set by a continuous real function; the definition above is what that phrase means.

Metric spaces are the first supply of examples, and there the Urysohn function can be written down rather than constructed. Let (X, d) be a metric space (section 2.6); its one-point sets are closed, since a metric space is Hausdorff, so the standing hypothesis of definition 5.4.1 is met. Let $A \subseteq X$ be closed and let $x \notin A$. If $A = \emptyset$ the constant function 0 is a Urysohn function for (x, A), so assume $A \neq \emptyset$. Write $d(y, A) = \inf_{a \in A} d(y, a)$ for the distance from a point to the set. This function is continuous, indeed 1-Lipschitz: for $y, z \in X$ and any $a \in A$ the triangle inequality gives $d(y, a) \leq d(y, z) + d(z, a)$, and taking the infimum over a on both sides yields $d(y, A) \leq d(y, z) + d(z, A)$; exchanging y and z gives $|d(y, A) - d(z, A)| \leq d(y, z)$. Now set

$$f(y) = \frac{d(y, x)}{d(y, x) + d(y, A)}$$

The denominator never vanishes: it is zero only when $d(y, x) = 0$ and $d(y, A) = 0$, that is, only when $y = x$ and x lies in the closure of A, which is A itself, contradicting $x \notin A$. So f is a quotient of continuous real functions with nonvanishing denominator, hence continuous, and it takes values in $[0, 1]$ because the numerator is one of the two nonnegative summands in the denominator. Finally $f(x) = 0$, and $f(a) = 1$ for $a \in A$ because $d(a, A) = 0$ there. Every metric space is therefore completely regular. The same conclusion follows from proposition 4.2.5 together with the implication recorded above, but that route produces no formula for the function.

The reason for isolating this axiom rather than working with normality is its behaviour under the two constructions of section 2.1 and section 2.2. Normality is stable under neither, and the discussion after the next proof exhibits its failure for subspaces. Complete regularity is stable under both, and the proof is short in each case because the axiom asks for a function, and functions restrict along inclusions and compose with projections.

Proposition 5.4.2: Complete Regularity Passes to Subspaces and Products

1. If X is completely regular and $Y \subseteq X$ carries the subspace topology of section 2.1, then Y is completely regular.
2. If $\{X_j\}_{j \in J}$ is a family of completely regular spaces, then the product $\prod_{j \in J} X_j$ with the product topology of definition 2.2.13 is completely regular.

***Proof*:**

1. Write $\iota \colon Y \to X$ for the inclusion. One-point sets are closed in Y: for $y \in Y$ the set $\{y\} = \{y\} \cap Y$ is the intersection with Y of a set closed in X, hence closed in the subspace topology. Now let $y \in Y$ and let $B \subseteq Y$ be closed in Y with $y \notin B$. By the definition of the subspace topology there is a closed $A \subseteq X$ with $B = A \cap Y$. Since $y \in Y$ and $y \notin B$, we have $y \notin A$. Complete regularity of X gives a continuous $f \colon X \to [0, 1]$ with $f(y) = 0$ and $f \equiv 1$ on A. The restriction $f \circ \iota \colon Y \to [0, 1]$ is continuous, being a composite of continuous maps; it sends y to 0, and it takes the value 1 on B because $B \subseteq A$. So $f \circ \iota$ is a Urysohn function for (y, B) in Y, and Y is completely regular.
2. Write $X = \prod_{j \in J} X_j$ and let $\pi_j \colon X \to X_j$ be the coordinate projections. Each π_j is

continuous: for U open in X_j the preimage $\pi_j^{-1}(U)$ is the basic open set of definition 2.2.13 whose j-th slot is U and whose other slots are the full factors. (Compare theorem 2.2.6, which identifies the product topology as the coarsest one making the projections continuous.) One-point sets are closed in X: for $x = (x_j)_{j\in J}$ we have

$$\{x\} = \bigcap_{j\in J} \pi_j^{-1}\big(\{x_j\}\big)$$

an intersection of preimages of closed sets under continuous maps, hence closed.

Now let $A \subseteq X$ be closed with $x \notin A$. If $A = \emptyset$ the constant function 0 is a Urysohn function for (x, A), so assume $A \neq \emptyset$. The complement $X \setminus A$ is open and contains x, so by definition 2.2.13 there is a basic open set $\prod_{j\in J} U_j$ with

$$x \in \prod_{j\in J} U_j \subseteq X \setminus A$$

where U_j is open in X_j for every j and $U_j = X_j$ for every j outside a finite set $F \subseteq J$. The set F is nonempty, since $F = \emptyset$ would force $\prod_j U_j = X$ and hence $A = \emptyset$.

For each $j \in F$ the set $X_j \setminus U_j$ is closed in X_j and does not contain x_j, so complete regularity of X_j gives a continuous $f_j : X_j \to [0,1]$ with $f_j(x_j) = 0$ and $f_j \equiv 1$ on $X_j \setminus U_j$. Define

$$f : X \to [0,1], \qquad f(y) = \max_{j\in F} f_j\big(\pi_j(y)\big)$$

a maximum over a finite nonempty set, so it is well defined. To see that f is continuous, use the subbasis of $[0,1]$ consisting of the sets $[0,a)$ and $(a,1]$; every open subset of $[0,1]$ is a union of finite intersections of these. A maximum is less than a exactly when every term is, and greater than a exactly when some term is, so

$$f^{-1}\big([0,a)\big) = \bigcap_{j\in F} (f_j \circ \pi_j)^{-1}\big([0,a)\big), \qquad f^{-1}\big((a,1]\big) = \bigcup_{j\in F} (f_j \circ \pi_j)^{-1}\big((a,1]\big)$$

The first is a finite intersection of open sets and the second a union of open sets, each $f_j \circ \pi_j$ being continuous, so both are open and f is continuous.

Finally $f(x) = \max_{j\in F} f_j(x_j) = 0$. Let $y \in A$. Then $y \notin \prod_j U_j$, so $\pi_j(y) \notin U_j$ for some $j \in J$; that index lies in F, since $U_j = X_j$ for $j \notin F$. For this j we get $f_j(\pi_j(y)) = 1$, whence $f(y) = 1$. Thus f is a Urysohn function for (x, A), and X is completely regular, as we sought to show.

The two parts together say that the class of completely regular spaces is closed under the operations that build new spaces out of old in section 2.1 and section 2.2, which is exactly the stability the embedding theorem needs: it presents a space as a subspace of a product, and the presentation is only useful if the class is not left in the process.

The contrast with normality is sharp enough to be exhibited concretely. On the Sorgenfrey line $\mathbb{R}_l$, with basis the half-open intervals $[a,b)$, each basic set $[a,b)$ is also closed, since its complement $(-\infty,a) \cup [b,\infty)$ is a union of basic sets. One-point sets are closed there, since $\mathbb{R}_l$ is Hausdorff: for

$x < y$ the basic sets $[x, y)$ and $[y, y+1)$ are disjoint neighbourhoods. Given $x \in \mathbb{R}_l$ and a closed A with $x \notin A$, choose $b > x$ with $[x, b) \subseteq \mathbb{R}_l \setminus A$ and let f be the function taking the value 0 on $[x, b)$ and 1 elsewhere; f is continuous because $[x, b)$ is clopen, and it is a Urysohn function for (x, A). So $\mathbb{R}_l$ is completely regular, and by part (2) so is the Sorgenfrey plane $\mathbb{R}_l \times \mathbb{R}_l$, which example 4.1.9 records as a space that is not normal. That same space settles the subspace question as well: being completely regular it embeds in a cube by theorem 5.4.3 below, and a cube is compact by theorem 3.2.44 and Hausdorff, hence normal by lemma 4.2.1, so a non-normal space can sit as a subspace of a normal one. Complete regularity is therefore strictly weaker than normality with closed one-point sets, and by proposition 5.4.2 it is inherited by subspaces and by products, whereas normality, as the Sorgenfrey plane inside a cube shows, is not inherited by subspaces.

Complete regularity is stated one point and one closed set at a time. The embedding theorem reads the whole family of Urysohn functions as a single map, taking as coordinates all continuous functions $X \to [0, 1]$ at once. Continuity of that map is automatic; injectivity uses the axiom to separate two points; and the one substantive point is that the map is open onto its image, which uses the axiom again, now to separate a point from the complement of an open set. The converse direction costs nothing, since proposition 5.4.2 has already made the class of completely regular spaces closed under the two operations involved.

Theorem 5.4.3: Embedding in a Cube

A topological space X is completely regular if and only if X is homeomorphic to a subspace of a cube $[0, 1]^J$, that is, of a product of copies of $[0, 1]$ indexed by some set J and carrying the product topology of definition 2.2.13.

When X is completely regular one may take $J = C(X, [0, 1])$, the set of all continuous functions from X to $[0, 1]$, and the homeomorphism onto its image to be the *evaluation map* $e \colon X \to [0, 1]^J$ determined by

$$\pi_f\big(e(x)\big) = f(x) \qquad \text{for all } x \in X,\ f \in J \tag{5.4}$$

***Proof*:**
Step 1: a subspace of a cube is completely regular. The interval $[0, 1]$ is metrizable, as a subspace of $\mathbb{R}$ with the metric $|s - t|$, so it is normal by proposition 4.2.5, and its one-point sets are closed because a metric space is Hausdorff. Applying theorem 4.3.1 to $\{t\}$ and a disjoint closed set shows $[0, 1]$ is completely regular. By proposition 5.4.2(2) the cube $[0, 1]^J$ is completely regular for every index set J, and by proposition 5.4.2(1) so is every subspace of it. Complete regularity is defined purely in terms of the topology, so it is preserved by homeomorphism; hence a space homeomorphic to a subspace of a cube is completely regular. This proves the "if" direction.

For the converse, let X be completely regular, put $J = C(X, [0, 1])$, and let $e \colon X \to [0, 1]^J$ be given by (5.4), so that $e(x)$ is the point of the cube whose f-th coordinate is $f(x)$.

Step 2: e is continuous and injective. For continuity it suffices to check preimages of basic open sets. Let $\prod_{f \in J} U_f$ be a basic open set of definition 2.2.13, with U_f open in $[0, 1]$ for each f and $U_f = [0, 1]$ for every f outside a finite set F. By (5.4) a point x satisfies $e(x) \in \prod_f U_f$ exactly when $f(x) \in U_f$ for every $f \in F$, the conditions at the remaining coordinates being vacuous.

Hence

$$e^{-1}\Big(\prod_{f\in J} U_f\Big) = \bigcap_{f\in F} f^{-1}(U_f)$$

a finite intersection of open subsets of X, since each $f \in J$ is continuous. So e is continuous.

For injectivity, let $x \neq y$ in X. One-point sets are closed, so $\{y\}$ is a closed set not containing x, and complete regularity gives a continuous $f\colon X \to [0,1]$ with $f(x) = 0$ and $f(y) = 1$. This f is a member of J, and $\pi_f(e(x)) = 0 \neq 1 = \pi_f(e(y))$ by (5.4), so $e(x) \neq e(y)$.

Step 3: e is a homeomorphism onto its image. Give $Z = e(X)$ the subspace topology from $[0,1]^J$. By Step 2 the map $e\colon X \to Z$ is a continuous bijection, so it remains to show it is open, that is, that $e(U)$ is open in Z for every open $U \subseteq X$. Fix such a U and a point $z \in e(U)$, say $z = e(x)$ with $x \in U$. The set $X \setminus U$ is closed and does not contain x, so complete regularity provides $f \in J$ with $f(x) = 0$ and $f \equiv 1$ on $X \setminus U$. Put

$$W = \pi_f^{-1}\big([0,1)\big) \cap Z$$

Since $[0,1)$ is open in $[0,1]$ and π_f is continuous, $\pi_f^{-1}([0,1))$ is open in the cube, so W is open in Z. It contains z, because $\pi_f(z) = f(x) = 0 < 1$ by (5.4). And $W \subseteq e(U)$: a point of W has the form $e(y)$ for some $y \in X$, and $\pi_f(e(y)) = f(y) < 1$ forces $y \notin X \setminus U$, that is, $y \in U$, so $e(y) \in e(U)$. Thus every point of $e(U)$ has an open neighbourhood in Z contained in $e(U)$, so $e(U)$ is open in Z.

A continuous open bijection is a homeomorphism, so $e\colon X \to Z$ is a homeomorphism onto the subspace Z of $[0,1]^J$, completing the proof.

The theorem converts a separation axiom into a concrete position: the completely regular spaces are exactly the subspaces of cubes. The index set is large, since for nonempty X the set $J = C(X,[0,1])$ contains all the constant functions and so has at least the cardinality of $[0,1]$, but nothing smaller is claimed and nothing smaller is needed. What matters is that the target is a product of intervals, because the interval is compact and theorem 3.2.44 makes the product compact however large J is. A completely regular space is thus a subspace of a compact Hausdorff space, and conversely every subspace of a compact Hausdorff space is completely regular: a compact Hausdorff space is normal by lemma 4.2.1 with closed one-point sets, hence completely regular, and proposition 5.4.2(1) passes this to subspaces. The two descriptions of the class coincide.

That equivalence is what makes the next definition possible. If X sits inside a compact Hausdorff space, then taking the closure of X there produces a compact Hausdorff space in which X is dense, and the cube of theorem 5.4.3 is a canonical choice of ambient space, depending on nothing but X itself.

Definition 5.4.4: Stone-Čech Compactification

Let X be a completely regular space, let $J = C(X, [0,1])$, and let $e\colon X \to [0,1]^J$ be the evaluation embedding of theorem 5.4.3. The *Stone-Čech compactification* of X is the closure of the image,

$$\beta X = \overline{e(X)} \subseteq [0,1]^J$$

equipped with the subspace topology, together with the map $e\colon X \to \beta X$.

The definition needs three verifications, all of them already available. First, $[0,1]^J$ is compact: each factor $[0,1]$ is compact by the Heine-Borel theorem, and an arbitrary product of compact spaces is compact by theorem 3.2.44. This is the point at which the Tychonoff theorem does work no finite argument replaces, since J contains a distinct constant function for each $t \in [0,1]$ and is therefore uncountable for every nonempty X. Second, $[0,1]^J$ is Hausdorff: if $s \neq t$ in the cube then $s_j \neq t_j$ for some j, and choosing disjoint open $U \ni s_j$, $V \ni t_j$ in $[0,1]$ makes $\pi_j^{-1}(U)$ and $\pi_j^{-1}(V)$ disjoint open sets separating s from t. Third, βX is closed in the cube, being a closure, so it is compact by theorem 3.2.7 and Hausdorff as a subspace of a Hausdorff space. And $e(X)$ is dense in βX: the closure of $e(X)$ inside the subspace βX is $\overline{e(X)} \cap \beta X = \beta X$.

So βX is a compact Hausdorff space containing a dense homeomorphic copy of X, which is a compactification of X in the sense of definition 3.2.37, except in the degenerate case where X is already compact. In that case $e(X)$ is compact and therefore closed in the Hausdorff cube by theorem 3.2.16, so $\beta X = e(X)$ and nothing new is added: a compact Hausdorff space is its own Stone-Čech compactification.

The construction is canonical but opaque. Nothing in the definition explains why one should index by *all* of $C(X, [0,1])$ rather than by some subfamily, and nothing yet distinguishes βX from the many other compactifications a space admits. The following theorem answers both objections at once. It says that maps out of βX into compact Hausdorff spaces are the same thing as maps out of X, which pins βX down up to homeomorphism and so makes the choice of index set immaterial.

Theorem 5.4.5: Universal Property of the Stone-Čech Compactification

Let X be completely regular with Stone-Čech compactification $e\colon X \to \beta X$. For every compact Hausdorff space K and every continuous map $g\colon X \to K$ there is a unique continuous map $\tilde{g}\colon \beta X \to K$ with

$$\tilde{g} \circ e = g$$

***Proof*:**
Step 1: uniqueness. Suppose $h_1, h_2\colon \beta X \to K$ are continuous with $h_1 \circ e = h_2 \circ e$, and let

$$E = \{p \in \beta X \mid h_1(p) = h_2(p)\}$$

be their equalizer. The set E is closed. Indeed, let $p \notin E$, so $h_1(p) \neq h_2(p)$; since K is Hausdorff there are disjoint open sets $U \ni h_1(p)$ and $V \ni h_2(p)$, and then $h_1^{-1}(U) \cap h_2^{-1}(V)$ is an open neighbourhood of p on which h_1 and h_2 take values in disjoint sets, hence never agree. So the complement of E is open. By hypothesis E contains $e(X)$, which is dense in βX, so E is a closed set containing a dense set, giving $E = \beta X$ and $h_1 = h_2$. Both hypotheses were used, and each is needed: density of $e(X)$ to reach all of βX, and Hausdorffness of K to know that an equalizer is closed.

Step 2: K is a closed subspace of a cube. Being compact Hausdorff, K is normal by lemma 4.2.1, and its one-point sets are closed because it is Hausdorff; applying theorem 4.3.1 to a point and a disjoint closed set shows K is completely regular. By theorem 5.4.3 there is an index set I, namely $I = C(K, [0,1])$, and a homeomorphism κ of K onto the subspace $\kappa(K)$ of $[0,1]^I$. Compactness is preserved by homeomorphism, so $\kappa(K)$ is compact, and the cube is Hausdorff by the argument recorded after definition 5.4.4; hence $\kappa(K)$ is closed in $[0,1]^I$ by theorem 3.2.16.

Step 3: a map between the two cubes. For each $i \in I$ set

$$g_i = \pi_i \circ \kappa \circ g \colon X \to [0,1]$$

a composite of continuous maps, so $g_i \in J = C(X, [0,1])$. Define $G \colon [0,1]^J \to [0,1]^I$ by reading off, in coordinate i, the coordinate of the source indexed by g_i:

$$G\big((t_f)_{f\in J}\big) = (t_{g_i})_{i\in I}$$

so that $\pi_i \circ G = \pi_{g_i}$ for every $i \in I$. This map is continuous. Let $\prod_{i\in I} U_i$ be a basic open set of $[0,1]^I$, with $U_i = [0,1]$ for every i outside a finite set F. A point t satisfies $G(t) \in \prod_i U_i$ exactly when $t_{g_i} \in U_i$ for every $i \in F$, so

$$G^{-1}\Big(\prod_{i\in I} U_i\Big) = \bigcap_{i\in F} \pi_{g_i}^{-1}(U_i)$$

a finite intersection of open sets, each projection being continuous.

The map G was built so as to carry e to g. For $x \in X$ and $i \in I$, equation (5.4) gives

$$\pi_i\big(G(e(x))\big) = \pi_{g_i}\big(e(x)\big) = g_i(x) = \pi_i\big(\kappa(g(x))\big)$$

Two points of $[0,1]^I$ with the same coordinate in every slot are equal, so $G(e(x)) = \kappa(g(x))$ for every x, that is, $G \circ e = \kappa \circ g$.

Step 4: restriction to βX. A continuous map sends the closure of a set into the closure of its image: for continuous G and any S, the preimage $G^{-1}(\overline{G(S)})$ is closed and contains S, hence contains $\overline{S}$, which says $G(\overline{S}) \subseteq \overline{G(S)}$. Taking $S = e(X)$ and using Step 3,

$$G(\beta X) = G\big(\overline{e(X)}\big) \subseteq \overline{G(e(X))} = \overline{\kappa(g(X))} \subseteq \kappa(K)$$

the last inclusion because $\kappa(K)$ is a closed set containing $\kappa(g(X))$, by Step 2. So the restriction of G to βX takes its values in $\kappa(K)$, and it is continuous into $\kappa(K)$: it is continuous into the cube as a restriction of a continuous map, and an open set of the subspace $\kappa(K)$ has the form $V \cap \kappa(K)$ with V open in $[0,1]^I$, whose preimage under $G|_{\beta X}$ is simply the preimage of V, since $G|_{\beta X}$ takes all of its values in $\kappa(K)$. Define

$$\tilde{g} = \kappa^{-1} \circ \big(G|_{\beta X}\big) \colon \beta X \to K$$

a composite of continuous maps, hence continuous. For $x \in X$,

$$\tilde{g}\big(e(x)\big) = \kappa^{-1}\big(G(e(x))\big) = \kappa^{-1}\big(\kappa(g(x))\big) = g(x)$$

so $\tilde{g} \circ e = g$. Existence and uniqueness are both established, completing the proof.

The property characterizes βX, and the standard argument explains why the opaque construction is nonetheless canonical. Suppose $e'\colon X \to Z$ is another map into a compact Hausdorff space with the same extension property. Applying the property of βX to e' gives a unique continuous $u\colon \beta X \to Z$ with $u \circ e = e'$, and applying the property of Z to e gives a unique continuous $v\colon Z \to \beta X$ with $v \circ e' = e$. Then $v \circ u\colon \beta X \to \beta X$ satisfies $(v \circ u) \circ e = e$, and so does the identity; uniqueness in theorem 5.4.5 applied to $K = \beta X$ and $g = e$ forces $v \circ u = \mathrm{id}_{\beta X}$, and symmetrically $u \circ v = \mathrm{id}_Z$. So u is a homeomorphism compatible with the two embeddings, and it is the only one. Whatever else βX is, it does not depend on the index set J beyond the fact that J was large enough.

The property also justifies calling βX the *largest* compactification. Let Y be any compactification of X in the sense of definition 3.2.37: a compact Hausdorff space containing X as a dense subspace. The inclusion $X \to Y$ is a continuous map into a compact Hausdorff space, so theorem 5.4.5 extends it to a continuous $q\colon \beta X \to Y$ restricting to the identity on X. The image $q(\beta X)$ is compact, since any open cover of it pulls back along q to an open cover of the compact space βX and a finite subcover pushes forward; so $q(\beta X)$ is closed in Y by theorem 3.2.16, and it contains the dense set X, whence $q(\beta X) = Y$. Every compactification of X is therefore a continuous image of βX under a map fixing X. At the other extreme sits the one-point compactification of definition 3.2.37, available exactly for the noncompact locally compact Hausdorff spaces of definition 3.2.34; it adds a single point, which is as few as any compactification of a noncompact space can add, while by the argument just given there is nothing larger than βX. A further consequence comes from taking $K = [a, b]$: every bounded continuous function $X \to \mathbb{R}$ extends uniquely to a continuous function on βX, so the bounded continuous functions of X all become continuous functions on a compact space.

The countable discrete space is the simplest input for which the construction produces something new, and what it produces is large enough to show how far βX can sit from anything one could draw.

Example 5.4.6: The Stone-Čech Compactification of $\mathbb{N}$

Give $\mathbb{N}$ the discrete topology, so that every subset is open and every function out of $\mathbb{N}$ is continuous. Then $\mathbb{N}$ is completely regular: one-point sets are closed, and for $x \in \mathbb{N}$ and a closed A with $x \notin A$ the indicator function χ_A, equal to 1 on A and 0 off A, is continuous and is a Urysohn function for the pair. So $\beta\mathbb{N}$ is defined, with index set $J = C(\mathbb{N}, [0, 1])$, which here is the set of all functions $\mathbb{N} \to [0, 1]$. Identify $\mathbb{N}$ with its image $e(\mathbb{N}) \subseteq \beta\mathbb{N}$ throughout.

Every point of $\mathbb{N}$ is isolated in $\beta\mathbb{N}$. Fix $n \in \mathbb{N}$ and let $f = \chi_{\{n\}} \in J$, so $f(n) = 1$ and $f(m) = 0$ for $m \neq n$. The set $W = \pi_f^{-1}\big((\tfrac{1}{2}, 1]\big) \cap \beta\mathbb{N}$ is open in $\beta\mathbb{N}$ and contains n, while $W \cap \mathbb{N} = \{n\}$ by (5.4). Now an open set is contained in the closure of its intersection with a dense set: if $p \in W$ and O is an open neighbourhood of p, then $O \cap W$ is a nonempty open subset of $\beta\mathbb{N}$ and so meets the dense set $\mathbb{N}$, whence every neighbourhood of p meets $W \cap \mathbb{N}$. Therefore $W \subseteq \overline{W \cap \mathbb{N}} = \overline{\{n\}} = \{n\}$, the last equality because $\beta\mathbb{N}$ is Hausdorff. So $W = \{n\}$ is open. Consequently $\mathbb{N}$ is an open discrete subspace of $\beta\mathbb{N}$, and the *remainder* $\beta\mathbb{N} \setminus \mathbb{N}$ is closed in $\beta\mathbb{N}$, hence compact by theorem 3.2.7.

The remainder is nonempty. If $\beta\mathbb{N}$ equalled $\mathbb{N}$, then $\mathbb{N}$ would be compact, being homeomorphic to a compact space. But the cover of $\mathbb{N}$ by its one-point subsets is an open cover with no finite subcover, since $\mathbb{N}$ is infinite. So $\beta\mathbb{N} \setminus \mathbb{N} \neq \emptyset$.

No sequence of natural numbers converges to a point of the remainder. Let $n_1, n_2, \ldots$ be natural numbers, let $p \in \beta\mathbb{N} \setminus \mathbb{N}$, and suppose $n_k \to p$. First, no value can repeat infinitely often: a constant subsequence with value n converges to n, and a subsequence of a convergent sequence has the same limit, so n and p would both be limits of one sequence in the Hausdorff space $\beta\mathbb{N}$, which is impossible because disjoint neighbourhoods of two candidate limits cannot both contain a tail. Every value therefore occurs finitely often, so discarding repetitions leaves a subsequence of pairwise distinct terms, still converging to p; replacing the original sequence by it, assume the n_k are distinct. Let $A = \{n_1, n_3, n_5, \dots\}$ be the odd-indexed terms and let $f = \chi_A \in J$. Applying theorem 5.4.5 with $K = [0,1]$ extends f to a continuous $\tilde{f}\colon \beta\mathbb{N} \to [0,1]$ with $\tilde{f}(n) = f(n)$ for every $n \in \mathbb{N}$. Then $\tilde{f}(n_k) \to \tilde{f}(p)$, since for any neighbourhood V of $\tilde{f}(p)$ the preimage $\tilde{f}^{-1}(V)$ is a neighbourhood of p and so contains n_k for all large k. But the terms being distinct, $\tilde{f}(n_k) = f(n_k)$ alternates between 1 and 0 and converges to nothing. So $n_k \not\to p$ after all.

$\beta\mathbb{N}$ is not metrizable. Suppose it were, and pick p in the nonempty remainder. In a metric space every point of the closure of a set is the limit of a sequence from that set, so some sequence in the dense set $\mathbb{N}$ would converge to p, which the previous paragraph forbids. Hence $\beta\mathbb{N}$ carries no metric.

Size. The construction bounds the cardinality directly. Since J is the set of all functions $\mathbb{N} \to [0,1]$, its cardinality is $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$, and $\beta\mathbb{N}$ is a subset of $[0,1]^J$, giving

$$|\beta\mathbb{N}| \leq \left(2^{\aleph_0}\right)^{2^{\aleph_0}} = 2^{2^{\aleph_0}}$$

This bound is attained.[a]

Comparison with the one-point compactification. The space $\mathbb{N}$ is locally compact Hausdorff in the sense of definition 3.2.34, since every one-point set is a compact neighbourhood of its point, and its one-point compactification in the sense of definition 3.2.37 can be exhibited directly. Let $\mathbb{N}^+ = \{0\} \cup \{2^{-n} \mid n \in \mathbb{N}\}$ carry the subspace topology from $\mathbb{R}$, and identify $n \in \mathbb{N}$ with 2^{-n} and the added point ∞ with 0. Each point 2^{-n} is isolated in $\mathbb{N}^+$, so that identification is a homeomorphism of the discrete space $\mathbb{N}$ onto the proper subspace $\{2^{-n} \mid n \in \mathbb{N}\}$, whose closure in $\mathbb{N}^+$ is all of $\mathbb{N}^+$; and $\mathbb{N}^+$ is compact, because a member of an open cover containing 0 omits only finitely many of the points 2^{-n}, while it is Hausdorff and metrizable as a subspace of $\mathbb{R}$. So $\mathbb{N}^+$ is a compactification of $\mathbb{N}$ adding one point, and it is countable and metrizable. The surjection $q\colon \beta\mathbb{N} \to \mathbb{N}^+$ produced above collapses the whole remainder to the added point: for $n \in \mathbb{N}$ the set $\{n\}$ is open in $\mathbb{N}^+$, so $q^{-1}(\{n\})$ is open in $\beta\mathbb{N}$ and meets $\mathbb{N}$ exactly in $\{n\}$, and the closure argument used above for isolated points gives $q^{-1}(\{n\}) \subseteq \overline{\{n\}} = \{n\}$. Hence $q^{-1}(\mathbb{N}) = \mathbb{N}$ and $q^{-1}(\infty) = \beta\mathbb{N} \setminus \mathbb{N}$.

[a] That $|\beta\mathbb{N}| = 2^{2^{\aleph_0}}$ was proved by Pospíšil in 1937; the lower bound is the harder half and is not needed here.

The example measures the gap between the two ends of the range on a single space. Compactifying $\mathbb{N}$ minimally costs one point and keeps the space countable and metrizable; compactifying it maximally costs $2^{2^{\aleph_0}}$ points, destroys metrizability, and produces a remainder no sequence from $\mathbb{N}$ can reach. The extra points carry the extension property. A bounded sequence of reals is a continuous map $g\colon \mathbb{N} \to [a,b]$, so theorem 5.4.5 supplies a continuous $\tilde{g}\colon \beta\mathbb{N} \to [a,b]$; a point p of the remainder then assigns to every bounded sequence the value $\tilde{g}(p)$, a limit along p that always exists and agrees with the ordinary limit whenever that exists. The one-point compactification assigns no such value, since a continuous extension to $\mathbb{N}^+$ exists only for the sequences that already converge.

Exercise 5.4.1

1. Show that normality with closed points implies complete regularity, that complete regularity implies regularity, and that the first implication is strict, citing a witness already in the text.
2. Show that $\beta X = X$ if and only if X is compact Hausdorff.
3. Show that $|\beta\mathbb{N}| \leq 2^{2^{\aleph_0}}$, and deduce from compactness alone that $\beta\mathbb{N} \setminus \mathbb{N}$ is nonempty.
4. Show that a continuous $f \colon X \to Y$ between completely regular spaces extends to $\beta f \colon \beta X \to \beta Y$, and that this makes β a functor.
5. Show that $\beta[0, 1) \neq [0, 1]$, so the Stone-Čech compactification is not the "smallest" one.

6

Fundamental Group

Connectedness captured the idea of a space that cannot be broken apart, and path-connectedness strengthened it to the requirement that any two points be joined by a path (the topologist's sine curve marks the gap between the two). The idea driving this chapter generalizes path-connectedness from points to loops: instead of asking which points can be joined, ask which loops can be contracted to a point continuously. For example, on a sphere every loop collapses to a single point, while on a torus a loop running around the hole cannot. That is a purely topological distinction, and the rest of the chapter makes it precise.

From now on, for notational simplicity, the convention is that

$$I = [0, 1]$$

6.1 Homotopy and Path-Homotopy

There is a lot of little pieces that need to fit together to see the use of the fundamental group. First, here's the basic ontology for this area:

Definition 6.1.1: Homotopy

Let X, Y be topological spaces and let f, f' be continuous maps from X to Y. Then f is *homotopic* to f' if there is a continuous map $F : X \times I \to Y$ such that

$$F(x, 0) = f(x), \qquad F(x, 1) = f'(x)$$

F is called a homotopy between f and f', written $f \simeq f'$

So two maps are homotopic when one can be continuously deformed into the other. Note that the topology on $X \times I$ combines the topology of X with the Euclidean topology on I, and it is the interval coordinate that carries the deformation. Note also that what is deformed is the *map*, not the underlying space: homotopy is a relation between *functions*, defined on the function space. Thinking about the graphs of those functions is where the intuition of continuously deforming shapes comes in.

Example 6.1.2: Homotopy

1. Let $f, g : \mathbb{R} \to \mathbb{R}^2$ with $f(x) = (x, x^3)$ and $g(x) = (x, e^x)$. Then $F : \mathbb{R} \times I \to \mathbb{R}^2$,
$$F(x,t) = (x, (1-t)x^3 + te^x)$$
Is a homotopy. $F(x, 0) = (x, x^3)$, and $F(x, 1) = (x, e^x)$, the way is easily checked to be continuous. Thus
$$f \simeq g$$
2. Take $f : I \to I$ with $x \mapsto x^2$. Then
$$F(x,t)((1-t)x^2 + tx)$$
is a homotopy. More generally, x^2 can be any polynomial, and this homotopy will work.
3. Take $f : \mathbb{R} \to \mathbb{R}$ be a continuous function, and $g : \mathbb{R} \to \mathbb{R}$, $g(x) = 0$. Then:
$$F : \{\mathbb{R}\} \times I \to \mathbb{R} \qquad F(x,t) = (1-t)f(x)$$
is a continuous function, with $F(x, 0) = f(x)$ and $F(x, 1) = 0$. Thus, every continuous function on $\mathbb{R}$ is homotopic to the constant function.

Reading the graphs, the point $\{0\}$ with the trivial topology gives shape of $\mathbb{R}$ with the Euclidean topology can be continuously deformed to on another.

A map homotopic to a constant deserves a name:

Definition 6.1.3: nullhomotopic

If f is homotopic to a constant map, then f is *nullhomotopic*

Nullhomotopy says that the map can be continuously deformed to a constant, so that it carries no information a loop could detect; in that sense it is topologically trivial[1].

The homotopies above all used the straight-line formula $(1-t)a + tb$, the standard way of joining two points by a path. Homotopies of that form get their own name.

[1] not necessarily trivial in the most general sense, because compactness, normality, connectedness, and all the other properties might still be present in this "trivial" case, but it's trivial when it comes to trying to form to bending it around

Definition 6.1.4: Straight-Line Homotopy

If F is a homotopy such that it's construction is:

$$F(x,t) = ((1-t)f(x) + tg(x))$$

then it is a Straight-line homotopy

The notation is deliberately that of an equivalence relation ($\sim$), and that is exactly how two homotopic paths should be read: *equivalent up to continuous deformation.*

Lemma 6.1.5: Equivalence Relation With Homotopy

For fixed X and Y, homotopy is an equivalence relation on the set of continuous maps $X \to Y$.

***Proof*:**
The three properties in turn. Throughout, F denotes a homotopy, so a continuous map $X \times I \to Y$ with the stated values at $t = 0$ and $t = 1$.

1. *Reflexivity.* Given f, set $F(x,t) = f(x)$ for all t. This is the composite of f with the projection $X \times I \to X$, hence continuous, and $F(x,0) = F(x,1) = f(x)$. So $f \simeq f$.
2. *Symmetry.* Let F be a homotopy from f to f' and put $F'(x,t) = F(x, 1-t)$. This is F composed with the homeomorphism $(x,t) \mapsto (x, 1-t)$ of $X \times I$, hence continuous, and it runs from f' to f. So $f' \simeq f$.
3. *Transitivity.* Let F be a homotopy from f to f' and G one from f' to f''. Define
$$H(x,t) = \begin{cases} F(x,2t) & t \leq \frac{1}{2} \\ G(x,2t-1) & t \geq \frac{1}{2} \end{cases}$$
The two sets $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$ are closed in $X \times I$ and cover it, each formula is continuous on its piece, and at $t = \frac{1}{2}$ both give $f'(x)$. By lemma 1.4.6 the map H is continuous, and $H(x,0) = f(x)$ with $H(x,1) = f''(x)$. So $f \simeq f''$.

All three hold, so $\simeq$ is an equivalence relation.

Being an equivalence relation, homotopy partitions maps into equivalence classes:

Definition 6.1.6: Equivalence Class Of homotopies

Write $[f]$ for the equivalence class of f with respect to $\simeq$

Homotopy is the doorway to homotopy groups, homology, exact sequences and the Hopf fibration. The homotopies used here are all "one-dimensional", in the sense that the deforming parameter ranges over a single interval; replacing that interval by a cube of higher dimension is what produces the higher homotopy groups, and that is where algebraic topology begins. The restriction is on the probes, not on the spaces being probed.

6.1.1 Path-Homotopy

The natural one-dimensional object to work with then is a path:

Definition 6.1.7: Path-Homotopic

Two paths $f, \tilde{f}$ in X are *path-homotopic* if

1. these have the same initial and final points: $f(0) = f'(0)$, $f(1) = f'(1)$
2. There exists a continuous function $F : I \times I \to X$ such that

$$F(s,0) = f(s), F(s,1) = f'(s), F(0,t) = x_0, f(1,t) = x_1$$

   where $f(0) = f'(0) = x_0$ and $x_1 = f(1) = f'(1)$. Written

$$f \simeq_p f'$$

   F is called a *path-homotopy*

The interval parameter is often read as time: the deformation has one unit of time in which to be carried out, and it must be continuous throughout.

Example 6.1.8: More Homotopies

1. All the maps $I \to I$ given by $x \mapsto x^n$ are path-homotopic. In fact, any path such that $f(0) = g(0) = 0$, $f(1) = g(1) = 1$ will be path-homotopic by the straight-line homotopy
2. More generally, if C is a convex set, then any two paths with the same initial and ending points are path-homotopic by the straight-line homotopy
3. The straight-line homotopy is not the only one available.
4. "Are there paths that are not homotopic?". That can be the case too. Take $X = \mathbb{R}^2 - \{0\}$ and

$$f(s) = (\cos(\pi s), \sin(\pi s)) \qquad g(s) = (\cos(\pi s), 2\sin(\pi s))$$

   then these two functions are path homotopic by the straight-line homotopy, as the picture shows:

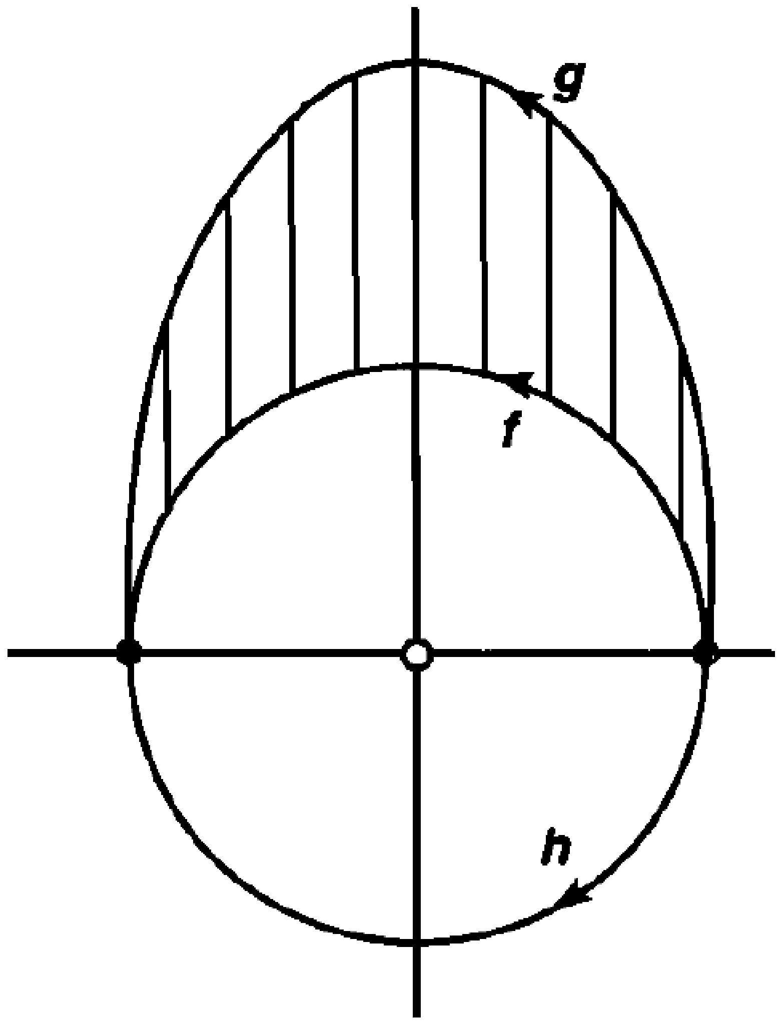

Figure 6.1: Same initial and end points, but one path passes on the other side of the missing point.

In general, if the point 0 were not missing, there would be:

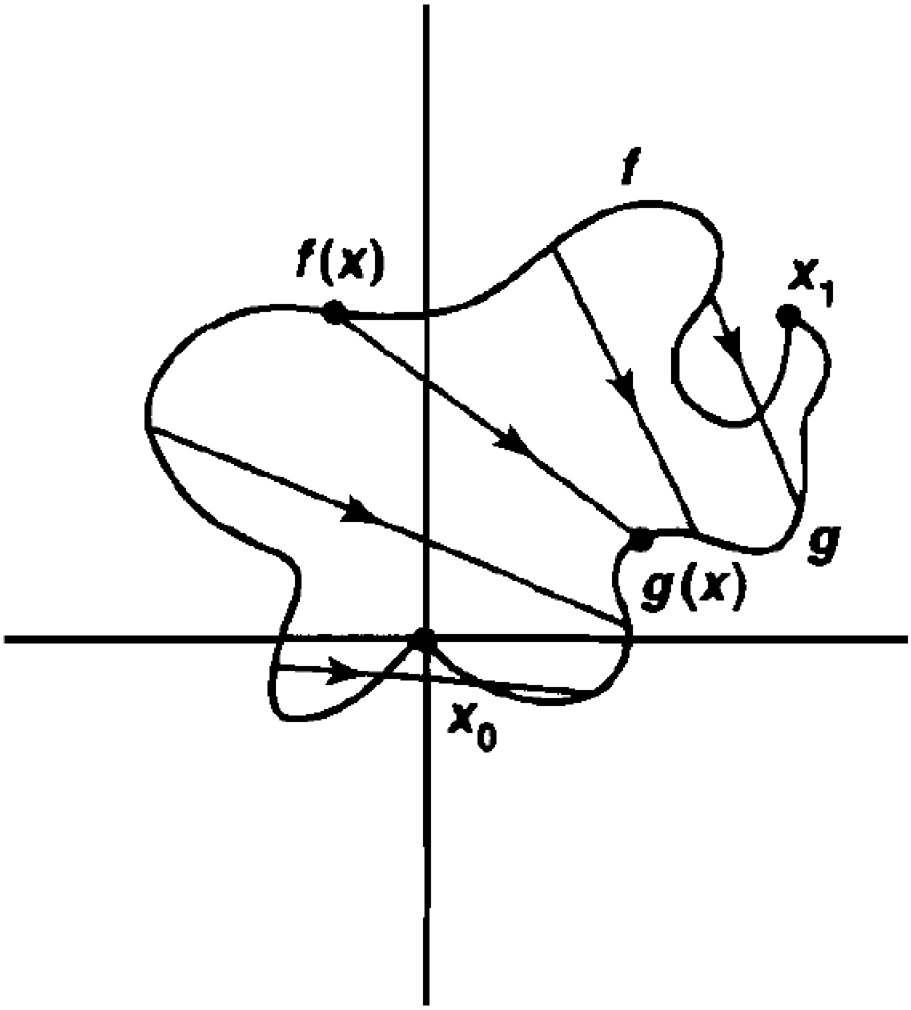

Figure 6.2: With the point restored, the space is convex and the two paths are homotopic.

However, the path $h(s) = (\cos(\pi s), -\sin(\pi s))$ is *not* path-homotopic. In the visual, this is the h path. Passing a continuous path *through* the point 0 introduces a discontinuity. The full argument is laborious at this stage, so being satisfied with this "hole" intuition is useful.

Note that the equivalence relation constructions works just as well on path-homotopies, just more book-keeping. Because of the lower dimension, the two do *not* define the same equivalence classes, so the relevant equivalence

Definition 6.1.9: Equivalence Class Of Path

If f is a path in X, write $[f]$ for the equivalence class of f with respect to $\simeq_p$

Example 6.1.10: Homotopy Equivalence

Recall all polynomials $x \mapsto x^n$? from earlier? These are in the same equivalence class

$$[f] = [g]$$

And since $F'(x,t) = (x, (t-1)x + t0)$ is also a path-homotopy,

$$[id_0] = [f] = [g]$$

meaning it is also *nullhomotopic*

6.1.11 Note This is an equivalence class, but it will not necessarily form a quotient space: the equivalence is formed on the *functions*, not on the points of the space. One should need to define a topology on the paths in order to define a quotient topology on that function space.

6.1.2 The Groupoid of Paths

As the note says, this collection of equivalence classes need not carry a topology. It does, however, carry *algebraic* structure, and the rest of the section builds it. The first ingredients are the structures: identities and a binary operation:

Definition 6.1.12: identity path

For $x \in X$, write e_x for the constant path $e_x : I \to X$ with $e_x(t) = x$ for all $t \in I$

In the notation of the preceding discussion, the identity element attached to the point x_0 is e_{x_0}.

Next, a way of relating paths. The natural candidate is to run one path after another. But arbitrary paths cannot be concatenated, since the endpoint of one need not be the initial point of the next, and the result would be discontinuous. Concatenation therefore carries a restriction:

Definition 6.1.13: combining paths Operators: $*$

If f, g are paths in X and $f(1) = g(0)$, define $f * g$ by

$$(f * g)(t) = \begin{cases} f(2t) & t \in [0, \frac{1}{2}] \\ g(2t-1) & t \in [\frac{1}{2}, 1] \end{cases}$$

Example 6.1.14: Paths

Try thinking of any two paths in $\mathbb{R}^2$ and combining them into 1.

Lemma 6.1.15: Well-defined: $*$ Preserves Equivalence Relation

Let f, f' be paths from x_0 to x_1 and g, g' paths from x_1 to x_2. If $f \simeq_p f'$ and $g \simeq_p g'$, then

$$(f * g) \simeq_p (f' * g')$$

***Proof*:**
Let F be a path homotopy from f to f' and G one from g to g'. The construction mirrors the definition of $*$ itself: traverse the first homotopy at double speed, then the second. Define

$$H(x,t) = \begin{cases} F(2x,t) & x \in \left[0, \frac{1}{2}\right] \\ G(2x-1,t) & x \in \left[\frac{1}{2}, 1\right] \end{cases}$$

the split being on the path variable x, not on the deformation parameter t.

This is well defined at $x = \frac{1}{2}$: there the two formulas give $F(1,t)$ and $G(0,t)$, which are x_1 and x_1 respectively, because F is a path homotopy and so fixes the terminal point of f, and G likewise fixes the initial point of g. The two closed strips $[0, \frac{1}{2}] \times I$ and $[\frac{1}{2}, 1] \times I$ cover $I \times I$, each formula is continuous on its strip as a composite of continuous maps, and they agree on the overlap, so H is continuous by lemma 1.4.6.

Finally $H(x,0) = (f*g)(x)$ and $H(x,1) = (f'*g')(x)$ by construction, and H holds the endpoints fixed, since $H(0,t) = F(0,t) = x_0$ and $H(1,t) = G(1,t) = x_2$. So H is a path homotopy from $f * g$ to $f' * g'$, as we sought to show.

Consequently

$$[f] * [g] = [f * g]$$

With the operation in hand, the question is which group axioms it satisfies. Recall that $*$ is defined only on compatible paths, where the endpoint of one matches the start of the next. That restriction has to be carried through every statement below.

Theorem 6.1.16: Properties Of $*$

1. (Associativity) Assume f, g, h are paths in X, $f(1) = g(0)$, $g(1) = h(0)$. Then

$$([f] * [g]) * [h] = [f] * ([g] * [h])$$

2. (identity) If f is a path in X with $f(0) = x_0$ and $f(1) = x_1$, then

$$[f] * [e_{x_1}] = [f] = [e_{x_0}] * [f]$$

3. (invertible) Given a path f in X, define $\tilde{f}(t) = f(1-t)$. Then

$$[f] * [\tilde{f}] = [e_{f(0)}]$$
$$[\tilde{f}] * [f] = [e_{f(1)}]$$

***Proof*:**
Clauses (1) and (2) are reparametrisation identities: the two sides traverse the same paths in the same order and differ only in when the switches occur, so the homotopy between them slides the breakpoints linearly. Clause (3) is not of that kind and gets its own construction.

(1) Associativity. Both $(f * g) * h$ and $f * (g * h)$ traverse f, then g, then h, differing only in when the switches occur: the first uses the breakpoints $\frac{1}{4}, \frac{1}{2}$ and the second $\frac{1}{2}, \frac{3}{4}$. Let $\gamma\colon [0,1] \to [0,1]$ be the piecewise-linear map carrying $0, \frac{1}{4}, \frac{1}{2}, 1$ to $0, \frac{1}{2}, \frac{3}{4}, 1$; it is continuous, increasing and fixes the endpoints, and $\big(f * (g * h)\big) \circ \gamma = (f * g) * h$. The straight-line homotopy $H(s,t) = \big(f * (g * h)\big)\big((1-t)\gamma(s) + ts\big)$ is then continuous, starts at $(f * g) * h$, ends at $f * (g * h)$, and fixes both endpoints since $\gamma(0) = 0$ and $\gamma(1) = 1$. So the two classes agree.

(2) Identity. The path $f * e_{x_1}$ is f run at double speed and then held at x_1, that is, $f \circ \delta$ where δ carries $0, \frac{1}{2}, 1$ to $0, 1, 1$ piecewise-linearly. The same straight-line homotopy $H(s,t) = f\big((1-t)\delta(s) + ts\big)$ joins $f * e_{x_1}$ to f and fixes the endpoints. The argument for $e_{x_0} * f$ is symmetric.

(3) Inverses. Define

$$H(s,t) = \begin{cases} f\big(2s(1-t)\big) & s \leq \frac{1}{2} \\ f\big(2(1-s)(1-t)\big) & s \geq \frac{1}{2} \end{cases}$$

The two formulas agree at $s = \frac{1}{2}$, both giving $f(1-t)$, and each is continuous on its closed strip, so H is continuous by lemma 1.4.6. At $t = 0$ it is $f * \tilde{f}$; at $t = 1$ it is constant at $f(0)$; and $H(0,t) = H(1,t) = f(0)$ for every t, so the endpoints are fixed. Hence $[f] * [\tilde{f}] = [e_{f(0)}]$. Applying this to $\tilde{f}$, whose reverse is f, gives $[\tilde{f}] * [f] = [e_{\tilde{f}(0)}] = [e_{f(1)}]$, completing the proof.

As the construction shows, the identity is not unique. So $*$ falls short of being a binary operation: it is not a function but a *partial function*. Two arbitrary paths cannot be combined, only those whose endpoints match, as the definition of $*$ required. Restricting to such paths does give a genuine binary operation, but only on that restricted domain; there is no way to read $*$ as an operation on the whole of $P(X)$.

There is a name for a "group" built on a partial function, and hence carrying several identities depending on how the partial function is restricted:

Definition 6.1.17: Groupoid

A *groupoid* is a set G with a *partial function* $* : G \times G \to G$ *and* a unary function $^{-1}$: $G \to G$ which then satisfy the group axioms (associativity, inverse, identity). The difference from a group is that:

1. (inverse) $a^{-1} * a$ and $a * a^{-1}$ is always defined
2. if $a * b$, is defined then $a * b * b^{-1} = a$ and $a^{-1} * a * b = b$ is also defined

How exactly the partial function $*$ is "partial" depends on the context; there is no one way of defining this partial function. The inverse is taken as a unary operation precisely because the identity is no longer unique, as shown in theorem 6.1.16.

Example 6.1.18: Groupoid

Taking $(\mathbb{R}, *, ^{-1})$ as the groupoid with $*$ as defined and $^{-1}(f) = \tilde{f}$ where $\tilde{f} = f(1-x)$, the reverse of f, gives a groupoid.

With the algebraic structure named, its elements can be identified:

Definition 6.1.19: Path Space

Let $P(X) = \{$equivalence classes of paths $[f]\}$, the *path space*

there is a natural map $P(X) \to X^2$, $[f] \mapsto (f(0), f(1))$. This map is surjective if and only if X is path connected. The groupoid above can therefore be written as

$$(P(X), *, ^{-1})$$

6.1.3 Loops and the Fundamental Group

There is still no unique identity and no total operation, so this is not yet a group. The obstruction is that arbitrary paths cannot be combined. Requiring the initial and end points to coincide removes it, which motivates one further restriction, to the paths that *loop*.

Definition 6.1.20: Loop

Given $x_0 \in X$, a *loop based at* x_0 is a path which begins and ends at x_0, that is

$$f : I \to X, \qquad f(0) = f(1) = x_0$$

Restricting to loops has a geometric payoff. In a plane with a point removed, a loop around that point cannot be contracted to a constant map, whereas in the whole plane every loop can. Which loops refuse to contract is the concept the rest of the chapter turns on.

With this definition the algebra restricts to the subset of $P(X)$ whose initial and end points agree.

Definition 6.1.21: Fundamental group at x_0

Let
$$\pi_1(X, x_0) = \{[f] | f(0) = f(1) = x_0\}$$
be the Path space with loops at x_0. Then $\pi_1(X, x_0)$ is a group under $*$! That is, if $[f], [g] \in \pi_1(X, x_0)$ then $[f] * [g] \in \pi_1(X, x_0)$, and $(\pi_1(X, x_0), *)$ is a group.

Note that $\pi_1(X, x_0) \subseteq P(X)$. The group $\pi_1(X, x_0)$ is the *fundamental group* of X *based at* x_0, or the *first homotopy group*

6.1.22 Note Nothing so far forces $\pi_1(X, x_0)$ to be trivial. It can contain many distinct classes of loops, and telling the trivial case from the rich one is precisely what makes the construction worth having.

The notation $\pi(X)$ is sometimes used for the groupoid of X.

The subscript invites an obvious question: if there is a first homotopy group, is there a second, or an nth? There is, written $\pi_n(X, x_0)$, and it is assembled from maps of the n-sphere into X rather than from maps of the interval. Those groups belong to homotopy theory and are not treated here.

Example 6.1.23: Fundamental Group of Euclidean Space

Take $\mathbb{R}^n$ with the Euclidean topology and any $x_0 \in \mathbb{R}^n$. The straight-line homotopy contracts every loop at x_0 to the constant path e_{x_0}, so the group has a single element:
$$\pi_1(\mathbb{R}^n, x_0) = \{[e_{x_0}]\}$$
The same argument runs in any convex subset of $\mathbb{R}^n$, since the segment joining a point of a loop to x_0 stays inside a convex set. Every convex space therefore has trivial fundamental group.

The fundamental group need not be abelian. Figure 6.3 shows the obstruction: in a plane with two points removed, a loop around the first puncture followed by a loop around the second is not path homotopic to the two run in the opposite order, since no homotopy between them can be swept across a missing point.

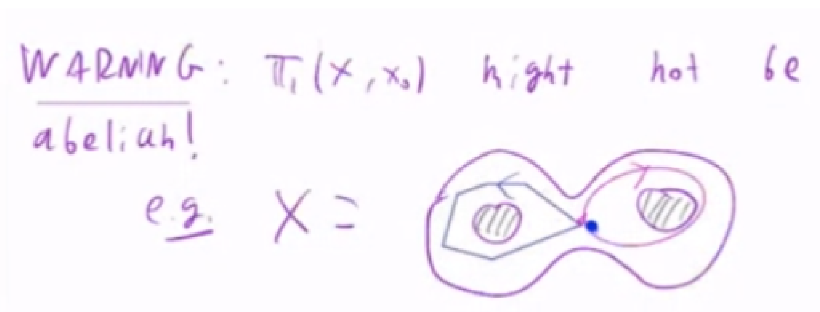

Figure 6.3: Two loops in a twice-punctured plane. The two orders of composition are not path homotopic, so the fundamental group is not abelian.

The definition attaches a group to each point of X, which raises the question of how those groups compare. Were they unrelated from point to point, π_1 would be recording something about the basepoint rather than about the space, and would be useless as an invariant. What happens instead is that any path from one basepoint to another induces an isomorphism between the two groups. The isomorphism *type* of π_1 therefore depends only on the path component containing the basepoint, which is why path components are the right unit of study.

Here is the isomorphism.

Proposition 6.1.24: Group Isomorphism Between Fundamental Groups

Let α be a path in X with $\alpha(0) = x_0$ and $\alpha(1) = x_1$, and define

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad \hat{\alpha}([f]) = [\tilde{\alpha}] * [f] * [\alpha]$$

where $\tilde{\alpha}$ is the reverse of α. Then $\hat{\alpha}$ is a group isomorphism.

The recipe reads left to right: travel backwards along α from x_1 to x_0, run the loop, then travel forwards again. Consecutive endpoints match at each junction, so the composite is a loop based at x_1. In the special case $x_0 = x_1$ with α itself a loop, $\hat{\alpha}$ is conjugation by $[\alpha]$, hence an inner automorphism of $\pi_1(X, x_0)$.

***Proof*:**
The argument is algebraic, and every manipulation below is licensed by theorem 6.1.16. For the homomorphism property, let $[f], [g] \in \pi_1(X, x_0)$. Then

$$\begin{aligned}
\hat{\alpha}([f] * [g]) = \hat{\alpha}([f * g]) &= [\tilde{\alpha}] * [f * g] * [\alpha] \\
&= [\tilde{\alpha}] * [f * e_{x_0} * g] * [\alpha] \\
&= [\tilde{\alpha}] * [f * \alpha * \tilde{\alpha} * g] * [\alpha] \\
&= ([\tilde{\alpha}] * [f] * [\alpha]) * ([\tilde{\alpha}] * [g] * [\alpha]) \\
&= \hat{\alpha}([f]) * \hat{\alpha}([g])
\end{aligned}$$

the second line inserting the constant path at x_0 and the third replacing it by $[\alpha] * [\tilde{\alpha}]$.

For bijectivity, note that $\tilde{\alpha}$ runs from x_1 to x_0 and so determines a map $\hat{\tilde{\alpha}} : \pi_1(X, x_1) \to \pi_1(X, x_0)$ by the same formula. Composing the two in either order and cancelling $[\alpha] * [\tilde{\alpha}] = [e_{x_0}]$ and $[\tilde{\alpha}] * [\alpha] = [e_{x_1}]$ gives

$$\hat{\alpha}(\hat{\tilde{\alpha}}([f])) = [f], \qquad \hat{\tilde{\alpha}}(\hat{\alpha}([g])) = [g]$$

for every $[f] \in \pi_1(X, x_1)$ and $[g] \in \pi_1(X, x_0)$. So $\hat{\alpha}$ is a bijective homomorphism, and therefore an isomorphism, as claimed.

One caution before the consequence. The isomorphism $\hat{\alpha}$ depends on the path α, not on the pair of basepoints alone. Two paths α, β from x_0 to x_1 satisfy $\hat{\beta} \circ \hat{\alpha}^{-1} =$ conjugation by $[\tilde{\beta} * \alpha] \in \pi_1(X, x_1)$, so all choices of path agree exactly when the group is abelian. There is thus no canonical isomorphism between the groups at two distinct basepoints, only one canonical up to conjugation.

That caveat aside, the consequence is what licenses the phrase "*the* fundamental group" of a space.

Corollary 6.1.25: Path-Connectedness And Isomorphism

If X is path-connected, $\pi_1(X, x_0)$ is isomorphic to $\pi_1(X, x_1)$ for any $x_0, x_1 \in X$

***Proof*:**
Let $x_0, x_1 \in X$. Since X is path-connected there is a path α from x_0 to x_1, and proposition 6.1.24 attaches to it an isomorphism $\pi_1(X, x_0) \to \pi_1(X, x_1)$. As x_0 and x_1 were arbitrary, all the fundamental groups of X are isomorphic, as we sought to show.

Two path-connected spaces with non-isomorphic fundamental groups are therefore not homeomorphic. The converse fails: matching fundamental groups do not force a homeomorphism, and $\mathbb{R}$ against $\mathbb{R}^2$ is the standard counterexample. Both have trivial fundamental group, yet $\mathbb{R} \setminus \{0\}$ is disconnected while $\mathbb{R}^2 \setminus \{0\}$ is path-connected, so no homeomorphism can exist. The invariant separates spaces; it does not classify them.

The simplest value it can take deserves a name.

Definition 6.1.26: Simply-Connected

A space X is *simply-connected* if it is path-connected and $\pi_1(X, x_0)$ is the trivial group for some, and hence by corollary 6.1.25 every, $x_0 \in X$.

Triviality of π_1 says that loops carry no information, and in a simply-connected space that collapse propagates to paths in general: an entire homotopy class is pinned down by where it starts and where it ends.

Lemma 6.1.27: Path Space Of A Simply-Connected Space

If X is simply connected, the endpoint map

$$\pi : P(X) \to X^2, \qquad \pi([f]) = (f(0), f(1))$$

is a bijection. Equivalently, a path homotopy class in a simply-connected space is determined by its endpoints.

***Proof*:**
First, any two paths sharing their endpoints are path homotopic. Let α, β be paths from x_0 to x_1. Then $\alpha * \tilde{\beta}$ is a loop at x_0, so $[\alpha] * [\tilde{\beta}] \in \pi_1(X, x_0)$, and that group is trivial by hypothesis:

$$[\alpha] * [\tilde{\beta}] = [e_{x_0}]$$

Composing on the right with $[\beta]$ and applying theorem 6.1.16,

$$[\alpha] = [\alpha] * [e_{x_1}] = [\alpha] * [\tilde{\beta}] * [\beta] = [e_{x_0}] * [\beta] = [\beta]$$

The map π is well defined, since a path homotopy holds both endpoints fixed and so every representative of a class shares the same pair. It is injective by the paragraph above: if $\pi([\alpha]) = \pi([\beta])$ then α and β agree at both endpoints, hence $[\alpha] = [\beta]$. It is surjective because a simply-connected space is path-connected, so any $(x_0, x_1) \in X^2$ is joined by some path α, and that path satisfies $\pi([\alpha]) = (x_0, x_1)$. A well-defined injection that is onto is a bijection, which is what was wanted.

6.1.4 Induced Homomorphisms and Functoriality

An invariant is only useful if it is functorial: a continuous map between spaces should induce a map between their fundamental groups, and composites should correspond to composites. Failing that, the group attached to X would say nothing about how X relates to anything else, and no argument could pass from a statement about spaces to a statement about groups. The correspondence does exist, and it sends continuous maps to group homomorphisms.

Definition 6.1.28: induced homomorphic function

Suppose $h : X \to Y$ is continuous, with $h(x_0) = y_0$ (i.e. $h(X, x_0) \to (Y, y_0)$) . Then define

$$h_* : \pi_1(x, x_0) \to \pi_1(Y, y_0) \qquad h_*([\alpha]) = [h \circ \alpha]$$

6.1.29 Trivia The definition asks for more than continuity: h must also carry the chosen basepoint to the chosen basepoint. Categorically, the ambient category is therefore not **Top** but the category of *pointed* spaces, whose objects are pairs (X, x_0) and whose morphisms are the basepoint-preserving continuous maps.

Proposition 6.1.30: Properties Of h_*

1. h_* is a group homomorphism
2. If $i : (X, x_0) \to (X, x_0)$ is the identity map, then i_* is the identity homomorphism
3. The assignment is functorial: if $f : (X, x_0) \to (Y, y_0)$ and $g : (Y, y_0) \to (Z, z_0)$ are continuous and basepoint-preserving, then

$$(g \circ f)_* = g_* \circ f_*$$

Statements (2) and (3) together say that π_1 is a functor from the category of pointed spaces to the category **Grp** of groups.

***Proof*:**
Each claim in turn.

1. First, h_* is well defined. If F is a path homotopy from α to β, then $h \circ F$ is continuous as a composite of continuous maps, and it holds the endpoints fixed because F does, so it is a path homotopy from $h \circ \alpha$ to $h \circ \beta$. For the homomorphism property, observe that $h \circ (\alpha * \beta)$ and $(h \circ \alpha) * (h \circ \beta)$ are literally the same function: each sends t to $h(\alpha(2t))$ for

$t \leq \frac{1}{2}$ and to $h(\beta(2t-1))$ for $t \geq \frac{1}{2}$. Hence

$$\begin{aligned} h_*([\alpha] * [\beta]) &= h_*([\alpha * \beta]) \\ &= [h \circ (\alpha * \beta)] \\ &= [(h \circ \alpha) * (h \circ \beta)] \\ &= [h \circ \alpha] * [h \circ \beta] \\ &= h_*([\alpha]) * h_*([\beta]) \end{aligned}$$

2. $i_*([\alpha]) = [i \circ \alpha] = [\alpha]$ for every class, so i_* is the identity homomorphism.
3. Composition of functions is associative, so for every $[\alpha] \in \pi_1(X, x_0)$,

$$\begin{aligned} (g \circ f)_*([\alpha]) &= [(g \circ f) \circ \alpha] \\ &= [g \circ (f \circ \alpha)] \\ &= g_*([f \circ \alpha]) \\ &= g_*(f_*([\alpha])) \end{aligned}$$

and two homomorphisms agreeing on every element are equal.

This establishes all three statements.

Functoriality is what turns π_1 from a construction into an invariant. A homeomorphism h carrying x_0 to y_0 has a continuous inverse, so by (2) and (3) the homomorphism $(h^{-1})_*$ inverts h_*; homeomorphic pointed spaces have isomorphic fundamental groups. Read in the other direction, groups that fail to be isomorphic certify that no homeomorphism exists.

Exercise 6.1.1

1. Show that any two maps into a convex subset of $\mathbb{R}^n$ are homotopic, and that homotopy of maps $X \to Y$ is not the same as path homotopy even when $X = I$.
2. Show that $*$ is not associative on paths, but is associative on path homotopy classes, and identify which part of theorem 6.1.16 carries this.
3. Show that $\pi_1(X, x_0)$ is trivial for X convex, and that $\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$.
4. Show that in a convex subset of $\mathbb{R}^n$ the straight-line homotopy makes any two paths with the same endpoints path homotopic, and observe that this is exactly the conclusion of lemma 6.1.27 for such a set.

6.2 Covering Spaces and the Circle

Every fundamental group computed so far has been trivial, because convex spaces are as far as the present machinery reaches. Producing a non-trivial answer requires a way of unwinding loops, and the device that does it is the covering space.

6.2.1 Covering spaces

The idea is to replace a space by one lying above it, in which loops can be unrolled and their winding read off. Applied to the circle it yields the first non-trivial fundamental group in this book. Covering spaces also return in complex analysis, where they organize the branches of multivalued functions and underlie the construction of Riemann surfaces [8].

Definition 6.2.1: Evenly Covered and Slices

Let $p : E \to B$ be a continuous surjective map. An open set $U \subseteq B$ is *evenly covered* by p if the inverse image $p^{-1}(U)$ can be written as a union of disjoint open sets V_α in E such that for each α the restriction of p to V_α is a homeomorphism of V_α onto U. The collection $\{V_\alpha\}$ is the partition of $p^{-1}(U)$ into *slices*.

The picture is of a stack of copies of U sitting above it, each mapped down onto U without distortion. A map for which every point of the base admits such a neighbourhood gets its own name.

Definition 6.2.2: Covering Map

Let $p : E \to B$ be continuous and surjective. If every point b of B has a neighborhood U that is evenly covered by p, then p is called a *covering map*, and E is said to be a *covering space* of B.

Example 6.2.3: Covering Maps

1. The identity map $i : X \to X$ is trivially a covering map: every open U is evenly covered by the single slice U itself.

2. Let $X \times \{1, \ldots, n\}$ be n disjoint copies of X, with the second factor discrete, and let $p(x, i) = x$. Any open $U \subseteq X$ has $p^{-1}(U) = \bigcup_i U \times \{i\}$, a disjoint union of open sets each mapped homeomorphically onto U. This is the picture to keep in mind for covering spaces in general:

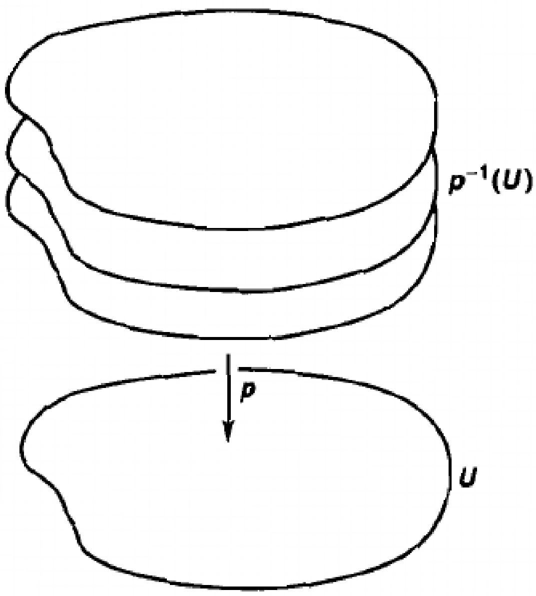

Figure 6.4: Covering space example

3. More generally, the projection $\pi_X : X \times Y \to X$ is a covering map whenever Y carries the discrete topology. Discreteness is what makes each $U \times \{y\}$ open, and hence a legitimate slice; for non-discrete Y the preimage $U \times Y$ does not break into disjoint open pieces mapping bijectively onto U.

Two consequences of the definition are worth isolating before any harder example, since both get used repeatedly.

Corollary 6.2.4: properties of Covering Maps

Let $p : E \to B$ be a covering map.

1. For every $y \in B$, the subspace $p^{-1}(y)$ has the discrete topology
2. p is an open map

***Proof*:**

1. Fix $y \in B$ and choose a neighbourhood U of y evenly covered by p, with slices $\{V_\alpha\}$. Let $e \in p^{-1}(y)$. Since the slices partition $p^{-1}(U) \ni e$, there is exactly one α with $e \in V_\alpha$, and since $p|_{V_\alpha}$ is injective, $V_\alpha \cap p^{-1}(y) = \{e\}$. So $\{e\}$ is open in the subspace $p^{-1}(y)$, being the intersection of $p^{-1}(y)$ with an open set of E. Every singleton of $p^{-1}(y)$ is therefore open, which is the definition of the discrete topology.
2. Let $A \subseteq E$ be open; the claim is that $p(A)$ is open. By definition 1.3.4 it suffices to find,

around each $x \in p(A)$, an open set contained in $p(A)$. Pick such an x and a neighbourhood U of x evenly covered by p, with slices $\{V_\alpha\}$. Since $x \in p(A)$ there is some $y \in A$ with $p(y) = x$, and $y \in p^{-1}(U)$, so $y \in V_\beta$ for some β. Then $A \cap V_\beta$ is open in E and contains y. As $p|_{V_\beta}$ is a homeomorphism onto U, it is an open map, so $p(A \cap V_\beta)$ is open in U, hence open in B because U is. Finally $x = p(y) \in p(A \cap V_\beta) \subseteq p(A)$, giving the required neighbourhood.

Both statements follow.

The examples so far cover disconnected spaces, which is the easy case: the slices are handed over by the disjoint pieces. The interesting covering spaces sit over connected bases, where the sheets must be produced rather than assumed. The circle is the fundamental instance.

Example 6.2.5: The Real Line Covers The Circle

The map $p : \mathbb{R} \to S^1$ given by

$$p(x) = (\cos(2\pi x), \sin(2\pi x))$$

is a covering map. Geometrically it wraps the line around the circle infinitely often: each interval $[n, n+1]$ makes exactly one full circuit.

***Proof*:**

Let $U = \left\{(x, y) \in S^1 \middle| x > 0\right\}$, an open half of the circle. A point $p(t)$ lies in U exactly when $\cos(2\pi t) > 0$, that is, when $2\pi t$ lies in $(-\frac{\pi}{2} + 2\pi n, \frac{\pi}{2} + 2\pi n)$ for some $n \in \mathbb{Z}$. Hence

$$p^{-1}(U) = \bigcup_{n\in\mathbb{Z}} V_n, \qquad V_n = \left(n - \frac{1}{4},\ n + \frac{1}{4}\right)$$

a union of disjoint open intervals, since consecutive intervals are separated by the points $n + \frac{1}{2}$. Visually:

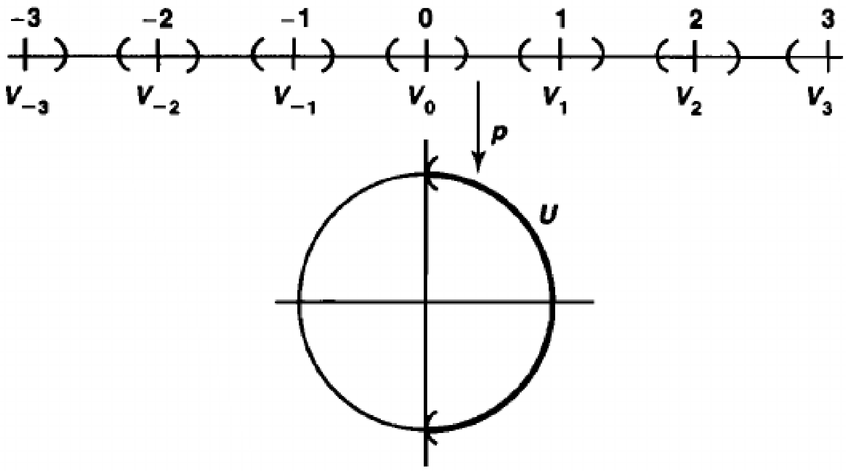

Figure 6.5: The preimage under p of the open half-circle $x > 0$.

It remains to show that each $p|_{V_n}$ is a homeomorphism onto U. Pass to the closures: $\overline{V_n} = [n - \frac{1}{4}, n + \frac{1}{4}]$ is compact by corollary 3.2.25, and p carries it onto $\overline{U} = \{(x, y) \in S^1 | x \geq 0\}$. That restriction is injective, because cos and sin determine $2\pi t$ modulo 2π and the interval $\overline{V_n}$ has length $\frac{1}{2} < 1$; it is surjective onto $\overline{U}$ because every angle in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is realised. So $p|_{\overline{V_n}}$ is a continuous bijection from a compact space to the Hausdorff space $\overline{U}$, and therefore a homeomorphism by corollary 3.2.20. A homeomorphism restricts to a homeomorphism on any subset, and $p(V_n) = U$, so $p|_{V_n} : V_n \to U$ is a homeomorphism as required. Thus U is evenly covered.

The same argument applies verbatim to the three remaining half-circles $x < 0$, $y > 0$ and $y < 0$, using the corresponding intervals of length $\frac{1}{2}$. These four open sets cover S^1, so every point of S^1 has an evenly covered neighbourhood, and p is a covering map.

6.2.2 Properties of Covering Maps

The natural questions come next: does restricting the domain of a covering map leave a covering map, does a product of covering maps remain one, and is there a cheaper condition equivalent to the definition.

The last of these deserves care, because covering maps are in particular *local homeomorphisms*: around every $e \in E$ there is a neighbourhood U such that $p|_U$ is a homeomorphism onto an open subset of B. It is tempting to think the converse holds, since in the circle example the real work went into producing homeomorphisms on the slices, and overlapping sheets could simply be merged. The converse fails, for a reason that is easy to miss.

Example 6.2.6: Local Homeomorphism That's Not a Covering Map
Take the map of example 6.2.5 and restrict its domain to the positive reals:

$$p : \mathbb{R}_+ \to S^1 \qquad p(x) = (\cos(2\pi x), \sin(2\pi x))$$

This is still surjective and still a local homeomorphism, and it is even evenly covering for *most* points. Consider, though, the point $b_0 = (1, 0)$ and any open neighbourhood U of it. The preimages behave as before except for the slice nearest the origin, which is truncated to $(0, \epsilon)$ for some ϵ depending on U. Restricting p to that piece gives not a homeomorphism onto U but an *embedding*: a map that becomes a homeomorphism only after the codomain is cut down to its image. The image of $(0, \epsilon)$ is an arc starting at b_0 and running to $p(\epsilon)$, and it misses b_0 itself, so it is a proper subset of U.

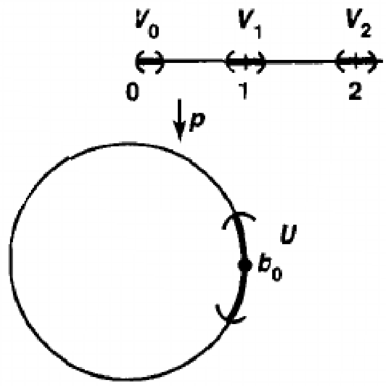

Figure 6.6: The truncated slice over a neighbourhood of b_0. The figure does not draw the embedding itself; the relevant open set is the arc beginning at b_0.

A cleaner example makes the same point with no trigonometry. Let

$$E = ([0,1] \times \{0\}) \cup ((0,1) \times \{1\}), \qquad B = [0,1]$$

with p the projection to the first coordinate, which is surjective because of the first piece. Around $0 \in B$ take any basic neighbourhood $[0,\epsilon)$. Its preimage splits into $[0,\epsilon) \times \{0\}$ and $(0,\epsilon) \times \{1\}$, and the second of these maps onto $(0,\epsilon)$ rather than onto $[0,\epsilon)$. No neighbourhood of 0 is evenly covered. The failure is the same in both cases: a slice can be an embedding onto a proper subset of U rather than a homeomorphism onto U.

The example is a warning about shrinking the domain of a covering map: an arbitrary subspace of E need not cover anything. Restricting the *base* is a different matter, provided the domain is cut down to the full preimage.

Proposition 6.2.7: Subspace Of Covering Space

Let $p : E \to B$ be a covering map. If B_0 is a subspace of B, and if $E_0 = p^{-1}(B_0)$, then the map $p_0 : E_0 \to B_0$ obtained by restricting p is a covering map

***Proof*:**
Let $b_0 \in B_0$. Since p is a covering map there is an open $U \ni b_0$ in B that is evenly covered in the sense of definition 6.2.1, so $p^{-1}(U) = \bigsqcup_\alpha V_\alpha$ with each V_α open in E and $p|_{V_\alpha} : V_\alpha \to U$ a homeomorphism. Then $U \cap B_0$ is an open neighbourhood of b_0 in the subspace B_0 (section 2.1), and

$$p_0^{-1}(U \cap B_0) = p^{-1}(U) \cap E_0 = \bigsqcup_\alpha (V_\alpha \cap E_0)$$

a disjoint union of sets open in E_0. For each α the restriction of $p|_{V_\alpha}$ to $V_\alpha \cap E_0$ is a continuous bijection onto $U \cap B_0$, with continuous inverse the restriction of $(p|_{V_\alpha})^{-1}$; a restriction of a homeomorphism to a subspace is a homeomorphism onto its image, and that image is exactly $U \cap B_0$ because $V_\alpha \cap E_0 = (p|_{V_\alpha})^{-1}(U \cap B_0)$. So $U \cap B_0$ is evenly covered by p_0. As b_0 was arbitrary, p_0 is a covering map, completing the proof.

Products behave as well as one could hope.

Proposition 6.2.8: Product Of Covering Maps

If $p : E \to B$ and $p' : E' \to B'$ are covering maps, then

$$p \times p' : E \times E' \to B \times B'$$

is a covering map.

***Proof*:**
Let $(b, b') \in B \times B'$. Choose neighbourhoods $U \ni b$ and $U' \ni b'$ evenly covered by p and p' respectively, with slices $\{V_\alpha\}$ and $\left\{V'_\beta\right\}$. Then $U \times U'$ is a basic open neighbourhood of (b, b') in the product topology, and

$$(p \times p')^{-1}(U \times U') = p^{-1}(U) \times p'^{-1}(U') = \bigcup_{\alpha,\beta} \left(V_\alpha \times V'_\beta\right)$$

Each $V_\alpha \times V'_\beta$ is open in $E \times E'$, and the sets are pairwise disjoint: two of them with $(\alpha, \beta) \neq (\gamma, \delta)$ differ in at least one coordinate, where the corresponding slices are already disjoint. Finally $p \times p'$ restricted to $V_\alpha \times V'_\beta$ is the product of the two homeomorphisms $p|_{V_\alpha}$ and $p'|_{V'_\beta}$, hence a homeomorphism onto $U \times U'$. So $U \times U'$ is evenly covered, and since (b, b') was arbitrary, $p \times p'$ is a covering map.

Example 6.2.9: Covering The Torus And The Figure Eight

1. Taking p of example 6.2.5 in each factor gives

   $$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$$

   a covering map by proposition 6.2.8. Its base is the torus, so the plane covers the torus, with the integer lattice $\mathbb{Z}^2$ sitting over the basepoint.

2. Inside the torus sit the two circles $S^1 \times \{b_0\}$ and $\{b_0\} \times S^1$, whose union B_0 is a figure eight. Taking $E_0 = (p \times p)^{-1}(B_0)$, which is the grid of horizontal and vertical lines through the integer lattice, proposition 6.2.7 makes the restriction $E_0 \to B_0$ a covering map. The figure eight admits several essentially different covers: one may unwrap a single circle and leave the other intact, unwrap both as in the grid just described, or pass to the universal cover, which turns out to be an infinite tree.

6.2.3 Fundamental group of a circle

The machinery is now in place to compute $\pi_1(S^1)$, and the answer is the first non-trivial one in this book. The intuition to carry through the argument is a winding count. A loop on the circle that travels once around subtends a total angle of 2π; the constant loop subtends 0; nothing in between is possible without failing to close up. The available totals are therefore $2k\pi$ for $k \in \mathbb{Z}$, and the theorem to be proved is that these counts are exactly the path homotopy classes. Put physically, the question is how many ways a piece of string can be wound around a circle with both ends tacked at one point, counted up to the deformations a homotopy allows.

The examples above all display a space covered by a larger one projecting down, but nothing so far explains how to move information between the two. The device that does it lifts a map on the base to a map on the cover.

Definition 6.2.10: Lifting

Let $p : E \to B$ be a continuous function and let $f : X \to B$ be a continuous map. A *lifting* of f is a continuous map $\tilde{f} : X \to E$ such that $f = p \circ \tilde{f}$, that is, such that the diagram

$$\begin{tikzcd} & E \arrow[d, "p"] \\ X \arrow[ur, "\tilde{f}"] \arrow[r, "f"'] & B \end{tikzcd}$$

commutes.

The name is literal: where f sends x to a point of B, the lift sends x to one of the points sitting above it in E. For the covering of example 6.2.5 the picture is a corkscrew, with the lift choosing a height on the spiral over each point of the circle.

Lifts matter because a covering map admits them for paths, and does so uniquely once a starting height is chosen.

Lemma 6.2.11: paths can be lifted

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$, and let $f : [0, 1] \to B$ be a path with $f(0) = b_0$. Then f has a unique lifting to a path $\tilde{f}$ in E with $\tilde{f}(0) = e_0$.

***Proof*:**
Existence. Cover B by open sets each of which is evenly covered by p; this is possible by definition 6.2.2. Their preimages under f form an open cover of $[0, 1]$, which is a compact metric space by theorem 3.2.24, so lemma 3.2.31 supplies a Lebesgue number. Choose a subdivision

$$0 = s_0 < s_1 < \cdots < s_n = 1$$

fine enough that each $[s_{i-1}, s_i]$ has diameter below that number. Then each $f([s_{i-1}, s_i])$ lies in a single evenly covered open set U_i.

Define $\tilde{f}$ by induction on i, starting with $\tilde{f}(0) = e_0$. Suppose $\tilde{f}$ has been defined and is continuous on $[0, s_{i-1}]$. The point $\tilde{f}(s_{i-1})$ lies in $p^{-1}(U_i)$, hence in exactly one slice V of the partition of $p^{-1}(U_i)$. Set

$$\tilde{f}(s) = (p|_V)^{-1}(f(s)) \qquad \text{for } s \in [s_{i-1}, s_i]$$

which is legitimate because $p|_V$ is a homeomorphism onto U_i and $f(s) \in U_i$. This is continuous on $[s_{i-1}, s_i]$ as a composite of continuous maps, and it agrees with the previously defined value at s_{i-1}, since $\tilde{f}(s_{i-1})$ is the unique point of V above $f(s_{i-1})$. The two closed pieces $[0, s_{i-1}]$ and $[s_{i-1}, s_i]$ therefore glue to a continuous map on $[0, s_i]$ by lemma 1.4.6. After n steps $\tilde{f}$ is defined and continuous on all of $[0, 1]$, and $p \circ \tilde{f} = f$ holds on each piece by construction.

Uniqueness. Let $\tilde{f}$ and $\tilde{g}$ both lift f with $\tilde{f}(0) = \tilde{g}(0) = e_0$, and use the same subdivision. Suppose the two agree on $[0, s_{i-1}]$, which holds for $i = 1$. On $[s_{i-1}, s_i]$ both take values in $p^{-1}(U_i)$, and both are continuous, so their images lie in the slices; since $[s_{i-1}, s_i]$ is connected by theorem 3.1.42 and the slices are disjoint open sets, each of $\tilde{f}$ and $\tilde{g}$ maps the whole interval into a single slice. The two slices agree, both being the one containing the common value at s_{i-1}. On that slice p is injective, and $p \circ \tilde{f} = f = p \circ \tilde{g}$, so $\tilde{f} = \tilde{g}$ on $[s_{i-1}, s_i]$. Induction gives agreement on $[0, 1]$, and the lift is unique, as claimed.

The same subdivision argument, run in two variables, lifts homotopies.

Lemma 6.2.12: Homotopies Can Be Lifted

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$, and let $F : I \times I \to B$ be continuous with $F(0, 0) = b_0$. Then there exists a unique lift $\tilde{F} : I \times I \to E$ with $\tilde{F}(0, 0) = e_0$.

***Proof*:**
Uniqueness first, since it also pins down the construction. Any lift is determined on $\{0\} \times I$ and on each $I \times \{s\}$ by lemma 6.2.11, applied to the corresponding path in B with the starting height already fixed. Since every point of $I \times I$ lies on such a path, at most one lift with $\tilde{F}(0, 0) = e_0$ exists.

For existence, cover B by evenly covered open sets and pull back along F to get an open cover of $I \times I$, which is compact by theorem 3.2.27 and metric. Take a Lebesgue number from lemma 3.2.31 and choose subdivisions

$$0 = s_0 < \cdots < s_m = 1, \qquad 0 = t_0 < \cdots < t_n = 1$$

fine enough that each rectangle $R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ has diameter below it, so that $F(R_{ij})$ lies in a single evenly covered set U_{ij}.

Order the rectangles left to right along the bottom row, then left to right along the next row, and so on, and define $\tilde{F}$ on them in that order, beginning with $\tilde{F}(0, 0) = e_0$. At each stage the part of ∂R_{ij} on which $\tilde{F}$ is already defined, namely the union of the left and bottom edges where those meet earlier rectangles, is connected: it is a single point, one edge, or two edges meeting at a corner. That set maps into $p^{-1}(U_{ij})$, and being connected it lands in a single slice V by the argument used above. Define

$$\tilde{F}(x) = (p|_V)^{-1}(F(x)) \qquad \text{for } x \in R_{ij}$$

This agrees with the values already assigned on the shared edges, since those values lie in V and $p|_V$ is injective. Each R_{ij} is closed, and finitely many of them cover $I \times I$, so repeated application of lemma 1.4.6, one rectangle at a time in the chosen order, assembles the pieces into a continuous $\tilde{F}$ on $I \times I$ with $p \circ \tilde{F} = F$, as required.

Lifting a homotopy is not yet enough; what the fundamental group needs is that the lift of a *path* homotopy is again a path homotopy, so that the construction descends to homotopy classes.

Lemma 6.2.13: Path Homotopies can be lifted

In the situation of lemma 6.2.12, if F is a path homotopy then so is $\tilde{F}$.

***Proof*:**
Let F be a path homotopy from f to g, so that $F(\{0\} \times I) = \{b_0\}$ and $F(\{1\} \times I) = \{b_1\}$ are single points. Consider the restriction of $\tilde{F}$ to $\{0\} \times I$. It is continuous and lands in $p^{-1}(b_0)$, which is discrete by corollary 6.2.4. A continuous map from the connected space $\{0\} \times I$ to a discrete space is constant, so $\tilde{F}$ is constant on the left edge. The same argument on $\{1\} \times I$ makes $\tilde{F}$ constant on the right edge. Hence $\tilde{F}$ holds both endpoints fixed throughout the deformation and is a path homotopy, which is what was needed.

The payoff is that lifting is well defined on path homotopy classes.

Lemma 6.2.14: Path-Homotopic Paths Have Equal Lift Endpoints

Let f, g be paths in B from b_0 to b_1 with $f \simeq_p g$, and let $\tilde{f}, \tilde{g}$ be their lifts beginning at e_0. Then $\tilde{f}(1) = \tilde{g}(1)$, and $\tilde{f} \simeq_p \tilde{g}$.

***Proof*:**
Let F be a path homotopy from f to g and let $\tilde{F}$ be its lift with $\tilde{F}(0,0) = e_0$, which is a path homotopy by lemma 6.2.13. Its restriction to $I \times \{0\}$ is a lift of f starting at e_0, hence equals $\tilde{f}$ by the uniqueness in lemma 6.2.11; likewise the restriction to $I \times \{1\}$ equals $\tilde{g}$. So $\tilde{F}$ is a path homotopy from $\tilde{f}$ to $\tilde{g}$, and being a path homotopy it fixes the terminal point, giving $\tilde{f}(1) = \tilde{g}(1)$ as claimed.

6.2.15 Note The lifts $\tilde{f}$ and $\tilde{g}$ are *paths*, not loops, even when f and g are loops: a lift of a loop returns to the fibre over the basepoint, but not necessarily to the point it started from. That gap is exactly the information the next definition extracts.

Since lifting respects path homotopy, the terminal point of a lift depends only on the class of the loop downstairs. That assignment is the bridge between the fundamental group of B and the fibre over the basepoint.

Definition 6.2.16: Lifting Correspondence

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Given $[f] \in \pi_1(B, b_0)$, let $\tilde{f}$ be the lifting of f with $\tilde{f}(0) = e_0$, and define

$$\varphi : \pi_1(B, b_0) \to p^{-1}(b_0), \qquad \varphi([f]) = \tilde{f}(1)$$

Then φ is the *lifting correspondence* derived from p and e_0.

The map is well defined on classes by lemma 6.2.14, and it does land in $p^{-1}(b_0)$, since $p(\tilde{f}(1)) = f(1) = b_0$.

6.2.17 Note The basepoint e_0 is part of the data, not an afterthought: the fibre over b_0 typically has many points, and each one gives a different φ. Choosing another e_0 permutes the values without changing the qualitative conclusions below, in the same way that changing x_0 in $\pi_1(X, x_0)$ changes the group only up to isomorphism.

The covering spaces of interest are path-connected, and that hypothesis alone already forces φ to see the whole fibre.

Proposition 6.2.18: Lifting Correspondence and Path-Connected Spaces

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. If E is path-connected, then the lifting correspondence is surjective. If E is also simply-connected, then the lifting correspondence is bijective.

***Proof*:**
Surjectivity. Assume E is path-connected and let $e_1 \in p^{-1}(b_0)$. Choose a path f_E in E with $f_E(0) = e_0$ and $f_E(1) = e_1$, and set $f = p \circ f_E$, a path in B. Then f_E is a lift of f starting at e_0, and

$$f(0) = p(f_E(0)) = p(e_0) = b_0$$
$$f(1) = p(f_E(1)) = p(e_1) = b_0$$

so f is in fact a loop at b_0 and $[f] \in \pi_1(B, b_0)$. By the uniqueness in lemma 6.2.11 the lift used to compute φ is f_E itself, whence $\varphi([f]) = f_E(1) = e_1$. Every point of the fibre is hit, so φ is surjective.

Injectivity. Assume in addition that E is simply connected, and suppose $\varphi([f]) = \varphi([g])$ for loops f, g at b_0. Their lifts $\tilde{f}, \tilde{g}$ both start at e_0 and, by hypothesis, both end at the same point of the fibre. Two paths in a simply-connected space with matching endpoints are path homotopic by lemma 6.1.27, so there is a path homotopy $\tilde{F}$ from $\tilde{f}$ to $\tilde{g}$. Composing with p gives a continuous map $p \circ \tilde{F}$ which is a path homotopy from $p \circ \tilde{f} = f$ to $p \circ \tilde{g} = g$: it is continuous as a composite, and it holds the endpoints fixed because $\tilde{F}$ does and p is a function. Hence $[f] = [g]$.

Surjective and injective together give a bijection, as claimed.

The proposition converts a question about loops in B, which is hard, into a question about a fibre in E, which is often just a matter of counting points.

Example 6.2.19: Covering Spaces Properties

1. Let $B = E = S^1$, and take $p : E \to B$ $z \mapsto z^2$, and take $b_0 = e_0 = (1, 0)$. Visually:

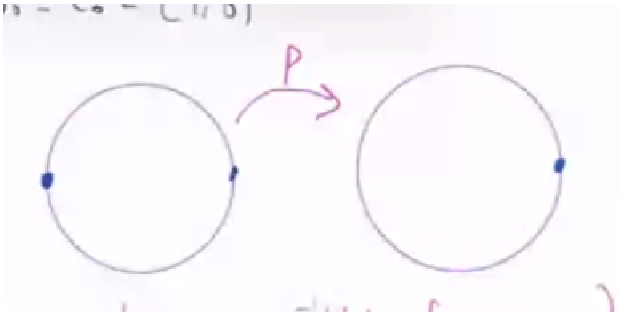

Figure 6.7: visual example of pre-image of b_0

Then $\varphi : \pi_1(S^1, b_0) \to p^{-1}(b_0) = \{(1,0), (-1,0)\}$.

Two loops make the point. First the constant loop $e_{b_0} : [0,1] \to S^1$ with $e_{b_0}(t) = b_0$. Its lift starting at e_0 is the constant path e_{e_0}, since that path is continuous, starts at e_0, and projects to e_{b_0}; uniqueness in lemma 6.2.11 makes it the lift. So

$$\varphi([e_{b_0}]) = e_{e_0}(1) = e_0 = (1,0)$$

Second, let

$$f : I \to S^1, \qquad f(t) = (\cos(2\pi t), \sin(2\pi t))$$

the loop that traverses the circle once. Since p squares the angle, the lift starting at e_0 traverses only half:

$$\tilde{f} : I \to S^1, \qquad \tilde{f}(t) = (\cos(\pi t), \sin(\pi t))$$

Then $\tilde{f}(1) = (-1,0)$, so $\varphi([f]) = (-1,0)$, which incidentally exhibits the surjectivity of φ directly. The two values of φ differ, so $[f] \neq [e_{b_0}]$: the loop f is not path homotopic to a constant, and this is the first non-contractible loop established in this book.

2. Let X be the figure eight of example 6.2.9 and let b_0 be the point where the circles meet. Take f traversing the left circle counter-clockwise and g traversing the right circle clockwise, so $[f], [g] \in \pi_1(X, b_0)$. Computing φ on a product uses only the definition: to lift $f * g$ from e_0, lift f from e_0, then lift g from wherever that lift ended. Let E be the following covering space:

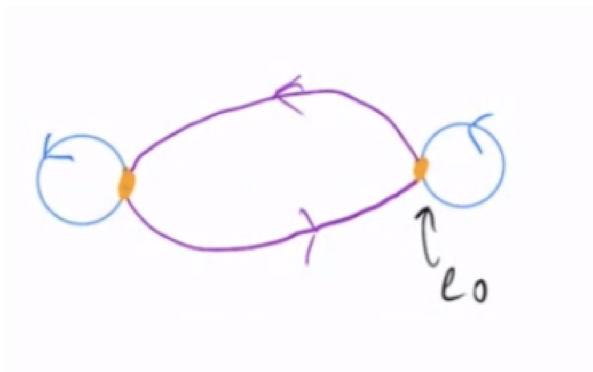

Figure 6.8: Lifting Space

and take e_0 as in the image. Then

$$\varphi([f]) = e_1 \tag{6.1}$$
$$\varphi([g]) = e_0 \tag{6.2}$$
$$\varphi([f] * [g]) = e_1 \tag{6.3}$$
$$\varphi([g] * [f]) = e_1 \tag{6.4}$$

This separates $[f]$ from $[g]$ but says nothing about the order of the product, since both orders give e_1. That is a limitation of this particular cover, not of the method: φ is surjective but need not be injective, so it reports only part of the structure. A larger cover reports more.

3. Same X and same b_0, with a different covering space:

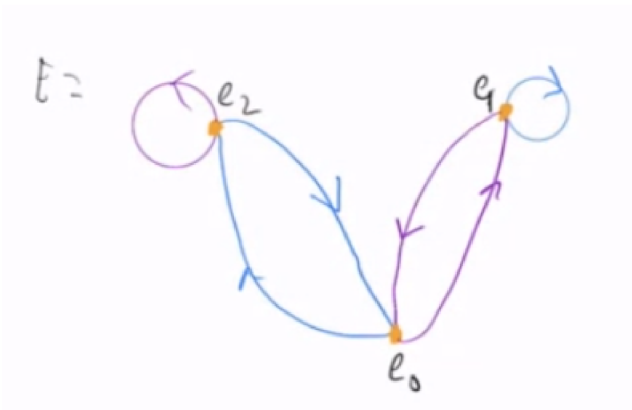

Figure 6.9: Different Lifting Space

Now,

$$\varphi([f] * [g]) = e_1 \tag{6.5}$$
$$\varphi([g] * [f]) = e_2 \tag{6.6}$$

The two products land in different points of the fibre, so $[f] * [g] \neq [g] * [f]$ and $\pi_1(X, b_0)$ is not abelian. This is the concrete instance of the non-commutativity announced with fig. 6.3.

The circle can now be settled outright.

Theorem 6.2.20: Fundamental Groups Of The Circle

The fundamental group of S^1 is isomorphic to $(\mathbb{Z}, +)$.

***Proof*:**
Take the covering map $p : \mathbb{R} \to S^1$ of example 6.2.5, given by $p(t) = (\cos(2\pi t), \sin(2\pi t))$. It satisfies

$$p(r) = p(s) \iff r - s \in \mathbb{Z}$$

since cos and sin agree at two arguments precisely when those arguments differ by a multiple of 2π. Let $b_0 = (1, 0)$, so $p^{-1}(b_0) = \mathbb{Z}$, and take $e_0 = 0$. The lifting correspondence is then a map

$$\varphi : \pi_1(S^1, b_0) \to \mathbb{Z}$$

The space $\mathbb{R}$ is path-connected and, being convex, simply connected by example 6.1.23, so φ is a bijection by proposition 6.2.18. It remains to show that φ is a homomorphism, that is, that $\varphi([f] * [g]) = \varphi([f]) + \varphi([g])$.

Let f, g be loops at b_0 with lifts $\tilde{f}, \tilde{g}$ starting at 0, and write $k = \tilde{f}(1) \in \mathbb{Z}$, so $\varphi([f]) = k$. Define $\tilde{g}_2(t) = \tilde{g}(t) + k$, a continuous path in $\mathbb{R}$. Because k is an integer,

$$p(\tilde{g}_2(t)) = p(\tilde{g}(t) + k) = p(\tilde{g}(t)) = g(t)$$

so $\tilde{g}_2$ is also a lift of g, translated so as to begin where $\tilde{f}$ ends: $\tilde{g}_2(0) = k = \tilde{f}(1)$. The two paths therefore concatenate, and $\tilde{f} * \tilde{g}_2$ is defined. It is a lift of $f * g$, since applying p to it gives f on $[0, \frac{1}{2}]$ and g on $[\frac{1}{2}, 1]$, which is $f * g$; and it starts at 0. By uniqueness in lemma 6.2.11 it is *the* lift used to compute $\varphi([f * g])$. Hence

$$\begin{aligned}\varphi([f] * [g]) = \varphi([f * g]) &= (\tilde{f} * \tilde{g}_2)(1) \\ &= \tilde{g}_2(1) \\ &= \tilde{g}(1) + k = \varphi([g]) + \varphi([f])\end{aligned}$$

so φ is a bijective homomorphism onto $\mathbb{Z}$, and $\pi_1(S^1, b_0) \cong (\mathbb{Z}, +)$ as claimed. The translation step is what the figure records:

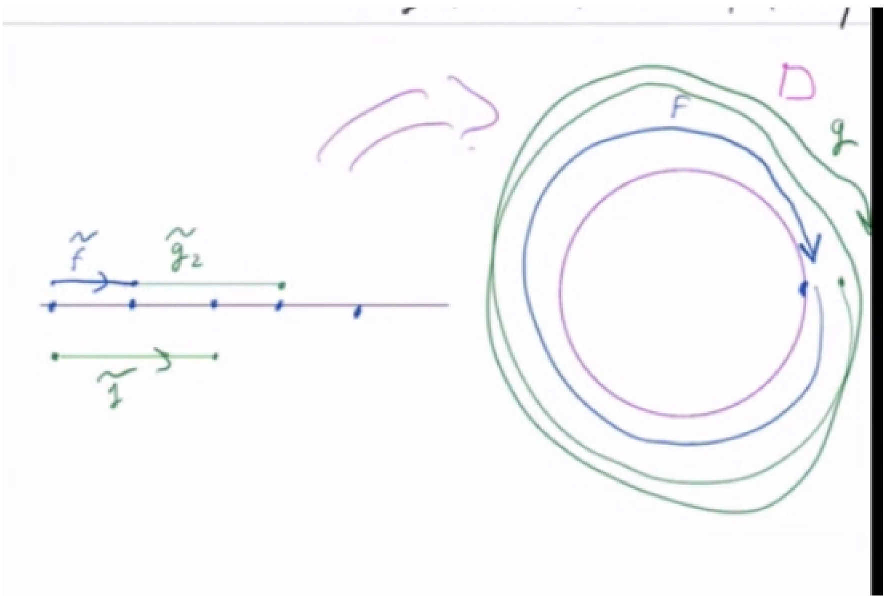

Figure 6.10: Translating the lift of g so that it begins where the lift of f ends.

Since $\mathbb{Z}$ is abelian, so is $\pi_1(S^1)$, which matches the picture: two loops on a circle can always be slid past one another, because there is nothing for them to catch on. On the figure eight there is, and the previous example confirms that the group there is not abelian. The homomorphism φ is what turns that geometric intuition into a computation, and the general version of the correspondence, in which the fibre is identified with a coset space rather than with a group, appears in section 6.3.

6.2.4 Further computations

The circle is the seed from which most elementary computations grow. Collected here is what the machinery of this chapter yields, together with what it does not.

Space	**Fundamental group**	**Status here**
$\mathbb{R}^n$, and any convex set	trivial	proved, example 6.1.23
S^1	$\mathbb{Z}$	proved, theorem 6.2.20
Figure eight	non-abelian	proved, example 6.2.19
Torus $S^1 \times S^1$	$\mathbb{Z} \times \mathbb{Z}$	stated
Figure eight	free on two generators	stated
S^n for $n \geq 2$	trivial	stated

The three stated entries are true and are not proved here. The torus follows from the circle together with the fact that π_1 of a product is the product of the π_1's, a short argument from the universal property of the product topology which is left aside only because nothing later depends on it. The finer description of the figure eight needs the Seifert-van Kampen theorem, and the spheres are cleanest with the same machinery; both belong to algebraic topology proper rather than to this book, and neither is used anywhere below. The pattern is worth naming: covering spaces compute π_1 well whenever a good cover can be seen, and systematic tools take over when one cannot.

There is also a genuine surprise at the other end of the size scale. A topological space with four points can have fundamental group $\mathbb{Z}$: take $X = \{1, 2, 3, 4\}$ with open sets generated by $\{1\}$, $\{2\}$, $\{1, 2, 3\}$ and $\{1, 2, 4\}$, the *pseudocircle*. It is not Hausdorff and not homeomorphic to S^1, yet $\pi_1(X) \cong \mathbb{Z}$. The proof identifies a covering space of X by an infinite "fence" of alternating open and closed points and shows that fence is simply connected, which needs more about finite spaces than is developed here; the statement is recorded as a pointer, not used.

Exercise 6.2.1

1. Show that a covering map is a local homeomorphism and an open map, and that the converse fails, citing example 6.2.6.
2. Show that the fibres of a covering map over a connected base all have the same cardinality.
3. Compute the lifting correspondence φ for the covering $p\colon S^1 \to S^1$, $z \mapsto z^n$, and compute $p_*\pi_1(S^1, b_0)$ as a subgroup of $\pi_1(S^1, b_0) \cong \mathbb{Z}$.
4. Show that φ is surjective when the total space is path-connected and injective when it is simply connected.
5. Deduce from theorem 6.2.20 that S^1 is not simply connected and that $\mathbb{R}$ and S^1 are not homeomorphic.
6. Show that $\pi_1(X, x_0)$ depends on the basepoint when X is not path-connected, so that the hypothesis of corollary 6.1.25 cannot be dropped.

6.3 The Lifting Criterion

The lifting results so far handle two domains and no others: lemma 6.2.11 lifts a path, lemma 6.2.12 lifts a square, and both lift unconditionally, with no hypothesis on the covering and none on the map beyond continuity. What makes those cases free is that I and $I \times I$ carry no loops that could obstruct anything. An arbitrary domain does. A continuous map $f: Y \to B$ pushes the loops of Y into B, and a lift of f would have to push them into E compatibly, which the loops of B need not permit. This section shows that this is the *only* obstruction, and that it is measured by a containment between two subgroups of $\pi_1(B, b_0)$.

Throughout, $p: E \to B$ is a covering map with $b_0 \in B$ and $e_0 \in p^{-1}(b_0)$, and the domain of the map to be lifted is written Y with basepoint y_0. For a path α, the reverse path is written α^{-1}, so $\alpha^{-1}(t) = \alpha(1-t)$; theorem 6.1.16 writes it $\tilde{\alpha}$, but the tilde is used throughout for lifts and the bar for closure, so neither is available.

Over an evenly covered neighbourhood a lift must select one slice out of many, and there is nothing in definition 6.2.10 to say that the selection made over one part of the domain constrains the selection made over another. Two lifts of the same map could then agree on part of Y and separate elsewhere. Over a connected domain neither happens, and a lift is pinned down by its value at a single point. Every construction in this section defines a lift by making one arbitrary choice at the basepoint, so this is the fact that has to come first.

Theorem 6.3.1: Uniqueness of Lifts

Let $p: E \to B$ be a covering map, let Y be connected, and let $f: Y \to B$ be continuous. If $\tilde{f}_1, \tilde{f}_2: Y \to E$ are lifts of f in the sense of definition 6.2.10 which agree at one point of Y, then $\tilde{f}_1 = \tilde{f}_2$.

***Proof*:**
Let

$$A = \{y \in Y \mid \tilde{f}_1(y) = \tilde{f}_2(y)\}$$

and let $y_1 \in A$ be the point at which the two lifts are assumed to agree. The argument shows that A and its complement are both open; connectedness then forces $A = Y$.

Fix $y \in Y$ and set up the apparatus used in both steps. Since p is a covering map, the point $f(y)$ has a neighbourhood U evenly covered by p (definition 6.2.2), and $p^{-1}(U)$ is the union of disjoint open slices $\{V_\alpha\}$, each carried homeomorphically onto U by p (definition 6.2.1). Both $\tilde{f}_1(y)$ and $\tilde{f}_2(y)$ lie in $p^{-1}(U)$, since $p(\tilde{f}_i(y)) = f(y) \in U$, so each lies in exactly one slice: say $\tilde{f}_1(y) \in V_1$ and $\tilde{f}_2(y) \in V_2$. Because the lifts are continuous,

$$N = \tilde{f}_1^{-1}(V_1) \cap \tilde{f}_2^{-1}(V_2)$$

is an open neighbourhood of y.

Step 1: A is open. Suppose $y \in A$. Then $\tilde{f}_1(y) = \tilde{f}_2(y)$, and this single point lies in exactly one slice, so $V_1 = V_2$; call it V. Let $y' \in N$. Then $\tilde{f}_1(y')$ and $\tilde{f}_2(y')$ both lie in V, and

$$p(\tilde{f}_1(y')) = f(y') = p(\tilde{f}_2(y'))$$

Since p restricted to V is a homeomorphism onto U, it is injective on V, so $\tilde{f}_1(y') = \tilde{f}_2(y')$. Hence $N \subseteq A$.

Step 2: $Y \setminus A$ is open. Suppose $y \notin A$. Were $V_1 = V_2$, the injectivity of p on that common slice together with $p(\tilde{f}_1(y)) = f(y) = p(\tilde{f}_2(y))$ would give $\tilde{f}_1(y) = \tilde{f}_2(y)$, contrary to $y \notin A$. So $V_1 \neq V_2$, and distinct slices are disjoint. For $y' \in N$ we have $\tilde{f}_1(y') \in V_1$ and $\tilde{f}_2(y') \in V_2$, two disjoint sets, so $\tilde{f}_1(y') \neq \tilde{f}_2(y')$. Hence $N \subseteq Y \setminus A$.

The set A is therefore open and closed in Y, and it is nonempty because $y_1 \in A$. A connected space has no such subset other than the whole space (section 3.1), so $A = Y$ and the two lifts agree everywhere, as we sought to show.

The hypothesis on Y is connectedness alone: no separation axiom, no local hypothesis, and nothing about E or about the map f. What the theorem denies is a lift that changes its mind, and the proof locates the reason. Over an evenly covered U the slices are open and disjoint, so agreement and disagreement are both stable under small movement of the point y, and a connected domain cannot accommodate a set that is stable in both directions.

The covering $p\colon \mathbb{R} \to S^1$ used to compute $\pi_1(S^1)$ in theorem 6.2.20 makes the content concrete. Suppose Y is connected with a point y_0, and let $\tilde{f}\colon Y \to \mathbb{R}$ be a lift of a map $f\colon Y \to S^1$. For each $n \in \mathbb{Z}$ the translate $\tilde{f} + n$ is again a lift, because $p(x+n) = p(x)$ for integral n. These are the only ones. Given any lift $\tilde{g}$, the two values $\tilde{g}(y_0)$ and $\tilde{f}(y_0)$ lie in the same fibre $p^{-1}(f(y_0))$, and p identifies two reals exactly when they differ by an integer, so $n = \tilde{g}(y_0) - \tilde{f}(y_0)$ lies in $\mathbb{Z}$; the lifts $\tilde{g}$ and $\tilde{f} + n$ agree at y_0, and theorem 6.3.1 identifies them. The set of lifts of f, when it is nonempty, is thus a copy of $\mathbb{Z}$, indexed by the value taken at a single point of Y.

Uniqueness settles what a lift is once it exists. Whether one exists is a separate question, and one direction of the answer is immediate. If $\tilde{f}$ lifts f, then f factors as $p \circ \tilde{f}$, and applying the induced-homomorphism construction of definition 6.1.28 to that factorization confines the image of f_* inside the image of p_*. So the containment

$$f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(E, e_0))$$

is a necessary condition, obtained from nothing but functoriality. The theorem below is the statement that it is also sufficient, which is not formal at all: it has to produce a map $Y \to E$ out of an inclusion of subgroups.

Theorem 6.3.2: The Lifting Criterion

Let $p\colon E \to B$ be a covering map with $p(e_0) = b_0$. Let Y be connected and locally path-connected, let $y_0 \in Y$, and let $f\colon Y \to B$ be continuous with $f(y_0) = b_0$. Then a lift $\tilde{f}\colon Y \to E$ of f with $\tilde{f}(y_0) = e_0$ exists if and only if

$$f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(E, e_0))$$

where f_* and p_* are the induced homomorphisms of definition 6.1.28. When it exists, the lift is unique.

***Proof*:**
Two observations about lifted paths are used repeatedly. Both follow from the uniqueness half of lemma 6.2.11 together with the identities

$$p \circ (\sigma_1 * \sigma_2) = (p \circ \sigma_1) * (p \circ \sigma_2), \qquad p \circ \sigma^{-1} = (p \circ \sigma)^{-1}$$

which are immediate from the definition of concatenation (definition 6.1.13) and of the reverse path.

1. Let γ_1, γ_2 be paths in B with $\gamma_1(1) = \gamma_2(0)$, let $\tilde{\gamma}_1$ be the lift of γ_1 beginning at e, and let $\tilde{\gamma}_2$ be the lift of γ_2 beginning at $\tilde{\gamma}_1(1)$. Then $\tilde{\gamma}_1 * \tilde{\gamma}_2$ is the lift of $\gamma_1 * \gamma_2$ beginning at e.
2. Let γ be a path in B and $\tilde{\gamma}$ its lift beginning at e. Then $\tilde{\gamma}^{-1}$ is the lift of γ^{-1} beginning at $\tilde{\gamma}(1)$.

In each case the exhibited path is continuous, projects to the asserted path under p, and starts at the asserted point, so uniqueness identifies it with the lift.

Necessity. Suppose $\tilde{f}$ is a lift of f with $\tilde{f}(y_0) = e_0$. Then $\tilde{f}$ carries y_0 to e_0 and p carries e_0 to b_0, so both are maps of pointed spaces and the functoriality clause of proposition 6.1.30 applies to the factorization $f = p \circ \tilde{f}$, giving $f_* = p_* \circ \tilde{f}_*$. Hence for every $[\alpha] \in \pi_1(Y, y_0)$,

$$f_*([\alpha]) = p_*\big(\tilde{f}_*([\alpha])\big) \in p_*(\pi_1(E, e_0))$$

which is the asserted containment. Neither hypothesis on Y was used here.

Sufficiency, Step 1: Y is path-connected. Since Y is locally path-connected and is an open subset of itself, theorem 3.1.40 makes every path component of Y open. The complement of one path component is the union of the others, hence open, so every path component is also closed. Since Y is connected and contains y_0, there is exactly one, and Y is path-connected in the sense of definition 3.1.25.

Step 2: the pointwise definition. Let $y \in Y$ and choose a path α in Y from y_0 to y, available by Step 1. Then $f \circ \alpha$ is a path in B from b_0 to $f(y)$, and lemma 6.2.11 provides its unique lift $\widetilde{f \circ \alpha}$ beginning at e_0. Define

$$\tilde{f}(y) = \widetilde{f \circ \alpha}(1) \tag{6.7}$$

The formula is forced rather than chosen. If $g \colon Y \to E$ is any lift of f with $g(y_0) = e_0$, then $g \circ \alpha$ is continuous, satisfies $p \circ (g \circ \alpha) = f \circ \alpha$, and begins at e_0, so it is *the* lift of $f \circ \alpha$ beginning at e_0, and $g(y) = (g \circ \alpha)(1) = \widetilde{f \circ \alpha}(1)$. So (6.7) is the only candidate; what remains is to see that it defines anything at all, and that what it defines is continuous.

Step 3: independence of the path, where the subgroup hypothesis enters. Let α and β be two paths in Y from y_0 to y, and write $h = f \circ \alpha$ and $k = f \circ \beta$, both paths in B from b_0 to $f(y)$. The concatenation $\gamma = \alpha * \beta^{-1}$ is a loop in Y based at y_0, and

$$f \circ \gamma = h * k^{-1}$$

is a loop in B based at b_0 whose path-homotopy class is $f_*([\gamma])$. By hypothesis $f_*([\gamma])$ lies in $p_*(\pi_1(E, e_0))$, so there is a loop δ in E based at e_0 with

$$p \circ \delta \simeq_p h * k^{-1}$$

Both sides are loops based at b_0, so lemma 6.2.14 says that their lifts beginning at e_0 have the same endpoint. The lift of $p \circ \delta$ beginning at e_0 is δ itself, by the uniqueness half of lemma 6.2.11, and $\delta(1) = e_0$. Hence the lift σ of $h * k^{-1}$ beginning at e_0 satisfies $\sigma(1) = e_0$.

Now unwind σ. Let $\tilde{h}$ be the lift of h beginning at e_0 and μ the lift of k^{-1} beginning at $\tilde{h}(1)$. By observation (1), $\tilde{h} * \mu$ is the lift of $h * k^{-1}$ beginning at e_0, so $\sigma = \tilde{h} * \mu$ and therefore $\mu(1) = \sigma(1) = e_0$. By observation (2) applied to μ, the reverse μ^{-1} is the lift of $(k^{-1})^{-1} = k$ beginning at $\mu(1) = e_0$, so $\mu^{-1} = \tilde{k}$, the lift of k beginning at e_0. Reading off endpoints,

$$\tilde{k}(1) = \mu^{-1}(1) = \mu(0) = \tilde{h}(1)$$

so α and β assign the same value in (6.7), and $\tilde{f}$ is a well-defined map $Y \to E$.

Two properties are now immediate. First, $p(\tilde{f}(y)) = p(\widetilde{f \circ \alpha}(1)) = (f \circ \alpha)(1) = f(y)$, so $\tilde{f}$ is a set-theoretic lift of f. Second, taking α to be the constant path e_{y_0} of definition 6.1.12 makes $f \circ \alpha$ the constant path e_{b_0}, whose lift beginning at e_0 is the constant path e_{e_0}, so $\tilde{f}(y_0) = e_0$.

Step 4: continuity, where local path-connectedness enters. Fix $y \in Y$. Let U be a neighbourhood of $f(y)$ evenly covered by p, let V be the slice of $p^{-1}(U)$ containing $\tilde{f}(y)$, and let $q = (p|_V)^{-1} : U \to V$, a homeomorphism by definition 6.2.1. The set $f^{-1}(U)$ is open and contains y; let N be the path component of $f^{-1}(U)$ containing y. By theorem 3.1.40, N is open in Y. The claim is that

$$\tilde{f}|_N = q \circ f|_N$$

Let $y' \in N$ and choose a path δ' from y to y' lying in N, which exists because N is a path component. Fix a path α in Y from y_0 to y. Then $\alpha * \delta'$ runs from y_0 to y', so by Step 3 it computes $\tilde{f}(y')$: the value is the endpoint of the lift of

$$f \circ (\alpha * \delta') = (f \circ \alpha) * (f \circ \delta')$$

beginning at e_0. By observation (1) that lift is $\widetilde{f \circ \alpha} * \nu$, where ν is the lift of $f \circ \delta'$ beginning at $\widetilde{f \circ \alpha}(1) = \tilde{f}(y)$. The path δ' lies in $N \subseteq f^{-1}(U)$, so $f \circ \delta'$ lies in U and the composite $q \circ f \circ \delta'$ is defined; it is continuous, satisfies $p \circ (q \circ f \circ \delta') = f \circ \delta'$, and begins at $q(f(y)) = \tilde{f}(y)$, the last equality because $\tilde{f}(y)$ lies in V and projects to $f(y)$. Uniqueness in lemma 6.2.11 gives $\nu = q \circ f \circ \delta'$, whence

$$\tilde{f}(y') = \nu(1) = q(f(\delta'(1))) = q(f(y'))$$

which is the claim. So $\tilde{f}$ agrees with the continuous map $q \circ f$ on the open set N, and is therefore continuous at y. As y was arbitrary, $\tilde{f}$ is continuous, and it is a lift of f carrying y_0 to e_0.

Uniqueness is theorem 6.3.1, applicable because Y is connected and any two such lifts agree at y_0, completing the proof.

The criterion converts a question about maps into a question about subgroups. The covering p contributes one subgroup of $\pi_1(B, b_0)$, namely $p_*(\pi_1(E, e_0))$, and it is the same subgroup no matter which f is being tested; the map f contributes another, $f_*(\pi_1(Y, y_0))$; and f lifts exactly when the second sits inside the first. The subgroup $p_*(\pi_1(E, e_0))$ records which loops of B open up into paths when traced in E, since a loop lies in it precisely when its lift beginning at e_0 closes up again. A map lifts when everything it can wrap around B is already unwrapped upstairs.

A computation with the double covering of the circle shows the arithmetic. In complex notation, identifying $(x, y) \in \mathbb{R}^2$ with $x + iy$ so that $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$, the covering of theorem 6.2.20 is $x \mapsto e^{2\pi ix}$. The map $p_2 \colon S^1 \to S^1$, $z \mapsto z^2$, is also a covering map: for $w \in S^1$ the set $U = S^1 \setminus \{w\}$ is evenly covered, because $p_2^{-1}(U)$ is S^1 with the two square roots of w deleted, a disjoint union of two open half-circles, each of which p_2 carries homeomorphically onto U by doubling the angle. Every point of S^1 lies in such a U, since w may be any pointo ther than the one under consideration. Take basepoints $b_0 = e_0 = 1$. Under the isomorphism $\pi_1(S^1, 1) \cong \mathbb{Z}$ of theorem 6.2.20, the integer n corresponds to the class of the loop $\omega_n(t) = e^{2\pi int}$, since the lift of ω_n beginning at 0 is $t \mapsto nt$. The generator $[\omega_1]$ satisfies $p_2 \circ \omega_1 = \omega_2$, so p_{2*} is multiplication by 2 and

$$p_{2*}(\pi_1(S^1, 1)) = 2\mathbb{Z}$$

The domain S^1 is connected and locally path-connected, every point having arbitrarily small arc neighbourhoods, so the criterion applies to any map out of it. The map $f(z) = z^4$ has f_* equal to multiplication by 4, so $f_*(\pi_1(S^1, 1)) = 4\mathbb{Z} \subseteq 2\mathbb{Z}$ and f lifts; the lift is $z \mapsto z^2$, as one checks directly. The map $g(z) = z^3$ has $g_*(\pi_1(S^1, 1)) = 3\mathbb{Z}$, which is not contained in $2\mathbb{Z}$ because 3 is odd, so g admits no lift carrying 1 to 1. It admits none at all: if $\tilde{g}$ were any lift, then $-\tilde{g}$ would be one too, since $(-\tilde{g})^2 = \tilde{g}^2 = g$, and $\tilde{g}(1)$ lies in $p_2^{-1}(1) = \{1, -1\}$, so one of $\tilde{g}$ and $-\tilde{g}$ carries 1 to 1, which the criterion has just forbidden. Odd powers do not factor through the double cover, and the criterion sees exactly that.

Local path-connectedness of Y deserves a second look, because it is used in two steps and for two different reasons, and the second is not the reason one might expect. In Step 1 it upgrades the stated hypothesis of connectedness to path-connectedness, and that use is necessary but unremarkable: without paths from y_0 there is nothing for Steps 2 and 3 to work with, and hypothesising a path-connected Y outright would dispense with this appeal. Once the paths are in hand, Steps 2 and 3 define $\tilde{f}$ at every point of Y and never appeal to it again: path-connectedness of Y supplies the paths, the subgroup hypothesis makes the answer independent of which path is chosen, and the resulting function satisfies $p \circ \tilde{f} = f$ on the nose. What fails without local path-connectedness is continuity. The formula $\tilde{f} = q \circ f$ of Step 4 is verified on the path component N of $f^{-1}(U)$ containing y, and that is all the argument can reach, since the only information about $\tilde{f}(y')$ available is the one carried along a path from y to y'. If N is not a neighbourhood of y, the identity $\tilde{f} = q \circ f$ holds on a set with empty interior and says nothing about the behaviour of $\tilde{f}$ near y. Local path-connectedness is precisely the hypothesis making N open, by theorem 3.1.40, and it is not automatic: the closed topologist's sine curve is connected and has two path components, the oscillating curve and the limiting vertical segment, and the second of these is not open, since every neighbourhood of one of its points meets the curve at points no path from it can reach.

The cheapest way to satisfy the subgroup condition is to have no loops to test. A simply connected domain contributes the trivial subgroup, which is contained in everything, and the criterion then imposes no condition whatever on the map.

Corollary 6.3.3: Lifting from a Simply Connected Space

Let $p\colon E \to B$ be a covering map with $p(e_0) = b_0$. Let Y be simply connected (definition 6.1.26) and locally path-connected, and let $y_0 \in Y$. Then every continuous $f\colon Y \to B$ with $f(y_0) = b_0$ has a unique lift $\tilde{f}\colon Y \to E$ with $\tilde{f}(y_0) = e_0$.

***Proof*:**
A simply connected space is path-connected by definition, hence connected by lemma 3.1.28, so Y satisfies the hypotheses of theorem 6.3.2. Its fundamental group $\pi_1(Y, y_0)$ is trivial, and f_* is a group homomorphism by the homomorphism clause of proposition 6.1.30, so $f_*(\pi_1(Y, y_0))$ is the trivial subgroup of $\pi_1(B, b_0)$, which is contained in $p_*(\pi_1(E, e_0))$. The criterion supplies the lift and its uniqueness, completing the proof.

Convex subsets of $\mathbb{R}^n$ are the standard supply of such domains: they are path-connected, every loop contracts along the straight-line homotopy so π_1 is trivial, and each point has arbitrarily small convex, hence path-connected, neighbourhoods. Taking $Y = I$ and $Y = I \times I$ recovers the existence and uniqueness assertions of lemma 6.2.11 and lemma 6.2.12, though not new proofs of them, and not the remaining clause of lemma 6.2.12 asserting that the lift of a path homotopy is again a path homotopy: the construction of the lift in theorem 6.3.2 ran through both lemmas, and the corollary delivers existence and uniqueness and nothing else. The gain is the domains those lemmas do not reach: every continuous map into B from a disc, from a cube, or from $\mathbb{R}^n$ itself lifts, whatever B and whatever the covering.

The corollary raises the question of what happens when the domain is not simply connected. The smallest failure is the identity map of the circle, and working it out confirms that the criterion detects a real obstruction rather than an artifact of the hypotheses.

Example 6.3.4: Failure of the Criterion for the Identity Map of the Circle

Let $p\colon \mathbb{R} \to S^1$ be the covering map $p(x) = (\cos 2\pi x, \sin 2\pi x)$ used to compute $\pi_1(S^1)$ in theorem 6.2.20, let $b_0 = (1, 0)$, and let $e_0 = 0 \in p^{-1}(b_0) = \mathbb{Z}$. Take $Y = S^1$ with $y_0 = b_0$, and let $f = \mathrm{id}_{S^1}$ be the identity map.

Every hypothesis of theorem 6.3.2 holds. The domain S^1 is connected and locally path-connected, since each point has arbitrarily small arc neighbourhoods and an arc is path-connected, and $f(y_0) = b_0$.

The two subgroups. By theorem 6.2.20, $\pi_1(S^1, b_0) \cong (\mathbb{Z}, +)$. The identity map induces the identity homomorphism by the identity clause of proposition 6.1.30, so

$$f_*(\pi_1(S^1, b_0)) = \pi_1(S^1, b_0) \cong \mathbb{Z}$$

On the other side, $\pi_1(\mathbb{R}, 0)$ is trivial: if γ is a loop in $\mathbb{R}$ based at 0, then $H(s,t) = (1-t)\gamma(s)$ is continuous, satisfies $H(s, 0) = \gamma(s)$ and $H(s, 1) = 0$, and fixes the endpoints since $H(0, t) = H(1, t) = 0$, so γ is path-homotopic to the constant path at 0. A homomorphism out of the trivial group has trivial image, so

$$p_*(\pi_1(\mathbb{R}, 0)) = \{[e_{b_0}]\}$$

The subgroup $f_*(\pi_1(S^1, b_0))$ is therefore infinite while $p_*(\pi_1(\mathbb{R}, 0))$ has one element, and the containment demanded by theorem 6.3.2 fails. Every nontrivial class of $\pi_1(S^1, b_0)$ is an obstruction, the class of the loop $\omega(t) = (\cos 2\pi t, \sin 2\pi t)$ among them.

No lift exists. The criterion already forbids one, but the obstruction can be exhibited directly, which also shows that the failure does not depend on the choice of e_0. Suppose $\tilde{f}\colon S^1 \to \mathbb{R}$ were continuous with $p \circ \tilde{f} = \mathrm{id}_{S^1}$, and let $n = \tilde{f}(b_0)$, necessarily an element of $p^{-1}(b_0) = \mathbb{Z}$. The composite $\tilde{f} \circ \omega$ is a path in $\mathbb{R}$ with

$$p \circ (\tilde{f} \circ \omega) = (p \circ \tilde{f}) \circ \omega = \omega$$

and initial point $\tilde{f}(\omega(0)) = \tilde{f}(b_0) = n$, so it is the lift of ω beginning at n. That lift is $t \mapsto n + t$, since this path is continuous, begins at n, and satisfies $p(n + t) = \omega(t)$; uniqueness in lemma 6.2.11 leaves no other. Evaluating at $t = 1$ gives $\tilde{f}(\omega(1)) = n + 1$. But $\omega(1) = \omega(0) = b_0$, so $\tilde{f}(b_0) = n$ and $\tilde{f}(b_0) = n + 1$ at once, which is absurd. No lift of the identity exists, for any choice of basepoint in the fibre.

The obstruction is the class $[\omega]$ and nothing else. The covering $p\colon \mathbb{R} \to S^1$ unwinds that class, so $p_*(\pi_1(\mathbb{R}, 0))$ is the trivial subgroup, as small as a subgroup can be, and for a connected, locally path-connected domain theorem 6.3.2 then reads: a map into S^1 lifts through p exactly when it kills every loop of its domain.

Exercise 6.3.1

1. Show that theorem 6.3.1 fails without connectedness of the domain, and exhibit two distinct lifts agreeing at a point.
2. Verify theorem 6.3.2 for $Y = S^1$, $B = S^1$, p the n-fold cover with $n \geq 2$, and f the identity.
3. Show that a map from a simply-connected, locally path-connected space always lifts.
4. Show that the lifting criterion depends on the image subgroup, by exhibiting f with $f_*\pi_1(Y) \subseteq p_*\pi_1(E)$ that lifts, and f violating the containment that does not.
5. Show that a lift of a map from a connected domain is determined by its value at one point, and that when Y is simply connected and locally path-connected the number of lifts equals the number of points of $p^{-1}(b_0)$.

6.4 The Universal Cover and the Classification of Coverings

The lifting criterion (theorem 6.3.2) reads off, from one subgroup of $\pi_1(B, b_0)$, exactly which maps into B lift to a given covering of it. The subgroup turns out to record the covering itself: it determines the covering up to isomorphism, and every subgroup of $\pi_1(B, b_0)$ arises from one. Both halves of that statement rest on a single covering, the one whose total space is simply connected, and the first task is to find the condition on B under which such a covering exists.

That condition can be read off before anything is built. Suppose $p\colon E \to B$ is a covering map whose total space is simply connected, let $U \subseteq B$ be an open set evenly covered by p (definition 6.2.1), let $x \in U$, and fix a slice V of $p^{-1}(U)$ together with the point $\tilde{x} \in V$ satisfying $p(\tilde{x}) = x$. Given a loop

f in U based at x, the composite

$$\tilde{f} = (p|_V)^{-1} \circ f$$

is a loop in E based at $\tilde{x}$ with $p \circ \tilde{f} = f$, since $p|_V$ is a homeomorphism of V onto U. Because E is simply connected (definition 6.1.26) there is a path homotopy F from $\tilde{f}$ to the constant path $e_{\tilde{x}}$ (definition 6.1.12), and then $p \circ F$ is a path homotopy in B from f to e_x. So the class of f dies in $\pi_1(B, x)$: the homomorphism $\iota_* \colon \pi_1(U, x) \to \pi_1(B, x)$ induced (definition 6.1.28) by the inclusion $\iota \colon U \to B$ is trivial.

The loop f was contracted in B, not in U, and nothing in the argument contracts it inside U. What a simply connected covering forces is exactly this weaker statement, and it is the weaker statement that gets a name.

Definition 6.4.1: Semilocally Simply Connected

A topological space B is *semilocally simply connected* if every $b \in B$ has an open neighbourhood U such that the homomorphism

$$\iota_* \colon \pi_1(U, b) \to \pi_1(B, b)$$

induced (definition 6.1.28) by the inclusion $\iota \colon U \to B$ is trivial, that is, carries every element of $\pi_1(U, b)$ to the identity of $\pi_1(B, b)$.

The prefix records how much weaker this is than the condition it resembles. A space is *locally simply connected* when every point has a neighbourhood base of simply connected open sets, so that loops in the small sets die inside those sets; semilocal simple connectivity asks only that they die by the time they reach B, and asks it of one neighbourhood per point rather than of a base. The argument above produces the semilocal condition and no more, so it is the semilocal condition that a theorem asserting existence of a simply connected covering can hope to assume. Note also that the definition asks nothing of a simply connected B: there the target group is trivial and any U whatsoever will do.

The hypothesis is not vacuous. A space can carry, in every neighbourhood of a point, loops that survive in the whole space, and the standard example is an infinite shrinking family of circles sharing a single point.

Example 6.4.2: The Hawaiian Earring

For $n \geq 1$ let

$$C_n = \{(x, y) \in \mathbb{R}^2 \mid (x - 1/n)^2 + y^2 = 1/n^2\}$$

be the circle of radius $1/n$ centred at $(1/n, 0)$, and let $\mathcal{H} = \bigcup_{n \geq 1} C_n$ carry the subspace topology from $\mathbb{R}^2$. Every C_n passes through the origin $0 = (0, 0)$, and two of them meet nowhere else: subtracting the equations $x^2 + y^2 = 2x/n$ and $x^2 + y^2 = 2x/m$ for $m \neq n$ gives $x = 0$ and hence $y = 0$.

1. *$\mathcal{H}$ is not semilocally simply connected at* 0. Fix n and let $r_n \colon \mathcal{H} \to C_n$ be the identity on C_n and the constant 0 on every other circle. To see that r_n is continuous, note first that C_n is closed in $\mathbb{R}^2$, hence in $\mathcal{H}$, and that $\bigcup_{m \neq n} C_m$ is closed in $\mathbb{R}^2$ as well: a point y outside it is nonzero, say $\|y\| = d > 0$, and every point of C_m lies within distance $2/m$ of the origin, hence at distance at least $d - 2/m$ from y; so only the finitely many indices m with $2/m \geq d/2$ can put a point of C_m within distance $d/2$ of y, and a small enough ball about y misses every C_m with $m \neq n$. Since the circles meet only at 0, the complement in $\mathcal{H}$ of $\bigcup_{m \neq n} C_m$ is $C_n \setminus \{0\}$,

which is therefore open in $\mathcal{H}$. Now let $W \subseteq C_n$ be open in C_n. If $0 \notin W$ then W is open in $C_n \setminus \{0\}$ and hence in $\mathcal{H}$, and $r_n^{-1}(W) = W$. If $0 \in W$ then $r_n^{-1}(W) = W \cup \bigcup_{m \neq n} C_m$, whose complement in $\mathcal{H}$ is $C_n \setminus W$, a closed subset of the closed set C_n. Either way the preimage is open.

The inclusion $j\colon C_n \to \mathcal{H}$ satisfies $r_n \circ j = \mathrm{id}_{C_n}$, so functoriality of the induced homomorphisms (proposition 6.1.30) gives $(r_n)_* \circ j_* = \mathrm{id}$ on $\pi_1(C_n, 0)$, and j_* is injective. The affine homeomorphism $y \mapsto ny - (1,0)$ of $\mathbb{R}^2$ carries C_n onto the unit circle S^1, so $\pi_1(C_n, 0) \cong \pi_1(S^1, (-1,0)) \cong \mathbb{Z}$ by proposition 6.1.30, corollary 6.1.25, and theorem 6.2.20. Let ℓ_n be a loop at 0 generating $\pi_1(C_n, 0)$.

Let U be any open neighbourhood of 0 in $\mathcal{H}$. It contains every point of $\mathcal{H}$ within some distance $\delta > 0$ of the origin, hence contains C_n whenever $2/n < \delta$, that circle lying within distance $2/n$ of the origin. For such an n the inclusion $C_n \to \mathcal{H}$ factors through U, so the image of $\pi_1(U, 0) \to \pi_1(\mathcal{H}, 0)$ contains $j_*[\ell_n] \neq 1$. No neighbourhood of 0 has trivial induced homomorphism.

2. *A simply connected space that is not locally simply connected.* Let $a = (0,0,1) \in \mathbb{R}^3$, identify $\mathbb{R}^2$ with $\mathbb{R}^2 \times \{0\}$, and let

$$K = \{(1-t)x + ta \mid x \in \mathcal{H},\ t \in I\}$$

be the union of the segments joining a to the points of the earring. Every point of K lies on such a segment, so the segment from a to any point of K lies in K: the set is star-shaped about a. A loop g in K based at a is therefore path homotopic to the constant path e_a by $G(t,u) = (1-u)g(t) + ua$, which fixes the basepoint because $g(0) = g(1) = a$; and K is path-connected, again by the segments. So $\pi_1(K, a)$ is trivial and K is simply connected by corollary 6.1.25, hence semilocally simply connected.

A point $y \in K$ other than a is $(1-t)x + ta$ for a unique $t = y_3 < 1$ and a unique $x \in \mathcal{H}$, so

$$\rho\colon K \setminus \{a\} \to \mathcal{H}, \qquad \rho(y) = \frac{(y_1, y_2)}{1 - y_3}$$

is continuous and restricts to the identity on $\mathcal{H}$. Let $W_0 = \{y \in K \mid y_3 < 1/2\}$, an open neighbourhood of the origin in K that omits a, and let $V \subseteq W_0$ be any neighbourhood of the origin in K. As in (1), V contains every point of K within some distance $\delta > 0$ of the origin, so $C_n \subseteq \mathcal{H} \subseteq K$ lies in V once $2/n < \delta$; and $r_n \circ \rho|_V : V \to C_n$ restricts to the identity on C_n, so the inclusion $C_n \to V$ induces an injection on fundamental groups and $\pi_1(V, 0) \neq 1$. No neighbourhood of the origin contained in W_0 is simply connected.

The first computation shows that $\mathcal{H}$ admits no covering whose total space is simply connected, since such a covering would force the induced homomorphisms to be trivial. The second shows that the semilocal condition is strictly the weaker one and that the stronger version would be the wrong hypothesis to impose: K is its own simply connected covering under the identity map, yet no small neighbourhood of its origin is simply connected. Away from constructions of this kind the hypothesis costs nothing. An open $U \subseteq \mathbb{R}^n$ satisfies it because each of its points has a ball neighbourhood, and the straight-line homotopy contracts every loop in a ball inside the ball itself; the same argument, run in a chart, shows that every topological manifold in the sense of chapter 5 is semilocally simply connected.

A covering whose total space is simply connected has a property that explains what it is for: it lies above all the others.

Definition 6.4.3: Universal Cover

Let B be a topological space. A covering map $p\colon E \to B$ (definition 6.2.2) is a *universal cover* of B if E is simply connected in the sense of definition 6.1.26, that is, path-connected with trivial fundamental group.

The circle supplies the first instance. The map $p\colon \mathbb{R} \to S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$, used to compute $\pi_1(S^1)$ in theorem 6.2.20, is a covering map whose total space is convex, so the straight-line homotopy contracts every loop in $\mathbb{R}$ and $\mathbb{R}$ is simply connected. It is a universal cover of S^1, and example 6.4.9 places it at the bottom of the complete list of coverings of the circle.

Two observations turn the definition into the factoring property that names it. The first is recorded separately, since the deck-transformation theory of section 6.5 needs it as well.

Lemma 6.4.4: Coverings Inherit Local Path-Connectedness

Let $p\colon E \to B$ be a covering map with B locally path-connected. Then E is locally path-connected.

***Proof*:**
Let $p\colon E \to B$ be a covering map with B locally path-connected, let $e \in E$, and let W be an open set containing e. Choose an evenly covered open $U \ni p(e)$ and the slice V of $p^{-1}(U)$ containing e. Then $p(W \cap V)$ is open in B, since $p|_V$ is a homeomorphism onto the open set U, so the path component U' of $p(e)$ in $p(W \cap V)$ is open (theorem 3.1.40); and $(p|_V)^{-1}(U')$ is then an open path-connected set with $e \in (p|_V)^{-1}(U') \subseteq W$. So E is locally path-connected whenever B is, as we sought to show.

The second observation is the factoring itself, and it is what justifies the word *universal*.

Proposition 6.4.5: The Universal Cover Factors Through Every Covering

Let B be path-connected and locally path-connected, let $p\colon E \to B$ be a universal cover with $p(e_0) = b_0$, and let $q\colon E' \to B$ be any covering map with E' path-connected and $q(e_0') = b_0$. Then there is a continuous surjection $h\colon E \to E'$ with $h(e_0) = e_0'$ and $q \circ h = p$.

***Proof*:**
Let B be path-connected and locally path-connected, let $p\colon E \to B$ be a universal cover with $p(e_0) = b_0$, and let $q\colon E' \to B$ be any covering map with E' path-connected and $q(e_0') = b_0$. The space E is path-connected by hypothesis and locally path-connected by lemma 6.4.4, and $p_*(\pi_1(E, e_0))$ is the trivial subgroup, so it is contained in $q_*(\pi_1(E', e_0'))$; the lifting criterion (theorem 6.3.2) therefore produces a continuous $h\colon E \to E'$ with $h(e_0) = e_0'$ and $q \circ h = p$. This h is surjective. Indeed, given $e' \in E'$, choose a path σ in E' from e_0' to e'; then $q \circ \sigma$ is a path in B starting at b_0, which lifts to a path τ in E starting at e_0 (lemma 6.2.11), and $h \circ \tau$ is a lift of $q \circ \sigma$ to E' starting at e_0'. Uniqueness of lifts (theorem 6.3.1) gives $h \circ \tau = \sigma$, so $h(\tau(1)) = e'$. Every path-connected covering of B is thus a continuous surjective image of the universal cover, compatibly with the projections to B, completing the proof.

This is the sense in which one covering covers them all.

Where such an E could come from is dictated by the same lifting theory. In a simply connected E two paths σ, τ starting at e_0 and ending at the same point satisfy $[\sigma] * [\tau^{-1}] = [e_{e_0}]$, hence $[\sigma] = [\tau]$ by the groupoid identities of theorem 6.1.16, where τ^{-1} denotes the reverse path $t \mapsto \tau(1-t)$ (written $\tilde{\tau}$ in that theorem; in this section the tilde is reserved for lifts). So a point of E is determined by the path class from e_0 that reaches it, and p sends that point to the endpoint of the class. The construction below takes this description as the definition of E and manufactures a topology in which it is true.

Theorem 6.4.6: Existence of a Universal Cover

Let B be path-connected (definition 3.1.25), locally path-connected and semilocally simply connected (definition 6.4.1), and fix $b_0 \in B$. Then B has a universal cover: there is a covering map $p\colon E \to B$ with E simply connected.

***Proof*:**
Call an open set $U \subseteq B$ *admissible* if it is path-connected and every loop in U is path homotopic *in* B to a constant path. The second condition says exactly that $\iota_*\colon \pi_1(U, u) \to \pi_1(B, u)$ is trivial for every $u \in U$. Note at once that an open path-connected subset of an admissible set is admissible, a loop in the subset being a loop in the larger set.

Step 1: the admissible sets form a basis for B. Let $b \in B$. Semilocal simple connectivity supplies an open $U_0 \ni b$ with $\pi_1(U_0, b) \to \pi_1(B, b)$ trivial. Let U be the path component of b in U_0, which is open by theorem 3.1.40 and path-connected. Let g be a loop in U based at some $u \in U$ and choose a path σ in U from b to u. Then $\sigma * g * \sigma^{-1}$ is a loop in U_0 based at b, so $[\sigma] * [g] * [\sigma^{-1}] = [e_b]$ in $\pi_1(B, b)$, and multiplying by $[\sigma^{-1}]$ on the left and $[\sigma]$ on the right gives $[g] = [e_u]$ in B by theorem 6.1.16. So U is admissible.

Now let W be open with $b \in W$. Take an admissible $U \ni b$ as above and let V be the path component of b in $U \cap W$, open by theorem 3.1.40. Then V is an open path-connected subset of the admissible set U, hence admissible, and $b \in V \subseteq W$. The admissible sets are a basis.

Step 2: the set E and the map p. Let

$$E = \{[\gamma] \mid \gamma \text{ a path in } B \text{ with } \gamma(0) = b_0\}$$

be the set of path homotopy classes (definition 6.1.7) of paths in B issuing from b_0, let $\tilde{b}_0 = [e_{b_0}]$ be the class of the constant path, and define

$$p\colon E \to B, \qquad p([\gamma]) = \gamma(1)$$

which is well defined because path homotopic paths share their endpoints, and surjective because B is path-connected. For an admissible U and a class $[\gamma] \in E$ with $\gamma(1) \in U$, put

$$U_{[\gamma]} = \{[\gamma] * [\delta] \mid \delta \text{ a path in } U \text{ with } \delta(0) = \gamma(1)\}$$

a subset of E depending only on the class $[\gamma]$ and on U. Since the constant path at $\gamma(1)$ is a path in U, the class $[\gamma]$ lies in $U_{[\gamma]}$.

Step 3: these sets form a basis for a topology on E. The key point is that a set $U_{[\gamma]}$ is unchanged if $[\gamma]$ is replaced by any of its members:

$$[\gamma'] \in U_{[\gamma]} \implies U_{[\gamma']} = U_{[\gamma]} \tag{6.8}$$

Indeed, write $[\gamma'] = [\gamma] * [\delta]$ with δ a path in U. Any member $[\gamma'] * [\delta']$ of $U_{[\gamma']}$ equals $[\gamma] * [\delta * \delta']$ with $\delta * \delta'$ a path in U, so $U_{[\gamma']} \subseteq U_{[\gamma]}$. Also $[\gamma] = [\gamma'] * [\delta^{-1}]$ by theorem 6.1.16, with δ^{-1} a path in U, so $[\gamma] \in U_{[\gamma']}$ and the same computation gives $U_{[\gamma]} \subseteq U_{[\gamma']}$.

The sets $U_{[\gamma]}$ cover E, since $\gamma(1)$ lies in some admissible set by Step 1 and $[\gamma] \in U_{[\gamma]}$. Let $[\eta] \in U_{[\gamma]} \cap V_{[\beta]}$ with U, V admissible. By (6.8) these two sets are $U_{[\eta]}$ and $V_{[\eta]}$. Choose by Step 1 an admissible W with $\eta(1) \in W \subseteq U \cap V$. A path in W is a path in U and a path in V, so

$$[\eta] \in W_{[\eta]} \subseteq U_{[\eta]} \cap V_{[\eta]}$$

The two basis conditions hold, and the sets $U_{[\gamma]}$ generate a topology on E in which they are open.

Step 4: p is a covering map and every admissible U is evenly covered. First, $p(U_{[\gamma]}) = U$: the inclusion $\subseteq$ holds because $(\gamma * \delta)(1) = \delta(1) \in U$, and given $u \in U$ the path connectedness of U supplies a path δ in U from $\gamma(1)$ to u, whence $p([\gamma] * [\delta]) = u$. So p carries basic open sets to open sets and is an open map. It is continuous: if $[\gamma] \in p^{-1}(W)$ with W open, choose by Step 1 an admissible U with $\gamma(1) \in U \subseteq W$, and then $U_{[\gamma]} \subseteq p^{-1}(U) \subseteq p^{-1}(W)$.

Fix an admissible U. Every class in $p^{-1}(U)$ lies in its own $U_{[\gamma]}$, so $p^{-1}(U)$ is the union of the sets $U_{[\gamma]}$ with $\gamma(1) \in U$, and by (6.8) two of these sets are equal or disjoint. It remains to see that p is injective on each of them. Suppose $[\gamma] * [\delta_1]$ and $[\gamma] * [\delta_2]$ lie in $U_{[\gamma]}$ with $\delta_1(1) = \delta_2(1)$. Then $\delta_1 * \delta_2^{-1}$ is a loop in U, so admissibility makes it path homotopic in B to a constant path, and theorem 6.1.16 gives $[\delta_1] = [\delta_2]$ as classes of paths in B, whence $[\gamma] * [\delta_1] = [\gamma] * [\delta_2]$. So $p|_{U_{[\gamma]}}$ is a continuous open bijection onto U, that is, a homeomorphism. The sets $U_{[\gamma]}$ are the slices of $p^{-1}(U)$ in the sense of definition 6.2.1, and since the admissible sets cover B, the map p is a covering map (definition 6.2.2).

Step 5: E is path-connected, and the lift of a path is visible. For a path γ in B with $\gamma(0) = b_0$ and $s \in I$ let $\gamma_s(t) = \gamma(st)$, a path from b_0 to $\gamma(s)$, and define

$$\tilde{\gamma}\colon I \to E, \qquad \tilde{\gamma}(s) = [\gamma_s]$$

Then $\tilde{\gamma}(0) = [e_{b_0}] = \tilde{b}_0$, $\tilde{\gamma}(1) = [\gamma]$, and $p(\tilde{\gamma}(s)) = \gamma_s(1) = \gamma(s)$, so $\tilde{\gamma}$ lifts γ (definition 6.2.10) once continuity is known.

Fix $s_0 \in I$. By (6.8) every basic open set containing $\tilde{\gamma}(s_0)$ has the form $U_{[\gamma_{s_0}]}$ with U admissible and $\gamma(s_0) \in U$, so it suffices to place $\tilde{\gamma}(s)$ in that set for s near s_0. The set $\gamma^{-1}(U)$ is open in I and contains s_0, so it contains an interval J around s_0 in I. For $s \in J$ let $\delta(t) = \gamma\big((1-t)s_0 + ts\big)$, a path from $\gamma(s_0)$ to $\gamma(s)$ lying in U, since the parameters $(1-t)s_0 + ts$ all lie in the interval J. Both γ_s and $\gamma_{s_0} * \delta$ have the form $\gamma \circ \varphi$ for a continuous $\varphi\colon I \to I$ with $\varphi(0) = 0$ and $\varphi(1) = s$, namely $\varphi_1(t) = st$ and

$$\varphi_2(t) = \begin{cases} 2s_0 t & t \in [0, \frac{1}{2}] \\ (2-2t)s_0 + (2t-1)s & t \in [\frac{1}{2}, 1] \end{cases}$$

and $H(t,u) = \gamma\big((1-u)\varphi_1(t) + u\varphi_2(t)\big)$ is then a path homotopy from γ_s to $\gamma_{s_0} * \delta$: it is continuous, its parameter stays in I by convexity, and it fixes $t = 0$ and $t = 1$ because φ_1 and φ_2 agree there. Hence $\tilde{\gamma}(s) = [\gamma_{s_0}] * [\delta] \in U_{[\gamma_{s_0}]}$ for all $s \in J$, and $\tilde{\gamma}$ is continuous at s_0.

Since every class $[\gamma] \in E$ is joined to $\tilde{b}_0$ by the path $\tilde{\gamma}$, the space E is path-connected.

Step 6: E is simply connected. Let f be a loop in E based at $\tilde{b}_0$ and put $\gamma = p \circ f$, a loop in B based at b_0. Both f and the path $\tilde{\gamma}$ of Step 5 are lifts of γ starting at $\tilde{b}_0$, so $f = \tilde{\gamma}$ by theorem 6.3.1. Evaluating at 1 gives

$$\tilde{b}_0 = f(1) = \tilde{\gamma}(1) = [\gamma]$$

so γ is path homotopic in B to the constant path e_{b_0}. Lift that path homotopy to E starting at $\tilde{b}_0$ (lemmas 6.2.12 and 6.2.13). A path homotopy holds its initial point fixed, so the lift is a path homotopy whose two edges are the lift of γ starting at $\tilde{b}_0$, namely f, and the lift of e_{b_0} starting at $\tilde{b}_0$, which is the constant path at $\tilde{b}_0$. Hence $\pi_1(E, \tilde{b}_0)$ is trivial, and with Step 5 this makes E simply connected (definition 6.1.26), completing the proof.

The proof delivers more than the statement, and the extra will be used. The total space is the set of path classes issuing from b_0, the projection is evaluation at the endpoint, the evenly covered sets are the admissible ones, and by Step 5 the lift of a path γ starting at b_0 to the path starting at $\tilde{b}_0$ is $s \mapsto [\gamma_s]$, so that it *ends at* $[\gamma]$. The lifting correspondence of definition 6.2.16 is therefore the identity map in this model: a loop class $[\gamma] \in \pi_1(B, b_0)$ is sent to the point $[\gamma]$ of the fibre $p^{-1}(b_0)$. The fibre over b_0 is a copy of $\pi_1(B, b_0)$, which is the observation the classification theorem is built on.

Before that, the covering just constructed should be seen to be the only one of its kind. Proposition 6.4.5 produces a map between any two universal covers in either direction; running the uniqueness of lifts on the two composites shows that the maps are mutually inverse.

Corollary 6.4.7: Uniqueness of the Universal Cover

Let B be path-connected and locally path-connected, let $b_0 \in B$, and let $p\colon E \to B$ and $p'\colon E' \to B$ be universal covers of B with $p(e_0) = p'(e_0') = b_0$. Then there is exactly one homeomorphism $h\colon E \to E'$ with

$$h(e_0) = e_0', \qquad p' \circ h = p$$

***Proof*:**
Both E and E' are path-connected by definition 6.1.26 and locally path-connected by lemma 6.4.4, and their fundamental groups are trivial, so

$$p_*(\pi_1(E, e_0)) \subseteq p'_*(\pi_1(E', e_0')), \qquad p'_*(\pi_1(E', e_0')) \subseteq p_*(\pi_1(E, e_0))$$

both hold. The lifting criterion (theorem 6.3.2) applied to p with target covering p' gives a continuous $h\colon E \to E'$ with $h(e_0) = e_0'$ and $p' \circ h = p$, and applied to p' with target covering p gives a continuous $h'\colon E' \to E$ with $h'(e_0') = e_0$ and $p \circ h' = p'$.

The composite $h' \circ h\colon E \to E$ satisfies $p \circ (h' \circ h) = p' \circ h = p$, so it is a lift of p along p, and it fixes e_0. The identity of E is another such lift. Since E is path-connected and hence connected (lemma 3.1.28), theorem 6.3.1 forces $h' \circ h = \mathrm{id}_E$. The same argument gives $h \circ h' = \mathrm{id}_{E'}$, so h

is a homeomorphism. Any map with the two displayed properties is a lift of p along p' sending e_0 to e_0', so theorem 6.3.1 makes it equal to h, as we sought to show.

Uniqueness is what licenses the article in "the universal cover". It is uniqueness up to a specified isomorphism, not equality: the space E is pinned down only after a point of the fibre over b_0 has been chosen, and the choice matters, since h is the unique map carrying e_0 to the chosen e_0' and there are as many choices as there are points in the fibre. That fibre is a copy of $\pi_1(B, b_0)$, so the ambiguity in "the" universal cover is measured by the fundamental group itself. Section 6.5 takes up the self-maps this ambiguity produces.

The universal cover now becomes a machine for producing all the others. A point of the model built above is a path class from b_0, and a loop class $[\alpha] \in \pi_1(B, b_0)$ acts on such a point by prefixing the loop, $[\gamma] \mapsto [\alpha] * [\gamma]$, without disturbing the endpoint and hence without disturbing the projection. Collapsing the orbits of a subgroup H produces a space between the universal cover and B, and the covering it defines has H for its subgroup. Together with the rigidity supplied by the lifting criterion this accounts for every path-connected covering of B, and does so in the exact form of a Galois correspondence: subgroups on one side, coverings on the other.

Before stating it, a word on the equivalence being classified. Two covering maps $p\colon E \to B$ and $p'\colon E' \to B$ of the same base are *isomorphic* if there is a homeomorphism $h\colon E \to E'$ with $p' \circ h = p$; if basepoints $e_0 \in p^{-1}(b_0)$ and $e_0' \in (p')^{-1}(b_0)$ have been chosen and h can be taken with $h(e_0) = e_0'$, the isomorphism is *basepoint-preserving*. The composite of two such homeomorphisms and the inverse of one are again of this form, so both are equivalence relations on the coverings of B.

Theorem 6.4.8: Classification of Coverings

Let B be path-connected, locally path-connected and semilocally simply connected, and fix $b_0 \in B$.

1. For every subgroup H of $\pi_1(B, b_0)$ there are a covering map $p_H\colon E_H \to B$ with E_H path-connected and a point $e_H \in p_H^{-1}(b_0)$ with
$$(p_H)_*\big(\pi_1(E_H, e_H)\big) = H$$
2. Let $p\colon E \to B$ and $p'\colon E' \to B$ be covering maps with E and E' path-connected, and let $e_0 \in p^{-1}(b_0)$, $e_0' \in (p')^{-1}(b_0)$. Then the two coverings are isomorphic by a basepoint-preserving isomorphism if and only if
$$p_*\big(\pi_1(E, e_0)\big) = p'_*\big(\pi_1(E', e_0')\big)$$
Consequently $(p, e_0) \mapsto p_*(\pi_1(E, e_0))$ induces a bijection from the basepoint-preserving isomorphism classes of path-connected coverings of B onto the set of subgroups of $\pi_1(B, b_0)$.
3. With p, p', e_0, e_0' as in (2), the coverings are isomorphic, with no condition on basepoints, if and only if $p_*(\pi_1(E, e_0))$ and $p'_*(\pi_1(E', e_0'))$ are conjugate subgroups of $\pi_1(B, b_0)$. Consequently the isomorphism classes of path-connected coverings of B correspond bijectively to the conjugacy classes of subgroups of $\pi_1(B, b_0)$.

***Proof*:**
Throughout, E and E' are locally path-connected by lemma 6.4.4, and one fact about lifts is used repeatedly. For a covering $p\colon E \to B$ with $p(e_0) = b_0$ and a loop α in B based at b_0, write $\tilde{\alpha}$ for the lift of α starting at e_0 (lemma 6.2.11). Then

$$[\alpha] \in p_*\big(\pi_1(E, e_0)\big) \iff \tilde{\alpha} \text{ is a loop} \tag{6.9}$$

For if $\tilde{\alpha}(1) = e_0$ then $[\alpha] = p_*[\tilde{\alpha}]$; and conversely, if $[\alpha] = p_*[f]$ for a loop f in E based at e_0, then α is path homotopic to $p \circ f$, whose lift starting at e_0 is f by theorem 6.3.1, so $\tilde{\alpha}(1) = f(1) = e_0$ by lemma 6.2.14.

1. Let $q\colon \widetilde{B} \to B$ be the universal cover constructed in theorem 6.4.6, in the model recorded after its proof: the points of $\widetilde{B}$ are the path homotopy classes of paths in B issuing from b_0, the projection is $q([\gamma]) = \gamma(1)$, the basepoint is $\tilde{b}_0 = [e_{b_0}]$, and for each admissible U, that is, each open path-connected $U \subseteq B$ all of whose loops are path homotopic in B to constant paths, the sets $U_{[\gamma]}$ are the slices of $q^{-1}(U)$.

   The group $\pi_1(B, b_0)$ acts on $\widetilde{B}$ by

   $$[\alpha] \cdot [\gamma] = [\alpha] * [\gamma]$$

   which is defined because $\alpha(1) = b_0 = \gamma(0)$, and which is an action by theorem 6.1.16. It preserves the fibres of q, since $(\alpha * \gamma)(1) = \gamma(1)$, and each $[\alpha]$ acts as a homeomorphism: the bijection $[\gamma] \mapsto [\alpha] * [\gamma]$ carries the basic set $U_{[\gamma]}$ onto $U_{[\alpha]*[\gamma]}$, and its inverse, the action of $[\alpha]^{-1}$, does likewise. The action is also free, since $[\alpha] * [\gamma] = [\gamma]$ gives $[\alpha] = [e_{b_0}]$ after multiplication by $[\gamma^{-1}]$ on the right.

   Let E_H be the set of H-orbits with the quotient topology (definition 2.4.3) and $\pi\colon \widetilde{B} \to E_H$ the orbit map; put $e_H = \pi(\tilde{b}_0)$ and $p_H(\pi([\gamma])) = \gamma(1)$. This is well defined, because points of an orbit share their endpoint, and continuous, because $\pi^{-1}(p_H^{-1}(W)) = q^{-1}(W)$ is open for every open $W \subseteq B$. It is surjective because q is, and E_H is path-connected because $\widetilde{B}$ is and π is continuous and surjective. Moreover π is an open map: for $A \subseteq \widetilde{B}$ open, $\pi^{-1}(\pi(A)) = \bigcup_{h \in H} h \cdot A$ is a union of open sets.

   Let $U \subseteq B$ be admissible and let $\{V_\lambda\}$ be the slices of $q^{-1}(U)$. Each $h \in H$ permutes the slices, since $h \cdot U_{[\gamma]} = U_{h\cdot[\gamma]}$, so

   $$p_H^{-1}(U) = \pi\big(q^{-1}(U)\big) = \bigcup_\lambda \pi(V_\lambda)$$

   is a union of open sets. Two of them are equal or disjoint: if $\pi(V_\lambda)$ meets $\pi(V_\mu)$ then $h \cdot V_\lambda$ meets V_μ for some $h \in H$, so $h \cdot V_\lambda = V_\mu$ because distinct slices are disjoint, and then $\pi(V_\lambda) = \pi(V_\mu)$. Each $p_H|_{\pi(V_\lambda)}$ is a bijection onto U: it is surjective because $q(V_\lambda) = U$, and if two points of $\pi(V_\lambda)$ have the same image, representatives $x, y \in V_\lambda$ of them satisfy $q(x) = q(y)$ and hence $x = y$, because $q|_{V_\lambda}$ is injective. It is continuous, and it is open: for $A \subseteq \pi(V_\lambda)$ open, $p_H(A) = q(\pi^{-1}(A) \cap V_\lambda)$ is open because q is. So $p_H|_{\pi(V_\lambda)}$ is a homeomorphism onto U, the set U is evenly covered, and since the admissible sets cover B, the map p_H is a covering map.

   Finally, let α be a loop in B based at b_0. The lift of α to $\widetilde{B}$ starting at $\tilde{b}_0$ ends at $[\alpha]$, by Step 5 of the proof of theorem 6.4.6, and composing it with π gives a lift of α to E_H starting at e_H, which is *the* lift by theorem 6.3.1. That lift is a loop precisely when $[\alpha]$

and $\tilde{b}_0 = [e_{b_0}]$ lie in the same H-orbit, that is, when $[\alpha] = [\eta] * [e_{b_0}] = [\eta]$ for some $[\eta] \in H$, that is, when $[\alpha] \in H$. By (6.9) this says $(p_H)_*(\pi_1(E_H, e_H)) = H$.

2. Suppose first that $h\colon E \to E'$ is a homeomorphism with $p' \circ h = p$ and $h(e_0) = e'_0$. Functoriality (proposition 6.1.30) gives $p_* = p'_* \circ h_*$, and h_* is an isomorphism of $\pi_1(E, e_0)$ onto $\pi_1(E', e'_0)$ because h^{-1} induces its inverse. Hence the two images in $\pi_1(B, b_0)$ agree.

   Suppose conversely that the images agree. Both E and E' are path-connected and locally path-connected, so the lifting criterion (theorem 6.3.2) applies twice: from $p_*(\pi_1(E, e_0)) \subseteq p'_*(\pi_1(E', e'_0))$ it produces $h\colon E \to E'$ with $p' \circ h = p$ and $h(e_0) = e'_0$, and from the reverse inclusion it produces $h'\colon E' \to E$ with $p \circ h' = p'$ and $h'(e'_0) = e_0$. As in corollary 6.4.7, the composites $h' \circ h$ and $h \circ h'$ are lifts of p and of p' fixing the respective basepoints, so theorem 6.3.1 identifies them with the identity maps. Thus h is a basepoint-preserving isomorphism.

   The displayed subgroup therefore depends only on the basepoint-preserving isomorphism class and separates distinct classes, so the induced map to subgroups is well defined and injective; part (1) makes it surjective.

3. Two preliminaries. First, changing the basepoint in the fibre conjugates the subgroup. Let $e_1 \in p^{-1}(b_0)$, let τ be a path in E from e_0 to e_1, and let $\alpha = p \circ \tau$, a loop in B based at b_0. The change-of-basepoint isomorphism $\hat{\tau}\colon \pi_1(E, e_0) \to \pi_1(E, e_1)$ of proposition 6.1.24 sends $[f]$ to $[\tau^{-1}] * [f] * [\tau]$, and applying p_* to this gives $[\alpha]^{-1} * p_*[f] * [\alpha]$, since p carries concatenations to concatenations and τ^{-1} to α^{-1}. As $\hat{\tau}$ is onto,

$$p_*\big(\pi_1(E, e_1)\big) = [\alpha]^{-1}\, p_*\big(\pi_1(E, e_0)\big)\, [\alpha]$$

   Second, every conjugate arises this way: given $[\alpha] \in \pi_1(B, b_0)$, lift α to a path τ starting at e_0 (lemma 6.2.11) and take $e_1 = \tau(1)$, which lies in $p^{-1}(b_0)$.

   Now suppose $h\colon E \to E'$ is a homeomorphism with $p' \circ h = p$. Put $e'_1 = h(e_0)$, a point of $(p')^{-1}(b_0)$. By the argument of (2), $p_*(\pi_1(E, e_0)) = p'_*(\pi_1(E', e'_1))$, and by the first preliminary the latter is conjugate to $p'_*(\pi_1(E', e'_0))$. Conversely, suppose $p'_*(\pi_1(E', e'_0)) = [\alpha]^{-1} p_*(\pi_1(E, e_0))[\alpha]$ for some $[\alpha] \in \pi_1(B, b_0)$. The second preliminary supplies $e_1 \in p^{-1}(b_0)$ with $p_*(\pi_1(E, e_1)) = [\alpha]^{-1} p_*(\pi_1(E, e_0))[\alpha]$, so $p_*(\pi_1(E, e_1)) = p'_*(\pi_1(E', e'_0))$ and part (2), applied with the basepoints e_1 and e'_0, produces an isomorphism $E \to E'$.

   Since conjugate subgroups are exactly those arising from different choices of basepoint in a fixed covering, the bijection of (2) descends to a bijection between isomorphism classes and conjugacy classes, completing the proof.

The correspondence reverses inclusions. The trivial subgroup belongs to the universal cover, which lies above every other covering by the factoring property established after definition 6.4.3; the whole group $\pi_1(B, b_0)$ belongs to the identity covering $\mathrm{id}\colon B \to B$, which lies below every other. Between the two extremes the construction in part (1) makes the ordering explicit: if $H \subseteq H'$ then each H-orbit in $\widetilde{B}$ is contained in a single H'-orbit, so $E_{H'}$ is a quotient of E_H by a map commuting with the projections to B. Every covering in the family is a quotient of the single universal cover, and the subgroup attached to it records which identifications were made.

The two ways of counting the classification are worth separating. Basepoints remembered, the invariant is a subgroup and the classes are in bijection with all subgroups; basepoints forgotten, the invariant is only a conjugacy class, because moving the basepoint around the fibre conjugates the subgroup by the loop traced below. When $\pi_1(B, b_0)$ is abelian the two counting agree, and the coverings of the circle are the smallest instance where the whole picture can be written out.

Example 6.4.9: The Coverings of the Circle

Regard S^1 as the unit circle in $\mathbb{C}$ with basepoint $b_0 = 1$. It is path-connected and locally path-connected, and it is semilocally simply connected for the strongest possible reason: each point lies in an open arc, an arc is homeomorphic to an open interval of $\mathbb{R}$, and the straight-line homotopy contracts every loop in an interval inside the interval. So theorem 6.4.8 applies, and by theorem 6.2.20 the group to be classified is $\pi_1(S^1, b_0) \cong \mathbb{Z}$. Its subgroups are the trivial subgroup and the subgroups $n\mathbb{Z}$ for $n \geq 1$: a nontrivial subgroup contains a least positive element n, and division with remainder puts every element of it in $n\mathbb{Z}$. Since $\mathbb{Z}$ is abelian, conjugacy classes of subgroups are subgroups, so the two classifications of theorem 6.4.8 coincide.

Fix the isomorphism $\pi_1(S^1, b_0) \cong \mathbb{Z}$ concretely. For $k \in \mathbb{Z}$ let $\omega_k(t) = e^{2\pi i k t}$, a loop at b_0. Under the covering $p\colon \mathbb{R} \to S^1$, $p(t) = e^{2\pi i t}$, of theorem 6.2.20, the lift of ω_k starting at 0 is $t \mapsto kt$, which ends at k, so the lifting correspondence (definition 6.2.16) sends $[\omega_k]$ to k and $\pi_1(S^1, b_0) = \{[\omega_k] \mid k \in \mathbb{Z}\}$.

Two families of coverings are at hand.

1. The map $p\colon \mathbb{R} \to S^1$ itself. Its total space is convex, hence simply connected, so p is a universal cover and $p_*(\pi_1(\mathbb{R}, 0))$ is the trivial subgroup.

2. For $n \geq 1$ the map $p_n\colon S^1 \to S^1$, $p_n(z) = z^n$. It is a covering map: for an open half-circle $U = \left\{e^{i\theta} \mid \theta_0 < \theta < \theta_0 + \pi\right\}$ the preimage $p_n^{-1}(U)$ is the union over $k = 0, \ldots, n-1$ of the arcs
$$A_k = \left\{e^{i\theta} \mid \tfrac{\theta_0 + 2\pi k}{n} < \theta < \tfrac{\theta_0 + \pi + 2\pi k}{n}\right\}$$
which are disjoint because each has angular width π/n while consecutive ones start $2\pi/n$ apart, and p_n restricted to A_k is a continuous open bijection onto U, hence a homeomorphism. Such U cover S^1. For the subgroup, $p_n \circ \omega_k = \omega_{nk}$, so $(p_n)_*[\omega_k] = [\omega_{nk}]$ and
$$(p_n)_*\big(\pi_1(S^1, b_0)\big) = n\mathbb{Z}$$

Every subgroup of $\mathbb{Z}$ has thus been realized, so by theorem 6.4.8 every path-connected covering of S^1 is isomorphic to exactly one of
$$\mathbb{R} \to S^1, \qquad p_1 = \mathrm{id}, \qquad p_2, \qquad p_3, \qquad \ldots$$
and the list is complete. The fibres record the indices: $p_n^{-1}(1)$ is the set of n-th roots of unity, of size $n = [\mathbb{Z} : n\mathbb{Z}]$, while $p^{-1}(1) = \mathbb{Z}$ is infinite, matching the infinite index of the trivial subgroup.

The circle exhibits the correspondence in its simplest nontrivial form, and it also shows what the correspondence is for. The coverings were not found by inspection and then checked against the subgroups; the subgroups of $\mathbb{Z}$ were listed first, and the classification then guarantees that the list above omits nothing. Any path-connected covering of S^1 other than the universal one is p_n for exactly one $n \geq 1$, and that n is the index in $\mathbb{Z}$ of the subgroup attached to it.

Exercise 6.4.1

1. Show that the hawaiian earring of example 6.4.2 is not semilocally simply connected, and

hence has no universal cover.

2. Let B be path-connected and locally path-connected. Show that a universal cover, when it exists, is unique up to homeomorphism over the base, and that its fibre is in bijection with $\pi_1(B, b_0)$.

3. Identify all path-connected covering spaces of S^1 up to isomorphism, using theorem 6.4.8, and explain why for S^1 this is the same as identifying the connected ones.

4. Show that a simply-connected, locally path-connected space is its own universal cover.

5. Show that the subgroup $p_*\pi_1(E, e)$ changes by conjugation when the basepoint e moves within the fibre.

6.5 Deck Transformations

Theorem 6.4.8 matches the connected coverings of a space against the subgroups of its fundamental group. A single covering also carries symmetries of its own: homeomorphisms of the total space that shuffle each fibre and leave the projection untouched. This section identifies the group those symmetries form, and finds it assembled out of exactly the subgroup the classification attaches to the covering. Throughout, $p\colon E \to B$ is a covering map with $b_0 \in B$ and $e_0 \in p^{-1}(b_0)$, and for a path u the reverse path $t \mapsto u(1-t)$ is written u^{-1}, the tilde being reserved for lifts.

The covering $p\colon \mathbb{R} \to S^1$ used to compute $\pi_1(S^1)$ in theorem 6.2.20 is unchanged by the translations $x \mapsto x + n$ with $n \in \mathbb{Z}$: each translation permutes the points of every fibre of p, and p composed with it is again p. Nothing about the circle entered that description, and nothing beyond it should be asked of a symmetry of a covering. Such a symmetry is a homeomorphism of the total space that the projection cannot detect.

Definition 6.5.1: Deck Transformation

Let $p\colon E \to B$ be a covering map (definition 6.2.2). A *deck transformation* of p is a homeomorphism $h\colon E \to E$ satisfying

$$p \circ h = p$$

The collection of all deck transformations of p is written $\operatorname{Deck}(E/B)$.

The equation $p \circ h = p$ says that h moves each point inside its own fibre: $p(h(e)) = p(e)$, so $h(p^{-1}(b)) \subseteq p^{-1}(b)$ for every $b \in B$, with equality because h^{-1} satisfies the same equation. A deck transformation is therefore a simultaneous permutation of all the fibres, continuous in the total space and invisible in the base.

Under composition these permutations form a group. The identity id_E qualifies. If $p \circ h = p$ and $p \circ k = p$ then $p \circ (h \circ k) = (p \circ h) \circ k = p \circ k = p$, so the composite qualifies. Composing $p \circ h = p$ on the right with h^{-1} gives $p = p \circ h^{-1}$, so inverses qualify. Associativity is associativity of composition of maps. Thus $\operatorname{Deck}(E/B)$ is a subgroup of the group of self-homeomorphisms of E, and it acts on each fibre $p^{-1}(b)$ by permutations.

One deck group can be computed on sight. Let X be nonempty and connected, let $F = \{1, \dots, n\}$ carry the discrete topology, and let $q\colon X \times F \to X$ be the projection, a covering map by example 6.2.3.

A deck transformation h of q preserves the first coordinate, so $h(x,i) = (x, \sigma(x,i))$ for a function σ into F. For each fixed i the map $x \mapsto \sigma(x,i)$ is the composite of h with the projection onto the discrete factor, hence continuous, and X is connected, so it is constant; write $\sigma(i)$ for its value. Then $h(x,i) = (x, \sigma(i))$, and such an h is a homeomorphism exactly when σ is a bijection of F, its inverse being $(x,j) \mapsto (x, \sigma^{-1}(j))$. So $h \mapsto \sigma$ identifies $\mathrm{Deck}\,((X \times F)/X)$ with the symmetric group on n letters: the sheets may be permuted arbitrarily and independently of one another.

That independence is what a connected total space forbids. Take $n \geq 3$ in the computation just made: the bijection exchanging 2 and 3 and fixing every other sheet gives a deck transformation that fixes the point $(x,1)$ and still moves $(x,2)$, because no path in $X \times F$ joins those two sheets. When E is path-connected the equation $p \circ h = p$ exhibits h as a lift of the map p through the covering p, with id_E as a competing lift, and lifts of a map on a connected domain cannot part ways once they have agreed anywhere.

Proposition 6.5.2: Deck Transformations Act Freely

Let $p\colon E \to B$ be a covering map with E path-connected, and let $h \in \mathrm{Deck}\,(E/B)$. If $h(e) = e$ for a single point $e \in E$, then $h = \mathrm{id}_E$. Consequently two deck transformations that agree at one point of E are equal, and $\mathrm{Deck}\,(E/B)$ acts freely on every fibre of p.

***Proof*:**
The space E is path-connected, hence connected by lemma 3.1.28. Both h and id_E are continuous maps $E \to E$ satisfying $p \circ h = p$ and $p \circ \mathrm{id}_E = p$, so both are lifts of the map $p\colon E \to B$ through the covering p in the sense of definition 6.2.10. They agree at e. By theorem 6.3.1, two lifts of a map from a connected space that agree at one point agree everywhere, so $h = \mathrm{id}_E$.

If $h_1, h_2 \in \mathrm{Deck}\,(E/B)$ satisfy $h_1(e) = h_2(e)$, then $h_2^{-1} \circ h_1$ is a deck transformation fixing e, hence the identity by the previous paragraph, and $h_1 = h_2$. In particular the only element of $\mathrm{Deck}\,(E/B)$ fixing a point of a fibre $p^{-1}(b)$ is the identity, which is freeness of the action on that fibre, as we sought to show.

Freeness is a rigidity statement: the single value $h(e_0)$ determines h on all of E. It follows that

$$\mathrm{Deck}\,(E/B) \longrightarrow p^{-1}(b_0), \qquad h \longmapsto h(e_0)$$

is injective, so the deck group is no larger than the fibre, and the whole question of its size becomes the question of which points of the fibre are values of this map. At one extreme the map hits only e_0 and the deck group is trivial; at the other it is onto and the covering has as much symmetry as its fibre permits.

Definition 6.5.3: Regular Covering

Let $p\colon E \to B$ be a covering map with E path-connected, and let $b_0 \in B$. The covering is *regular*, or *normal*, if $\mathrm{Deck}\,(E/B)$ acts transitively on the fibre $p^{-1}(b_0)$: for every pair of points $e, e' \in p^{-1}(b_0)$ there is a deck transformation h with $h(e) = e'$.

Regularity of the covering $p\colon \mathbb{R} \to S^1$, $p(x) = e^{2\pi i x}$, can be read off at once. The fibre over the basepoint 1 is the set $\mathbb{Z} \subseteq \mathbb{R}$, and the translation $x \mapsto x + (m-k)$ is a deck transformation carrying k to m, so the action on that fibre is transitive. This computation is carried out in full in example 6.5.6.

The definition names one basepoint, and it need not have. First, B is path-connected: it is the image of the path-connected space E under the continuous surjection p, and composing a path in E with p produces a path in B between the images of its endpoints. Now suppose $\mathrm{Deck}\,(E/B)$ acts transitively on $p^{-1}(b_0)$, let $b \in B$, and let e, e' be two points of $p^{-1}(b)$. Choose a path u in B from b to b_0 and lift it by lemma 6.2.11 twice, once to the path $\tilde{u}_1$ with $\tilde{u}_1(0) = e$ and once to the path $\tilde{u}_2$ with $\tilde{u}_2(0) = e'$. Both lifts end in $p^{-1}(b_0)$, so transitivity there supplies $h \in \mathrm{Deck}\,(E/B)$ with $h(\tilde{u}_1(1)) = \tilde{u}_2(1)$. The path $h \circ \tilde{u}_1$ is again a lift of u, since $p \circ h \circ \tilde{u}_1 = p \circ \tilde{u}_1 = u$, and it agrees with $\tilde{u}_2$ at the parameter value 1; as I is connected, theorem 6.3.1 forces $h \circ \tilde{u}_1 = \tilde{u}_2$, and evaluating at 0 gives $h(e) = e'$. Transitivity on one fibre is therefore transitivity on all of them.

The second name, "normal", is not yet justified. It comes from a reformulation of the definition in which the fibre disappears and only the subgroup $p_*(\pi_1(E, e_0))$ remains, and that reformulation is a by-product of the computation of the deck group itself, so it is deferred until after theorem 6.5.4.

What that computation needs is a criterion for the map $h \mapsto h(e_0)$ to reach a prescribed point e_1 of the fibre. The lifting criterion answers questions of this shape, since a deck transformation carrying e_0 to e_1 is precisely a lift of p through p with the basepoint e_1 prescribed, and its answer is a comparison of the subgroups $p_*(\pi_1(E, e_0))$ and $p_*(\pi_1(E, e_1))$. Those two subgroups turn out to be conjugate, so the criterion becomes a requirement that a conjugation fix a subgroup, and the elements performing such conjugations form the normalizer. Recall that for a subgroup H of a group G the *normalizer* of H in G is

$$N(H) = \left\{ g \in G \;\middle|\; g^{-1} H g = H \right\}$$

a subgroup of G containing H, and the largest subgroup of G in which H is normal.

Theorem 6.5.4: The Deck Group as a Quotient of a Normalizer

Let $p\colon E \to B$ be a covering map with E path-connected and B locally path-connected. Fix $b_0 \in B$ and $e_0 \in p^{-1}(b_0)$, write

$$H = p_*(\pi_1(E, e_0)) \subseteq \pi_1(B, b_0)$$

for the image of the induced homomorphism (definition 6.1.28), and let $N(H)$ denote the normalizer of H in $\pi_1(B, b_0)$. Then

$$\mathrm{Deck}\,(E/B) \cong N(H)/H$$

***Proof*:**

Write $\varphi\colon \pi_1(B, b_0) \to p^{-1}(b_0)$ for the lifting correspondence of definition 6.2.16, so that $\varphi([\gamma]) = \tilde{\gamma}(1)$ where $\tilde{\gamma}$ is the lift of the loop γ with $\tilde{\gamma}(0) = e_0$; it is surjective by proposition 6.2.18, E being path-connected.

Step 0: the total space is locally path-connected. Let $e \in E$ and let W be an open set containing e. Choose a neighbourhood U of $p(e)$ evenly covered by p and let V be the slice containing e (definition 6.2.1), so that $p|_V : V \to U$ is a homeomorphism. The set $V \cap W$ is open in V, so $p(V \cap W)$ is open in U and hence in B, and it contains $p(e)$. Since B is locally path-connected there is a path-connected open set U' with $p(e) \in U' \subseteq p(V \cap W)$, and then $(p|_V)^{-1}(U')$ is an open path-connected set containing e and contained in W. So the path-connected open sets form a neighbourhood base at each point of E, and theorem 6.3.2 is available with E as the domain.

Step 1: the subgroups attached to the points of the fibre are conjugates of H. Let $g = [\gamma] \in \pi_1(B, b_0)$ and put $e_1 = \varphi(g)$, so that the lift $\tilde{\gamma}$ is a path in E from e_0 to e_1 with $p \circ \tilde{\gamma} = \gamma$. By proposition 6.1.24 the map sending $[\tilde{f}]$ to $[\tilde{\gamma}^{-1} * \tilde{f} * \tilde{\gamma}]$ is an isomorphism $\pi_1(E, e_0) \to \pi_1(E, e_1)$. Composition with p distributes over concatenation and commutes with reversal, so

$$p \circ (\tilde{\gamma}^{-1} * \tilde{f} * \tilde{\gamma}) = \gamma^{-1} * (p \circ \tilde{f}) * \gamma$$

and, using associativity and $[\gamma^{-1}] = [\gamma]^{-1}$ from theorem 6.1.16, the class of the right-hand side is $g^{-1} p_*([\tilde{f}]) g$. As $[\tilde{f}]$ runs over $\pi_1(E, e_0)$ the classes $[\tilde{\gamma}^{-1} * \tilde{f} * \tilde{\gamma}]$ run over all of $\pi_1(E, e_1)$, whence

$$p_*(\pi_1(E, e_1)) = g^{-1} H g \tag{6.10}$$

Every point of $p^{-1}(b_0)$ is of the form $\varphi(g)$, so (6.10) describes the subgroup attached to each point of the fibre.

Step 2: which points of the fibre are reached. Let $e_1 \in p^{-1}(b_0)$. The claim is that a deck transformation h with $h(e_0) = e_1$ exists if and only if $p_*(\pi_1(E, e_1)) = H$.

Suppose first that such an h exists. It is a homeomorphism, so h_* and $(h^{-1})_*$ are mutually inverse by the functoriality of proposition 6.1.30, and $h_* \colon \pi_1(E, e_0) \to \pi_1(E, e_1)$ is an isomorphism. The same functoriality gives

$$p_*(\pi_1(E, e_1)) = p_*\big(h_*(\pi_1(E, e_0))\big) = (p \circ h)_*(\pi_1(E, e_0)) = p_*(\pi_1(E, e_0)) = H$$

Conversely suppose $p_*(\pi_1(E, e_1)) = H$. Apply theorem 6.3.2 to the continuous map $p \colon E \to B$ on the path-connected, locally path-connected space E, lifting it through the covering p with the basepoint e_0 sent to e_1: the criterion asks for $p_*(\pi_1(E, e_0)) \subseteq p_*(\pi_1(E, e_1))$, which holds with equality, and produces a continuous $h \colon E \to E$ with $p \circ h = p$ and $h(e_0) = e_1$. Exchanging the roles of e_0 and e_1 produces a continuous $k \colon E \to E$ with $p \circ k = p$ and $k(e_1) = e_0$. Then $k \circ h$ and $h \circ k$ are lifts of p through p fixing e_0 and e_1 respectively, so both equal id_E by theorem 6.3.1 compared against the lift id_E. Hence h is a homeomorphism with inverse k, that is, $h \in \mathrm{Deck}\,(E/B)$.

Combining this with (6.10), a point $e_1 = \varphi(g)$ of the fibre is $h(e_0)$ for some deck transformation h exactly when $g^{-1} H g = H$, that is, exactly when $g \in N(H)$.

Step 3: the homomorphism. For $g \in N(H)$ let $\Psi(g)$ be the deck transformation with $\Psi(g)(e_0) = \varphi(g)$, which exists by Step 2 and is unique by proposition 6.5.2. To see that $\Psi \colon N(H) \to \mathrm{Deck}\,(E/B)$ is a homomorphism, take $g = [\gamma]$ and $g' = [\delta]$ in $N(H)$ with lifts $\tilde{\gamma}, \tilde{\delta}$ starting at e_0. The path $\Psi(g) \circ \tilde{\delta}$ satisfies $p \circ \Psi(g) \circ \tilde{\delta} = p \circ \tilde{\delta} = \delta$, so it is a lift of δ, and it starts at $\Psi(g)(e_0) = \tilde{\gamma}(1)$. Hence the concatenation $\tilde{\gamma} * (\Psi(g) \circ \tilde{\delta})$ is defined (definition 6.1.13), and composing it with p gives γ on the first half of the interval and δ on the second, so it is a lift of $\gamma * \delta$ beginning at e_0. Uniqueness of path lifts (lemma 6.2.11) identifies it as *the* lift, so

$$\varphi(gg') = \big(\tilde{\gamma} * (\Psi(g) \circ \tilde{\delta})\big)(1) = \Psi(g)\big(\tilde{\delta}(1)\big) = \Psi(g)\big(\Psi(g')(e_0)\big)$$

The deck transformations $\Psi(gg')$ and $\Psi(g) \circ \Psi(g')$ therefore agree at e_0, and proposition 6.5.2 makes them equal.

Step 4: kernel and image. By proposition 6.5.2 again, $\Psi(g) = \mathrm{id}_E$ if and only if $\varphi(g) = e_0$, so the kernel of Ψ is the set of $g \in N(H)$ whose lift closes up. If $\varphi([\gamma]) = e_0$ then $\tilde{\gamma}$ is a loop at e_0 and $[\gamma] = p_*([\tilde{\gamma}]) \in H$. Conversely if $[\gamma] = p_*([\tilde{f}])$ for a loop $\tilde{f}$ at e_0, then γ is path-homotopic to $p \circ \tilde{f}$, whose lift at e_0 is $\tilde{f}$; path-homotopic loops have lifts with the same terminal point by lemma 6.2.14, so $\varphi([\gamma]) = \tilde{f}(1) = e_0$. Thus $\ker \Psi = H$.

For surjectivity, let $h \in \mathrm{Deck}\,(E/B)$ and put $e_1 = h(e_0)$, a point of $p^{-1}(b_0)$ because $p(h(e_0)) = p(e_0) = b_0$. Choose g with $\varphi(g) = e_1$, possible since φ is surjective. Step 2 gives $p_*(\pi_1(E, e_1)) = H$, and (6.10) turns this into $g^{-1}Hg = H$, so $g \in N(H)$. Now $\Psi(g)$ and h agree at e_0, hence are equal by proposition 6.5.2. So Ψ is a surjective homomorphism from $N(H)$ onto $\mathrm{Deck}\,(E/B)$ with kernel H, and the first isomorphism theorem gives $N(H)/H \cong \mathrm{Deck}\,(E/B)$, completing the proof.

The theorem locates the deck group inside the fundamental group of the base, and it locates it twice over. The subgroup H records how much of $\pi_1(B, b_0)$ is already visible upstairs, and the loops in H contribute nothing: their lifts close up and the associated deck transformation is the identity. The normalizer records how much of $\pi_1(B, b_0)$ moves the fibre in a way the covering can absorb, and loops outside $N(H)$ carry e_0 to a point whose subgroup is a different conjugate of H, so no homeomorphism over B can make the move. What survives is the quotient of the second by the first.

The reformulation promised after definition 6.5.3 now follows. Since every point of $p^{-1}(b_0)$ is $\varphi(g)$ for some g, Step 2 says that the values of the injection $h \mapsto h(e_0)$ are exactly the points $\varphi(g)$ with $g \in N(H)$, so that map is onto the fibre if and only if $N(H) = \pi_1(B, b_0)$. Being onto is the same as transitivity: if $h_1(e_0) = e$ and $h_2(e_0) = e'$ then $h_2 \circ h_1^{-1}$ carries e to e', and conversely transitivity produces some h with $h(e_0) = e_1$ for each e_1. So, under the hypotheses of the theorem,

$$p \text{ is regular} \iff N(H) = \pi_1(B, b_0) \iff H \trianglelefteq \pi_1(B, b_0)$$

which is where the alternative name comes from, and in that case the theorem reads $\mathrm{Deck}\,(E/B) \cong \pi_1(B, b_0)/H$. Combining freeness with transitivity, the deck group of a regular covering is in bijection with each fibre, so the number of sheets equals the order of $\pi_1(B, b_0)/H$, which is the index of H in $\pi_1(B, b_0)$. Regularity is thus a condition on the subgroup rather than on the space: the covering that theorem 6.4.8 attaches to a normal subgroup is regular, and the one attached to a subgroup that is not normal is not. When $\pi_1(B, b_0)$ is abelian every subgroup is normal and the condition is no condition at all. The circle is such a base: its n-fold covering has H the subgroup $n\mathbb{Z}$ of $\mathbb{Z}$, so the theorem predicts a deck group of order n, and example 6.5.6 exhibits it as the group of rotations by nth roots of unity.

Regularity also says that B is recoverable from E together with its deck group. A covering map is open (corollary 6.2.4) and surjective, so a subset $W \subseteq B$ with $p^{-1}(W)$ open has $W = p(p^{-1}(W))$ open, and p presents B as a quotient of E in the sense of definition 2.4.3. The orbits of $\mathrm{Deck}\,(E/B)$ are contained in the fibres of p, since a deck transformation preserves each fibre, and the two families coincide exactly when the action on every fibre is transitive, that is, exactly when the covering is regular. For a regular covering, then, the map induced on the orbit space $E/\mathrm{Deck}\,(E/B)$ is a bijection onto B; it is continuous because p is, and open because a set open in the orbit space has open preimage Ω in E, and its image under the induced map is $p(\Omega)$, open because p is. So a regular covering is the passage from a space to its quotient by a free action of the deck group.

Pushing H to the smallest it can be pushes the deck group to the largest it can be, and the subgroup criterion becomes vacuous.

Corollary 6.5.5: The Deck Group of the Universal Cover

Let B be locally path-connected and let $p\colon E \to B$ be a universal cover (definition 6.4.3), with $b_0 \in B$ and $e_0 \in p^{-1}(b_0)$. Then p is regular and

$$\mathrm{Deck}\,(E/B) \cong \pi_1(B, b_0)$$

***Proof*:**
The total space E is simply connected, so it is path-connected and $\pi_1(E, e_0)$ is the trivial group (definition 6.1.26); hence $H = p_*(\pi_1(E, e_0))$ is the trivial subgroup of $\pi_1(B, b_0)$. The trivial subgroup is normal in any group, so p is regular and $N(H) = \pi_1(B, b_0)$. Theorem 6.5.4 applies, its hypotheses being met, and yields

$$\mathrm{Deck}\,(E/B) \cong N(H)/H = \pi_1(B, b_0)/\{1\} \cong \pi_1(B, b_0)$$

completing the proof.

The corollary converts homotopy classes of loops into homeomorphisms. An element of $\pi_1(B, b_0)$ is an equivalence class of maps into B, an object with no action on anything; the corresponding deck transformation is a single homeomorphism of E, and the group law is composition of maps. The isomorphism is not canonical, being manufactured from the lifting correspondence at the chosen point e_0 of the fibre, but its existence is what makes the fundamental group usable as a group of symmetries. Together with the quotient description above it presents B as $E/\pi_1(B, b_0)$: a space admitting a universal cover is the orbit space of a simply connected space under a free action of its own fundamental group.

Both halves of the theory can be watched at work on the circle, where the coverings are known and the deck groups can be computed by hand.

Example 6.5.6: Deck Transformations of the Coverings of the Circle
Regard S^1 as the unit circle $\{z \in \mathbb{C} \mid |z| = 1\}$ and let $b_0 = 1$ be its basepoint. It is path-connected, being the image of $\mathbb{R}$ under the continuous map $x \mapsto e^{2\pi i x}$. Every point of S^1 has a neighbourhood homeomorphic to an open interval of $\mathbb{R}$, and open intervals are path-connected, so S^1 is locally path-connected; the same neighbourhoods make it semilocally simply connected, since under such a homeomorphism the straight-line homotopy contracts every loop in the neighbourhood to a constant inside that neighbourhood, so a fortiori inside S^1. All three hypotheses of theorem 6.4.8 are therefore in force, and the results of this section apply to the coverings of S^1. The subgroups of $\pi_1(S^1, 1) \cong \mathbb{Z}$ are the trivial one and the subgroups $n\mathbb{Z}$ with $n \geq 1$, so by that theorem the two families below account for every connected covering of the circle (compare example 6.4.9).

1. Take the covering map $p\colon \mathbb{R} \to S^1$ used in theorem 6.2.20, written in the present notation as $p(x) = e^{2\pi i x}$, for which $p(x) = p(y)$ holds precisely when $x - y \in \mathbb{Z}$. Let h be a deck transformation. Then $p(h(x)) = p(x)$ for every x, so the continuous function

$$d\colon \mathbb{R} \to \mathbb{R}, \qquad d(x) = h(x) - x$$

takes values in $\mathbb{Z}$. Its image is a connected subset of $\mathbb{Z}$, and $\mathbb{Z}$ is discrete in $\mathbb{R}$ because

$\{n\} = \mathbb{Z} \cap (n - \frac{1}{2}, n + \frac{1}{2})$, so the image is a single point n and $h(x) = x + n$ for all x. Conversely each translation $t_n(x) = x + n$ is a homeomorphism of $\mathbb{R}$ with $p \circ t_n = p$. Hence

$$\operatorname{Deck}(\mathbb{R}/S^1) = \{t_n \mid n \in \mathbb{Z}\}$$

and since $t_n \circ t_m = t_{n+m}$ and $t_n = t_m$ forces $n = m$ on evaluation at 0, the map $n \mapsto t_n$ is an isomorphism $\mathbb{Z} \to \operatorname{Deck}(\mathbb{R}/S^1)$. The fibre $p^{-1}(1)$ is $\mathbb{Z} \subseteq \mathbb{R}$ and t_{m-k} carries k to m, so the action on it is transitive and the covering is regular, as the normality of every subgroup of an abelian group predicts.

The deck group just computed also determines $\pi_1(S^1)$. The space $\mathbb{R}$ is path-connected and convex, so the straight-line homotopy contracts every loop in it to a constant path and $\pi_1(\mathbb{R}, 0)$ is trivial; thus $\mathbb{R}$ is simply connected in the sense of definition 6.1.26 and p is a universal cover of S^1. Corollary 6.5.5 then gives

$$\pi_1(S^1, 1) \cong \operatorname{Deck}(\mathbb{R}/S^1) \cong \mathbb{Z}$$

which is theorem 6.2.20. The route is independent of the original computation: it uses the uniqueness of lifts, the lifting criterion and the convexity of $\mathbb{R}$, and never asks which integer a given loop represents.

2. Fix $n \geq 1$, write E for a second copy of S^1 so that the two circles stay apart in the notation, and take $p_n \colon E \to S^1$ with $p_n(z) = z^n$. This is a covering map. Given $w = e^{i\alpha} \in S^1$, let U be the complement of the point $-w$, an open arc consisting of the points $e^{i\psi}$ with $\alpha - \pi < \psi < \alpha + \pi$. A point $z = e^{i\theta}$ lies in $p_n^{-1}(U)$ exactly when $n\theta$ lies in that range modulo 2π, that is, exactly when θ lies in one of the n disjoint open arcs

$$A_k = \left\{ e^{i\theta} \;\middle|\; \frac{\alpha - \pi + 2\pi k}{n} < \theta < \frac{\alpha + \pi + 2\pi k}{n} \right\}, \qquad k = 0, \ldots, n-1$$

each of angular length $2\pi/n$. On A_k the map p_n is a continuous bijection onto U with inverse $e^{i\psi} \mapsto e^{i(\psi + 2\pi k)/n}$, and this inverse is continuous because ψ may be chosen as a continuous function on the arc U, which omits the point $-w$. So p_n restricts to a homeomorphism of each A_k onto U, and U is evenly covered.

Let h be a deck transformation of p_n. From $h(z)^n = z^n$ it follows that $(h(z)/z)^n = 1$, so the function $z \mapsto h(z)/z$, continuous because multiplication and inversion are continuous on $\mathbb{C} \setminus \{0\}$, takes values in the set μ_n of nth roots of unity. That set is finite, hence discrete, and E is connected, so the function is a constant $\zeta \in \mu_n$ and $h(z) = \zeta z$ is a rotation. Conversely multiplication by any ζ with $\zeta^n = 1$ is a homeomorphism of E satisfying $(\zeta z)^n = z^n$. Hence

$$\operatorname{Deck}(E/S^1) = \{z \mapsto \zeta z \mid \zeta^n = 1\} \cong \mu_n \cong \mathbb{Z}/n\mathbb{Z}$$

the isomorphism sending ζ to the rotation by ζ, and μ_n being cyclic of order n generated by $e^{2\pi i/n}$.

Theorem 6.5.4 predicts this answer from the subgroup. Under the isomorphism $\pi_1(S^1, 1) \cong \mathbb{Z}$ of theorem 6.2.20, which is the lifting correspondence of the covering p of item (1), the generator is the class of the loop $\omega(t) = e^{2\pi i t}$; the same holds for $\pi_1(E, 1)$, since E is another copy of the circle and the loop ω lives in it as well. The image $p_n \circ \omega$ is the loop $t \mapsto e^{2\pi i n t}$, whose lift through p starting at 0 is $t \mapsto nt$, so it corresponds to the integer n. Hence $H = (p_n)_*(\pi_1(E, 1))$ corresponds to $n\mathbb{Z}$. The group $\mathbb{Z}$ is abelian, so $N(H) = \mathbb{Z}$, the covering is regular, and

$$\mathrm{Deck}\,(E/S^1) \cong N(H)/H \cong \mathbb{Z}/n\mathbb{Z}$$

in agreement with the direct computation. The number of sheets is the index $[\mathbb{Z} : n\mathbb{Z}] = n$, which is the cardinality of the deck group, as regularity requires.

Read together with the classification, the two computations show what the theory gives. A covering of B is determined by a subgroup of $\pi_1(B, b_0)$; the coverings attached to normal subgroups are those carrying a group action with quotient B; and the group acting is the quotient of the fundamental group by that subgroup. Questions about coverings have become questions about subgroups, and the passage in the other direction manufactures a space from a subgroup.

Exercise 6.5.1

1. Compute $\mathrm{Deck}\,(E/B)$ for the n-fold cover $S^1 \to S^1$ and for $\mathbb{R} \to S^1$, and check both against theorem 6.5.4.
2. Show that for a covering with path-connected total space, a deck transformation fixing one point is the identity, and that the deck group acts freely. Why does the hypothesis matter?
3. Show that for a regular covering, $\mathrm{Deck}\,(E/B)$ acts transitively on each fibre, and that the quotient $E/\mathrm{Deck}\,(E/B)$ is homeomorphic to B.
4. Exhibit a covering that is not regular, and compute its deck group to show it is smaller than the fibre.
5. Show that for the universal cover, $\mathrm{Deck}\,(E/B) \cong \pi_1(B, b_0)$, and identify the isomorphism concretely for $\mathbb{R} \to S^1$.

6.6 Retractions and Contractions

The computation of $\pi_1(S^1)$ is not an end in itself. Two classical theorems fall straight out of it, and the route to both runs through a single notion: that of collapsing a space onto a subspace while leaving that subspace alone. Retractions appeared already among the examples of quotient maps, and their interaction with π_1 is direct, which is what makes them useful.

Definition 6.6.1: Retraction

If $A \subseteq X$, a *retraction* of X onto A is a continuous map $r : X \to A$ such that $r|_A$ is the identity on A. When such an r exists, A is called a *retract* of X.

Example 6.6.2: Retraction

1. If $A = \{a\}$ is a single point, the constant map $r(x) = a$ is a retraction.
2. If $X = Y \times Z$ and $A = Y \times \{z_0\}$, then $r(y, z) = (y, z_0)$ is a retraction.

3. Let X be the annulus $\{x \in \mathbb{R}^2 \mid 1 \le \|x\| \le 2\}$ and A the inner circle $\|x\| = 1$. Then $r(x) = x/\|x\|$ pushes the annulus onto A and fixes it pointwise, so A is a retract of X.
4. $X = \mathbb{R}$ and $A = \mathbb{R}_{\ge 0}$, with $r(t) = |t|$.

The point of the notion is that a retraction forces an inequality between fundamental groups: the retract cannot have more loops than the space it sits inside.

Lemma 6.6.3: Induced Maps And Retractions

Let A be a retract of X and let $a_0 \in A$. Then the induced map $j_* : \pi_1(A, a_0) \to \pi_1(X, a_0)$ is injective, where $j : A \to X$ is the inclusion.

***Proof*:**
Let $r : X \to A$ be a retraction, so that $r \circ j = \mathrm{id}_A$, the two maps agreeing because r fixes A pointwise. Applying proposition 6.1.30,

$$r_* \circ j_* = (r \circ j)_* = (\mathrm{id}_A)_* = \mathrm{id}_{\pi_1(A,a_0)}$$

so for every $[f] \in \pi_1(A, a_0)$ we have $r_*(j_*([f])) = [f]$. A map with a left inverse is injective: if $j_*([f]) = j_*([g])$, applying r_* gives $[f] = [g]$. Hence j_* is injective, as claimed.

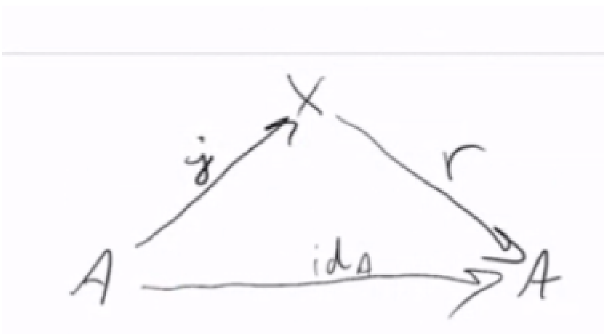

Figure 6.11: The retraction r supplies a left inverse to the map induced by inclusion.

Reading the lemma backwards turns it into an obstruction: a subspace with a bigger fundamental group than its ambient space cannot be a retract. The disc and its boundary circle are exactly such a pair.

Theorem 6.6.4: No Retraction Of The Disc Onto Its Boundary

Let B^2 be the closed unit disc in $\mathbb{R}^2$ and S^1 its boundary circle. There is no retraction of B^2 onto S^1.

***Proof*:**
Suppose one existed. Fix $a_0 \in S^1$. By lemma 6.6.3 the inclusion would induce an injection

$$j_* : \pi_1(S^1, a_0) \to \pi_1(B^2, a_0)$$

The domain is $\mathbb{Z}$ by theorem 6.2.20, which is infinite. The codomain is trivial: B^2 is convex, so example 6.1.23 applies and $\pi_1(B^2, a_0)$ has one element. No injection from an infinite set into a one-element set exists, so the retraction cannot exist either, which is the contradiction wanted.

Exercise 6.6.1

1. Show that a retract of a Hausdorff space is closed, and that a retract of a connected space is connected.
2. Show that S^1 is a retract of $\mathbb{R}^2 \setminus \{0\}$ but not of B^2.
3. Show that if A is a retract of X and X is simply connected, then A is simply connected.
4. Show that a one-point set is a retract of any space, and that this makes the injectivity in lemma 6.6.3 vacuous there.

6.7 Brouwer's Fixed Point Theorem

The obstruction just established is the whole content of the next theorem. A continuous self-map of the disc with no fixed point would supply enough geometric data to build the retraction that cannot exist.

Theorem 6.7.1: Brouwer Fixed Point Theorem

If $f : B^2 \to B^2$ is continuous, then there exists $x \in B^2$ with $f(x) = x$.

***Proof*:**
Suppose for contradiction that $f(x) \neq x$ for every $x \in B^2$. The plan is to build a retraction of B^2 onto S^1, contradicting theorem 6.6.4.

Geometrically: since x and $f(x)$ are distinct points of the disc, they determine a line. Follow the ray from $f(x)$ through x until it leaves the disc, and let $g(x)$ be the point where it meets the boundary. The construction is well defined only because f has no fixed point; where $f(x) = x$ there is no ray to follow, and it is precisely at such a point that any formula for g would break down.

Formally, define $G : B^2 \times \mathbb{R}_{\geq 0} \to \mathbb{R}^2$ by

$$G(x, r) = x + r\,(x - f(x))$$

which traces the ray as r increases. Expanding the squared norm,

$$\|G(x, r)\|^2 = \|x\|^2 + 2r\,x \cdot (x - f(x)) + r^2\,\|x - f(x)\|^2$$

Setting this equal to 1 gives a quadratic in r with coefficients

$$a = \|x - f(x)\|^2, \qquad b = 2\,x \cdot (x - f(x)), \qquad c = \|x\|^2 - 1$$

Here $a > 0$ for every x, since f has no fixed point, and $c \leq 0$ because x lies in the disc. Hence $b^2 - 4ac \geq b^2 \geq 0$ and the unique nonnegative root is

$$r(x) = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \frac{-2\,x \cdot (x - f(x)) + \sqrt{4\,(x \cdot (x - f(x)))^2 + 4\,\|x - f(x)\|^2 \left(1 - \|x\|^2\right)}}{2\,\|x - f(x)\|^2}$$

Nonnegativity holds because $\sqrt{b^2 - 4ac} \geq \sqrt{b^2} = |b| \geq -b$.

Now set $g(x) = G(x, r(x))$, which lies in S^1 by the choice of $r(x)$. The map g is continuous: f is continuous by hypothesis, the expressions for a, b, c are built from it by sums, products and inner products, the denominator $2a$ never vanishes, and the square root is continuous on $[0, \infty)$; proposition 1.2.11 assembles these into continuity of r, hence of g.

Finally, g fixes the boundary. If $\|x\| = 1$ then $c = 0$, so the root reduces to $r(x) = (-b + |b|)/2a$. Moreover

$$b = 2\left(\|x\|^2 - x \cdot f(x)\right) = 2\,(1 - x \cdot f(x)) \geq 0$$

by the Cauchy-Schwarz inequality, since $x \cdot f(x) \leq \|x\|\,\|f(x)\| \leq 1$. So $|b| = b$, giving $r(x) = 0$ and $g(x) = G(x, 0) = x$.

Thus $g : B^2 \to S^1$ is continuous and restricts to the identity on S^1, that is, a retraction of the disc onto its boundary. This contradicts theorem 6.6.4, so f must have a fixed point after all.

Exercise 6.7.1

1. Show that theorem 6.7.1 fails for the open disc and for the circle, and say in each case what goes wrong.
2. Show that Brouwer's theorem holds for any space homeomorphic to B^2, and deduce it for the square $[0, 1]^2$.
3. Show that a continuous $f\colon [0, 1] \to [0, 1]$ has a fixed point, directly from theorem 3.1.23, and note that this is the one-dimensional case.
4. Show that the fixed point in Brouwer's theorem need not be unique, and that the fixed-point *set* does not vary continuously with f.
5. Show that Brouwer's theorem implies that every continuous $v\colon B^2 \to \mathbb{R}^2$ with $v(x) \cdot x < 0$ for all $x \in S^1$, a vector field pointing inward on the boundary, vanishes somewhere.
6. Show that theorem 6.7.1 and theorem 6.6.4 are equivalent, by deriving each from the other.

7

Topological Groups

A group and a topological space are, on their own, unrelated structures; a *topological group* is what results when the two are required to be compatible, so that the group operations respect the topology. The requirement is light (multiplication and inversion are asked only to be continuous), yet its consequences run through the whole chapter. Translation by a fixed element is a homeomorphism, so any two points of the group are topologically indistinguishable: the space is *homogeneous*, and its local structure is already determined by the neighbourhoods of the identity. Most of what follows extracts consequences from this single observation.

Homogeneity makes the separation axioms of chapter 4 degenerate. For a group, the weakest of them, T_0, already forces the Hausdorff property, so a topological group is Hausdorff as soon as its points are topologically distinct. Homogeneity also tames the quotient construction of section 2.4: the quotient G/H carries a group structure compatible with the quotient topology exactly when H is normal, and it is Hausdorff exactly when H is closed, so the algebraic and the topological conditions on H line up. Connectedness, developed in section 3.1, enters through the identity component, a closed normal subgroup whose quotient is totally disconnected; connected groups and totally disconnected groups are the two poles between which the general theory is organized. These are the point-set foundations on which the profinite groups of Galois theory and the Lie groups of geometry are later built.

The chapter treats only the topological side of the subject. Haar measure, the convolution algebra $L^1(G)$, and the unitary representations that make locally compact groups the setting for abstract harmonic analysis belong to the theory of unitary representations, taken up in the companion volume devoted to it; here the locally compact groups of section 7.7 are studied as spaces, not yet as measure spaces.

7.1 Definitions and Examples

A topological group is a set carrying two structures at once, a group law and a topology, coupled by a single compatibility require: the algebraic operations must be continuous. This section fixes that definition, records the small equivalence that lets one check compatibility with a single map rather than two, and stocks the reader with the examples that recur throughout the chapter. The separation behaviour of these groups is subtle and is treated on its own in section 7.4; nothing here presupposes any separation axiom.

The group operations are the multiplication $m\colon G\times G\to G$, $(x,y)\mapsto xy$, and the inversion $\iota\colon G\to G$, $x\mapsto x^{-1}$. Continuity of m is measured against the product topology on $G\times G$ (section 2.2); this is the only reasonable topology to put on the domain, and the choice is what forces the two variables to be controlled jointly rather than one at a time.

Definition 7.1.1: Topological Group

A *topological group* is a group G equipped with a topology such that

1. the multiplication $m\colon G\times G\to G$, $(x,y)\mapsto xy$, is continuous when $G\times G$ carries the product topology, and
2. the inversion $\iota\colon G\to G$, $x\mapsto x^{-1}$, is continuous.

No separation axiom is imposed.

The definition asks continuity of two maps, but a single map records both. The composite $(x,y)\mapsto xy^{-1}$ folds multiplication and inversion together, and its continuity is equivalent to that of the pair. This is the form one most often checks in practice, since a candidate group frequently comes with a natural formula for xy^{-1}.

Proposition 7.1.2: Single-Map Criterion

Let G be a group with a topology. Then G is a topological group if and only if the map

$$d\colon G\times G\to G,\qquad (x,y)\mapsto xy^{-1}$$

is continuous.

***Proof*:**
Suppose first that G is a topological group, so that m and ι are continuous. The map d factors as

$$G\times G\xrightarrow{\mathrm{id}_G\times\iota} G\times G\xrightarrow{m} G$$

where $\mathrm{id}_G\times\iota$ sends (x,y) to (x,y^{-1}). The map $\mathrm{id}_G\times\iota$ is continuous because a map into a product is continuous exactly when its two coordinate functions are (section 2.2), and here the coordinates are the projection $(x,y)\mapsto x$ and the composite $(x,y)\mapsto y\mapsto y^{-1}$, both continuous. Composing with the continuous map m shows that $d=m\circ(\mathrm{id}_G\times\iota)$ is continuous.

Conversely, suppose d is continuous. Setting $x=e$ gives inversion back: the map $y\mapsto(e,y)\mapsto ey^{-1}=y^{-1}$ is the composite of the continuous inclusion $y\mapsto(e,y)$ with d, so ι is continuous. Continuity of the inclusion $y\mapsto(e,y)$ holds because its coordinates are the constant map e

and the identity. Knowing ι is continuous, recover multiplication by feeding an inverted second argument: for $x, y \in G$,

$$xy = x(y^{-1})^{-1} = d(x, y^{-1})$$

so $m = d \circ (\mathrm{id}_G \times \iota)$ is a composite of continuous maps and is therefore continuous. Thus both m and ι are continuous, and G is a topological group, completing the proof.

The criterion is the working tool. To certify that a set with a group law and a topology is a topological group, exhibit a single formula for xy^{-1} and check that formula is continuous; the two-map definition is then automatic. The examples below use it and the raw definition interchangeably, choosing whichever is cleaner for the case at hand.

Two extremes bracket the subject: the discrete topology, where continuity is free and the group structure is unconstrained, and the additive real line, where continuity is the familiar continuity of addition and subtraction from calculus.

Example 7.1.3: Basic Topological Groups
Each of the following is a topological group.

1. *Any group with the discrete topology.* Let G be an arbitrary group and give it the discrete topology, so every subset is open. Then $G \times G$ carries the discrete topology as well, and every map out of a discrete space is continuous. In particular m and ι are continuous, so G is a topological group. A group with the discrete topology is a *discrete group*; the finite groups and the integers $(\mathbb{Z}, +)$ are the standard instances.

2. *The additive reals* $(\mathbb{R}, +)$ *and* $(\mathbb{R}^n, +)$. On $\mathbb{R}$ with its usual topology, the subtraction map $d(x, y) = x - y$ is continuous, being a polynomial in the two coordinates. By proposition 7.1.2, $(\mathbb{R}, +)$ is a topological group. The same argument applies coordinatewise to $(\mathbb{R}^n, +)$: subtraction $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ is $(x, y) \mapsto x - y$, whose components $x_i - y_i$ are each continuous, so the map into the product $\mathbb{R}^n$ is continuous.

3. *The multiplicative reals* $(\mathbb{R}^\times, \cdot)$. Let $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$ with the subspace topology (section 2.1) and multiplication as the group law. The map $d(x, y) = xy^{-1} = x/y$ is continuous on $\mathbb{R}^\times \times \mathbb{R}^\times$, since it is a quotient of the continuous coordinate functions with nonvanishing denominator. Thus $(\mathbb{R}^\times, \cdot)$ is a topological group.

4. *The circle* $\mathbb{S}^1$. Realize the circle as the unit complex numbers $\mathbb{S}^1 = \{z \in \mathbb{C} \big| |z| = 1\}$ under multiplication, with the subspace topology from $\mathbb{C}$. Complex multiplication and complex inversion $z \mapsto z^{-1} = \overline{z}$ are continuous on $\mathbb{S}^1$, the latter because inversion agrees with conjugation on the unit circle. Hence $\mathbb{S}^1$ is a topological group. It is the first example whose underlying space is compact and connected, and it recurs as the target of characters throughout the analysis of commutative topological groups.

These four already display the range. Discreteness makes topology inert and leaves group theory untouched; $(\mathbb{R}^n, +)$ is the local model for the groups of geometry and analysis; $(\mathbb{R}^\times, \cdot)$ shows that removing a point to restore invertibility keeps everything continuous; and $\mathbb{S}^1$ is compact, so it behaves very differently from the line under limiting operations. The verifications all reduced to continuity of a rational formula, which is the pattern the single-map criterion is built to exploit.

A structural feature separates topological groups from bare topological spaces: the group can move any point to any other by a continuous map with a continuous inverse. Fixing an element g and translating produces such a self-homeomorphism, and so does inversion. These maps transport local information from one point to another, and in particular from the identity to everywhere.

Proposition 7.1.4: Translations and Inversion Are Homeomorphisms

Let G be a topological group and $g \in G$. Then the left translation $L_g : G \to G$, $x \mapsto gx$, the right translation $R_g : G \to G$, $x \mapsto xg$, and the inversion $\iota : G \to G$, $x \mapsto x^{-1}$, are homeomorphisms of G. Their inverses are $L_{g^{-1}}$, $R_{g^{-1}}$, and ι itself.

***Proof*:**
Consider first the left translation L_g. It is continuous: the map $x \mapsto (g, x)$ from G to $G \times G$ has coordinates the constant g and the identity, hence is continuous, and L_g is its composite with the continuous multiplication m, so $L_g = m \circ (x \mapsto (g, x))$ is continuous. As a group-theoretic identity, $L_g \circ L_{g^{-1}} = L_{gg^{-1}} = L_e = \mathrm{id}_G$ and likewise $L_{g^{-1}} \circ L_g = \mathrm{id}_G$, so L_g is a bijection with inverse $L_{g^{-1}}$. Since $g^{-1} \in G$ is again an element of the group, $L_{g^{-1}}$ is continuous by the same argument applied to g^{-1}. A continuous bijection with continuous inverse is a homeomorphism, so L_g is one.

The right translation R_g is handled identically, replacing $x \mapsto (g, x)$ by $x \mapsto (x, g)$ and using $R_g \circ R_{g^{-1}} = R_{g^{-1}g} = \mathrm{id}_G$; its inverse is $R_{g^{-1}}$.

For inversion, ι is continuous by definition 7.1.1, and $(x^{-1})^{-1} = x$ gives $\iota \circ \iota = \mathrm{id}_G$, so ι is its own inverse. A continuous map equal to its own two-sided inverse is a homeomorphism, so ι is one, as we sought to show.

The content is that G is *homogeneous*: for any two points x, y the left translation $L_{yx^{-1}}$ carries x to y and is a homeomorphism, so no point is topologically distinguished from another. A neighbourhood of one point is, up to translation, a neighbourhood of any other, which is why the topology of a topological group is controlled by the neighbourhoods of the identity alone. That principle is developed in section 7.2; the present proposition supplies the homeomorphisms it runs on. Inversion, meanwhile, is the symmetry that lets one pass between left and right versions of any statement, exchanging L_g with $R_{g^{-1}}$ under conjugation by ι.

The examples so far are commutative. The matrix groups supply the first noncommutative topological groups and, at the same time, the first whose continuity is not quite a one-line polynomial check, since inversion involves dividing by a determinant. The general linear group is the prototype.

Example 7.1.5: The General Linear Group $\mathrm{GL}_n(\mathbb{R})$

Let $\mathrm{GL}_n(\mathbb{R})$ denote the group of invertible $n \times n$ real matrices under multiplication. Identify the set of all $n \times n$ real matrices with $\mathbb{R}^{n^2}$ by listing the entries, and give $\mathrm{GL}_n(\mathbb{R})$ the subspace topology it inherits as a subset of $\mathbb{R}^{n^2}$.

First, $\mathrm{GL}_n(\mathbb{R})$ is an open subset of $\mathbb{R}^{n^2}$. The determinant $\det : \mathbb{R}^{n^2} \to \mathbb{R}$ is a polynomial in the matrix entries, hence continuous, and

$$\mathrm{GL}_n(\mathbb{R}) = \{A \in \mathbb{R}^{n^2} \mid \det A \neq 0\} = \overset{-1}{\det}\big(\mathbb{R} \setminus \{0\}\big)$$

is the preimage of the open set $\mathbb{R} \setminus \{0\}$ under a continuous map, so it is open. It is therefore a topological space in its own right, an open subspace of Euclidean space.

Multiplication is continuous. The (i, j) entry of a product AB is $\sum_{k=1}^{n} A_{ik} B_{kj}$, a polynomial in the $2n^2$ entries of A and B. Each entry of the product map $\mathrm{GL}_n(\mathbb{R}) \times \mathrm{GL}_n(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R})$ is thus continuous, so the map into the product space $\mathbb{R}^{n^2}$ is continuous, and it lands in the open subset $\mathrm{GL}_n(\mathbb{R})$.

Inversion is continuous. Cramer's rule expresses the inverse as

$$(A^{-1})_{ij} = \frac{(-1)^{i+j}\, M_{ji}(A)}{\det A}$$

where $M_{ji}(A)$ is the (j,i) minor of A, itself a polynomial in the entries of A. On $\mathrm{GL}_n(\mathbb{R})$ the denominator $\det A$ never vanishes, so each entry of A^{-1} is a rational function of the entries of A with nonvanishing denominator, hence continuous. Every entry of the inversion map is continuous, so the map into $\mathbb{R}^{n^2}$ is continuous, and again it lands in $\mathrm{GL}_n(\mathbb{R})$. By definition 7.1.1, $\mathrm{GL}_n(\mathbb{R})$ is a topological group, and it is noncommutative for $n \geq 2$.

The determinant does double duty: its nonvanishing carves out $\mathrm{GL}_n(\mathbb{R})$ as an open set, and the same nonvanishing is what makes the Cramer formula for the inverse continuous. This example is the entry point to Lie theory, where $\mathrm{GL}_n(\mathbb{R})$ and its closed subgroups are studied as smooth manifolds and the continuity established here is refined to smoothness; that analytic layer is developed in the series' companion volume on Lie groups.

Exercise 7.1.1

1. Use proposition 7.1.2 to verify that $GL_n(\mathbb{R})$ is a topological group, and that the upper-triangular invertible matrices form one.
2. Show that an infinite group with the co-finite topology is *not* a topological group, and that a finite one is.
3. Show that the product of two topological groups is a topological group, and that a subgroup with the subspace topology is one.
4. Show that if $A \subseteq G$ is open then AB is open for every $B \subseteq G$, and give closed A, B in a topological group whose product is not closed.
5. Show that if a topological group is locally compact at the identity then it is locally compact at every point.

7.2 Translations and Homogeneity

That each translation L_g is a homeomorphism was recorded when the structure maps were introduced (proposition 7.1.4). This section draws out its consequence: a topological group looks the same at every point, so its topology is stored entirely in the neighbourhoods of the identity. The section extracts a base at e, transports it to every point by translation, and isolates the one closure-theoretic tool (a symmetric neighbourhood that halves under multiplication) that the separation and subgroup arguments of the following sections will use repeatedly.

A space is called *homogeneous* when its self-homeomorphisms act transitively on points: given any two points, some homeomorphism of the space onto itself carries the first to the second. For a group the requisite homeomorphism is already at hand. To move x to y, left-multiply by yx^{-1}; this is the translation $L_{yx^{-1}}$, and translations are homeomorphisms.

Proposition 7.2.1: Homogeneity of a Topological Group

Let G be a topological group. For any two points $x, y \in G$ the left translation $L_{yx^{-1}}: G \to G$ is a homeomorphism carrying x to y. In particular G is a homogeneous space: its self-homeomorphisms act transitively. Consequently, if P is a topological property of a point that is preserved by homeomorphisms and P holds at one point of G, then P holds at every point of G.

***Proof*:**
Fix $x, y \in G$ and set $g = yx^{-1}$. By proposition 7.1.4 the left translation $L_g: G \to G$, $t \mapsto gt$, is a homeomorphism, with continuous inverse $L_{g^{-1}}$. It sends x to $gx = yx^{-1}x = y$, so it carries x to y. Since x and y were arbitrary, the self-homeomorphisms of G act transitively, which is the definition of a homogeneous space.

For the final claim, suppose P is a property of a point invariant under homeomorphisms and that P holds at some point $x_0 \in G$. Let $y \in G$ be arbitrary. The homeomorphism $L_{yx_0^{-1}}$ carries x_0 to y, and P is preserved by homeomorphisms, so P holds at y. As y was arbitrary, P holds at every point of G, as we sought to show.

The force of homogeneity is that a topological group has no distinguished points: any local question (whether the space is locally compact at a point, locally connected there, first countable there, metrizable near there) has the same answer at every point, because a translation moves the answer from one point to another without disturbing it. This is why local hypotheses on a topological group are always phrased at the identity alone. To say "G is locally compact" it suffices to produce one compact neighbourhood of e; translation then furnishes a compact neighbourhood of every point. The identity e is not special as a point of the space, since homogeneity forbids that, but it is distinguished as a *marker*, the one point whose local data are recorded. Everything in the following sections that reads "a neighbourhood of e" is implicitly a statement about every point, converted to a statement about one.

The property in question must be local and homeomorphism-invariant for the transport to work. Being isolated is such a property, and it already exposes how rigid homogeneity is.

Example 7.2.2: Discreteness Is All-or-Nothing
Let G be a topological group in which the identity e is an isolated point, meaning $\{e\}$ is open. The property "$\{x\}$ is open" is local and preserved by homeomorphisms. It holds at $x = e$, so by proposition 7.2.1 it holds at every $x \in G$: the translate $L_x(\{e\}) = \{x\}$ is the image of an open set under the homeomorphism L_x, hence open. Every singleton is therefore open, and G carries the discrete topology.

Thus a topological group is either discrete or has no isolated points at all; there is no middle case in which some points are isolated and others are not. Concretely, in $(\mathbb{R}, +)$ no point is isolated, so $\mathbb{R}$ is not discrete; in $\mathbb{Z}$ viewed with the subspace topology from $\mathbb{R}$, the point 0 is isolated (the interval $(-1, 1)$ meets $\mathbb{Z}$ only in 0), so $\mathbb{Z}$ is discrete. The dichotomy rules out any topology on $\mathbb{Z}$, compatible with the group operations, that isolates 0 without isolating every integer.

Homogeneity says the local topology is the same everywhere; the next result says how much of it a single point already records. Because every neighbourhood of x is the translate of a neighbourhood of e, and conversely, the neighbourhood filter at e determines the neighbourhood filter at every other point, and hence the whole topology. The following proposition makes this precise in the two forms used in practice: a criterion for openness phrased at e, and the transport of a neighbourhood base at e to a base at each point.

Proposition 7.2.3: The Topology Is Determined at the Identity

Let G be a topological group and let $\mathcal{N}$ denote the filter of neighbourhoods of the identity e.

1. A subset $U \subseteq G$ is open if and only if for every $g \in U$ the set $g^{-1}U$ belongs to $\mathcal{N}$, that is, $g^{-1}U$ is a neighbourhood of e.

2. If $\mathcal{B}$ is a neighbourhood base at e, then for each $g \in G$ the translated family $g\mathcal{B} = \{\, gV \,\big|\, V \in \mathcal{B} \,\}$ is a neighbourhood base at g. Consequently $\{\, gV \,\big|\, g \in G,\ V \in \mathcal{B} \,\}$ is a base for the topology of G.

***Proof*:**
Throughout, recall that for fixed g the left translation L_g and its inverse $L_{g^{-1}} = L_g^{-1}$ are homeomorphisms (proposition 7.1.4), so L_g and $L_{g^{-1}}$ carry open sets to open sets and neighbourhoods of a point to neighbourhoods of its image. Note that $L_{g^{-1}}(U) = g^{-1}U$ and $L_{g^{-1}}$ sends g to e.

Part (1). Suppose first that U is open and let $g \in U$. Then $g^{-1}U = L_{g^{-1}}(U)$ is open, being the image of the open set U under the homeomorphism $L_{g^{-1}}$; and it contains $L_{g^{-1}}(g) = e$. An open set containing e is a neighbourhood of e, so $g^{-1}U \in \mathcal{N}$.

Conversely, suppose $g^{-1}U \in \mathcal{N}$ for every $g \in U$. Fix $g \in U$. Since $g^{-1}U$ is a neighbourhood of e, its interior $\operatorname{int}(g^{-1}U)$ is an open set containing e. Applying the homeomorphism L_g, the set $L_g\big(\operatorname{int}(g^{-1}U)\big) = g\operatorname{int}(g^{-1}U)$ is open and contains $L_g(e) = g$; and it is contained in $g\,(g^{-1}U) = U$. Thus every point g of U has an open neighbourhood contained in U, so U is open.

Part (2). Fix $g \in G$. Each gV with $V \in \mathcal{B}$ is a neighbourhood of g: writing V as a neighbourhood of e, the homeomorphism L_g carries it to a neighbourhood $L_g(V) = gV$ of $L_g(e) = g$. It remains to check cofinality. Let W be an arbitrary neighbourhood of g. Then $g^{-1}W = L_{g^{-1}}(W)$ is a neighbourhood of $L_{g^{-1}}(g) = e$, so by the base property of $\mathcal{B}$ there is $V \in \mathcal{B}$ with $V \subseteq g^{-1}W$. Applying L_g gives $gV \subseteq g\,(g^{-1}W) = W$. Hence every neighbourhood of g contains a member of $g\mathcal{B}$, and $g\mathcal{B}$ is a neighbourhood base at g.

For the base claim we may assume each $V \in \mathcal{B}$ is open: the interiors $\{\operatorname{int}V \,\big|\, V \in \mathcal{B}\}$ again form a neighbourhood base at e (each $\operatorname{int}V$ is an open neighbourhood of e contained in V, and any neighbourhood of e contains some $V \supseteq \operatorname{int}V$), so replacing $\mathcal{B}$ by this base changes nothing in parts (1) and (2). With $\mathcal{B}$ open, every member gV is open, being $L_g(V)$. To see that $\{\, gV \,\big|\, g \in G,\ V \in \mathcal{B} \,\}$ is a base, let U be open and $h \in U$. Since $g\mathcal{B}$ with $g = h$ is a neighbourhood base at h and U is a neighbourhood of h, there is $V \in \mathcal{B}$ with $h \in hV \subseteq U$. Thus every open set is a union of members of the family, so it is a base for the topology of G, completing the proof.

The two parts say the same thing from opposite directions. Part (2) is the constructive form: to specify the topology of a group it is enough to hand over a neighbourhood base at e, and translation manufactures a base at every other point automatically. This is how topological groups are most often built: one prescribes the neighbourhoods of the identity and lets the group structure propagate them. Part (1) is the recognition form: to test whether a candidate set is open, one need not inspect it near every point but only ask, at each of its points g, whether sliding that point back to the identity by $L_{g^{-1}}$ lands the set inside a fixed neighbourhood of e. Both forms reduce a global topological question to data at the single point e, which is exactly what homogeneity licensed.

Example 7.2.4: The Neighbourhood Base at the Identity of $(\mathbb{R}, +)$

On the additive group $(\mathbb{R}, +)$ from example 7.1.3, the symmetric intervals $\mathcal{B} = \{\, (-\varepsilon, \varepsilon) \,|\, \varepsilon > 0 \,\}$ form a neighbourhood base at the identity 0: every neighbourhood of 0 contains some $(-\varepsilon, \varepsilon)$, and each $(-\varepsilon, \varepsilon)$ is itself a neighbourhood of 0. Here the group is written additively, so translation by g is the map $x \mapsto g + x$ and the translate of V is $g + V$.

By proposition 7.2.3(2), translating this base by g gives the family $g + \mathcal{B} = \{\, (g - \varepsilon, g + \varepsilon) \,|\, \varepsilon > 0 \,\}$, which is a neighbourhood base at g. These are the open intervals centred at g, and they are precisely the standard basic neighbourhoods of g in $\mathbb{R}$: the group structure reproduces the usual metric topology from the single datum of the intervals around 0. To test openness by proposition 7.2.3(1), a set $U \subseteq \mathbb{R}$ is open exactly when for each $g \in U$ the translate $-g + U = \{\, u - g \,|\, u \in U \,\}$ contains some $(-\varepsilon, \varepsilon)$, that is, when U contains the interval $(g - \varepsilon, g + \varepsilon)$; this recovers the familiar definition of an open subset of $\mathbb{R}$.

Having reduced the topology to the neighbourhoods of e, the remaining task is to control how those neighbourhoods interact with the group operations. In a metric space the basic manoeuvre is to halve a radius: given a ball of radius r, one works inside the ball of radius $r/2$, whose points combine to stay within distance r. A topological group has no metric to halve, but the continuity of multiplication supplies a substitute. The next definition and lemma isolate it.

Two features of a neighbourhood matter for the arguments to come. First, one often needs a neighbourhood together with its set of inverses, and it is convenient when the two coincide. Second, one needs a neighbourhood small enough that the product of any two of its points stays inside a prescribed target. The definition addresses the first feature.

Definition 7.2.5: Symmetric Neighbourhood

Let G be a topological group with identity e. For a subset $V \subseteq G$, write $V^{-1} = \{\, v^{-1} \,|\, v \in V \,\}$ for its image under inversion. A neighbourhood V of e is *symmetric* if $V = V^{-1}$, that is, $v \in V$ if and only if $v^{-1} \in V$.

For any neighbourhood U of e, the sets $U \cap U^{-1}$ and $U \cup U^{-1}$ are symmetric, since inversion is an involution and interchanges the two sets in each pair. Because inversion ι is a homeomorphism fixing e, the set $U^{-1} = \iota(U)$ is again a neighbourhood of e, so $U \cap U^{-1}$ is a symmetric neighbourhood of e contained in U. Symmetric neighbourhoods are therefore plentiful: every neighbourhood of e contains one.

Example 7.2.6: Symmetric Neighbourhoods in $(\mathbb{R}, +)$ and $\mathrm{GL}_n(\mathbb{R})$

In the additive group $(\mathbb{R}, +)$ inversion is $x \mapsto -x$, so $V^{-1} = -V$ and symmetry of a neighbourhood V of 0 means $-V = V$. The interval $(-\varepsilon, \varepsilon)$ is symmetric, whereas $(-1, 2)$ is not: its inverse is $(-2, 1)$, and the intersection $(-1, 2) \cap (-2, 1) = (-1, 1)$ is the symmetric neighbourhood the recipe above extracts from it.

In the matrix group $\mathrm{GL}_n(\mathbb{R})$ from example 7.1.5 inversion is $A \mapsto A^{-1}$, and a neighbourhood V of the identity matrix is symmetric when $A \in V$ forces $A^{-1} \in V$. Fix a submultiplicative matrix norm $\|\cdot\|$ with $\|I\| = 1$. The set $V = \{ A \in \mathrm{GL}_n(\mathbb{R}) \,\big|\, \|A - I\| < \frac{1}{2} \text{ and } \|A^{-1} - I\| < \frac{1}{2} \}$ is a neighbourhood of I, and it is symmetric by construction: the two defining inequalities are interchanged by $A \mapsto A^{-1}$, so $A \in V$ if and only if $A^{-1} \in V$.

The symmetric neighbourhood alone does not yet halve anything; for that one needs the product of two of its points to remain in a target set. Continuity of multiplication at the identity delivers this.

Lemma 7.2.7: Halving a Neighbourhood

Let G be a topological group and let U be a neighbourhood of the identity e. Then there exists a symmetric neighbourhood V of e with

$$VV \subseteq U$$

where $VV = \{ v_1 v_2 \,\big|\, v_1, v_2 \in V \}$. In particular $V \subseteq VV \subseteq U$, and $V^{-1} = V$.

***Proof*:**

Multiplication $m \colon G \times G \to G$ is continuous, and $m(e, e) = e \in U$. Since U is a neighbourhood of e, it contains an open set U' with $e \in U'$. By continuity of m at the point (e, e), the preimage $m^{-1}(U')$ is an open subset of $G \times G$ containing (e, e). By the definition of the product topology (section 2.2), a basic open neighbourhood of (e, e) has the form $W_1 \times W_2$ with W_1, W_2 open in G; hence there exist open sets W_1, W_2 with $e \in W_1$, $e \in W_2$, and

$$W_1 \times W_2 \subseteq m^{-1}(U')$$

Applying m, this says $w_1 w_2 \in U'$ for all $w_1 \in W_1$ and $w_2 \in W_2$; that is, $W_1 W_2 \subseteq U' \subseteq U$.

Set $W = W_1 \cap W_2$. Then W is an open neighbourhood of e with $WW \subseteq W_1 W_2 \subseteq U$, since $W \subseteq W_1$ and $W \subseteq W_2$. To make the neighbourhood symmetric, put

$$V = W \cap W^{-1}$$

As noted after definition 7.2.5, $W^{-1} = \iota(W)$ is an open neighbourhood of e because inversion ι is a homeomorphism fixing e; hence V is an open neighbourhood of e, and $V = V^{-1}$ because $V^{-1} = (W \cap W^{-1})^{-1} = W^{-1} \cap W = V$. Since $V \subseteq W$, we have $VV \subseteq WW \subseteq U$. Finally $e \in V$ gives $V = Ve \subseteq VV$, so $V \subseteq VV \subseteq U$, completing the proof.

This lemma is the group-theoretic replacement for halving a radius, and the analogy is exact. The target U plays the role of the ball of radius r; the symmetric V plays the role of the ball of radius $r/2$;

and the inclusion $VV \subseteq U$ is the triangle inequality $\frac{r}{2} + \frac{r}{2} = r$, rewritten multiplicatively. Symmetry, $V = V^{-1}$, is the counterpart of a ball being centred: it lets one pass freely between v and v^{-1} without leaving V, just as a centred ball treats x and $-x$ alike. The two properties combine into the estimate the following sections lean on: if a and b each lie within V of a point, then $a^{-1}b \in VV \subseteq U$, so a and b are within U of each other. This is precisely the manipulation a metric makes routine, now available in any topological group with no metric in sight.

The halving can be iterated, which is what turns a single application into a genuine substitute for a sequence of radii $r, r/2, r/4, \ldots$ tending to 0.

Example 7.2.8: Iterated Halving and Cubes of a Neighbourhood

Let U be a neighbourhood of e in a topological group G. Applying lemma 7.2.7 to U produces a symmetric V with $VV \subseteq U$. Applying it again to V produces a symmetric V' with $V'V' \subseteq V$; then

$$V'V'V'V' \subseteq VV \subseteq U$$

so V' is a symmetric neighbourhood whose fourfold product lands in U. Repeating n times yields, for any n, a symmetric neighbourhood $V^{(n)}$ of e with 2^n factors of $V^{(n)}$ contained in U. This is the exact analogue of subdividing a radius r into 2^n steps of size $r/2^n$.

A single application already gives control of triple products, which is the form the closure arguments use. Given U, take the symmetric V with $VV \subseteq U$, then a symmetric V_1 with $V_1V_1 \subseteq V$. Then

$$V_1V_1V_1 \subseteq V_1V_1V_1V_1 \subseteq VV \subseteq U$$

using $e \in V_1$ to insert a fourth factor. Thus every neighbourhood U of e contains a symmetric V_1 with $V_1V_1V_1 \subseteq U$; concretely in $(\mathbb{R}, +)$, starting from $U = (-1, 1)$ one may take $V = (-\frac{1}{2}, \frac{1}{2})$ and $V_1 = (-\frac{1}{4}, \frac{1}{4})$, and indeed $V_1 + V_1 + V_1 = (-\frac{3}{4}, \frac{3}{4}) \subseteq (-1, 1)$.

Exercise 7.2.1

1. Show that a topological group is discrete if and only if $\{e\}$ is open, and that a subgroup is discrete if and only if some neighbourhood of e meets it only in e.
2. Use proposition 7.2.3 to show that a topological group is first-countable if and only if e has a countable neighbourhood base.
3. Show that every neighbourhood U of e contains a symmetric neighbourhood V with $VV \subseteq U$, and that for each n there is a symmetric neighbourhood V_n of e whose 2^n-fold product $V_n \cdots V_n$ lies in U.
4. Show that a homomorphism of topological groups is continuous everywhere as soon as it is continuous at e.

7.3 Subgroups and Closures

The subsets that matter most in a topological group are the ones that respect both structures at once: the subgroups. This section fixes how a subgroup inherits a topology, and then asks how the two structures interact when a subgroup is enlarged to its closure or is already open. Three facts emerge, each a consequence of the continuity of the operations rather than of any special feature of the group. A subgroup is again a topological group; the closure of a subgroup is a subgroup, and stays normal if the subgroup was; and an open subgroup is automatically closed. The last of these, together with a symmetric neighbourhood argument, pins down when a discrete subgroup sits inside its ambient group as a closed set.

The starting point is that a subgroup carries a topology for free, from the ambient space, and that this topology is compatible with its group law.

Proposition 7.3.1: Subgroups Are Topological Groups

Let G be a topological group and $H \leq G$ a subgroup. Equipped with the subspace topology from G (section 2.1), H is a topological group.

***Proof*:**
The subspace topology makes the inclusion $j\colon H \to G$ continuous, and it is the coarsest topology with this property. Two maps must be shown continuous: the multiplication $m_H\colon H \times H \to H$ and the inversion $\iota_H\colon H \to H$ of the subgroup.

Multiplication. Since H is closed under the group law, m_H is the map $(h, h') \mapsto hh'$ obtained by restricting the ambient multiplication $m\colon G \times G \to G$ to $H \times H$ and observing that its values lie in H. Concretely, $j \circ m_H = m \circ (j \times j)$ as maps $H \times H \to G$. The product map $j \times j\colon H \times H \to G \times G$ is continuous because each factor j is, and m is continuous by hypothesis, so the composite $m \circ (j \times j)$ is continuous into G. Its image lies in H, and by the universal property of the subspace topology a map into H is continuous exactly when its composite with j is continuous into G. Hence m_H is continuous.

Inversion. The same argument applies to $\iota_H\colon H \to H$, $h \mapsto h^{-1}$. Here $j \circ \iota_H = \iota \circ j$, the composite of the continuous inclusion with the continuous ambient inversion ι, so $j \circ \iota_H$ is continuous into G with image in H; therefore ι_H is continuous. Both operations are continuous, so H is a topological group, as we sought to show.

The proposition says nothing surprising and everything necessary: it is the sanity check that the definition of a topological group is stable under passing to subgroups, so that every subgroup encountered from now on is silently a topological group in its own right. The mechanism is entirely formal, resting only on the two defining properties of the subspace topology, that the inclusion is continuous and that continuity of a map into H is tested by composing with the inclusion. Nothing about the group is used beyond closure of H under the operations.

Example 7.3.2: The Circle Inside the Nonzero Complexes

The unit circle $\mathbb{S}^1 = \{z \in \mathbb{C} \big| |z| = 1\}$ is a subgroup of the multiplicative group $\mathbb{C}^\times = (\mathbb{C} \setminus \{0\}, \cdot)$, since $|zw| = |z|\,|w|$ and $|z^{-1}| = |z|^{-1}$. Give $\mathbb{C}^\times$ its topology as a subspace of $\mathbb{C} \cong \mathbb{R}^2$. By proposition 7.3.1, $\mathbb{S}^1$ is a topological group in the subspace topology, and that subspace topology is exactly the usual topology of the circle: a basic open set is the intersection of $\mathbb{S}^1$ with an open disc in $\mathbb{C}^\times$, which is an open arc. Multiplication of unit complex numbers adds arguments, and this operation is continuous in the arc topology, matching the general statement. The same reasoning realizes the group μ_n of n-th roots of unity as a (finite, hence discrete) subgroup of $\mathbb{S}^1$.

Enlarging a subgroup to its closure produces a larger set, and the question is whether that larger set remains a subgroup. It does, and the reason is that the group operations are continuous: continuity forces a map to send the closure of a set into the closure of its image, which is enough to carry the subgroup axioms across to the closure. The one inclusion that does the work is $\overline{A}\,\overline{B} \subseteq \overline{AB}$, valid for any two subsets.

Proposition 7.3.3: The Closure of a Subgroup Is a Subgroup

Let G be a topological group and $H \leq G$ a subgroup. Then the closure $\overline{H}$ is a subgroup of G. If moreover $H \trianglelefteq G$, then $\overline{H} \trianglelefteq G$.

***Proof*:**
The argument rests on one inclusion, which is worth isolating. For any subsets $A, B \subseteq G$, write $AB = \{ab \big| a \in A,\ b \in B\}$ for their product. Then

$$\overline{A}\,\overline{B} \subseteq \overline{AB} \tag{7.1}$$

To see (7.1), recall that a continuous map f satisfies $f(\overline{S}) \subseteq \overline{f(S)}$ for every set S, since $f^{-1}(\overline{f(S)})$ is a closed set containing S and hence containing $\overline{S}$. Apply this to the continuous multiplication $m\colon G \times G \to G$ with $S = A \times B$. In the product topology $\overline{A \times B} = \overline{A} \times \overline{B}$ (section 2.2), and $m(A \times B) = AB$, so

$$\overline{A}\,\overline{B} = m\left(\overline{A} \times \overline{B}\right) = m\left(\overline{A \times B}\right) \subseteq \overline{m(A \times B)} = \overline{AB}$$

which is (7.1).

Closure under multiplication. Since H is a subgroup, $HH = H$. Taking $A = B = H$ in (7.1) gives $\overline{H}\,\overline{H} \subseteq \overline{HH} = \overline{H}$, so the product of any two elements of $\overline{H}$ again lies in $\overline{H}$.

Closure under inversion. The inversion $\iota\colon G \to G$ is a homeomorphism, being continuous with continuous inverse ι itself. A homeomorphism commutes with closure, so $\iota(\overline{H}) = \overline{\iota(H)}$. Now $\iota(H) = H^{-1} = H$ because H is a subgroup, whence $\overline{H}^{-1} = \iota(\overline{H}) = \overline{H}$: the closure is closed under inversion. Finally $e \in H \subseteq \overline{H}$, so $\overline{H}$ contains the identity. Thus $\overline{H}$ is a subgroup.

Normality. Suppose $H \trianglelefteq G$, and fix $g \in G$. Conjugation $c_g\colon x \mapsto gxg^{-1}$ is the composite $L_g \circ R_{g^{-1}}$ of two translations, each a homeomorphism, so c_g is a homeomorphism and again commutes with closure: $c_g(\overline{H}) = \overline{c_g(H)}$. Normality of H gives $c_g(H) = gHg^{-1} = H$, so $g\overline{H}g^{-1} = c_g(\overline{H}) = \overline{H}$. As $g \in G$ was arbitrary, $\overline{H}$ is normal, completing the proof.

The proposition lets one pass freely between a subgroup and its closure while staying inside the category of subgroups, a move used constantly: to build closed subgroups, one takes closures without fear of leaving the class, and normality survives the passage. The inclusion (7.1) is the entire content, and it is one instance of the recurring principle that continuity of the structure maps transports algebraic closure properties to topological closures. The same inclusion, applied to a single point $A = \{e\}$, will identify $\overline{\{e\}}$ as a normal subgroup and drive the separation theory of the next section.

Example 7.3.4: The Rationals in the Reals

Take $G = (\mathbb{R}, +)$ and the subgroup $H = \mathbb{Q}$ of rationals, written additively. The rationals are dense in $\mathbb{R}$, so $\overline{\mathbb{Q}} = \mathbb{R}$. Proposition 7.3.3 predicts that $\overline{\mathbb{Q}}$ is a subgroup, and indeed $\mathbb{R}$ is a subgroup of itself, the whole group. This is the extreme case, where the closure swallows all of G. A less degenerate instance is the cyclic subgroup $\mathbb{Z}\alpha = \{n\alpha \big| n \in \mathbb{Z}\}$ generated by an irrational α inside the circle $\mathbb{R}/\mathbb{Z}$: its closure is the entire circle, again a subgroup, whereas for rational $\alpha = p/q$ the subgroup $\mathbb{Z}\alpha$ is already the finite (hence closed) group of q-th roots of the identity. In each case the closure is a subgroup, exactly as the proposition asserts.

There is a second way a subgroup can interact with the topology, complementary to being closed: it can be open. Openness is a strong hypothesis for a subgroup, because a subgroup is a union of cosets and each coset is a translate of the subgroup. Translating an open set by a homeomorphism keeps it open, so if the subgroup is open then so is every one of its cosets, and the complement, being a union of the other cosets, is open as well.

Proposition 7.3.5: Open Subgroups Are Closed

Let G be a topological group and $H \leq G$ an open subgroup. Then H is closed.

***Proof*:**
The group G is the disjoint union of the left cosets of H: distinct cosets are disjoint, and every $g \in G$ lies in gH. The complement of H is therefore the union of all cosets other than H itself,

$$G \setminus H = \bigcup_{gH \neq H} gH$$

where the union runs over the nontrivial cosets. Each coset gH is the image $L_g(H)$ of the open set H under the left translation L_g, which is a homeomorphism, so gH is open. A union of open sets is open, so $G \setminus H$ is open, and hence H is closed, as we sought to show.

The proposition exposes a rigidity peculiar to groups: for a subgroup there is no room between open and closed, since being open forces being closed. The reason is homogeneity. A single open subgroup is copied by translation onto each of its cosets, tiling the whole group into open pieces of equal size, and H is one tile among a partition of the space into open tiles. The complement is a union of the remaining tiles, so it too is open. Read through the quotient, the statement says that an open subgroup has *discrete* quotient: the cosets are the points of G/H, each is open, so every point of G/H is open and the quotient topology is discrete. An open subgroup thus carves off a discrete quotient, and the coarser the group is near the identity, the fewer open subgroups it admits.

Example 7.3.6: Open Subgroups of the Reals and of a Discrete Group

In $(\mathbb{R}, +)$ the only open subgroup is $\mathbb{R}$ itself. Any subgroup containing an open interval around 0 contains an interval $(-\varepsilon, \varepsilon)$, hence all integer multiples of every point in it, hence all of $\mathbb{R}$ by the Archimedean property; and $\{0\}$ is not open. So proposition 7.3.5 says only that $\mathbb{R}$ is closed in $\mathbb{R}$, which is vacuous, and this reflects that $\mathbb{R}$ is connected and thus has no proper open-and-closed subgroup. The content of the proposition appears where a group has many open subgroups. In a discrete group every subgroup is open, and the proposition recovers the obvious fact that every subgroup is then also closed. The useful setting is intermediate, such as the additive group $\mathbb{Z}_p$ of

p-adic integers, where the subgroups $p^n\mathbb{Z}_p$ are open, and proposition 7.3.5 certifies each of them closed with discrete finite quotient $\mathbb{Z}_p/p^n\mathbb{Z}_p \cong \mathbb{Z}/p^n\mathbb{Z}$; these are the profinite groups of Galois theory.

The interplay between open and closed subgroups feeds directly into a question about discrete subgroups. A discrete subgroup need not be open, since the ambient topology can be far finer than the discrete one, and so the previous proposition does not apply to it. Nevertheless a discrete subgroup of a Hausdorff group is closed, and the proof isolates each of its points by a small symmetric neighbourhood, using the Hausdorff hypothesis to rule out any limit point outside the subgroup. The Hausdorff hypothesis is stated explicitly because the argument leans on it directly: it separates a putative limit point from the unique subgroup element sitting near it, and that separation is exactly what Hausdorffness provides. Without separation the conclusion fails, as the trivial subgroup $\{e\}$ shows in the indiscrete topology on any group with more than one element, where $\overline{\{e\}}$ is the whole group even though $\{e\}$ is discrete.

Corollary 7.3.7: Discrete Subgroups of Hausdorff Groups Are Closed

Let G be a Hausdorff topological group and $H \le G$ a discrete subgroup, meaning that H carries the discrete topology as a subspace of G. Then H is closed in G.

***Proof*:**
Discreteness of H means that the singleton $\{e\}$ is open in the subspace topology, so there is an open set U of G with

$$U \cap H = \{e\} \tag{7.2}$$

By lemma 7.2.7 choose a symmetric neighbourhood V of e, so $V = V^{-1}$, with $VV \subseteq U$. The claim is that $\overline{H} = H$, and since $H \subseteq \overline{H}$ always, it suffices to prove $\overline{H} \subseteq H$. Fix $x \in \overline{H}$.

Step 1: the neighbourhood xV of x meets H in at most one point. Suppose $h_1, h_2 \in xV \cap H$, say $h_1 = xv_1$ and $h_2 = xv_2$ with $v_1, v_2 \in V$. Then

$$h_2^{-1}h_1 = v_2^{-1}x^{-1}xv_1 = v_2^{-1}v_1 \in V^{-1}V = VV \subseteq U$$

using $V = V^{-1}$ for the equality $V^{-1}V = VV$. On the other hand $h_2^{-1}h_1 \in H$ because H is a subgroup, so $h_2^{-1}h_1 \in U \cap H = \{e\}$ by (7.2). Therefore $h_2^{-1}h_1 = e$, that is $h_1 = h_2$. So $xV \cap H$ has at most one element.

Step 2: the neighbourhood xV of x meets H. The translate $xV = L_x(V)$ is a neighbourhood of x, being the image of a neighbourhood of e under the homeomorphism L_x. Since $x \in \overline{H}$, every neighbourhood of x contains a point of H, so $xV \cap H \neq \emptyset$. Combined with Step 1, this gives $xV \cap H = \{h\}$ for a single element $h \in H$.

Step 3: the Hausdorff hypothesis forces $x = h$. Suppose to the contrary that $x \neq h$. Because G is Hausdorff, there is an open set N with $x \in N$ and $h \notin N$. Then $N \cap xV$ is again a neighbourhood of x, and it excludes h, so

$$(N \cap xV) \cap H \subseteq (xV \cap H) \setminus \{h\} = \emptyset$$

Thus the neighbourhood $N \cap xV$ of x contains no point of H, contradicting $x \in \overline{H}$. Therefore $x = h$, and in particular $x \in H$.

Every $x \in \overline{H}$ lies in H, so $\overline{H} = H$ and H is closed, completing the proof.

The corollary is the point-set foundation for treating discrete subgroups as closed subobjects, so that quotients by them, such as the tori $\mathbb{R}^n/\mathbb{Z}^n$ or the modular quotients built from lattices in a Lie group, inherit good separation properties from the ambient group. The proof is a template worth remembering: a discrete subgroup comes with a neighbourhood of the identity that isolates it, one shrinks that neighbourhood symmetrically so that translates overlap at most trivially, and Hausdorffness then prevents any stray limit point from hiding between an element of the subgroup and its would-be accumulation. The three ingredients, a symmetric shrinking neighbourhood, the subgroup structure, and the separation axiom, recur throughout the analysis of discrete subgroups.

Example 7.3.8: Lattices in Euclidean Space

The integer lattice $\mathbb{Z}^n$ is a discrete subgroup of the Hausdorff group $(\mathbb{R}^n, +)$: the open ball U of radius $\frac{1}{2}$ about the origin meets $\mathbb{Z}^n$ only in 0, so $\{0\}$ is open in the subspace topology and $\mathbb{Z}^n$ is discrete. By corollary 7.3.7, $\mathbb{Z}^n$ is closed in $\mathbb{R}^n$, which one also sees directly since its complement is a union of open unit cubes with the lattice points removed. Closedness is what makes the quotient $\mathbb{R}^n/\mathbb{Z}^n$, the n-torus, a Hausdorff topological group rather than a pathological quotient. Contrast this with the dense subgroup $\mathbb{Z} + \mathbb{Z}\sqrt{2}$ of $\mathbb{R}$: it is a subgroup, but it is *not* discrete, since every neighbourhood of 0 contains infinitely many of its points, and correspondingly it is not closed, its closure being all of $\mathbb{R}$. The corollary applies precisely to the discrete case, and its hypothesis of discreteness is exactly what the dense example fails.

Exercise 7.3.1

1. Show that the closure of a normal subgroup is normal, and that in a Hausdorff group the closure of an abelian subgroup is abelian.
2. Show that a subgroup of index 2 is normal, and that every coset of an open subgroup is clopen.
3. Show that $\mathbb{Z} \subseteq \mathbb{R}$ is closed and discrete, while $\mathbb{Q} \subseteq \mathbb{R}$ is dense and not discrete, and that $\mathbb{Z} + \mathbb{Z}\sqrt{2} \subseteq \mathbb{R}$ is dense.
4. Show that a discrete subgroup of a Hausdorff topological group is closed, and give a non-Hausdorff group with a discrete subgroup that is not closed.
5. Show that the connected component of e is contained in every open subgroup.

7.4 Separation

The separation axioms of chapter 4 form a strict hierarchy for a general space, and telling one level from the next can be delicate. On a topological group homogeneity flattens the hierarchy, dragging the weakest axioms up to the strongest. The task of this section is to make that collapse precise, tracing it back to a single object, the closure $\overline{\{e\}}$ of the identity.

The identity of a group is the natural place to test how well points can be told apart, because translation moves the test to e without loss. If two distinct points x and y cannot be separated, then e and $x^{-1}y$ cannot be separated either, so the failure of separation is entirely encoded in which points cluster around e. The set of such points is exactly $\overline{\{e\}}$, and its algebraic character is the first thing to establish.

Proposition 7.4.1: Closure of the Identity

Let G be a topological group. The closure $\overline{\{e\}}$ is a closed normal subgroup of G, and it is contained in every neighbourhood of e.

***Proof*:**
The singleton $\{e\}$ is a subgroup of G, and it is normal, since $geg^{-1} = e$ for every $g \in G$. By proposition 7.3.3, the closure of a subgroup is a subgroup and the closure of a normal subgroup is normal; applying this to $\{e\}$ shows that $\overline{\{e\}}$ is a normal subgroup of G. It is closed because a closure is always closed.

It remains to show that $\overline{\{e\}}$ is contained in every neighbourhood of e. Let U be a neighbourhood of e and let $x \in \overline{\{e\}}$. Membership in the closure means every neighbourhood of x meets $\{e\}$, that is, every neighbourhood of x contains e. The interior $\operatorname{int}(U)$ is an open neighbourhood of e, and since inversion ι is a homeomorphism, the set $V = \operatorname{int}(U)^{-1}$ is again an open neighbourhood of e. Applying the homeomorphism L_x to V yields the open neighbourhood xV of x. Because $x \in \overline{\{e\}}$, this neighbourhood contains e, so $e = xv$ for some $v \in V$. Then $x = v^{-1} \in V^{-1} = \operatorname{int}(U) \subseteq U$. Every point of $\overline{\{e\}}$ therefore lies in U, completing the proof.

The containment in the statement makes $\overline{\{e\}}$ a subset of every neighbourhood of e, hence of their intersection; the equivalence theorem below sharpens this into an identity of the two sets when G is Hausdorff. This subgroup measures the failure of the group to separate points near the identity. When it is trivial the group will turn out to be Hausdorff, and when it is all of G the topology is indiscrete. Because it is normal, one can always quotient it away, a construction the next section makes into the standard route to a Hausdorff group.

Example 7.4.2: The Trivial Topology on a Group
Let G be any group carrying the indiscrete topology, whose only open sets are $\emptyset$ and G. Both m and ι are continuous, since the preimage of an open set is open, so G is a topological group. The only closed sets are $\emptyset$ and G, so the smallest closed set containing e is G itself, giving $\overline{\{e\}} = G$. Consistently, the only neighbourhood of e is G, so the intersection of the neighbourhoods of e is G. At the opposite extreme, take G discrete: then $\{e\}$ is already closed, $\overline{\{e\}} = \{e\}$, and the intersection of the neighbourhoods of e is $\{e\}$ as well.

With $\overline{\{e\}}$ identified as the obstruction to separation, the collapse of the hierarchy can be stated as an equivalence. The list below runs from the weakest classical axiom, T_0, up through Hausdorffness, and then to two group-theoretic reformulations that hold only because the space is homogeneous. That a mere T_0 hypothesis on a topological group already forces the full Hausdorff conclusion is what distinguishes these spaces from arbitrary ones.

Theorem 7.4.3: Separation Criteria for Topological Groups

Let G be a topological group. The following are equivalent.

1. G is T_0.
2. G is T_1.
3. G is Hausdorff.
4. The singleton $\{e\}$ is closed.
5. The intersection of all neighbourhoods of e is $\{e\}$.

The proof runs as a cycle, (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) $\Rightarrow$ (1), so that each condition implies the next and the last closes back on the first. Only one link, (3) $\Rightarrow$ (4), holds for a general space: in any space a Hausdorff topology has closed singletons. The other four, (1) $\Rightarrow$ (2), (2) $\Rightarrow$ (3), (4) $\Rightarrow$ (5), and (5) $\Rightarrow$ (1), use the group structure in an essential way, through translation or the continuity of $(a, b) \mapsto a^{-1}b$; it is exactly these that fail for an arbitrary space, where T_1 does not force Hausdorff.

***Proof*:**
(1) $\Rightarrow$ (2). Assume G is T_0. It suffices to show that $\{e\}$ is closed, since translation then carries this to every point: for any g, the singleton $\{g\} = L_g(\{e\})$ is the image of a closed set under the homeomorphism L_g, hence closed, and a space in which every singleton is closed is T_1. To prove $\{e\}$ closed, fix $x \neq e$ and produce a neighbourhood of x excluding e, which shows $x \notin \overline{\{e\}}$. By T_0, some open set contains exactly one of e and x. If an open set W contains x but not e, then W is a neighbourhood of x avoiding e. If instead an open set U contains e but not x, then U^{-1} is open, since inversion ι is a homeomorphism, and xU^{-1} is an open neighbourhood of x; here $e \in xU^{-1}$ would give $x^{-1} \in U^{-1}$, hence $x \in U$, a contradiction, so xU^{-1} avoids e. In both cases x has a neighbourhood excluding e, so $x \notin \overline{\{e\}}$. As $x \neq e$ was arbitrary, $\overline{\{e\}} = \{e\}$, so $\{e\}$ is closed and G is T_1.

(2) $\Rightarrow$ (3). Assume G is T_1, so every singleton is closed; in particular $\{e\}$ is closed. Let $x \neq y$ in G. Then $x^{-1}y \neq e$, and since $\{e\}$ is closed, its complement $G \setminus \{e\}$ is open. The map $d\colon G \times G \to G$ sending $(a, b) \mapsto a^{-1}b$ is continuous, being the composite of ι on the first coordinate with m. Since $d(x, y) = x^{-1}y$ lies in the open set $G \setminus \{e\}$, continuity of d provides open neighbourhoods A of x and B of y with $d(A \times B) \subseteq G \setminus \{e\}$, that is, $a^{-1}b \neq e$ for all $a \in A$, $b \in B$. Equivalently $a \neq b$ whenever $a \in A$ and $b \in B$, so $A \cap B = \emptyset$. Thus x and y have disjoint open neighbourhoods, and G is Hausdorff (section 1.5).

(3) $\Rightarrow$ (4). Assume G is Hausdorff. In a Hausdorff space every singleton is closed: given $x \neq e$, disjoint open neighbourhoods U of x and V of e exist, so U is a neighbourhood of x missing e, whence $x \notin \overline{\{e\}}$. As x was arbitrary, $\overline{\{e\}} = \{e\}$, so $\{e\}$ is closed.

(4) $\Rightarrow$ (5). Assume $\{e\}$ is closed. Write I for the intersection of all neighbourhoods of e. Since $\{e\}$ is closed, $\overline{\{e\}} = \{e\}$, and by proposition 7.4.1 the set $\overline{\{e\}}$ is contained in every neighbourhood of e, so $\{e\} \subseteq I$. For the reverse inclusion, let $x \neq e$. Applying the homeomorphism L_x to the closed set $\{e\}$ shows $\{x\}$ is closed, so $G \setminus \{x\}$ is an open neighbourhood of e that excludes x;

hence $x \notin I$. Every point of I therefore equals e, giving $I \subseteq \{e\}$ and $I = \{e\}$.

(5) $\Rightarrow$ (1). Assume the intersection of all neighbourhoods of e is $\{e\}$. Let $x \neq y$. Then $x^{-1}y \neq e$, so by hypothesis $x^{-1}y$ fails to lie in some neighbourhood of e; choose an open neighbourhood U of e with $x^{-1}y \notin U$. The set xU is an open neighbourhood of x, and $y \notin xU$, since $y \in xU$ would give $x^{-1}y \in U$. Thus x has an open neighbourhood excluding y, which is the T_0 condition for the pair x, y. As the pair was arbitrary, G is T_0, completing the cycle and with it the proof.

The equivalence collapses the bottom of the separation tower onto a single condition, and it does so for a structural reason: homogeneity means a separation property need only be verified at e, and each of the five statements is a way of asserting that e can be separated from every other point. Condition (5) is the most useful in practice, because it is checked against a neighbourhood base at e rather than against all pairs of points. In the profinite groups of Galois theory, for instance, the open subgroups form such a base, and the group is Hausdorff exactly when their intersection is trivial, which is the separatedness built into the inverse-limit topology. From here on "Hausdorff topological group" may be used interchangeably with T_1, or with the requirement that $\{e\}$ be closed.

Example 7.4.4: The Circle and the Line with a Collapsed Coordinate

The circle group $\mathbb{S}^1 \subseteq \mathbb{C}^\times$, with its subspace topology, is Hausdorff, so by the theorem $\{1\}$ is closed, every singleton is closed, and the intersection of the neighbourhoods of 1 is $\{1\}$: the arcs about 1 shrink to the point. For a group where every condition of the theorem *fails*, take $G = \mathbb{R}^2$ under coordinatewise addition, but topologized by pulling the standard topology of $\mathbb{R}$ back along the projection $p\colon (s,t) \mapsto s$: the open sets of G are exactly the sets $U \times \mathbb{R}$ with $U \subseteq \mathbb{R}$ open. Addition and negation are continuous, since each acts on the first coordinate through the continuous operations of $\mathbb{R}$ and ignores the second, so G is a topological group. The closed sets are the sets $C \times \mathbb{R}$ with $C \subseteq \mathbb{R}$ closed, so the smallest closed set containing the identity $(0,0)$ is $\{0\} \times \mathbb{R}$, giving $\overline{\{(0,0)\}} = \{0\} \times \mathbb{R}$. This is a nontrivial closed normal subgroup, so $\{(0,0)\}$ is not closed and G is not T_1; the points $(0,0)$ and $(0,1)$ lie in exactly the same open sets and cannot be told apart even by the T_0 axiom. Every neighbourhood of $(0,0)$ contains $\{0\} \times \mathbb{R}$, and their intersection is precisely $\{0\} \times \mathbb{R}$, consistent with the containment proposition 7.4.1 guarantees. Collapsing the second coordinate by the quotient of the next section returns the Hausdorff group $\mathbb{R}$.

A Hausdorff group satisfies more than the axioms of the theorem: it separates a point from a closed set, not merely a point from a point. This is the regularity axiom T_3, and for a general space it is a genuine strengthening of Hausdorffness that must be imposed by hand. For a topological group it comes for free, and the reason is again translation together with the symmetric neighbourhoods of lemma 7.2.7: one shrinks a neighbourhood of e so that even its closure stays small.

Proposition 7.4.5: Hausdorff Groups are Regular

Every Hausdorff topological group is regular; that is, given a point x and a closed set A with $x \notin A$, there exist disjoint open sets $O_1 \ni x$ and $O_2 \supseteq A$.

***Proof*:**

Let $A \subseteq G$ be closed and $x \notin A$. Translating by $L_{x^{-1}}$, which is a homeomorphism, replaces A by the closed set $x^{-1}A$ and x by e, and $e \notin x^{-1}A$; a separation of e from $x^{-1}A$ translates back by L_x to a separation of x from A. So it suffices to separate e from a closed set $B = x^{-1}A$ not containing it.

Since B is closed and $e \notin B$, the complement $G \setminus B$ is an open neighbourhood of e. By lemma 7.2.7, there is a symmetric neighbourhood V of e, so $V = V^{-1}$, with

$$VV \subseteq G \setminus B$$

which says $VV \cap B = \emptyset$. Set $O_1 = \operatorname{int}(V)$, an open neighbourhood of e, and $O_2 = G \setminus \overline{V}$. The two steps below show that O_2 is an open set containing B and that it is disjoint from O_1, using that a symmetric V controls its own closure.

Step 1: the closure of V misses B. Suppose $b \in \overline{V} \cap B$. Since $b \in \overline{V}$, every neighbourhood of b meets V; in particular bV is a neighbourhood of b, being the image of the neighbourhood V of e under the homeomorphism L_b, so $bV \cap V \neq \emptyset$. Thus $bv_1 = v_2$ for some $v_1, v_2 \in V$, giving $b = v_2 v_1^{-1}$. Because V is symmetric, $v_1^{-1} \in V$, so $b \in VV \subseteq G \setminus B$, contradicting $b \in B$. Hence $\overline{V} \cap B = \emptyset$, so $B \subseteq G \setminus \overline{V} = O_2$.

Step 2: the two sets are disjoint. The set $O_2 = G \setminus \overline{V}$ is open, since $\overline{V}$ is closed, and it contains B by Step 1. Its complement is $\overline{V}$, so $O_2 \cap \overline{V} = \emptyset$, and $O_1 = \operatorname{int}(V) \subseteq V \subseteq \overline{V}$ gives $O_1 \cap O_2 = \emptyset$. Thus O_1 and O_2 are disjoint open sets with $e \in O_1$ and $B \subseteq O_2$. Translating back to x and A separates x from A, as we sought to show.

Because a Hausdorff group is T_1 by the equivalence above and regular by this proposition, it is T_3 in the strong sense that includes point-closedness, and so every Hausdorff topological group is automatically a T_3 space. The passage from "open neighbourhood U of e" to "open neighbourhood V with $\overline{V} \subseteq U$" is exactly the shrinking that regularity requires, and lemma 7.2.7 supplies it uniformly at e, after which homogeneity spreads it to every point. Iterating the construction yields neighbourhoods whose closures nest arbitrarily deep, which is the input to a stronger fact left to chapter 4: a Hausdorff topological group is completely regular, so that a continuous real-valued function separates any point from any disjoint closed set (a Urysohn function).

Exercise 7.4.1

1. Show that a topological group is Hausdorff if and only if $\{e\}$ is closed, and that it is T_0 if and only if it is Hausdorff.
2. Show that a Hausdorff topological group is regular, and identify where lemma 7.2.7 enters.
3. Show that a topological group in which $\overline{\{e\}} = G$ carries the trivial topology.
4. Show that a compact T_0 topological group is normal as a topological space, and name the two results that combine to prove it.

7.5 Quotient Groups

Passing to a quotient is the one construction where the group structure and the topology have to negotiate. Set-theoretically the coset space G/H is fixed the moment H is chosen; the question is which topology it should carry, and whether that topology is compatible with the group law inherited when H is normal. The quotient topology (section 2.4) is the forced choice, and this section checks that it does everything one wants: the projection is not merely continuous but open, the quotient of a topological group by a normal subgroup is again a topological group, and the separation of G/H is controlled by a single condition on H. The homogeneity that governs G itself reappears at the end, now attached to G/H even when H is not normal, as the notion of a homogeneous space.

Throughout, $H \leq G$ is a subgroup, $G/H = \{ gH : g \in G \}$ is the set of left cosets, and $\pi\colon G \to G/H$ is the map $g \mapsto gH$.

Definition 7.5.1: Quotient Topology on a Coset Space

Let $H \leq G$ be a subgroup of a topological group. The *coset space* G/H is the set of left cosets gH, and $\pi\colon G \to G/H$, $\pi(g) = gH$, is the *quotient map*. The *quotient topology* on G/H is the topology induced by π (section 2.4): a set $W \subseteq G/H$ is open exactly when $\pi^{-1}(W)$ is open in G. When $H \trianglelefteq G$ is normal, G/H carries its usual group structure, and it is then called the *quotient group*.

The definition asks nothing new: it simply installs the quotient topology from section 2.4 on the coset space, so that π becomes the universal continuous surjection out of G that is constant on cosets. A set downstairs is open precisely when its full preimage upstairs is open, which is the only topology making π continuous while identifying as many open sets as possible. What must be verified is not that this topology exists but that it interacts well with translation and, when H is normal, with the group operations, and that verification is the content of the results below. The preimage $\pi^{-1}(gH)$ of a single coset is the coset gH itself, viewed as a subset of G; this identification of points of G/H with subsets of G is used constantly.

Example 7.5.2: The Circle as a Quotient

Take $G = (\mathbb{R}, +)$ and $H = \mathbb{Z}$. The coset space $\mathbb{R}/\mathbb{Z}$ consists of the cosets $x + \mathbb{Z}$, and $\pi\colon \mathbb{R} \to \mathbb{R}/\mathbb{Z}$ sends x to its class modulo 1. A set $W \subseteq \mathbb{R}/\mathbb{Z}$ is open when $\pi^{-1}(W)$ is a union of the translates by integers of some open subset of $\mathbb{R}$, that is, a $\mathbb{Z}$-periodic open set. Wrapping the line around the circle, the map

$$x + \mathbb{Z} \mapsto (\cos 2\pi x, \sin 2\pi x)$$

is a bijection of $\mathbb{R}/\mathbb{Z}$ onto $\mathbb{S}^1 \subseteq \mathbb{R}^2$, and one checks directly that a $\mathbb{Z}$-periodic set is open in $\mathbb{R}$ if and only if its image is open in $\mathbb{S}^1$. So the quotient topology on $\mathbb{R}/\mathbb{Z}$ is exactly the circle's topology, and $\mathbb{R}/\mathbb{Z}$ recovers the compact group $\mathbb{S}^1$ as a coset space of the line.

The circle example already exhibits the phenomenon that makes quotients of topological groups better behaved than quotients of arbitrary spaces: the projection $\mathbb{R} \to \mathbb{R}/\mathbb{Z}$ carries open intervals to open arcs. This is a structural feature, and it comes from the fact that translating an open set by the elements of H produces an open set. The next result isolates this.

Proposition 7.5.3: The Quotient Map is Open

Let $H \leq G$ and let $\pi \colon G \to G/H$ be the quotient map. Then π is an open map: for every open $U \subseteq G$, the image $\pi(U)$ is open in G/H. Moreover

$$\pi^{-1}(\pi(U)) = UH = \bigcup_{h \in H} Uh \tag{7.3}$$

***Proof*:**
By the definition of the quotient topology, $\pi(U)$ is open in G/H if and only if $\pi^{-1}(\pi(U))$ is open in G, so it suffices to prove the identity (7.3) and then observe that its right-hand side is open.

First we identify $\pi^{-1}(\pi(U))$. An element $g \in G$ lies in $\pi^{-1}(\pi(U))$ exactly when $gH = uH$ for some $u \in U$, which holds exactly when $g \in uH$ for some $u \in U$, that is, when $g \in UH$. Writing the saturation coset by coset,

$$UH = \{\, uh : u \in U,\ h \in H \,\} = \bigcup_{h \in H} Uh$$

which proves (7.3).

It remains to see that UH is open. For each fixed $h \in H$ the right translation $R_h \colon x \mapsto xh$ is a homeomorphism of G (proposition 7.1.4), so $Uh = R_h(U)$ is open. A union of open sets is open, so $UH = \bigcup_{h \in H} Uh$ is open in G. Therefore $\pi^{-1}(\pi(U))$ is open, and $\pi(U)$ is open in G/H, completing the proof.

Openness of π, beyond its mere continuity, is what the constructions below rest on. The set UH appearing in (7.3) is the *saturation* of U: the smallest H-invariant set containing U, obtained by sliding U along all of H. Because each slide Uh is a translate of an open set, and translations are homeomorphisms, the saturation is again open, which is precisely where the group structure feeds the topology. The payoff is that π preserves open sets in both directions, so images and preimages of open sets are open, which is exactly the regularity one needs to transport the continuity of the multiplication upstairs to a continuity statement downstairs. Openness of π also fails for general quotient maps of topological spaces; it is special to the coset-space situation, and every structural result that follows leans on it.

With openness in hand, the compatibility of the group law with the topology on G/H follows almost formally, provided H is normal so that a group law exists at all.

Theorem 7.5.4: Quotients by Normal Subgroups

Let $H \trianglelefteq G$ be a normal subgroup of a topological group G, and give G/H the quotient topology. Then G/H is a topological group, and the quotient map $\pi \colon G \to G/H$ is a continuous, open, surjective homomorphism.

***Proof*:**
Since H is normal, the left cosets form a group under $(gH)(g'H) = (gg')H$, with identity eH and inversion $(gH)^{-1} = g^{-1}H$, and π is a group homomorphism by construction. Continuity of π is immediate from the definition of the quotient topology, and openness is proposition 7.5.3;

surjectivity is clear. It remains to prove that the multiplication

$$\bar{m}\colon G/H \times G/H \to G/H, \qquad (gH,\, g'H) \mapsto (gg')H$$

and the inversion $\bar{\iota}\colon G/H \to G/H$, $gH \mapsto g^{-1}H$, are continuous.

Multiplication. Let $m\colon G \times G \to G$ be the multiplication of G. The two projections fit into the square

$$\bar{m} \circ (\pi \times \pi) = \pi \circ m$$

as maps $G \times G \to G/H$, since both send (g, g') to $(gg')H$. Now $\pi \times \pi\colon G \times G \to G/H \times G/H$ is a surjective open map, because a product of open surjections is an open surjection: for open $U, U' \subseteq G$ the image $(\pi \times \pi)(U \times U') = \pi(U) \times \pi(U')$ is a basic open set of the product topology (section 2.2) by proposition 7.5.3, and every open subset of $G \times G$ is a union of such rectangles, so its image is open. A surjective open map is a quotient map, hence a set $W \subseteq G/H$ has $\bar{m}^{-1}(W)$ open if and only if $(\pi \times \pi)^{-1}(\bar{m}^{-1}(W))$ is open in $G \times G$. But

$$(\pi \times \pi)^{-1}(\bar{m}^{-1}(W)) = (\bar{m} \circ (\pi \times \pi))^{-1}(W) = (\pi \circ m)^{-1}(W) = m^{-1}(\pi^{-1}(W))$$

which is open because $\pi^{-1}(W)$ is open (continuity of π) and m is continuous. Therefore $\bar{m}^{-1}(W)$ is open whenever W is, so $\bar{m}$ is continuous.

Inversion. The same argument, with the inversion $\iota\colon G \to G$ in place of m and the open surjection π in place of $\pi \times \pi$, applies. From $\bar{\iota} \circ \pi = \pi \circ \iota$ and the fact that π is a quotient map, a set $W \subseteq G/H$ has $\bar{\iota}^{-1}(W)$ open if and only if $\pi^{-1}(\bar{\iota}^{-1}(W)) = \iota^{-1}(\pi^{-1}(W))$ is open, and the latter is open because $\pi^{-1}(W)$ is open and ι is continuous. Hence $\bar{\iota}$ is continuous.

Both structure maps are continuous, so G/H is a topological group, and π is a continuous open surjective homomorphism, as we sought to show.

The proof is worth reading as a template rather than a computation, because it shows how openness of π converts π into a quotient map in the topological sense and thereby lets one check continuity of a map *out of* G/H by lifting it to a map out of G. Continuity downstairs becomes continuity of the composite upstairs, which is already known. This is the general principle: a continuous map $f\colon G \to X$ that is constant on the cosets of H factors uniquely as $f = \bar{f} \circ \pi$ through a continuous map $\bar{f}\colon G/H \to X$, and $\bar{f}$ is continuous precisely because π is a quotient map. That factorization is the *universal property* of the quotient: G/H is the initial topological group receiving a continuous homomorphism from G that kills H, and every continuous homomorphism out of G trivial on H descends to it. When f is a continuous homomorphism with kernel exactly H, the descended map $\bar{f}$ is a continuous bijective homomorphism onto its image, the topological form of the first isomorphism theorem; it need not be a homeomorphism unless f is itself open onto its image.

Example 7.5.5: The Quotient $\mathrm{GL}_n(\mathbb{R})$ by Its Center

Let $G = \mathrm{GL}_n(\mathbb{R})$ and let H be its center, the subgroup of nonzero scalar matrices $\{\, \lambda I : \lambda \in \mathbb{R}^\times \,\}$, which is normal. By theorem 7.5.4 the projective linear group $G/H = \mathrm{PGL}_n(\mathbb{R})$ is a topological group, and $\pi\colon \mathrm{GL}_n(\mathbb{R}) \to \mathrm{PGL}_n(\mathbb{R})$ is a continuous open homomorphism. Concretely, a set of scalar-classes of matrices is open in $\mathrm{PGL}_n(\mathbb{R})$ exactly when the set of all matrices representing those classes is open in $\mathrm{GL}_n(\mathbb{R})$; and the determinant, which is not invariant under scaling, does *not* descend, whereas the vanishing locus of any homogeneous polynomial in the entries does, illustrating the universal property in action.

Continuity and openness settle the algebraic-topological compatibility, but they say nothing about separation: the quotient of a Hausdorff group can fail to be Hausdorff, and one wants to know exactly when it does not. The separation results of section 7.4 reduce every such question to a condition on the single point eH, and that reduction pins the answer to a property of H.

Proposition 7.5.6: Separation of the Quotient

Let $H \leq G$ be a subgroup of a topological group G, and give G/H the quotient topology.

1. G/H is Hausdorff if and only if H is closed in G.
2. G/H is discrete if and only if H is open in G.

***Proof*:**
Part (1). The point eH of G/H has $\pi^{-1}(\{eH\}) = H$, so $\{eH\}$ is closed in G/H if and only if its preimage H is closed in G, by the definition of the quotient topology. When H is normal, G/H is a topological group by theorem 7.5.4, and theorem 7.4.3 says a topological group is Hausdorff if and only if its identity is closed; combining these, G/H is Hausdorff if and only if $\{eH\}$ is closed if and only if H is closed.

For a general (not necessarily normal) subgroup H we argue directly. If G/H is Hausdorff, then points are closed, so $\{eH\}$ is closed and hence $H = \pi^{-1}(\{eH\})$ is closed. Conversely, suppose H is closed. Let $gH \neq g'H$ in G/H and set $a = g^{-1}g'$, so $a \notin H$. Since H is closed, $G \setminus H$ is an open neighbourhood of a. The map $\mu\colon G \times G \to G$, $(x, y) \mapsto xay$, is continuous and sends (e, e) to $a \in G \setminus H$, so $\mu^{-1}(G \setminus H)$ is an open neighbourhood of (e, e) and contains a product neighbourhood $V' \times V''$. Setting $V = V' \cap (V')^{-1} \cap V'' \cap (V'')^{-1}$ gives a symmetric neighbourhood of e (lemma 7.2.7) with

$$VaV \subseteq G \setminus H, \qquad \text{that is,} \quad VaV \cap H = \varnothing$$

Then $\pi(gV)$ and $\pi(g'V)$ are open neighbourhoods of gH and $g'H$ in G/H by proposition 7.5.3, and they are disjoint. Indeed, if they met, some gv and $g'v'$ with $v, v' \in V$ would satisfy $gvH = g'v'H$, hence $(gv)^{-1}(g'v') \in H$, that is,

$$v^{-1}g^{-1}g'v' = v^{-1}av' \in H$$

But $v^{-1} \in V$ by symmetry and $v' \in V$, so $v^{-1}av' \in VaV$, contradicting $VaV \cap H = \varnothing$. Hence $\pi(gV)$ and $\pi(g'V)$ separate gH from $g'H$, and G/H is Hausdorff.

Part (2). The singleton $\{eH\}$ is open in G/H if and only if $\pi^{-1}(\{eH\}) = H$ is open in G. Suppose H is open. For any gH the coset gH is open in G, being the translate $L_g(H)$ of the open set H, so its image $\{gH\} = \pi(gH)$ is open in G/H by proposition 7.5.3. Thus every singleton is open and G/H is discrete. Conversely, if G/H is discrete then $\{eH\}$ is open, so $H = \pi^{-1}(\{eH\})$ is open, completing the proof.

The first statement is the practical criterion one uses constantly: to build a Hausdorff quotient, quotient by a closed subgroup, and nothing more is required. It also explains why the non-Hausdorff quotients are exactly the ones obtained by collapsing a subgroup that is not already topologically closed, since the coset eH then cannot be separated from its limit points, which are the other cosets accumulating on it. The reduction is entirely local: separation of the whole quotient collapses to closedness of a single fiber, because homogeneity spreads the separation at eH to every pair of points. The second statement is the discrete counterpart: an open subgroup produces a discrete quotient, matching the fact from section 7.3 that open subgroups are also closed, so a discrete quotient is automatically Hausdorff. Between these extremes lie the interesting cases, closed but not open subgroups, whose quotients are Hausdorff but not discrete, as the circle $\mathbb{R}/\mathbb{Z}$ already shows.

Example 7.5.7: A Non-Hausdorff Quotient

Take $G = (\mathbb{R}, +)$ and $H = \mathbb{Q}$, the subgroup of rationals. Then $\mathbb{Q}$ is a dense subgroup of $\mathbb{R}$, so it is not closed: $\overline{\mathbb{Q}} = \mathbb{R}$. By proposition 7.5.6 the quotient $\mathbb{R}/\mathbb{Q}$ is not Hausdorff. In fact its topology is indiscrete. If $W \subseteq \mathbb{R}/\mathbb{Q}$ is nonempty and open, then $\pi^{-1}(W)$ is a nonempty open $\mathbb{Q}$-invariant subset of $\mathbb{R}$; being $\mathbb{Q}$-invariant it is invariant under translation by a dense set, and a nonempty open set invariant under a dense group of translations is all of $\mathbb{R}$, so $\pi^{-1}(W) = \mathbb{R}$ and $W = \mathbb{R}/\mathbb{Q}$. The only nonempty open set is the whole space, which is as far from Hausdorff as a topology can be, and the failure traces back exactly to the non-closedness of $\mathbb{Q}$.

Normality of H was used only to give G/H a group law; the topology and its separation properties never needed it. Dropping normality, the coset space G/H is no longer a group, but G still acts on it by left translation, and this action inherits all the good properties of the quotient. The resulting object, a space on which G acts transitively and continuously, is the natural home for the many geometric spaces that arise as G/H, and it deserves a name.

Definition 7.5.8: Homogeneous Space

Let G be a topological group and $H \le G$ a closed subgroup, not necessarily normal. The coset space G/H, with the quotient topology, is called a *homogeneous space* for G. The group G acts on G/H by left translation,

$$a \cdot (gH) = (ag)H \qquad (a \in G,\ gH \in G/H)$$

and this action $G \times G/H \to G/H$ is continuous and transitive. A *continuous action* of G on a space X is a continuous map $G \times X \to X$, $(a, x) \mapsto a \cdot x$, with $e \cdot x = x$ and $a \cdot (b \cdot x) = (ab) \cdot x$; it is *transitive* when for every pair $x, y \in X$ there is $a \in G$ with $a \cdot x = y$.

The two claims folded into the definition, that the left-translation action is continuous and transitive, are quick to verify and worth spelling out. Transitivity is immediate: given cosets gH and $g'H$, the element $a = g'g^{-1}$ satisfies $a \cdot (gH) = (g'g^{-1}g)H = g'H$, so G carries any coset to any other. Continuity is the same quotient-map argument used throughout the section. The action map factors as the outer route in

$$G \times G/H \xrightarrow{\alpha} G/H, \qquad \alpha \circ (\mathrm{id}_G \times \pi) = \pi \circ m'$$

where $m' \colon G \times G \to G$ is the multiplication and $\mathrm{id}_G \times \pi \colon G \times G \to G \times G/H$ is an open surjection (a product of an open surjection with the identity), hence a quotient map; since $\pi \circ m'$ is continuous, α is continuous. A closed H is required only so that G/H is Hausdorff (proposition 7.5.6), the setting in which homogeneous spaces are studied. The homogeneity of section 7.2 was the special case $H = \{e\}$, where $G/H = G$ acts on itself; the general definition says that every quotient looks the same from every point, because G moves any point to any other.

Example 7.5.9: Spheres as Homogeneous Spaces
The orthogonal group $\mathrm{O}(n+1)$ acts on the unit sphere $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$ by rotation, and this action is transitive: any unit vector can be rotated to any other. The stabilizer of the north pole $e_{n+1} = (0, \dots, 0, 1)$ is the subgroup of orthogonal matrices fixing that vector, a copy of $\mathrm{O}(n)$ acting on the orthogonal complement, and it is closed in $\mathrm{O}(n+1)$ as the preimage of the closed point e_{n+1} under the continuous evaluation $A \mapsto Ae_{n+1}$. Sending a coset $A\,\mathrm{O}(n)$ to the point Ae_{n+1} is then a continuous bijection $\mathrm{O}(n+1)/\mathrm{O}(n) \to \mathbb{S}^n$ intertwining the two actions, exhibiting the sphere as the homogeneous space $\mathbb{S}^n \cong \mathrm{O}(n+1)/\mathrm{O}(n)$. The single algebraic identity $\mathbb{S}^n = \mathrm{O}(n+1)/\mathrm{O}(n)$ encodes both the topology of the sphere and the symmetry group acting on it, which is precisely what the homogeneous-space viewpoint is for.

Exercise 7.5.1

1. Show that $\mathbb{R}/\mathbb{Z} \cong S^1$ as topological groups.
2. Show that the quotient map $G \to G/H$ is open but need not be closed.
3. Show that G/H is discrete if and only if H is open, and compact if G is compact.
4. Show that G/H is Hausdorff if and only if H is closed, and give a non-Hausdorff example.
5. Show that if H and G/H are connected then G is connected, and that the converse fails.
6. Show that $G/\overline{\{e\}}$ is a Hausdorff topological group, and that every continuous homomorphism from G to a Hausdorff group factors through it.

7.6 Connectedness and Total Disconnectedness

Homogeneity forces the connectedness structure of a topological group to be uniform across the space, and this section extracts the algebraic object that records it. The connected component of the identity is a subgroup, not merely a subset, so a group splits along a canonical exact sequence: a connected piece sitting inside, and a purely disconnected piece as the quotient. The disconnected quotients that arise this way are exactly the objects that carry the arithmetic of the later chapters, so pinning down their defining property closes the point-set theory of this chapter.

The starting point is the connected component of e. Connectedness of a subset was developed in section 3.1; the only new input is that in a group one component is distinguished, the one through the identity, and its translates cover the space.

Definition 7.6.1: Identity Component

Let G be a topological group with identity e. The *identity component* of G, written G°, is the connected component of e, that is, the union of all connected subsets of G that contain e.

By the general theory of components (section 3.1), G° is the largest connected subset of G containing e, and the components of G partition it into disjoint connected pieces. Because left translation L_g is a homeomorphism, it carries the component of e onto the component of $g \cdot e = g$; so every component of G is a translate gG° of the identity component, and the components are precisely the left cosets of G°. The topology of G near any point is thus a translated copy of the topology near e, and the identity component is the one representative from which every other component is recovered by sliding.

Example 7.6.2: The Identity Component of $\mathrm{GL}_n(\mathbb{R})$

The determinant $\det\colon \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times$ is continuous and surjective, and $\mathbb{R}^\times$ is disconnected into the two rays $(0,\infty)$ and $(-\infty,0)$. Hence $\mathrm{GL}_n(\mathbb{R})$ is disconnected: the two open sets

$$\mathrm{GL}_n^+(\mathbb{R}) = \{A \,|\, \det A > 0\}, \qquad \mathrm{GL}_n^-(\mathbb{R}) = \{A \,|\, \det A < 0\}$$

separate it. The identity I has $\det I = 1 > 0$, so $\mathrm{GL}_n(\mathbb{R})^\circ \subseteq \mathrm{GL}_n^+(\mathbb{R})$. In fact equality holds: $\mathrm{GL}_n^+(\mathbb{R})$ is path-connected, since any A with $\det A > 0$ can be joined to I by a continuous path of invertible matrices (row-reduce A toward I through elementary operations that never let the determinant vanish, keeping it positive throughout), and a path-connected set is connected. Therefore

$$\mathrm{GL}_n(\mathbb{R})^\circ = \mathrm{GL}_n^+(\mathbb{R})$$

the matrices of positive determinant. The other component is the coset $\mathrm{GL}_n^-(\mathbb{R}) = D \cdot \mathrm{GL}_n^+(\mathbb{R})$, where D is any fixed matrix of negative determinant, in accordance with the coset description above.

The example already exhibits the phenomenon this section formalizes: the identity component is not just a connected subset but a subgroup ($\mathrm{GL}_n^+(\mathbb{R})$ is closed under multiplication and inversion), it is normal (det is a homomorphism, so its sign class is conjugation-invariant), and the quotient $\mathrm{GL}_n(\mathbb{R})/\mathrm{GL}_n^+(\mathbb{R}) \cong \{\pm 1\}$ is a discrete two-point group, as disconnected as a group can be. The next result shows this is general.

Proposition 7.6.3: The Identity Component is a Closed Normal Subgroup

Let G be a topological group. Then G° is a closed normal subgroup of G.

***Proof*:**

Three properties must be checked: G° is a subgroup, it is normal, and it is closed.

Step 1: G° is a subgroup. It suffices to show that G° is closed under the map $(x,y) \mapsto xy^{-1}$ and contains e; the latter is immediate. Fix $g \in G^\circ$. Inversion $\iota\colon G \to G$ is a homeomorphism fixing e, so $\iota(G^\circ)$ is a connected set containing $\iota(e) = e$; by maximality of G° among connected sets through e, we have $\iota(G^\circ) \subseteq G^\circ$, that is, $g^{-1} \in G^\circ$. Next, left translation L_g is a homeomorphism, so $L_g(G^\circ) = gG^\circ$ is connected, and it contains $g \cdot e = g \in G^\circ$. Two connected sets that share the point g have connected union, so $G^\circ \cup gG^\circ$ is connected and contains e; by maximality again, $gG^\circ \subseteq G^\circ$. Thus for every $g, h \in G^\circ$ we have $gh \in gG^\circ \subseteq G^\circ$, and combined with $h^{-1} \in G^\circ$ this gives $gh^{-1} \in G^\circ$. Hence G° is a subgroup.

Step 2: G° is normal. Fix $g \in G$ and consider the conjugation map $c_g\colon x \mapsto gxg^{-1}$. It is the composite $L_g \circ R_{g^{-1}}$ of two homeomorphisms, hence a homeomorphism, and it fixes e since $geg^{-1} = e$. Therefore $c_g(G^\circ)$ is a connected set containing e, and by maximality $c_g(G^\circ) \subseteq G^\circ$, that is, $gG^\circ g^{-1} \subseteq G^\circ$ for every g. This holds for g and for g^{-1}, so in fact $gG^\circ g^{-1} = G^\circ$, and $G^\circ \trianglelefteq G$.

Step 3: G° is closed. The connected components of any topological space are closed, since the closure of a connected set is connected: if A is connected then $\overline{A}$ is connected, so $\overline{G^\circ}$ is a connected set containing e, and by maximality $\overline{G^\circ} \subseteq G^\circ$, forcing $\overline{G^\circ} = G^\circ$. Hence G° is closed, completing the proof.

The proof rests on a single mechanism used three times: a homeomorphism of G that fixes e must map the identity component into itself, because it sends connected-sets-through-e to connected-sets-through-e and G° is the maximal such set. Inversion, left translation followed by absorption, and conjugation are all instances. This is the algebraic dividend of homogeneity: the purely topological notion of "the component of e" is automatically stable under the group operations, so it hands back a subgroup for free. The normality in Step 2 is what makes the quotient G/G° a group, and Step 3 (closedness) is what makes that quotient Hausdorff.

Since G° is a closed normal subgroup, the quotient G/G° is a topological group by theorem 7.5.4, and it is Hausdorff because G° is closed. The next result identifies its connectedness structure: it has none. This is the sense in which G° captures *all* of the connectedness of G, leaving a quotient in which the only connected sets are points.

Proposition 7.6.4: The Component Quotient is Totally Disconnected

Let G be a topological group and let $\pi\colon G \to G/G^\circ$ be the quotient map. Then the only connected subsets of G/G° are the singletons and the empty set. In particular the identity component of G/G° is the trivial subgroup $\{\overline{e}\}$, where $\overline{e} = \pi(e)$.

***Proof*:**
Write $Q = G/G^\circ$. Since Q is homogeneous (translation by any element is a homeomorphism carrying the component of $\overline{e}$ onto the component of that element), it suffices to prove that the component of $\overline{e}$ in Q is $\{\overline{e}\}$: then every component of Q is a single point, and a set with two points, meeting two distinct components, cannot be connected.

Let C be the component of $\overline{e}$ in Q, and let $S = \pi^{-1}(C) \subseteq G$. Since $\pi(e) = \overline{e} \in C$, the set S contains G°. The claim is $S = G^\circ$; granting this, $C = \pi(S) = \pi(G^\circ) = \{\overline{e}\}$, as required, so the whole proof reduces to showing $S = G^\circ$.

Step 1: S is connected. The preimage of a connected set need not be connected in general, so this step uses the specific structure of π. The map π is open (proposition 7.5.3), and $S = \pi^{-1}(C)$ is saturated: it is a union of the cosets gG°, which are exactly the components of G. Suppose, for contradiction, that $S = A \sqcup B$ with A, B nonempty, disjoint, and open in the subspace S. Each of A, B is a union of full cosets gG°: a coset is connected, so it cannot meet both of the disjoint relatively open sets A and B, and it therefore lies entirely in one of them. Hence A and B are saturated, $A = \pi^{-1}(\pi(A))$ and $B = \pi^{-1}(\pi(B))$.

The images $\pi(A)$ and $\pi(B)$ separate C. First, $\pi(A)$ is open in C: writing $A = S \cap W$ with W open in G, the equality $\pi(A) = C \cap \pi(W)$ holds. The inclusion $\pi(A) \subseteq C \cap \pi(W)$ is immediate; for the reverse, if $q \in C \cap \pi(W)$ then $\pi^{-1}(q) \subseteq S$ (as $q \in C$), and some $x \in W$ has $\pi(x) = q$, so $x \in \pi^{-1}(q) \cap W \subseteq S \cap W = A$, giving $q \in \pi(A)$. Since π is open, $\pi(W)$ is open in Q, so $\pi(A) = C \cap \pi(W)$ is open in C; symmetrically $\pi(B)$ is open in C. They are nonempty (as A, B are), they cover C (as $A \cup B = S = \pi^{-1}(C)$), and they are disjoint (as A, B are saturated and disjoint, so no point of C has preimages in both). This separates C, contradicting its connectedness. Therefore S is connected.

Step 2: $S = G^\circ$. By Step 1, S is a connected subset of G containing e (since $G^\circ \subseteq S$). By the maximality of the identity component among connected sets through e, $S \subseteq G^\circ$. Combined with the inclusion $G^\circ \subseteq S$ already noted, $S = G^\circ$.

Thus $C = \pi(S) = \pi(G^\circ) = \{\bar{e}\}$, the component of $\bar{e}$ in Q is trivial, and by homogeneity every component of Q is a singleton, completing the proof.

Every topological group is therefore an extension of a totally disconnected group by a connected one: the closed normal subgroup G° is connected, and the quotient G/G° has no connectedness at all. Symbolically, the short exact sequence

$$1 \longrightarrow G^\circ \longrightarrow G \longrightarrow G/G^\circ \longrightarrow 1$$

separates the two regimes cleanly, and the study of topological groups splits along it into two nearly disjoint theories. The connected part G° is the province of Lie theory and the exponential map, where the group is locally coordinatized by a vector space; the quotient part G/G° is the province of arithmetic, built from finite and profinite data. For $\mathrm{GL}_n(\mathbb{R})$ the sequence reads $1 \to \mathrm{GL}_n^+(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R}) \to \{\pm 1\} \to 1$, the sign of the determinant. The name for the quotient regime is the following.

Definition 7.6.5: Totally Disconnected Group

A topological group G is *totally disconnected* if its only connected subsets are the empty set and the singletons, equivalently if $G^\circ = \{e\}$.

The two formulations agree by homogeneity: if $G^\circ = \{e\}$ then every component, being a translate of G°, is a single point, so no subset with two points can be connected; conversely if the only connected subsets are singletons then in particular the component of e is $\{e\}$. Proposition 7.6.4 says precisely that G/G° is totally disconnected for every G, so this class of groups is exactly what remains after the connected part is quotiented away. A totally disconnected group can still be far from discrete: total disconnectedness forbids nontrivial connected subsets, but it permits limit points, so the group may be perfect (no isolated points) while every point is its own component.

Example 7.6.6: The p-adic Integers as a Totally Disconnected Group

Fix a prime p and equip $\mathbb{Z}$ with the p-adic metric $d(x, y) = p^{-v}$, where p^v is the exact power of p dividing $x - y$ (and $d(x, x) = 0$). The completion of $(\mathbb{Z}, d)$ is a topological group $\mathbb{Z}_p$ under addition, in which a neighbourhood base at 0 is given by the subgroups $p^n\mathbb{Z}_p$ for $n \geq 0$. Each $p^n\mathbb{Z}_p$ equals the ball $\{x \in \mathbb{Z}_p : d(x, 0) \leq p^{-n}\}$, which is open because balls are open in the ultrametric d; it is also closed, being the complement of the union of the finitely many other cosets $a + p^n\mathbb{Z}_p$ with $0 \leq a < p^n$, each of which is open. A set that is both open and closed and contains a given point x but omits a point $y \neq x$ shows that no connected subset can contain both x and y: choose n large enough that $d(x, y) > p^{-n}$, so that the open-closed coset $x + p^n\mathbb{Z}_p$ contains x but not y, and it separates any prospective connected set through x and y into two nonempty relatively open pieces. Hence the only connected subsets of $\mathbb{Z}_p$ are singletons, and $\mathbb{Z}_p$ is totally disconnected. It is not discrete: 0 is a limit of the sequence $p, p^2, p^3, \ldots$, so no point is isolated. The group $\mathbb{Z}_p$ is thus a totally disconnected group that is compact (it is the limit of the finite quotients $\mathbb{Z}/p^n\mathbb{Z}$) yet has no isolated points, the antithesis of a connected Lie group.

The compact totally disconnected groups are the profinite groups of Galois theory: a profinite group is by construction a limit of finite discrete groups, and such a limit is compact, Hausdorff, and totally disconnected, with a neighbourhood base at the identity consisting of open normal subgroups of finite index. The p-adic integers $\mathbb{Z}_p$ of the example are the profinite completion of $\mathbb{Z}$ along the powers of p, and the absolute Galois group of a field, with its Krull topology, is the archetype: its open subgroups are the stabilizers of finite subextensions, and its total disconnectedness is the topological form of the fact that a field has arbitrarily small finite extensions. The connectedness dichotomy of this section is therefore the point-set gateway to that arithmetic theory, developed in full in the companion volumes on Galois cohomology and class field theory.

Exercise 7.6.1

1. Show that G°, the identity component, is a closed normal subgroup, and compute it for $GL_n(\mathbb{R})$ and for $\mathbb{Q}$.
2. Show that G/G° is totally disconnected, and that G° is open when G is locally connected.
3. Show that a totally disconnected topological group need not be discrete, using example 7.6.6.
4. Show that a connected topological group is generated by any neighbourhood of the identity, and deduce that a connected group having a neighbourhood of e whose elements commute pairwise is abelian. Show also that the converse of the first claim fails.

7.7 Locally Compact Groups

The separation and quotient theory of the preceding sections placed no require on the local size of the space. Adding one such requirement, local compactness, singles out the class of groups on which the analytic theory of the subject rests, and it is a requirement the neighbourhood structure at e controls completely. This section isolates that class, records the constructions that preserve it, and marks the door through which measure and representation theory enter.

A locally compact group is one that, as a space, has a compact neighbourhood of each point. Because translations are homeomorphisms and the space is homogeneous, it is enough to test this at the identity: a compact neighbourhood of e is carried by L_g to a compact neighbourhood of any g. So the condition reduces, as every local condition on a topological group does, to a single statement about the neighbourhoods of e.

Definition 7.7.1: Locally Compact Group

A topological group G is *locally compact* if its underlying space is Hausdorff and each point has a neighbourhood whose closure is compact. Equivalently, by homogeneity, if the identity e has a neighbourhood with compact closure.

The Hausdorff clause is a convention, adopted here once and kept throughout. Some authors define “locally compact” for arbitrary spaces and impose Hausdorffness separately; in the presence of a group law the two are almost always wanted together, since it is the Hausdorff locally compact groups that carry the invariant measure discussed at the end of this section, and a group that fails to be Hausdorff can be replaced by its largest Hausdorff quotient (section 1.5) without losing the analysis. Under this convention every locally compact group is in particular Hausdorff, so the separation results of section 1.5 and chapter 4 apply without further comment.

The two clauses of the definition agree because homogeneity moves a single compact neighbourhood everywhere. If C is a neighbourhood of e with $\overline{C}$ compact, then for any g the set $gC = L_g(C)$ is a neighbourhood of g, and $\overline{gC} = L_g(\overline{C})$ is compact as the continuous image of a compact set under the homeomorphism L_g; so a compact neighbourhood at e propagates to a compact neighbourhood at every point. The identity alone governs the whole group.

The immediate examples cover the groups already in hand and one that foreshadows the totally disconnected theory of section 3.1.

Example 7.7.2: Basic Locally Compact Groups
Each of the following is a locally compact group.

1. The additive group $\mathbb{R}^n$. The closed unit ball is a compact neighbourhood of the origin, by the Heine-Borel theorem, so $\mathbb{R}^n$ is locally compact; every closed bounded set is compact.

2. Any discrete group G. The singleton $\{e\}$ is open, hence an open neighbourhood of e, and it is its own closure and is compact (a one-point space is compact). Discreteness is the extreme case in which the compact neighbourhood is a point.

3. Any compact group G. The whole space G is a neighbourhood of e with compact closure $\overline{G} = G$, so compactness is local compactness in its strongest form. In particular the circle $\mathbb{S}$ and any finite group qualify.

4. The general linear group $\mathrm{GL}_n(\mathbb{R})$. It is an open subset of the space of $n \times n$ real matrices $\mathbb{R}^{n^2}$, being the preimage of $\mathbb{R} \setminus \{0\}$ under the continuous determinant map. A point $A \in \mathrm{GL}_n(\mathbb{R})$ therefore has an open ball B around it that lies inside $\mathrm{GL}_n(\mathbb{R})$, and a slightly smaller closed ball $B' \subseteq B$ is a compact neighbourhood of A; so $\mathrm{GL}_n(\mathbb{R})$ is locally compact.

These four span the range the definition is meant to capture: the continuous model $\mathbb{R}^n$, the discrete groups at the opposite pole, the compact groups where local compactness is global, and the matrix groups that are locally Euclidean without being either compact or discrete. A fifth kind of example sits outside the Euclidean picture entirely. The p-adic integers $\mathbb{Z}_p$, the limit of the finite quotients $\mathbb{Z}/p^n\mathbb{Z}$, form a group whose topology is at once compact and totally disconnected: it has no connected piece larger than a point, yet every point has arbitrarily small compact-open neighbourhoods, the cosets of the subgroups $p^n\mathbb{Z}_p$. Such totally disconnected locally compact groups are the topological shape of the profinite groups that organize Galois theory, and they are the reason the theory below is not merely a theory of manifolds; the connectedness apparatus of section 3.1 distinguishes them from the Lie-type examples above.

Local compactness is stable under the two constructions the previous sections built: passing to a closed subgroup, and passing to a Hausdorff quotient. Both preservations reduce to a fact about the underlying spaces, married to the subgroup and quotient theory already established.

Proposition 7.7.3: Closed Subgroups and Hausdorff Quotients

Let G be a locally compact group.

1. If $H \subseteq G$ is a closed subgroup, then H, with the subspace topology, is a locally compact group.

2. If $H \trianglelefteq G$ is a closed normal subgroup, then the quotient G/H is a locally compact group.

***Proof*:**

1. By proposition 7.3.1, H with the subspace topology is a topological group, and as a subspace of the Hausdorff space G it is Hausdorff. It remains to produce a compact neighbourhood of the identity in H. Since G is locally compact, e has a neighbourhood C in G with $\overline{C}$ compact; choose an open set U of G with $e \in U \subseteq C$. Then $U \cap H$ is an open neighbourhood of e in H, and its closure in H is $\overline{U \cap H} \cap H$, which is contained in $\overline{C} \cap H$. Now $\overline{C}$ is compact and H is closed in G, so $\overline{C} \cap H$ is a closed subset of the compact set $\overline{C}$ and is therefore compact. A closed subset of the compact set $\overline{C} \cap H$ is again compact, so the closure of $U \cap H$ in H is compact. Hence $U \cap H$ is a neighbourhood of the identity in H with compact closure, and H is locally compact.

2. Since $H \trianglelefteq G$ is a closed normal subgroup, theorem 7.5.4 makes G/H a topological group, and proposition 7.5.6 makes it Hausdorff, H being closed. Let $\pi\colon G \to G/H$ be the quotient map. By proposition 7.5.3, π is open, so it carries open neighbourhoods of e to open neighbourhoods of $\pi(e)$, the identity coset. Take a neighbourhood C of e in G with $\overline{C}$ compact, and an open set U with $e \in U \subseteq C$. Then $\pi(U)$ is an open neighbourhood of the identity of G/H, and $\pi(U) \subseteq \pi(\overline{C})$. The image $\pi(\overline{C})$ is the continuous image of the compact set $\overline{C}$ under π, hence compact, and a compact subset of the Hausdorff space G/H is closed. So $\pi(\overline{C})$ is a compact set containing the open neighbourhood $\pi(U)$ of the identity, whence the closure of $\pi(U)$ is a closed subset of the compact set $\pi(\overline{C})$ and is compact. Thus the identity of G/H has a neighbourhood with compact closure, and G/H is locally compact.

The two halves rest on the same mechanism read in opposite directions. A closed subgroup inherits compact closures because intersecting a compact set with a closed set leaves it compact; a Hausdorff quotient inherits them because the open quotient map pushes a compact closure forward to a compact set, and Hausdorffness makes that pushed-forward set closed. The closedness hypotheses are not decoration: a subgroup that is not closed need not be locally compact in the subspace topology (the rationals inside $\mathbb{R}$ are the standard failure, being neither closed nor locally compact), and G/H is not even Hausdorff, let alone locally compact, when H is not closed (proposition 7.5.6). With them, local compactness travels intact through the constructions of section 7.3 and section 7.5, so the category of locally compact groups is closed under exactly the operations one uses to build new groups from old.

This closure property is what makes local compactness the right standing hypothesis for the analytic continuation of the subject, taken up in the companion theory rather than here. The forward pointer below marks the transition, and it is the one place in this chapter where the group is studied as more than a space.

7.7.4 Toward Harmonic Analysis A locally compact Hausdorff group carries a nonzero measure that is invariant under left translation, the *Haar measure*, and this measure is unique up to a positive scalar. Its existence is the point at which the topology of G begins to do analytic work. Integration against Haar measure turns functions on G into a Banach space $L^1(G)$, whose multiplication is convolution; completing the picture with the adjoint structure gives the group C^*-algebra, and the representations of G by unitary operators on Hilbert space organize into the subject of abstract harmonic analysis. That subject (Haar measure, $L^1(G)$ and convolution, the unitary dual, and the Fourier theory of G) is developed in the companion volume on unitary representation theory, where local compactness is the running hypothesis throughout. In the present chapter such groups enter only as spaces: the structure recorded above is exactly the topological input those analytic constructions require, and no measure is built here.

Exercise 7.7.1

1. Show that a locally compact Hausdorff group is completely regular, and hence embeds in a cube.
2. Show that a closed subgroup of a locally compact group is locally compact, and that the quotient by a closed normal subgroup is too.
3. Show that $\mathbb{Q}$ is not locally compact, and that $\mathbb{Z}_p$ and $\mathbb{R}$ are.
4. Let K be a compact open subgroup of a topological group G. Show that for every $g \in G$ the subgroup $K \cap gKg^{-1}$ is open in K and of finite index in it.
5. Show that a discrete group is locally compact, and that it is compact if and only if it is finite.

8

What Next?

Point-set topology is unusual among graduate subjects in that it is almost never the destination. The definitions in this book were distilled precisely so that other subjects could be built on them, and a reader who has reached this point now holds the vocabulary that five other volumes of the series assume. This closing chapter is a map of where that vocabulary is spent.

Algebraic invariants. Chapter 6 computed one invariant, π_1, and used it to prove the Brouwer fixed point theorem in dimension two. The obvious question is what happens in higher dimensions, and the answer needs invariants that are not groups of loops: homology, cohomology and its ring structure, and the higher homotopy groups. That is the subject of *Algebraic Topology* [7], which takes the covering-space machinery developed here and builds functorial invariants on top of it. Hatcher's *Algebraic Topology* is the standard companion.

Putting calculus on a space. A topological manifold is a space that looks locally like $\mathbb{R}^n$; requiring in addition that the transition maps be smooth lets one differentiate. The payoff is transversality, Sard's theorem, degree theory, and de Rham cohomology, developed in *Differential Topology* [9]. Guillemin and Pollack's *Differential Topology* is the classic short introduction; Lee's *Introduction to Smooth Manifolds* the thorough one.

Topologies on spaces of functions. The weak topology of section 2.2 was a first sign that the useful topology on a space of functions is rarely one induced by a metric. Pushing that observation into infinite-dimensional linear algebra gives Banach and Hilbert spaces, the Hahn-Banach theorem, and the spectral theory of operators: *Functional Analysis* [10]. Rudin's *Functional Analysis* and Brezis' *Functional Analysis, Sobolev Spaces and Partial Differential Equations* are the two standard routes in.

Measuring a space. Compactness and the separation axioms are exactly the hypotheses under which a space admits enough continuous functions to support a measure. The Riesz-Markov theorem makes this precise, turning continuous functionals into measures, and *Measure Theory* [3] carries it out; its chapter on Haar measure takes section 7.7 as its point of departure. Folland's *Real Analysis* is the reference that keeps the topological hypotheses in view throughout.

When a space is also a group. Chapter 7 treated topological groups as objects. Requiring that they also be smooth manifolds gives Lie groups, where the interaction between the algebra and the geometry becomes rigid enough to classify: *Lie Groups* [13] develops the correspondence between a Lie group and its Lie algebra, the structure theory, and the representation theory of compact groups. It leans on the covering-space theory of chapter 6 throughout, since a connected Lie group is determined by its Lie algebra only up to a covering. Knapp's *Lie Groups Beyond an Introduction* is the standard reference.

More point-set topology. This book stops well short of the subject's own frontier. Uniform spaces, the Stone-Čech compactification and its universal property, descriptive set theory on Polish spaces, and continuum theory are all natural continuations. Munkres' *Topology* is the reference this book most often points at; Willard's *General Topology* and Engelking's *General Topology* go substantially further for a reader who wants the subject for its own sake.

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